DevLab_ICM20948

1.0.2

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Chapter 1

DevLab ICM20948 Arduino Library

Arduino driver for the module DevLab ICM-20948 9-axis IMU sensor.

This library is a DevLab adaptation derived from the 7Semi ICM20948 Arduino Library. The original MIT license notice from 7semi-solutions is preserved in `LICENSE`. See `NOTICE.md` for attribution and origin details.

The ICM-20948 integrates a 3-axis accelerometer, 3-axis gyroscope, and 3-axis magnetometer (AK09916), enabling motion tracking, orientation sensing, and navigation applications.

1.1 Features

- 9-axis motion sensing
  - 3-axis Accelerometer
  - 3-axis Gyroscope
  - 3-axis Magnetometer
- Configurable sensor ranges
  - Accel: ±2g / ±4g / ±8g / ±16g
  - Gyro: ±250 / ±500 / ±1000 / ±2000 dps
- Digital Low Pass Filter configurable(DLPF)
  - Noise reduction
  - Configurable bandwidth
- Adjustable sample rates
  - Accel: up to 1125 Hz
  - Gyro: up to 1100 Hz
- Internal I2C Master (for magnetometer)
  - AK09916 integration via SLV0/SLV4
- ISupports both communication interfaces
  - I^2C
  - SPI
- Temperature measurement

1.2 Connections / Wiring

The ICM-20948 supports both **I^{2}C** and **SPI** communication.

1.3 I^{2}C Connection

ICM-20948 Pin	MCU Pin	Notes
VCC	3.3V	5V only if board supports
GND	GND	Common ground
SDA	SDA	I^2C data
SCL	SCL	I^2C clock
AD0	GND/3V3	Address select

1.3.1 I^2C Notes

- Default I^2C address: 0x29
- Supported bus speeds:
  - 100 kHz
  - 400 kHz (recommended)

1.4 SPI Connection

ICM-20948 Pin	MCU Pin
VCC	3.3V
GND	GND
SCK	SCK
MOSI	MOSI
MISO	MISO
CS	GPIO

1.5 Installation

1.5.1 Arduino Library Manager

1. Open Arduino IDE
2. Go to Library Manager
3. Search for **DevLab ICM20948**
4. Click Install

1.5.2 Manual Installation

1. Download repository as ZIP
2. Arduino IDE → Sketch → Include Library → Add .ZIP Library

1.6 Library Overview

1.6.1 Initialize Sensor

- I2C

```
if (!imu.beginI2C(0x69, Wire, 400000))
{
  Serial.println("IMU init failed");
  while (1);
}
```

- SPI

```
if (!imu.beginSPI(CS_PIN, SPI, 1000000))
{
  Serial.println("SPI init failed");
  while (1);
}
```

1.6.2 Enable Sensors

```
imu.setSensors(true, true, true);
```

- accel → enable accelerometer
- gyro → enable gyroscope
- temp → enable temperatur

1.7 Reading Accelerometer

```
float ax, ay, az;
imu.readAccel(ax, ay, az);
```

- Output in g

1.8 Reading Gyroscope

```
float gx, gy, gz;
imu.readGyro(gx, gy, gz);
```

- Output in degrees/sec

1.9 Reading Temperature

```
float temp;
imu.readTemperature(temp);
```

- Output in °C

1.10 Reading Magnetometer

```
imu.initMag();

float mx, my, mz;
imu.readMag(mx, my, mz);
```

- Output in microtesla (μT)

1.11 Setting Accelerometer Scale

```
imu.setAccelScale(G_4);
```

- G_2, G_4, G_8, G_16

1.11.1 Setting Gyroscope Scale

```
imu.setGyroScale(DPS_2000);
```

1.11.2 Setting Sample Rate

```
imu.setAccelSampleRate(225);
imu.setGyroSampleRate(225);
```

1.11.3 DLPF Configuration

```
imu.setAccelDLPF(ACCEL_DLPFCFG_3, false);
imu.setDLPF(GYRO_DLPF_3, false);
```

1.11.4 Averaging Configuration

```
imu.setAccelAveraging(AVG_8);
```

1.12 Applications

- Robotics (obstacle detection)
- Drones (altitude sensing)
- Wearables (activity tracking)
- Navigation systems (IMU fusion)
- ndustrial motion sensing
- Gaming controllers

1.13 License

- MIT License
- Original work copyright: 2025 7semi-solutions
- DevLab adaptation maintains the original license notice and attribution.

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

Chapter 5

Class Documentation

5.1 BusIO_7Semi< Bus > Class Template Reference

`#include <BusIO_7Semi.h>`

Public Member Functions

- BusIO_7Semi (Bus &busRef)
- bool read (uint8_t reg, uint8_t &value)
- bool read (uint8_t reg, uint16_t &value)
- bool read (uint8_t reg, uint8_t *data, uint32_t len)
- bool write (uint8_t reg, uint8_t value)
- bool write (uint8_t reg, uint16_t value)
- bool write (uint8_t reg, const uint8_t *data, uint32_t len)
- bool readBits (uint8_t reg, uint8_t pos, uint8_t len, uint8_t &value)
- bool readBits (uint8_t reg, uint8_t pos, uint8_t len, uint16_t &value)
- bool writeBits (uint8_t reg, uint8_t pos, uint8_t len, uint8_t value)
- bool writeBits (uint8_t reg, uint8_t pos, uint8_t len, uint16_t value)
- bool readBit (uint8_t reg, uint8_t pos, uint8_t &value)
- bool readBit (uint8_t reg, uint8_t pos, uint16_t &value)
- bool writeBit (uint8_t reg, uint8_t pos, uint8_t value)
- bool writeBit (uint8_t reg, uint8_t pos, uint16_t value)

5.1.1 Detailed Description

template<typename Bus>
class BusIO_7Semi< Bus >

Definition at line 5 of file BusIO_7Semi.h.

5.1.2 Constructor & Destructor Documentation

5.1.2.1 BusIO_7Semi()

```
template<typename Bus >
BusIO_7Semi< Bus >::BusIO_7Semi (
            Bus & busRef )  [inline]
```

Constructor

- Stores interface reference

Definition at line 14 of file BusIO_7Semi.h.

5.1.3 Member Function Documentation

5.1.3.1 read() [1/3]

```
template<typename Bus >
bool BusIO_7Semi< Bus >::read (
            uint8_t reg,
            uint16_t & value )  [inline]
```

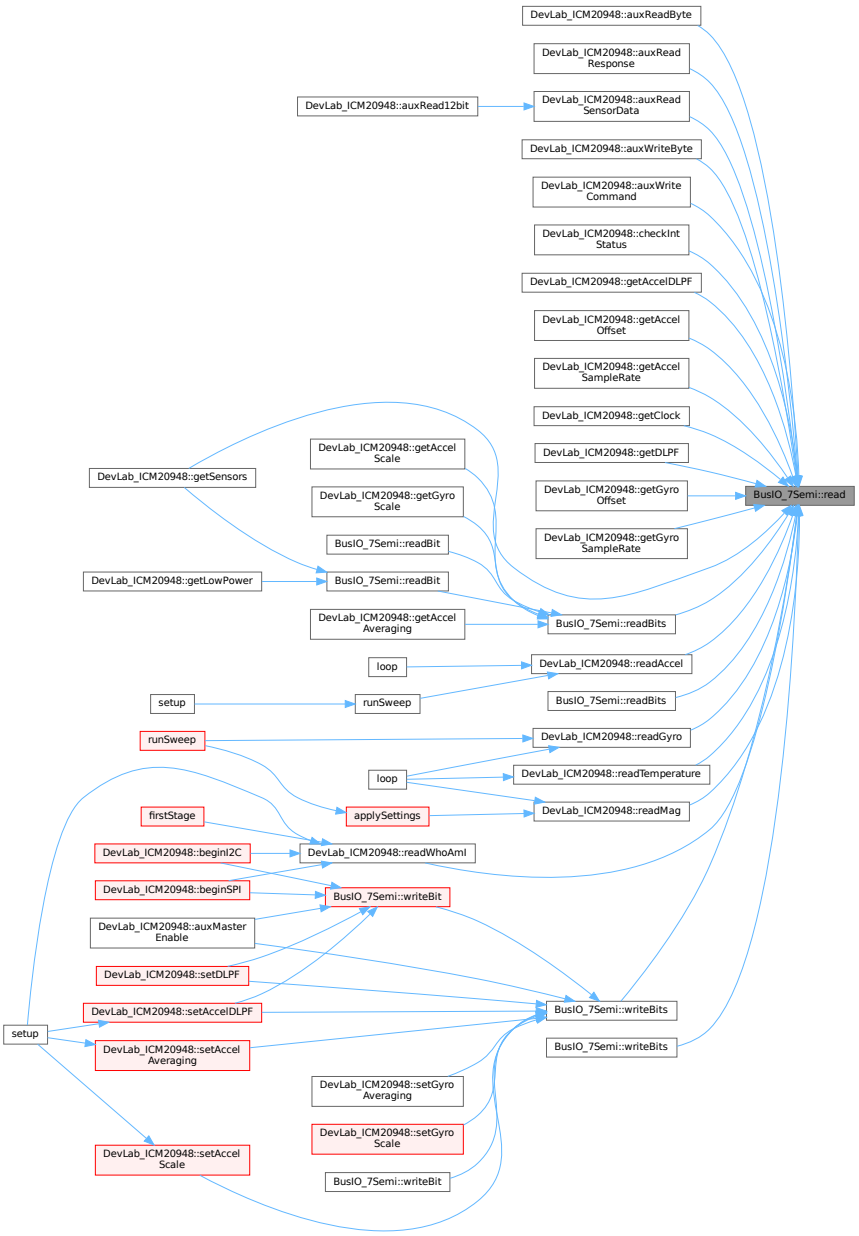
read 16-bit
Definition at line 23 of file BusIO_7Semi.h.

5.1.3.2 read() [2/3]

```
template<typename Bus >
bool BusIO_7Semi< Bus >::read (
            uint8_t reg,
            uint8_t & value )  [inline]
```

read 8-bit
Definition at line 17 of file BusIO_7Semi.h.

Here is the caller graph for this function:



5.1.3.3 read() [3/3]

```
template<typename Bus >
bool BusIO_7Semi< Bus >::read (
            uint8_t reg,
            uint8_t * data,
            uint32_t len )  [inline]
```

read burst
Definition at line 33 of file BusIO_7Semi.h.

5.1.3.4 readBit() [1/2]

```
template<typename Bus >
```

```
bool BusIO_7Semi< Bus >::readBit (
            uint8_t reg,
            uint8_t pos,
            uint16_t & value )  [inline]
```

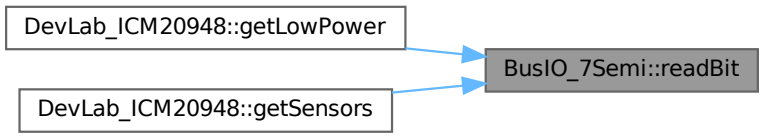
Definition at line 142 of file BusIO_7Semi.h.
Here is the call graph for this function:

5.1.3.5 readBit() [2/2]

```
template<typename Bus >
bool BusIO_7Semi< Bus >::readBit (
            uint8_t reg,
            uint8_t pos,
            uint8_t & value )  [inline]
```

Definition at line 137 of file BusIO_7Semi.h.
Here is the call graph for this function:

Here is the caller graph for this function:



5.1.3.6 readBits() [1/2]

```
template<typename Bus >
bool BusIO_7Semi< Bus >::readBits (
            uint8_t reg,
            uint8_t pos,
            uint8_t len,
            uint16_t & value )  [inline]
```

readBits (16-bit register)
Definition at line 78 of file BusIO_7Semi.h.

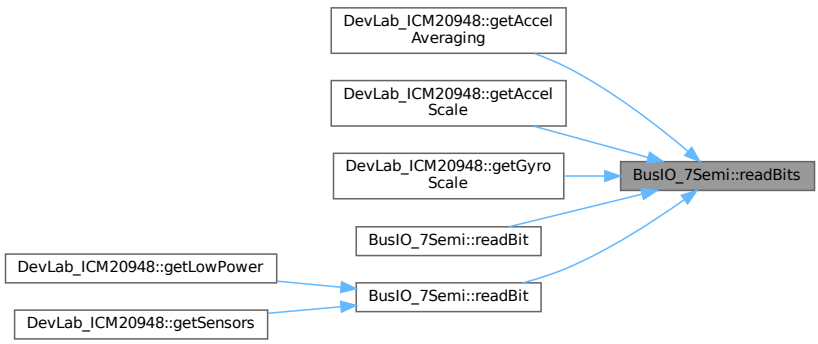
Here is the call graph for this function:

5.1.3.7 readBits() [2/2]

```
template<typename Bus >
bool BusIO_7Semi< Bus >::readBits (
            uint8_t reg,
            uint8_t pos,
            uint8_t len,
            uint8_t & value )  [inline]
```

readBits (8-bit register)
Definition at line 60 of file BusIO_7Semi.h.
Here is the call graph for this function:

Here is the caller graph for this function:



5.1.3.8 write() [1/3]

```
template<typename Bus >
bool BusIO_7Semi< Bus >::write (
```

```
            uint8_t reg,
            const uint8_t * data,
            uint32_t len )  [inline]
```

write burst
Definition at line 52 of file BusIO_7Semi.h.

5.1.3.9 write() [2/3]

```
template<typename Bus >
bool BusIO_7Semi< Bus >::write (
            uint8_t reg,
            uint16_t value )  [inline]
```

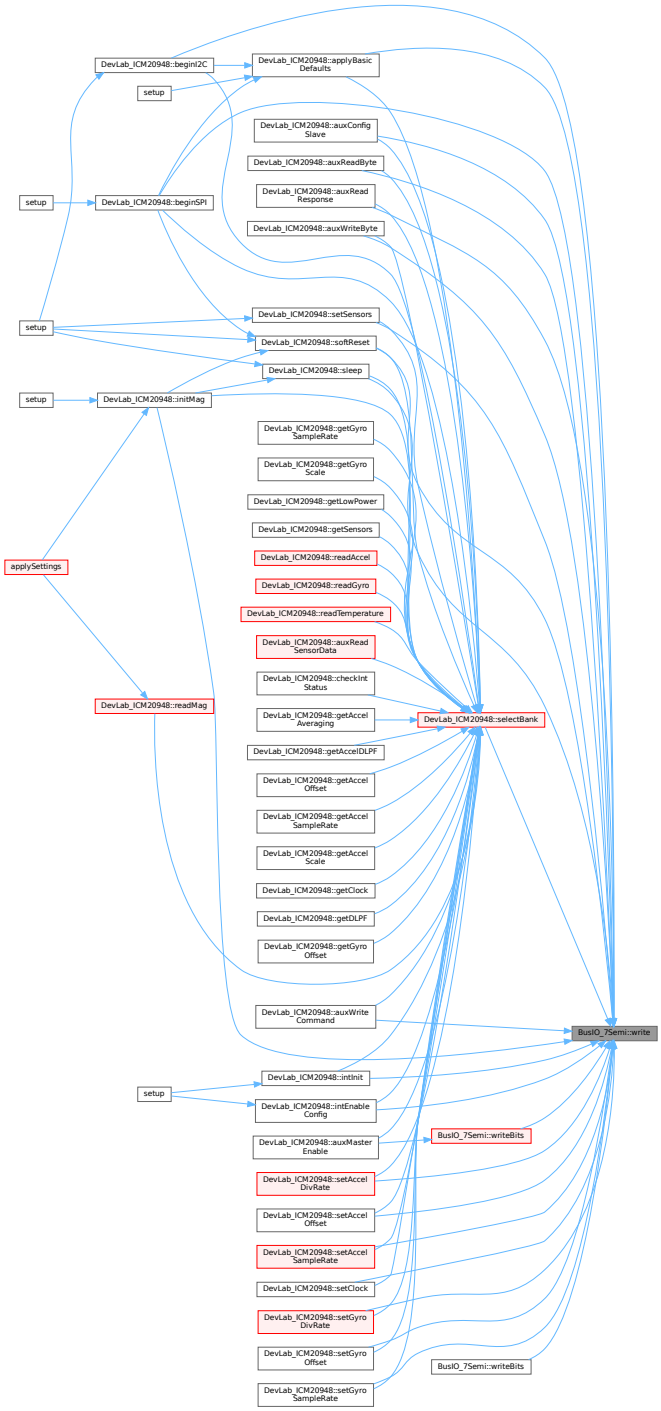
write 16-bit
Definition at line 45 of file BusIO_7Semi.h.

5.1.3.10 write() [3/3]

```
template<typename Bus >
bool BusIO_7Semi< Bus >::write (
            uint8_t reg,
            uint8_t value )  [inline]
```

write 8-bit
Definition at line 39 of file BusIO_7Semi.h.

Here is the caller graph for this function:

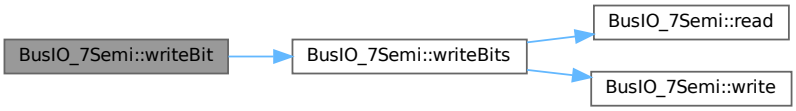


5.1.3.11 writeBit() [1/2]

```
template<typename Bus >
bool BusIO_7Semi< Bus >::writeBit (
            uint8_t reg,
            uint8_t pos,
            uint16_t value )  [inline]
```

Definition at line 151 of file BusIO_7Semi.h.

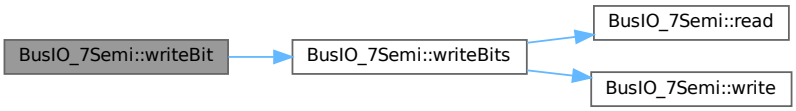
Here is the call graph for this function:



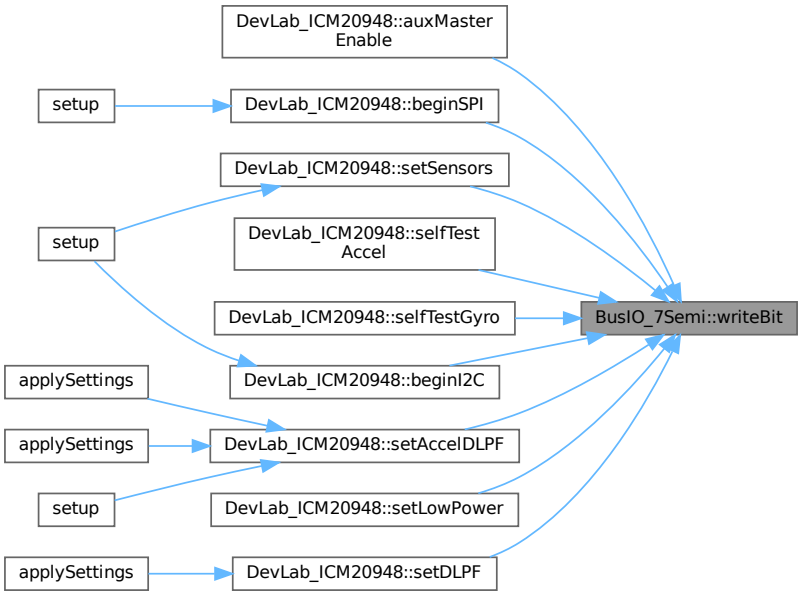
5.1.3.12 writeBit() [2/2]

```
template<typename Bus >
bool BusIO_7Semi< Bus >::writeBit (
            uint8_t reg,
            uint8_t pos,
            uint8_t value )  [inline]
```

Definition at line 147 of file BusIO_7Semi.h.
Here is the call graph for this function:



Here is the caller graph for this function:



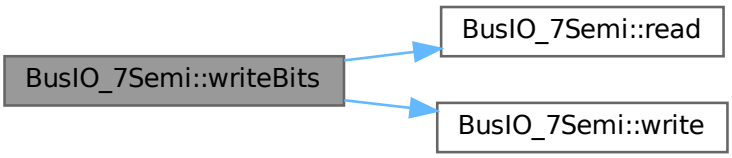
5.1.3.13 writeBits() [1/2]

```
template<typename Bus >
bool BusIO_7Semi< Bus >::writeBits (
            uint8_t reg,
            uint8_t pos,
            uint8_t len,
            uint16_t value )  [inline]
```

writeBits (16-bit register)

- Modifies specific bits in 16-bit register

Definition at line 120 of file BusIO_7Semi.h.
Here is the call graph for this function:



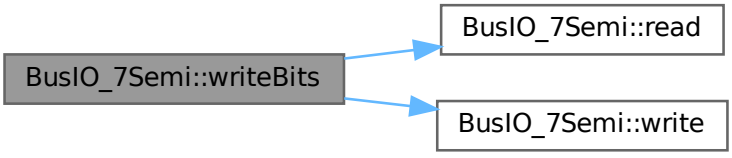
5.1.3.14 writeBits() [2/2]

```
template<typename Bus >
bool BusIO_7Semi< Bus >::writeBits (
            uint8_t reg,
            uint8_t pos,
            uint8_t len,
            uint8_t value )  [inline]
```

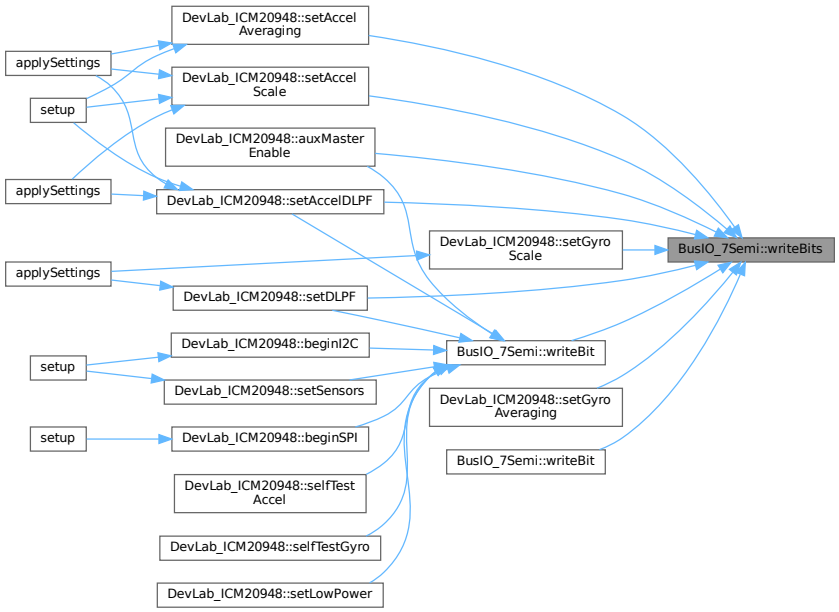
writeBits (8-bit register)

- Modifies specific bits in 8-bit register

Definition at line 98 of file BusIO_7Semi.h.
Here is the call graph for this function:



Here is the caller graph for this function:



The documentation for this class was generated from the following file:

- src/BusIO_7Semi.h

5.2 configDLPF Struct Reference

Gyroscope DLPF divider and timing configuration.

Public Attributes

- uint8_t dlpf_idx

  Index into the gyroDLPF lookup table.
- uint16_t div

  Sample-rate divider written to the IMU register.
- float nbw_hz

  Noise bandwidth in hertz for reporting.
- float odr_hz

  Output data rate in hertz for timing calculations.
- const char * nombre

  Text label printed with each captured sample.

5.2.1 Detailed Description

Gyroscope DLPF divider and timing configuration.
Accelerometer DLPF divider and timing configuration.
Definition at line 26 of file gyr_test.ino.

5.2.2 Member Data Documentation

5.2.2.1 div

`uint16_t configDLPF::div`
Sample-rate divider written to the IMU register.
Definition at line 30 of file gyr_test.ino.

5.2.2.2 dlpf_idx

`uint8_t configDLPF::dlpf_idx`
Index into the gyroDLPF lookup table.
Index into the accelDLPF lookup table.
Definition at line 28 of file gyr_test.ino.

5.2.2.3 nbw_hz

`float configDLPF::nbw_hz`
Noise bandwidth in hertz for reporting.
Definition at line 32 of file gyr_test.ino.

5.2.2.4 nombre

`const char * configDLPF::nombre`
Text label printed with each captured sample.
Definition at line 36 of file gyr_test.ino.

5.2.2.5 odr_hz

`float configDLPF::odr_hz`
Output data rate in hertz for timing calculations.
Definition at line 34 of file gyr_test.ino.
The documentation for this struct was generated from the following files:

- examples/gyr_test/gyr_test.ino
- examples/test/test.ino
- examples/test2/test2.ino

5.3 DevLab_ICM20948 Class Reference

`#include <DevLab_ICM20948.h>`

Public Member Functions

- DevLab_ICM20948 ()
- bool beginI2C (uint8_t address=0x69, TwoWire &i2cPort=Wire, uint32_t i2cSpeed=400000)
- bool beginSPI (uint8_t csPin, SPIClass &spiPort=SPI, uint32_t spiSpeed=1000000)
- bool readWhoAmI (uint8_t &whoAmI)
- bool selectBank (uint8_t bank)
- bool softReset ()
- bool sleep (bool en)
- bool applyBasicDefaults ()
- bool readAccel (float &x, float &y, float &z)
- bool readGyro (float &x, float &y, float &z)
- bool readTemperature (float &temperature)
- bool readMag (float &x, float &y, float &z)
- bool initMag ()
- bool setMagOpMode (ICM20948_Op_Mode opMode)

- bool setLowPower (bool enable)
- bool getLowPower (bool &enable)
- bool setClock (ICM20948_Clock_Source clock)
- bool getClock (uint8_t &clock)
- bool setGyroSampleRate (float sampleRate)
- bool getGyroSampleRate (float &sampleRate)
- bool setGyroDivRate (uint8_t divisor)
- bool setDLPF (ICM20948_Gyro_DLPF dlpf, bool bypass)
- bool getDLPF (uint8_t &dlpf, bool &bypass)
- bool setGyroScale (ICM20948_Gyro_FullScale fullScale)
- bool setGyroAveraging (ICM20948_Gyro_Average avg)
- bool getGyroScale (uint8_t &fullScale)
- bool selfTestGyro (bool x, bool y, bool z)
- bool setAccelDivRate (uint16_t divisor)
- bool setAccelSampleRate (uint16_t sampleRate)
- bool getAccelSampleRate (float &sampleRate)
- bool selfTestAccel (bool x, bool y, bool z)
- bool setAccelScale (ICM20948_Accel_FullScale fullScale)
- bool getAccelScale (uint8_t &fullScale)
- bool setAccelDLPF (uint8_t dlpf, bool bypass)
- bool getAccelDLPF (uint8_t &dlpf, bool &bypass)
- bool setAccelAveraging (ICM20948_Accel_Average avg)
- bool getAccelAveraging (uint8_t &avg)
- bool setSensors (bool accel_on, bool gyro_on, bool temp_on)
- bool getSensors (bool &accel_on, bool &gyro_on, bool &temp_on)
- bool setGyroOffset (uint16_t offsetX, uint16_t offsetY, uint16_t offsetZ)
- bool getGyroOffset (int16_t &offsetX, int16_t &offsetY, int16_t &offsetZ)
- bool setAccelOffset (int16_t offsetX, int16_t offsetY, int16_t offsetZ)
- bool getAccelOffset (int16_t &offsetX, int16_t &offsetY, int16_t &offsetZ)
- bool intInit (const ICM20948_IntPinConfig &cfg)
- bool intEnableConfig (const ICM20948_IntEnableConfig &cfg)
- bool checkIntStatus (ICM20948_IntStatus &status)
- bool auxMasterEnable (uint8_t clkFreq)
- bool auxWriteByte (uint8_t slaveAddr, uint8_t reg, uint8_t data)
- bool auxReadByte (uint8_t slaveAddr, uint8_t reg, uint8_t &data)
- bool auxWriteCommand (uint8_t slaveAddr, uint8_t cmd)
- bool auxConfigSlave (uint8_t slaveAddr, uint8_t reg, uint8_t numBytes)
- bool auxReadSensorData (uint8_t *buf, uint8_t len)
- bool auxReadResponse (uint8_t slaveAddr, uint8_t &data)
- bool auxRead12bit (uint16_t &raw)

5.3.1 Detailed Description

Class

- Manages ICM-20948 over I2C (SPI path disabled in this cut)
- Caches scale factors for LSB->physical conversions

Definition at line 172 of file DevLab_ICM20948.h.

5.3.2 Constructor & Destructor Documentation

5.3.2.1 DevLab_ICM20948()

```
DevLab_ICM20948::DevLab_ICM20948 ( )
```

- Construct with default I2C address; call begin() to attach bus

Definition at line 4 of file DevLab_ICM20948.cpp.

5.3.3 Member Function Documentation

5.3.3.1 applyBasicDefaults()

```
bool DevLab_ICM20948::applyBasicDefaults ( )
```

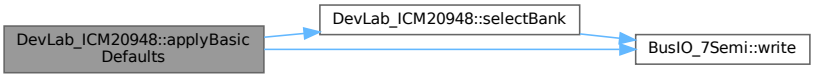
applyBasicDefaults

- Apply basic sensor configuration
- Configure power, filters, ranges, and sampling rates

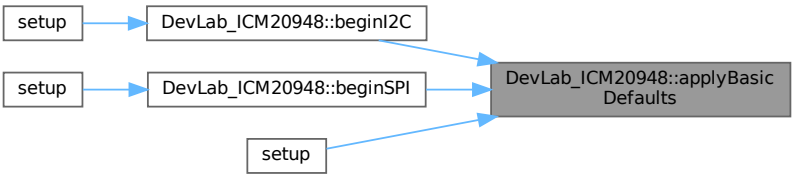
Returns:

- true → Configuration successful
- false → Any register write failed

Definition at line 181 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.2 auxConfigSlave()

```
bool DevLab_ICM20948::auxConfigSlave (
            uint8_t slaveAddr,
            uint8_t reg,
            uint8_t numBytes )
```

auxConfigSlave

- Configure SLV0 for automatic periodic reads from an auxiliary I2C slave
- The ICM20948 master will read numBytes starting at reg on every ODR cycle
- Data is deposited into EXT_SLV_SENS_DATA_00 and subsequent registers
- Call once during initialization; read results with auxReadSensorData

Parameters:

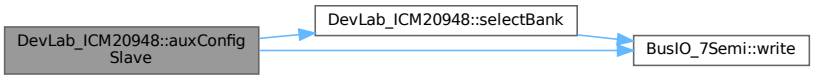
- slaveAddr: 7-bit I2C address of the auxiliary slave
- reg: starting register address to read from on the slave

- numBytes: number of bytes to read per cycle (1–15)

Returns:

- true → SLV0 configured successfully
- false → Operation failed

Definition at line 1374 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.3 auxMasterEnable()

```
bool DevLab_ICM20948::auxMasterEnable (
            uint8_t clkFreq )
```

auxMasterEnable

- Enable the ICM20948 internal I2C master for auxiliary bus
- Clears BYPASS_EN so the aux bus is controlled by the ICM
- Sets I2C_MST_EN in USER_CTRL
- Configures auxiliary master clock frequency

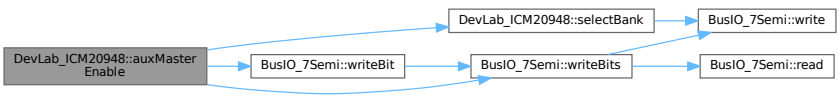
Parameters:

- clkFreq: clock divider value (e.g. 0x07 ≈ 345.6 kHz, 0x0D ≈ 400 kHz)

Returns:

- true → Master enabled successfully
- false → Operation failed

Definition at line 1260 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.4 auxRead12bit()

```
bool DevLab_ICM20948::auxRead12bit (
            uint16_t & raw )
```

auxRead12bit

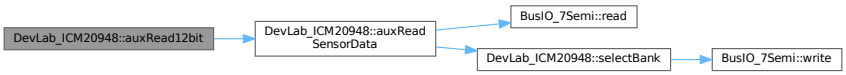
- Read a 12-bit value previously configured via auxConfigSlave (SLV0, numBytes = 2)

- Assumes little-endian packing: first byte = LSB, second byte = MSB
- Result is masked to 12 bits (0-4095)

Returns:

- true → Read successful
- false → Read failed

Definition at line 1424 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.5 auxReadByte()

```
bool DevLab_ICM20948::auxReadByte (
            uint8_t slaveAddr,
            uint8_t reg,
            uint8_t & data )
```

auxReadByte

- Read a single byte from a register of an auxiliary I2C slave via SLV4
- Generates a combined write-then-read transaction (reg byte + repeated START)
- Suitable for slaves that follow standard I2C register-read protocol

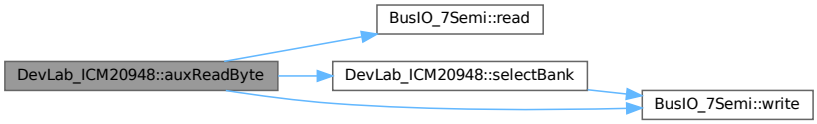
Note: slaves that need separate write/read transactions (e.g. command-response firmware) should use auxWrite↩ Command + auxReadResponse instead.
Parameters:

- slaveAddr: 7-bit I2C address of the auxiliary slave
- reg: register address to read from
- data: output byte received from slave

Returns:

- true → Read successful
- false → Timeout or NACK

Definition at line 1312 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.6 auxReadResponse()

```
bool DevLab_ICM20948::auxReadResponse (
            uint8_t slaveAddr,
            uint8_t & data )
```

auxReadResponse

- Read a single response byte from an auxiliary I2C slave via SLV4
- Generates a pure read transaction: [START, addr+R, 1 byte, STOP]
- No register byte is sent (REG_DIS); slave must already be ready to transmit
- Use after auxWriteCommand to complete a command-response exchange with slaves that require separate write and read transactions (e.g. PY32F0 firmware slaves)

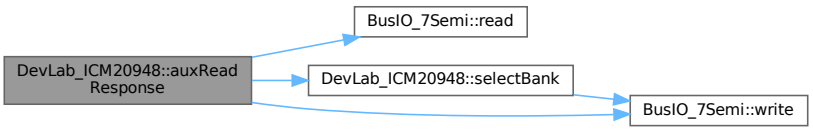
Parameters:

- slaveAddr: 7-bit I2C address of the auxiliary slave
- data: output byte received from slave

Returns:

- true → Response received successfully
- false → Timeout or NACK

Definition at line 1397 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.7 auxReadSensorData()

```
bool DevLab_ICM20948::auxReadSensorData (
            uint8_t * buf,
            uint8_t len )
```

auxReadSensorData

- Read bytes from EXT_SLV_SENS_DATA registers (BANK 0)
- Returns data collected by the ICM20948 master from SLV0–SLV3 on the last cycle
- Must call auxConfigSlave first to set up the automatic read

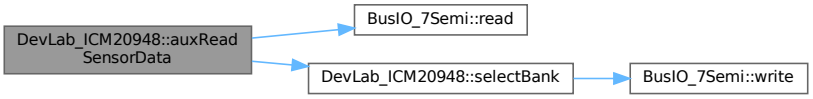
Parameters:

- buf: output buffer to store the received bytes
- len: number of bytes to read (must match numBytes configured in auxConfigSlave)

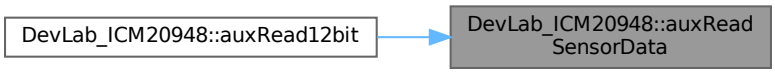
Returns:

- true → Read successful
- false → Bus error

Definition at line 1391 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.8 auxWriteByte()

```
bool DevLab_ICM20948::auxWriteByte (
            uint8_t slaveAddr,
            uint8_t reg,
            uint8_t data )
```

auxWriteByte

- Write a single byte to a register of an auxiliary I2C slave via SLV4
- Uses one-shot SLV4 transaction: polls SLV4_DONE, checks for NACK
- Suitable for configuration writes (not continuous streaming)

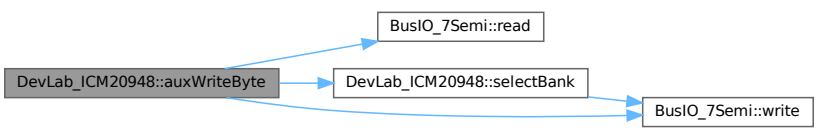
Parameters:

- slaveAddr: 7-bit I2C address of the auxiliary slave
- reg: target register address on the slave
- data: byte to write

Returns:

- true → Write acknowledged by slave
- false → Timeout or NACK

Definition at line 1281 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.9 auxWriteCommand()

```
bool DevLab_ICM20948::auxWriteCommand (
            uint8_t slaveAddr,
            uint8_t cmd )
```

auxWriteCommand

- Send a single command byte to an auxiliary I2C slave via SLV4
- Uses REG_DIS: generates [START, addr+W, cmd_byte, STOP] with no register prefix
- Designed for slaves that use a command-response protocol (no register addressing)

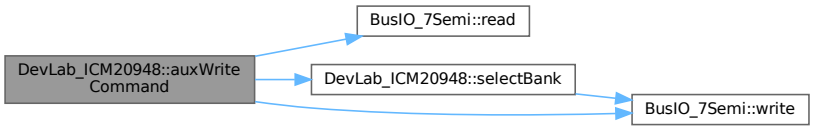
Parameters:

- slaveAddr: 7-bit I2C address of the auxiliary slave
- cmd: command byte to send

Returns:

- true → Command acknowledged by slave
- false → Timeout or NACK

Definition at line 1346 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.10 beginI2C()

```
bool DevLab_ICM20948::beginI2C (
            uint8_t address = 0x69,
            TwoWire & i2cPort = Wire,
            uint32_t i2cSpeed = 400000 )
```

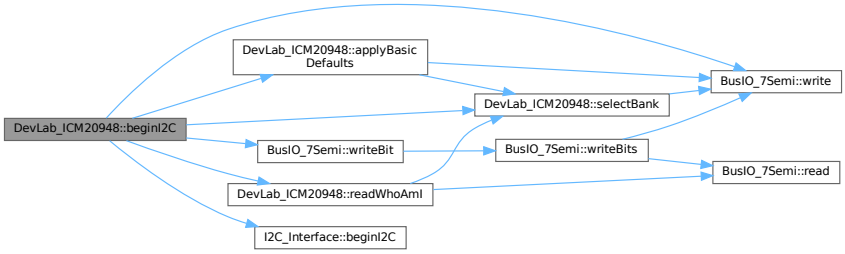
beginI2C

- Initialize ICM20948 using I2C interface
- Sets up communication layer (Interface + BusIO)
- Verifies device identity (WHO_AM_I)
- Configures basic power and clock settings

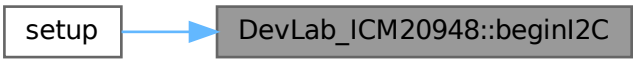
Returns:

- true → Device initialized successfully
- false → Communication or configuration failed

Definition at line 6 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



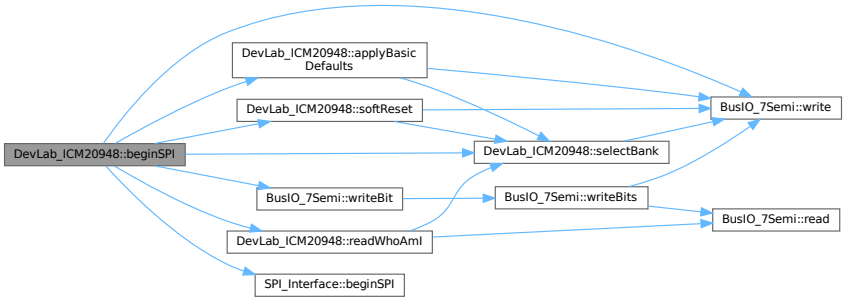
Here is the caller graph for this function:



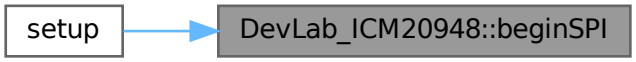
5.3.3.11 beginSPI()

```
bool DevLab_ICM20948::beginSPI (
            uint8_t csPin,
            SPIClass & spiPort = SPI,
            uint32_t spiSpeed = 1000000 )
```

Definition at line 60 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.12 checkIntStatus()

```
bool DevLab_ICM20948::checkIntStatus (
            ICM20948_IntStatus & status )
```

checkIntStatus

- Read and parse all four interrupt status registers in a single burst read
- Covers INT_STATUS (0x19), INT_STATUS_1 (0x1A), INT_STATUS_2 (0x1B), INT_STATUS_3 (0x1C) from BANK 0
- Reading clears the interrupt flags when clearMode = 0 in ICM20948_IntPinConfig
- Typically called inside the INT pin ISR or polled in the main loop

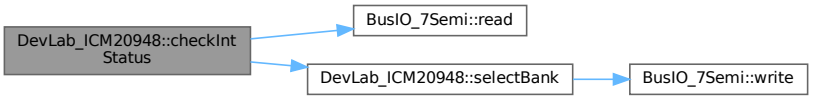
Parameters:

- status: ICM20948_IntStatus struct populated with the current interrupt flags

Returns:

- true → Status read successfully
- false → Operation failed

Definition at line 1231 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.13 getAccelAveraging()

```
bool DevLab_ICM20948::getAccelAveraging (
            uint8_t & avg )
```

getAccelAveraging

- Read accelerometer averaging / decimation setting (DEC3)

Flow:

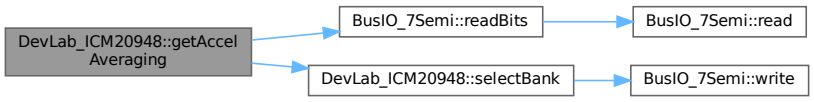
- Validate bus pointer

- Select USER BANK 2
- Read DEC3 bits [1:0]

Returns:

- true → Read successful
- false → Operation failed

Definition at line 988 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.14 getAccelDLPF()

```
bool DevLab_ICM20948::getAccelDLPF (
            uint8_t & dlpf,
            bool & bypass )
```

getAccelDLPF

- Read accelerometer DLPF configuration
- Returns filter setting and bypass state

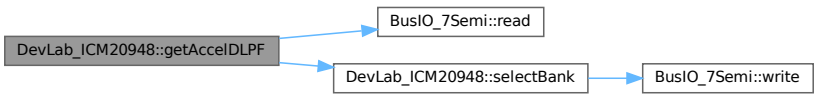
Flow:

- Validate bus pointer
- Select USER BANK 2
- Read ACCEL_CONFIG register
- Extract bypass and DLPF bits

Returns:

- true → Read successful
- false → Operation failed

Definition at line 926 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.15 getAccelOffset()

```
bool DevLab_ICM20948::getAccelOffset (
            int16_t & offsetX,
            int16_t & offsetY,
            int16_t & offsetZ )
```

getAccelOffset

- Read accelerometer offset values for X, Y, Z axes
- Reads offset registers from USER BANK 1

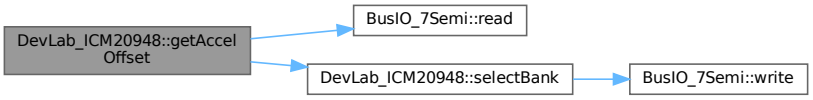
Flow:

- Validate bus pointer
- Select USER BANK 1
- Read offset registers
- Convert to signed values

Returns:

- true → Read successful
- false → Operation failed

Definition at line 1151 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.16 getAccelSampleRate()

```
bool DevLab_ICM20948::getAccelSampleRate (
            float & sampleRate )
```

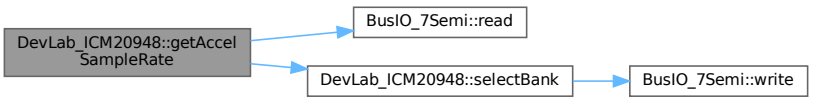
getAccelSampleRate

- Get accelerometer output data rate (ODR)
- Computes value from divider registers

Returns:

- true → Read successful
- false → Operation failed

Definition at line 796 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.17 getAccelScale()

```
bool DevLab_ICM20948::getAccelScale (
            uint8_t & fullScale )
```

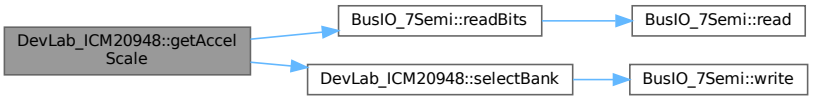
getAccelScale

- Get current accelerometer full-scale range (FS_SEL)

Returns:

- true → Read successful
- false → Operation failed

Definition at line 878 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.18 getClock()

```
bool DevLab_ICM20948::getClock (
            uint8_t & clock )
```

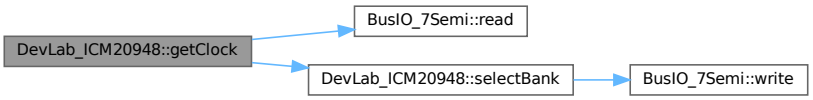
getClock

- Read current clock source (CLKSEL [2:0])

Returns:

- true → Read successful
- false → Operation failed

Definition at line 534 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.19 getDLPF()

```
bool DevLab_ICM20948::getDLPF (
            uint8_t & dlpf,
            bool & bypass )
```

getDLPF

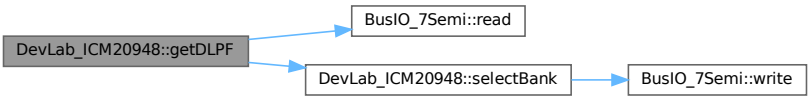
- Read gyroscope DLPF configuration

- Returns filter setting and bypass state

Returns:

- true → Read successful
- false → Operation failed

Definition at line 619 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.20 getGyroOffset()

```
bool DevLab_ICM20948::getGyroOffset (
            int16_t & offsetX,
            int16_t & offsetY,
            int16_t & offsetZ )
```

getGyroOffset

- Read gyroscope offset values for X, Y, Z axes
- Reads offset registers from USER BANK 2

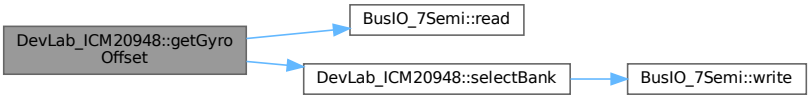
Flow:

- Validate bus pointer
- Select USER BANK 2
- Read offset registers
- Convert to signed values

Returns:

- true → Read successful
- false → Operation failed

Definition at line 1099 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.21 getGyroSampleRate()

```
bool DevLab_ICM20948::getGyroSampleRate (
            float & sampleRate )
```

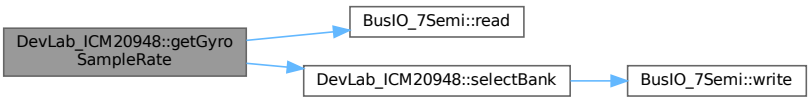
getGyroSampleRate

- Get current gyroscope sample rate
- Computes value from divider register

Returns:

- true → Read successful
- false → Operation failed

Definition at line 571 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.22 getGyroScale()

```
bool DevLab_ICM20948::getGyroScale (
            uint8_t & fullScale )
```

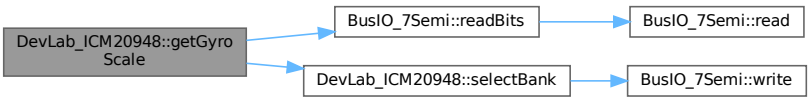
getGyroScale

- Get current gyroscope full-scale range (FS_SEL

Returns:

- true → Read successful
- false → Operation failed

Definition at line 665 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.23 getLowPower()

```
bool DevLab_ICM20948::getLowPower (
            bool & enable )
```

getLowPower

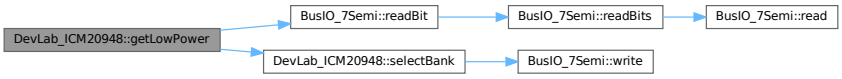
- Read low power mode status

- Returns state of LP_EN bit

Returns:

- true → Read successful
- false → Operation failed

Definition at line 502 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.24 getSensors()

```
bool DevLab_ICM20948::getSensors (
            bool & accel_on,
            bool & gyro_on,
            bool & temp_on )
```

getSensors

- Read accel, gyro, and temperature enable state

Flow:

- Validate bus pointer
- Select USER BANK 0
- Read PWR_MGMT_2 register
- Read TEMP_DIS bit
- Decode enable states

Returns:

- true → Read successful
- false → Operation failed

Definition at line 1039 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:

5.3.3.25 initMag()

```
bool DevLab_ICM20948::initMag ( )
```

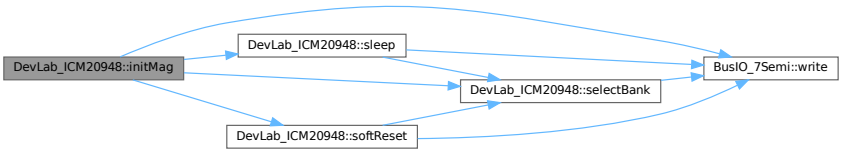
initMag

- Initialize AK09916 magnetometer using internal I2C master
- Verifies WHO_AM_I and enables continuous mode
- Configures SLV0 for automatic data read

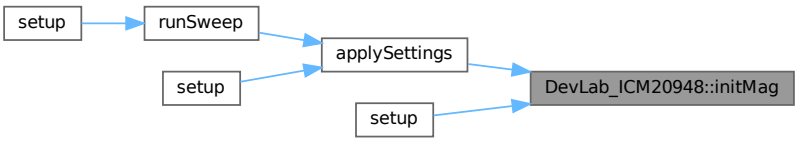
Returns:

- true → Initialization successful
- false → Operation failed

Definition at line 334 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.26 intEnableConfig()

```
bool DevLab_ICM20948::intEnableConfig (
            const ICM20948_IntEnableConfig & cfg )
```

intEnableConfig

- Enable or disable individual interrupt sources routed to the INT pin
- Writes INT_ENABLE (0x10), INT_ENABLE_1 (0x11), INT_ENABLE_2 (0x12), and INT_ENABLE_3 (0x13) in BANK 0
- Must call intInit() first to configure the pin electrical behavior

Parameters:

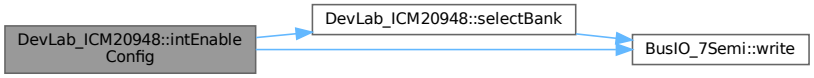
- cfg: ICM20948_IntEnableConfig struct with the desired interrupt sources enabled

Returns:

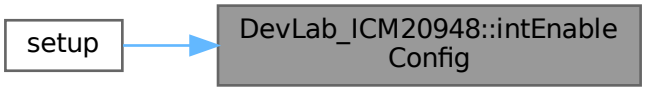
- true → All four registers written successfully

- false → Operation failed

Definition at line 1200 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.27 intInit()

```
bool DevLab_ICM20948::intInit (
            const ICM20948_IntPinConfig & cfg )
```

intInit

- Configure the physical behavior of the INT pin (INT_PIN_CFG register, BANK 0)
- Sets active level, drive mode (push-pull / open-drain), latch vs pulse mode, clear condition, FSYNC level, FSYNC interrupt enable, and I2C bypass

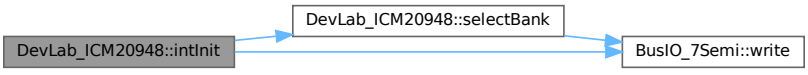
Parameters:

- cfg: ICM20948_IntPinConfig struct with the desired pin settings

Returns:

- true → Configuration written successfully
- false → Operation failed

Definition at line 1176 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:

5.3.3.28 readAccel()

```
bool DevLab_ICM20948::readAccel (
            float & x,
            float & y,
            float & z )
```

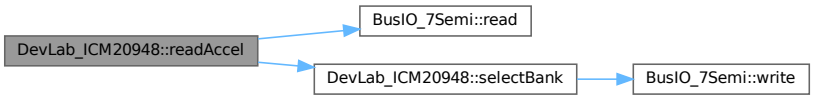
readAccel

- Read accelerometer data (X, Y, Z)
- Convert raw values to g using LSB scale

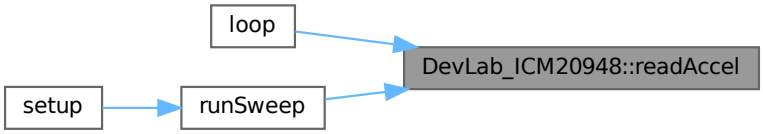
Returns:

- true → Read successful
- false → Read failed

Definition at line 238 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.29 readGyro()

```
bool DevLab_ICM20948::readGyro (
            float & x,
            float & y,
            float & z )
```

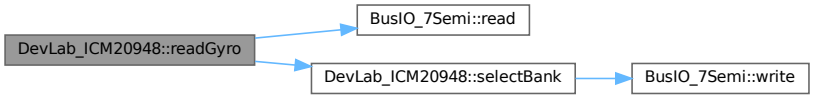
readGyro

- Read gyroscope data (X, Y, Z)
- Convert raw values to dps using LSB scale

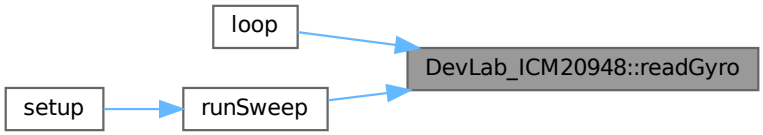
Returns:

- true → Read successful
- false → Read failed

Definition at line 258 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.30 readMag()

```
bool DevLab_ICM20948::readMag (
            float & x,
            float & y,
            float & z )
```

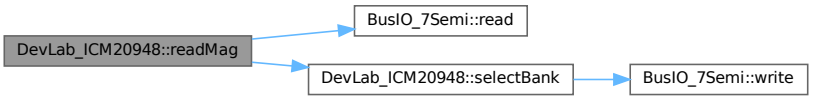
readMag

- Read magnetometer data via internal I2C master
- Data comes from EXT_SLV_SENS_DATA registers
- Convert raw values to μT

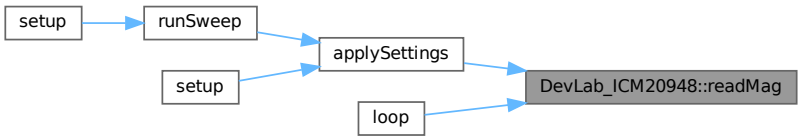
Returns:

- true → Read successful
- false → Read failed

Definition at line 310 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.31 readTemperature()

```
bool DevLab_ICM20948::readTemperature (
            float & temperature )
```

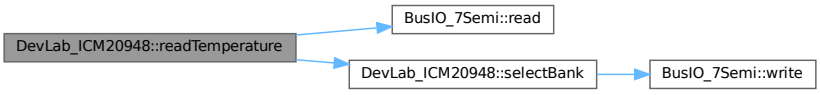
readTemperature

- Read temperature sensor
- Convert raw values to °C

Returns:

- true → Read successful
- false → Read failed

Definition at line 278 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:

5.3.3.32 readWhoAmI()

```
bool DevLab_ICM20948::readWhoAmI (
            uint8_t & whoAmI )
```

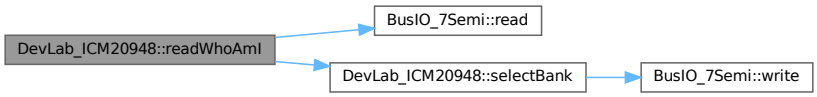
readWhoAmI

- Read WHO_AM_I register (device ID)
- Used during initialization to verify ICM20948
- Expected value: 0xEA

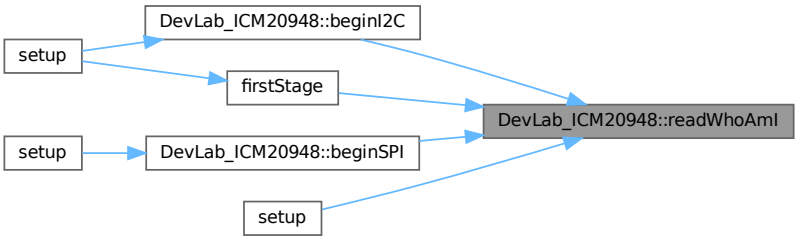
Returns:

- true → Read successful
- false → Operation failed

Definition at line 118 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.33 selectBank()

```
bool DevLab_ICM20948::selectBank (
            uint8_t bank )
```

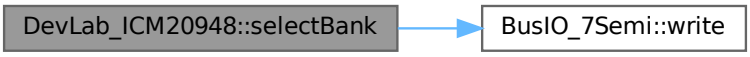
selectBank

- Select USER BANK (0–3)
- Used to access different register groups inside ICM20948

Returns:

- true → Bank switch successful
- false → Write failed

Definition at line 132 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.34 selfTestAccel()

```
bool DevLab_ICM20948::selfTestAccel (
            bool x,
            bool y,
            bool z )
```

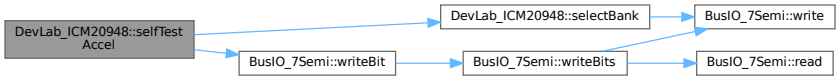
selfTestAccel

- Enable or disable accelerometer self-test per axis

Returns:

- true → Operation successful
- false → Operation failed

Definition at line 826 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.35 selfTestGyro()

```
bool DevLab_ICM20948::selfTestGyro (
            bool x,
            bool y,
            bool z )
```

selfTestGyro

- Enable or disable gyroscope self-test per axis

Returns:

- true → Operation successful
- false → Operation failed

Definition at line 683 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:

5.3.3.36 setAccelAveraging()

```
bool DevLab_ICM20948::setAccelAveraging (
            ICM20948_Accel_Average avg )
```

setAccelAveraging

- Configure accelerometer averaging / decimation (DEC3)
- Controls internal averaging of accel samples

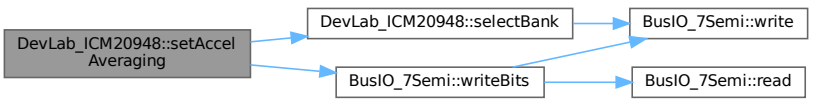
Flow:

- Validate bus pointer
- Select USER BANK 2
- Write DEC3 bits [1:0]

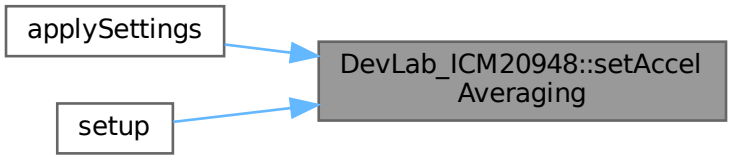
Returns:

- true → Operation successful
- false → Operation failed

Definition at line 952 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



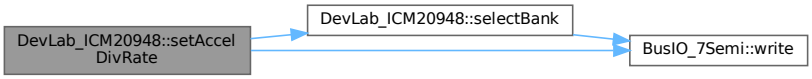
Here is the caller graph for this function:



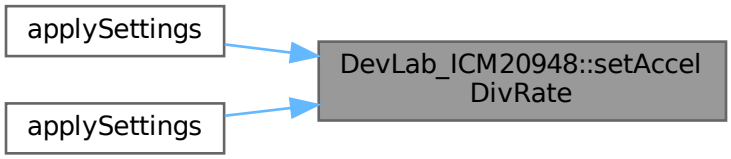
5.3.3.37 setAccelDivRate()

```
bool DevLab_ICM20948::setAccelDivRate (
            uint16_t divisor )
```

Definition at line 748 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.38 setAccelDLPF()

```
bool DevLab_ICM20948::setAccelDLPF (
            uint8_t dlpf,
            bool bypass )
```

setAccelDLPF

- Configure accelerometer digital low-pass filter (DLPF)
- Option to bypass filter (full bandwidth)

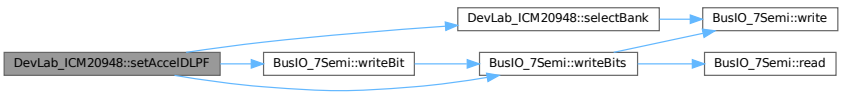
Flow:

- Validate bus pointer
- Select USER BANK 2
- Set or clear bypass bit
- Configure DLPF bits if not bypassed

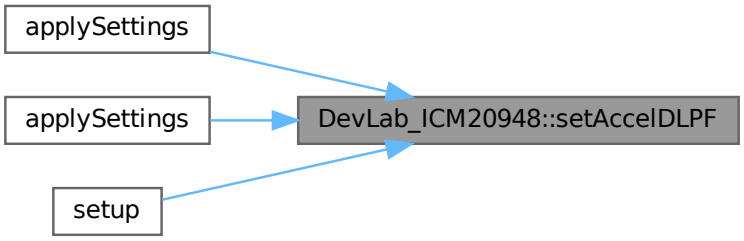
Returns:

- true → Operation successful
- false → Operation failed

Definition at line 897 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.39 setAccelOffset()

```
bool DevLab_ICM20948::setAccelOffset (
            int16_t offsetX,
            int16_t offsetY,
            int16_t offsetZ )
```

setAccelOffset

- Set accelerometer offset values for X, Y, Z axes
- Writes offset registers in USER BANK 1

Flow:

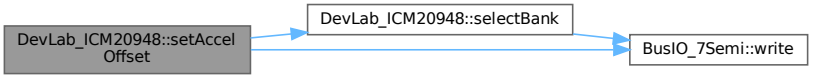
- Validate bus pointer
- Select USER BANK 1
- Pack offsets into byte array

- Write to offset registers

Returns:

- true → Operation successful
- false → Operation failed

Definition at line 1124 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.40 setAccelSampleRate()

```
bool DevLab_ICM20948::setAccelSampleRate (
            uint16_t sampleRate )
```

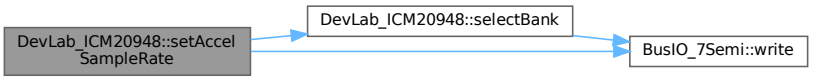
setAccelSampleRate

- Set accelerometer output data rate (ODR)
- Uses 1125 Hz base rate with 12-bit divider

Returns:

- true → Operation successful
- false → Operation failed

Definition at line 709 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:

5.3.3.41 setAccelScale()

```
bool DevLab_ICM20948::setAccelScale (
            ICM20948_Accel_FullScale fullScale )
```

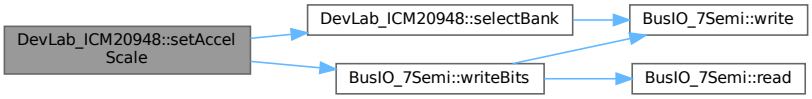
setAccelScale

- Set accelerometer full-scale range (FS_SEL)
- Controls measurement range (±2g, ±4g, ±8g, ±16g)

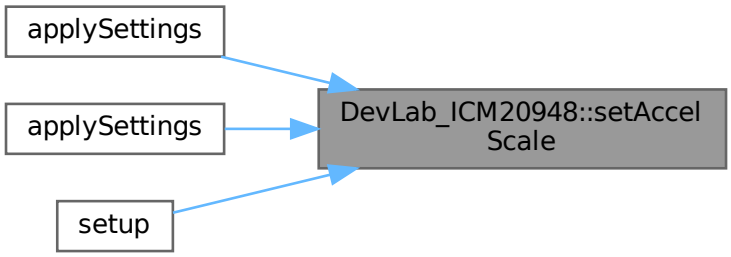
Returns:

- true → Operation successful
- false → Operation failed

Definition at line 857 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.42 setClock()

```
bool DevLab_ICM20948::setClock (
            ICM20948_Clock_Source clock )
```

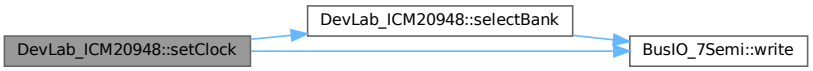
setClock

- Set clock source (CLKSEL [2:0])
- Selects internal clock for sensor operation

Returns:

- true → Operation successful
- false → Operation failed

Definition at line 520 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.43 setDLPF()

```
bool DevLab_ICM20948::setDLPF (
            ICM20948_Gyro_DLPF dlpf,
            bool bypass )
```

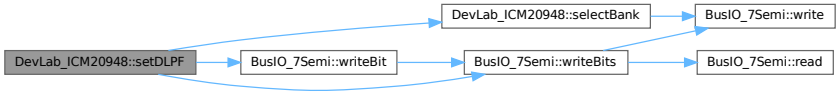
setDLPF

- Configure gyroscope digital low-pass filter (DLPF)
- Option to bypass filter (full bandwidth)

Returns:

- true → Operation successful
- false → Operation failed

Definition at line 593 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



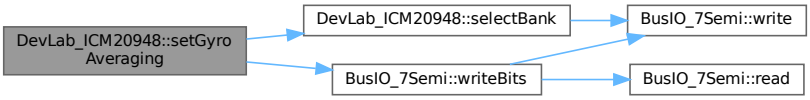
Here is the caller graph for this function:

5.3.3.44 setGyroAveraging()

```
bool DevLab_ICM20948::setGyroAveraging (
            ICM20948_Gyro_Average avg )
```

Definition at line 970 of file DevLab_ICM20948.cpp.

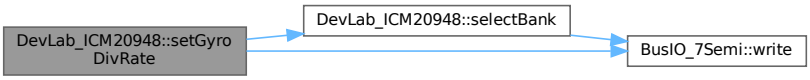
Here is the call graph for this function:



5.3.3.45 setGyroDivRate()

```
bool DevLab_ICM20948::setGyroDivRate (
            uint8_t divisor )
```

Definition at line 780 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:

5.3.3.46 setGyroOffset()

```
bool DevLab_ICM20948::setGyroOffset (
            uint16_t offsetX,
            uint16_t offsetY,
            uint16_t offsetZ )
```

setGyroOffset

- Set gyroscope offset values for X, Y, Z axes
- Writes offset registers in USER BANK 2

Flow:

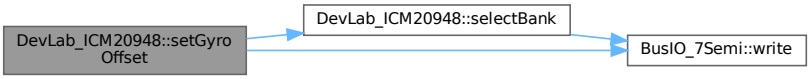
- Validate bus pointer
- Select USER BANK 2
- Pack offsets into byte array

- Write to offset registers

Returns:

- true → Operation successful
- false → Operation failed

Definition at line 1072 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.47 setGyroSampleRate()

```
bool DevLab_ICM20948::setGyroSampleRate (
            float sampleRate )
```

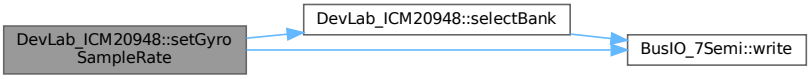
setGyroSampleRate

- Set gyroscope sample rate
- Uses internal divider (SRD) based on 1100 Hz base rate

Returns:

- true → Operation successful
- false → Invalid input or write failed

Definition at line 550 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.48 setGyroScale()

```
bool DevLab_ICM20948::setGyroScale (
            ICM20948_Gyro_FullScale fullScale )
```

setGyroScale

- Set gyroscope full-scale range (FS_SEL)
- Controls sensitivity (dps range)

Flow:

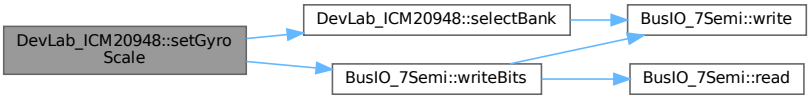
- Validate bus pointer
- Select USER BANK 2

- Write FS_SEL bits [2:1]

Returns:

- true → Operation successful
- false → Operation failed

Definition at line 644 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:

5.3.3.49 setLowPower()

```
bool DevLab_ICM20948::setLowPower (
            bool enable )
```

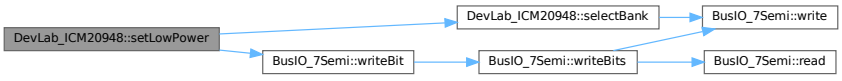
setLowPower

- Enable or disable low power mode
- Controls LP_EN bit in PWR_MGMT_1

Returns:

- true → Operation successful
- false → Operation failed

Definition at line 488 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



5.3.3.50 setMagOpMode()

```
bool DevLab_ICM20948::setMagOpMode (
            ICM20948_Op_Mode opMode )
```

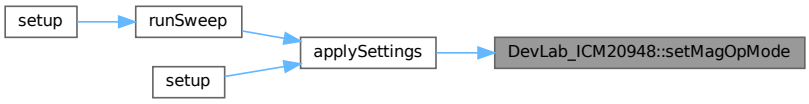
setMagOpMode

- Set magnetometer operation mode

Returns:

- true → Mode set successfully
- false → Write failed

Definition at line 410 of file DevLab_ICM20948.cpp.
Here is the caller graph for this function:



5.3.3.51 setSensors()

```
bool DevLab_ICM20948::setSensors (
            bool accel_on,
            bool gyro_on,
            bool temp_on )
```

setSensors

- Enable or disable accel, gyro, and temperature sensor
- Controls power gating via PWR_MGMT_2 and TEMP_DIS

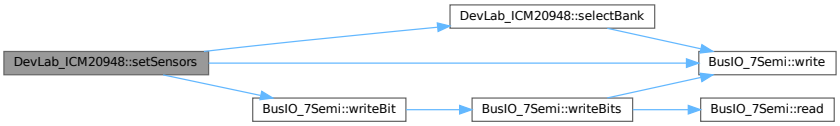
Flow:

- Validate bus pointer
- Select USER BANK 0
- Build PWR_MGMT_2 mask
- Configure temperature enable/disable

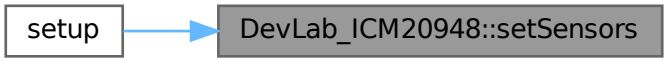
Returns:

- true → Operation successful
- false → Operation failed

Definition at line 1006 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.52 sleep()

```
bool DevLab_ICM20948::sleep (
            bool en )
```

sleep

- Enable or disable sleep mode
- Maintains clock source (CLKSEL = 1)

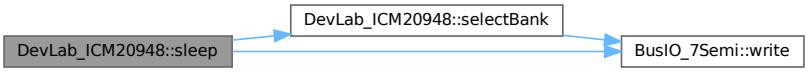
Flow:

- Select USER BANK 0
- Set or clear SLEEP bit

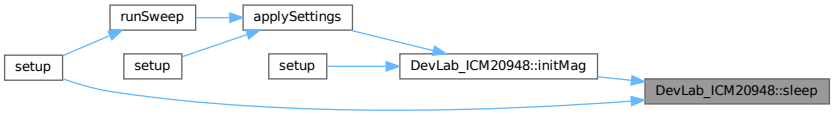
Returns:

- true → Operation successful
- false → Operation failed

Definition at line 164 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



5.3.3.53 softReset()

```
bool DevLab_ICM20948::softReset ( )
```

softReset

- Perform software reset of ICM20948
- Resets all internal registers to default state

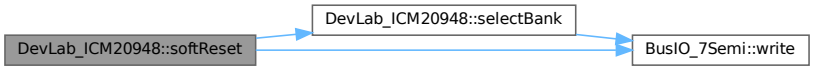
Flow:

- Select USER BANK 0
- Set DEVICE_RESET bit
- Wait for device to restart

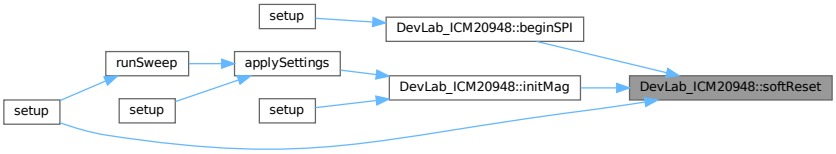
Returns:

- true → Reset successful
- false → Operation failed

Definition at line 144 of file DevLab_ICM20948.cpp.
Here is the call graph for this function:



Here is the caller graph for this function:



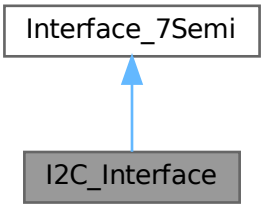
The documentation for this class was generated from the following files:

- src/DevLab_ICM20948.h
- src/DevLab_ICM20948.cpp

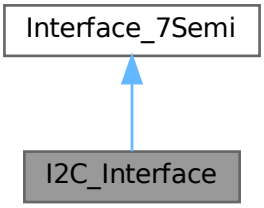
5.4 I2C_Interface Class Reference

```
#include <7Semi_I2C_Interface.h>
```

Inheritance diagram for I2C_Interface:



Collaboration diagram for I2C_Interface:



Public Member Functions

- bool beginI2C (uint8_t addr, TwoWire &wire, uint32_t speed, uint8_t sda=255, uint8_t scl=255)
- int8_t read (uint8_t reg, uint8_t *data, uint32_t len) override
- bool beginSPI (uint8_t, SPIClass &, uint32_t, uint8_t, uint8_t, uint8_t) override
- int8_t write (uint8_t reg, const uint8_t *data, uint32_t len) override

Public Member Functions inherited from Interface_7Semi

- virtual ~Interface_7Semi ()

Public Attributes

- TwoWire * i2c = nullptr
- uint8_t address = 0

Additional Inherited Members

Static Protected Member Functions inherited from Interface_7Semi

- static void delay_us (uint32_t us)

5.4.1 Detailed Description

Definition at line 5 of file 7Semi_I2C_Interface.h.

5.4.2 Member Function Documentation

5.4.2.1 beginI2C()

```
bool I2C_Interface::beginI2C (
            uint8_t address,
            TwoWire & wire,
            uint32_t speed,
            uint8_t sda = 255,
            uint8_t scl = 255 )  [inline], [virtual]
```

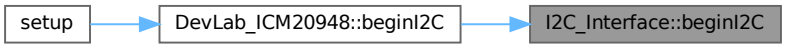
beginI2C()

- Initializes I2C peripheral
- Configures SDA / SCL pins if supported
- Sets communication clock

Returns:

- true → Initialization successful
- false → Initialization failed

Implements Interface_7Semi.
Definition at line 11 of file 7Semi_I2C_Interface.h.
Here is the caller graph for this function:



5.4.2.2 beginSPI()

```
bool I2C_Interface::beginSPI (
            uint8_t ,
            SPIClass & ,
            uint32_t ,
            uint8_t ,
            uint8_t ,
            uint8_t  )  [inline], [override], [virtual]
```

Implements Interface_7Semi.
Definition at line 69 of file 7Semi_I2C_Interface.h.

5.4.2.3 read()

```
int8_t I2C_Interface::read (
            uint8_t reg,
            uint8_t * data,
            uint32_t len )  [inline], [override], [virtual]
```

read()

- Reads data from device register

Parameters:

- reg : Register address

- data : Output buffer
- len : Number of bytes

Returns:

- 0 → Success
- <0 → Error

Implements Interface_7Semi.
Definition at line 32 of file 7Semi_I2C_Interface.h.

5.4.2.4 write()

```
int8_t I2C_Interface::write (
            uint8_t reg,
            const uint8_t * data,
            uint32_t len )  [inline], [override], [virtual]
```

write()

- Writes data to device register

Returns:

- 0 → Success
- <0 → Error

Implements Interface_7Semi.
Definition at line 75 of file 7Semi_I2C_Interface.h.

5.4.3 Member Data Documentation

5.4.3.1 address

```
uint8_t I2C_Interface::address = 0
```

Definition at line 9 of file 7Semi_I2C_Interface.h.

5.4.3.2 i2c

```
TwoWire* I2C_Interface::i2c = nullptr
```

Definition at line 8 of file 7Semi_I2C_Interface.h.
The documentation for this class was generated from the following file:

- src/7Semi_I2C_Interface.h

5.5 ICM20948_IntEnableConfig Struct Reference

```
#include <DevLab_ICM20948.h>
```

Public Attributes

- uint8_t wofEn: 1
- uint8_t womIntEn: 1
- uint8_t pllRdyEn: 1
- uint8_t dmpInt1En: 1
- uint8_t i2cMstIntEn: 1
- uint8_t rawDataRdyEn: 1
- uint8_t fifoOvfEn [5]
- uint8_t fifoWmEn [5]

5.5.1 Detailed Description

ICM20948_IntEnableConfig

- Selects which interrupt sources are routed to the INT pin
- Covers INT_ENABLE (0x10), INT_ENABLE_1 (0x11), INT_ENABLE_2 (0x12), INT_ENABLE_3 (0x13)
- FIFO overflow and watermark fields are per-channel arrays [0]–[4]
- Pass to intEnableConfig() to write all four registers in one call

Definition at line 122 of file DevLab_ICM20948.h.

5.5.2 Member Data Documentation

5.5.2.1 dmpInt1En

`uint8_t ICM20948_IntEnableConfig::dmpInt1En`
Definition at line 128 of file DevLab_ICM20948.h.

5.5.2.2 fifoOvfEn

`uint8_t ICM20948_IntEnableConfig::fifoOvfEn[5]`
Definition at line 135 of file DevLab_ICM20948.h.

5.5.2.3 fifoWmEn

`uint8_t ICM20948_IntEnableConfig::fifoWmEn[5]`
Definition at line 138 of file DevLab_ICM20948.h.

5.5.2.4 i2cMstIntEn

`uint8_t ICM20948_IntEnableConfig::i2cMstIntEn`
Definition at line 129 of file DevLab_ICM20948.h.

5.5.2.5 pllRdyEn

`uint8_t ICM20948_IntEnableConfig::pllRdyEn`
Definition at line 127 of file DevLab_ICM20948.h.

5.5.2.6 rawDataRdyEn

`uint8_t ICM20948_IntEnableConfig::rawDataRdyEn`
Definition at line 132 of file DevLab_ICM20948.h.

5.5.2.7 wofEn

`uint8_t ICM20948_IntEnableConfig::wofEn`
Definition at line 125 of file DevLab_ICM20948.h.

5.5.2.8 womIntEn

`uint8_t ICM20948_IntEnableConfig::womIntEn`
Definition at line 126 of file DevLab_ICM20948.h.
The documentation for this struct was generated from the following file:

- src/DevLab_ICM20948.h

5.6 ICM20948_IntPinConfig Struct Reference

`#include <DevLab_ICM20948.h>`

Public Attributes

- uint8_t activeLevel: 1
- uint8_t driveMode: 1
- uint8_t latchMode: 1
- uint8_t clearMode: 1
- uint8_t fsyncActLevel: 1
- uint8_t fsyncIntEn: 1
- uint8_t bypassEn: 1

5.6.1 Detailed Description

ICM20948_IntPinConfig

- Physical configuration of the INT pin (INT_PIN_CFG register, BANK 0)
- Controls electrical behavior and clear condition of the interrupt line
- Pass to intInit() to apply settings

Definition at line 103 of file DevLab_ICM20948.h.

5.6.2 Member Data Documentation

5.6.2.1 activeLevel

`uint8_t ICM20948_IntPinConfig::activeLevel`
Definition at line 105 of file DevLab_ICM20948.h.

5.6.2.2 bypassEn

`uint8_t ICM20948_IntPinConfig::bypassEn`
Definition at line 111 of file DevLab_ICM20948.h.

5.6.2.3 clearMode

`uint8_t ICM20948_IntPinConfig::clearMode`
Definition at line 108 of file DevLab_ICM20948.h.

5.6.2.4 driveMode

`uint8_t ICM20948_IntPinConfig::driveMode`
Definition at line 106 of file DevLab_ICM20948.h.

5.6.2.5 fsyncActLevel

`uint8_t ICM20948_IntPinConfig::fsyncActLevel`
Definition at line 109 of file DevLab_ICM20948.h.

5.6.2.6 fsyncIntEn

`uint8_t ICM20948_IntPinConfig::fsyncIntEn`
Definition at line 110 of file DevLab_ICM20948.h.

5.6.2.7 latchMode

`uint8_t ICM20948_IntPinConfig::latchMode`
Definition at line 107 of file DevLab_ICM20948.h.
The documentation for this struct was generated from the following file:

- src/DevLab_ICM20948.h

5.7 ICM20948_IntStatus Struct Reference

`#include <DevLab_ICM20948.h>`

Public Attributes

- uint8_t womInt: 1
- uint8_t pllRdyInt: 1
- uint8_t dmpInt1: 1
- uint8_t i2cMstInt: 1
- uint8_t rawDataRdy: 1
- uint8_t fifoOvf [5]
- uint8_t fifoWm [5]

5.7.1 Detailed Description

ICM20948_IntStatus

- Holds the parsed state of the four interrupt status registers
- Covers INT_STATUS (0x19), INT_STATUS_1 (0x1A), INT_STATUS_2 (0x1B), INT_STATUS_3 (0x1C)
- Populated by checkIntStatus() via a single 4-byte burst read
- Reading these registers clears the interrupt flags (when clearMode = 0 in IntPinConfig)

Definition at line 149 of file DevLab_ICM20948.h.

5.7.2 Member Data Documentation

5.7.2.1 dmpInt1

`uint8_t ICM20948_IntStatus::dmpInt1`
Definition at line 154 of file DevLab_ICM20948.h.

5.7.2.2 fifoOvf

`uint8_t ICM20948_IntStatus::fifoOvf[5]`
Definition at line 161 of file DevLab_ICM20948.h.

5.7.2.3 fifoWm

`uint8_t ICM20948_IntStatus::fifoWm[5]`
Definition at line 164 of file DevLab_ICM20948.h.

5.7.2.4 i2cMstInt

`uint8_t ICM20948_IntStatus::i2cMstInt`
Definition at line 155 of file DevLab_ICM20948.h.

5.7.2.5 pllRdyInt

`uint8_t ICM20948_IntStatus::pllRdyInt`
Definition at line 153 of file DevLab_ICM20948.h.

5.7.2.6 rawDataRdy

`uint8_t ICM20948_IntStatus::rawDataRdy`
Definition at line 158 of file DevLab_ICM20948.h.

5.7.2.7 womInt

`uint8_t ICM20948_IntStatus::womInt`

Definition at line 152 of file DevLab_ICM20948.h.

The documentation for this struct was generated from the following file:

- src/DevLab_ICM20948.h

5.8 icm_plat_t Struct Reference

`#include <ICM20948_platform.h>`

Public Attributes

- int(∗ read)(uint8_t reg, uint8_t ∗buf, size_t len, void ∗user)
- int(∗ write)(uint8_t reg, const uint8_t ∗buf, size_t len, void ∗user)
- uint8_t(∗ spi_reg_rw_bits)(uint8_t reg, int is_write)
- void(∗ delay_ms)(uint32_t ms)
- void ∗ user
- void(∗ spi_cs)(int level, void ∗user)
- uint8_t i2c_addr

5.8.1 Detailed Description

7Semi comment style

- Platform glue for bus IO, delays, and timing
- Implement these for your system (I2C or SPI)

Definition at line 13 of file ICM20948_platform.h.

5.8.2 Member Data Documentation

5.8.2.1 delay_ms

`void(* icm_plat_t::delay_ms) (uint32_t ms)`

- Millisecond delay

Definition at line 32 of file ICM20948_platform.h.

5.8.2.2 i2c_addr

`uint8_t icm_plat_t::i2c_addr`

- Device address (I2C) or ignored for SPI if you wrap inside user

Definition at line 45 of file ICM20948_platform.h.

5.8.2.3 read

`int(* icm_plat_t::read) (uint8_t reg, uint8_t *buf, size_t len, void *user)`

- Bus read: read `len` bytes from `reg` into `buf`
- Return 0 on success, nonzero on error

Definition at line 18 of file ICM20948_platform.h.

5.8.2.4 spi_cs

```
void(* icm_plat_t::spi_cs) (int level, void *user)
```

- If using SPI: set chip select active (1) or inactive (0)
- For I2C: leave as NULL

Definition at line 41 of file ICM20948_platform.h.

5.8.2.5 spi_reg_rw_bits

```
uint8_t(* icm_plat_t::spi_reg_rw_bits) (uint8_t reg, int is_write)
```

- Optional: SPI transfers may need register R/W bit mangling
- For I2C, leave as NULL

Definition at line 28 of file ICM20948_platform.h.

5.8.2.6 user

```
void* icm_plat_t::user
```

- User context pointer passed to read/write

Definition at line 36 of file ICM20948_platform.h.

5.8.2.7 write

```
int(* icm_plat_t::write) (uint8_t reg, const uint8_t *buf, size_t len, void *user)
```

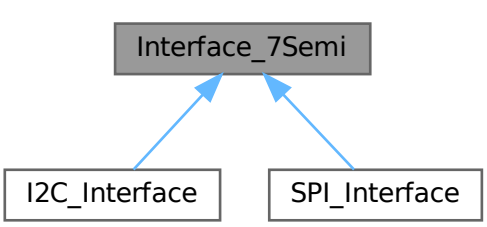
- Bus write: write `len` bytes from `buf` to `reg`
- Return 0 on success, nonzero on error

Definition at line 23 of file ICM20948_platform.h.
The documentation for this struct was generated from the following file:

- src/ICM20948_platform.h

5.9 Interface_7Semi Class Reference

`#include <7Semi_Interface.h>`
Inheritance diagram for Interface_7Semi:



Public Member Functions

- virtual ~Interface_7Semi ()
- virtual bool beginI2C (uint8_t address, TwoWire &wire, uint32_t speed, uint8_t sda=255, uint8_t scl=255)=0
- virtual bool beginSPI (uint8_t cs_pin, SPIClass &wire, uint32_t speed, uint8_t csk=255, uint8_t miso=255, uint8_t mosi=255)=0
- virtual int8_t read (uint8_t reg, uint8_t *data, uint32_t len)=0
- virtual int8_t write (uint8_t reg, const uint8_t *data, uint32_t len)=0

Static Protected Member Functions

- static void delay_us (uint32_t us)

5.9.1 Detailed Description

7Semi Universal Interface Layer

- Abstract communication layer for I2C / SPI
- Ensures sensor drivers remain hardware independent

Definition at line 15 of file 7Semi_Interface.h.

5.9.2 Constructor & Destructor Documentation

5.9.2.1 ~Interface_7Semi()

```
virtual Interface_7Semi::~Interface_7Semi ( )  [inline], [virtual]
```

Definition at line 18 of file 7Semi_Interface.h.

5.9.3 Member Function Documentation

5.9.3.1 beginI2C()

```
virtual bool Interface_7Semi::beginI2C (
            uint8_t address,
            TwoWire & wire,
            uint32_t speed,
            uint8_t sda = 255,
            uint8_t scl = 255 )  [pure virtual]
```

beginI2C()

- Initializes I2C peripheral
- Configures SDA / SCL pins if supported
- Sets communication clock

Returns:

- true → Initialization successful
- false → Initialization failed

Implemented in I2C_Interface, and SPI_Interface.

5.9.3.2 beginSPI()

```
virtual bool Interface_7Semi::beginSPI (
            uint8_t cs_pin,
            SPIClass & wire,
            uint32_t speed,
            uint8_t csk = 255,
            uint8_t miso = 255,
            uint8_t mosi = 255 )  [pure virtual]
```

Implemented in SPI_Interface, and I2C_Interface.

5.9.3.3 delay_us()

```
static void Interface_7Semi::delay_us (
            uint32_t us )  [inline], [static], [protected]
```

Delay wrapper
Definition at line 82 of file 7Semi_Interface.h.

5.9.3.4 read()

```
virtual int8_t Interface_7Semi::read (
            uint8_t reg,
            uint8_t * data,
            uint32_t len )  [pure virtual]
```

read()

- Reads data from device register

Parameters:

- reg : Register address
- data : Output buffer
- len : Number of bytes

Returns:

- 0 → Success
- <0 → Error

Implemented in I2C_Interface, and SPI_Interface.

5.9.3.5 write()

```
virtual int8_t Interface_7Semi::write (
            uint8_t reg,
            const uint8_t * data,
            uint32_t len )  [pure virtual]
```

write()

- Writes data to device register

Returns:

- 0 → Success
- <0 → Error

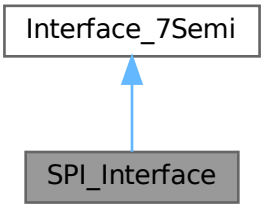
Implemented in I2C_Interface, and SPI_Interface.
The documentation for this class was generated from the following file:

- src/7Semi_Interface.h

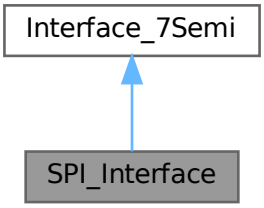
5.10 SPI_Interface Class Reference

```
#include <7Semi_SPI_Interface.h>
```

Inheritance diagram for SPI_Interface:



Collaboration diagram for SPI_Interface:



Public Member Functions

- bool beginSPI (uint8_t csPin, SPIClass &spiPort, uint32_t spiSpeed, uint8_t sck=255, uint8_t miso=255, uint8_t mosi=255) override
- bool beginI2C (uint8_t, TwoWire &, uint32_t, uint8_t, uint8_t) override
- int8_t read (uint8_t reg, uint8_t *data, uint32_t len) override
- int8_t write (uint8_t reg, const uint8_t *data, uint32_t len) override

Public Member Functions inherited from Interface_7Semi

- virtual ~Interface_7Semi ()

Public Attributes

- SPIClass * spi = nullptr
- uint8_t cs = 255
- uint32_t speed = 1000000

Additional Inherited Members

Static Protected Member Functions inherited from Interface_7Semi

- static void delay_us (uint32_t us)

5.10.1 Detailed Description

7Semi SPI Interface Implementation
Definition at line 7 of file 7Semi_SPI_Interface.h.

5.10.2 Member Function Documentation

5.10.2.1 beginI2C()

```
bool SPI_Interface::beginI2C (
            uint8_t address,
            TwoWire & wire,
            uint32_t speed,
            uint8_t sda,
            uint8_t scl )  [inline], [override], [virtual]
```

beginI2C()

- Initializes I2C peripheral
- Configures SDA / SCL pins if supported
- Sets communication clock

Returns:

- true → Initialization successful
- false → Initialization failed

Implements Interface_7Semi.
Definition at line 51 of file 7Semi_SPI_Interface.h.

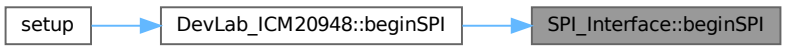
5.10.2.2 beginSPI()

```
bool SPI_Interface::beginSPI (
            uint8_t csPin,
            SPIClass & spiPort,
            uint32_t spiSpeed,
            uint8_t sck = 255,
            uint8_t miso = 255,
            uint8_t mosi = 255 )  [inline], [override], [virtual]
```

beginSPI()

- Initialize SPI interface
- Configure CS and SPI bus

Implements Interface_7Semi.
Definition at line 21 of file 7Semi_SPI_Interface.h.
Here is the caller graph for this function:



5.10.2.3 read()

```
int8_t SPI_Interface::read (
            uint8_t reg,
            uint8_t * data,
            uint32_t len )  [inline], [override], [virtual]
```

read()

- Read multiple bytes from register

Send register address (read)
Read data
Implements Interface_7Semi.
Definition at line 62 of file 7Semi_SPI_Interface.h.

5.10.2.4 write()

```
int8_t SPI_Interface::write (
            uint8_t reg,
            const uint8_t * data,
            uint32_t len )  [inline], [override], [virtual]
```

write()

- Write multiple bytes to register

Send register address (write)
Write data
Implements Interface_7Semi.
Definition at line 108 of file 7Semi_SPI_Interface.h.

5.10.3 Member Data Documentation

5.10.3.1 cs

```
uint8_t SPI_Interface::cs = 255
```

Definition at line 12 of file 7Semi_SPI_Interface.h.

5.10.3.2 speed

```
uint32_t SPI_Interface::speed = 1000000
```

Definition at line 13 of file 7Semi_SPI_Interface.h.

5.10.3.3 spi

```
SPIClass* SPI_Interface::spi = nullptr
```

Definition at line 11 of file 7Semi_SPI_Interface.h.
The documentation for this class was generated from the following file:

- src/7Semi_SPI_Interface.h

Chapter 6

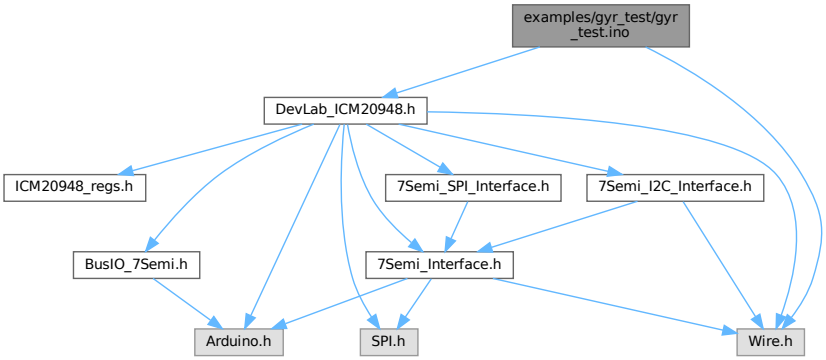
File Documentation

6.1 examples/gyr_test/gyr_test.ino File Reference

Gyroscope DLPF and full-scale sweep for the 7Semi ICM-20948.

```
#include <Wire.h>
#include <DevLab_ICM20948.h>
```

Include dependency graph for gyr_test.ino:



Classes

- struct configDLPF

  Gyroscope DLPF divider and timing configuration.

Macros

- #define SDA_PIN 6

  I2C SDA pin used by the example board.
- #define SCL_PIN 7

  I2C SCL pin used by the example board.
- #define I2C_FREQ 400000UL

  I2C bus speed in hertz.
- #define ICM_ADDR 0x69

  ICM-20948 I2C address selected by the AD0 pin.

Functions

- void printLinea ()

Print a separator line to Serial.

- void printDobleLinea ()

  Print a double separator line to Serial.

- void firstStage ()

  Verify the IMU identity register.

- bool applySettings (uint8_t fs_idx, const configDLPF &cfg)

  Apply one gyroscope full-scale and DLPF configuration.

- void runSweep ()

  Run the gyroscope validation sweep and print each captured sample.

- void setup ()

  Initialize I2C, reset/wake the IMU, enable gyro, and start the sweep.

- void loop ()

  Empty loop; this sketch runs the gyroscope sweep once in setup().

Variables

- const configDLPF configsDLPF [ ]
- const uint8_t N_DLPF = sizeof(configsDLPF) / sizeof(configsDLPF[0])

  Number of gyroscope DLPF configurations.

- const ICM20948_Gyro_FullScale gyroFSCfg [ ]

  Gyroscope full-scale ranges exercised by the sweep.

- const char * etiquetasGyroFSCfg [ ] = {"250", "500", "1000", "2000"}

  Text labels matching gyroFSCfg.

- const uint8_t N_FS = 4

  Number of gyroscope full-scale ranges.

- const ICM20948_Gyro_DLPF gyroDLPF [ ]

  Gyroscope DLPF enum values matching configsDLPF.

- const ICM20948_Gyro_Average gyroAVGCfg [ ]

  Gyroscope averaging settings available for experiments.

- const char * etiquetasGyroAVGCfg [ ] = {"1","2","4","8","16","32","64","128"}

  Text labels matching gyroAVGCfg.

- const uint8_t N_AVG = 8

  Number of gyroscope averaging settings.

- DevLab_ICM20948 imu

  Global ICM-20948 driver instance.

6.1.1 Detailed Description

Gyroscope DLPF and full-scale sweep for the 7Semi ICM-20948.
Applies each gyroscope filter/full-scale configuration over I2C and prints repeated samples for validation and noise comparison.
Definition in file gyr_test.ino.

6.1.2 Macro Definition Documentation

6.1.2.1 I2C_FREQ

```
#define I2C_FREQ 400000UL
```

I2C bus speed in hertz.
Definition at line 18 of file gyr_test.ino.

6.1.2.2 ICM_ADDR

```
#define ICM_ADDR 0x69
```

ICM-20948 I2C address selected by the AD0 pin.
Definition at line 20 of file gyr_test.ino.

6.1.2.3 SCL_PIN

```
#define SCL_PIN 7
```

I2C SCL pin used by the example board.
Definition at line 16 of file gyr_test.ino.

6.1.2.4 SDA_PIN

```
#define SDA_PIN 6
```

I2C SDA pin used by the example board.
Definition at line 14 of file gyr_test.ino.

6.1.3 Function Documentation

6.1.3.1 applySettings()

```
bool applySettings (
            uint8_t fs_idx,
            const configDLPF & cfg )
```

Apply one gyroscope full-scale and DLPF configuration.

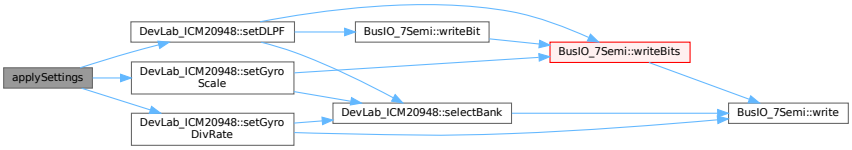
Parameters

fs_idx	Index into gyroFSCfg.
cfg	DLPF timing and divider configuration.

Returns

true if all configuration writes succeeded, otherwise false.

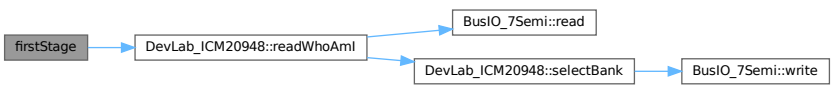
Definition at line 133 of file gyr_test.ino.
Here is the call graph for this function:



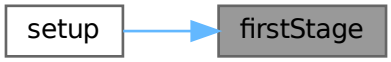
6.1.3.2 firstStage()

```
void firstStage ( )
```

Verify the IMU identity register.
Definition at line 117 of file gyr_test.ino.
Here is the call graph for this function:



Here is the caller graph for this function:



6.1.3.3 loop()

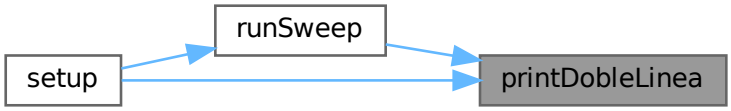
```
void loop ( )
```

Empty loop; this sketch runs the gyroscope sweep once in setup().
Definition at line 248 of file gyr_test.ino.

6.1.3.4 printDobleLinea()

```
void printDobleLinea ( )
```

Print a double separator line to Serial.
Definition at line 112 of file gyr_test.ino.
Here is the caller graph for this function:



6.1.3.5 printLinea()

```
void printLinea ( )
```

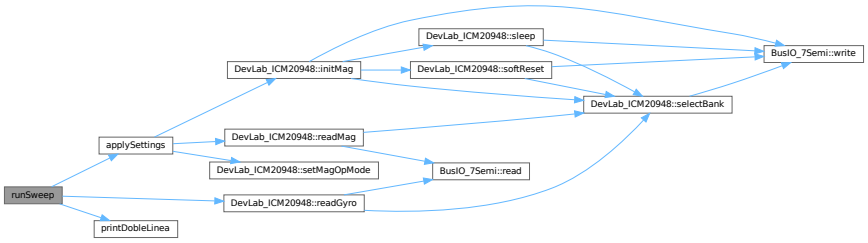
Print a separator line to Serial.
Definition at line 110 of file gyr_test.ino.

6.1.3.6 runSweep()

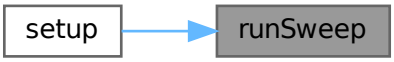
```
void runSweep ( )
```

Run the gyroscope validation sweep and print each captured sample.
Definition at line 158 of file gyr_test.ino.

Here is the call graph for this function:



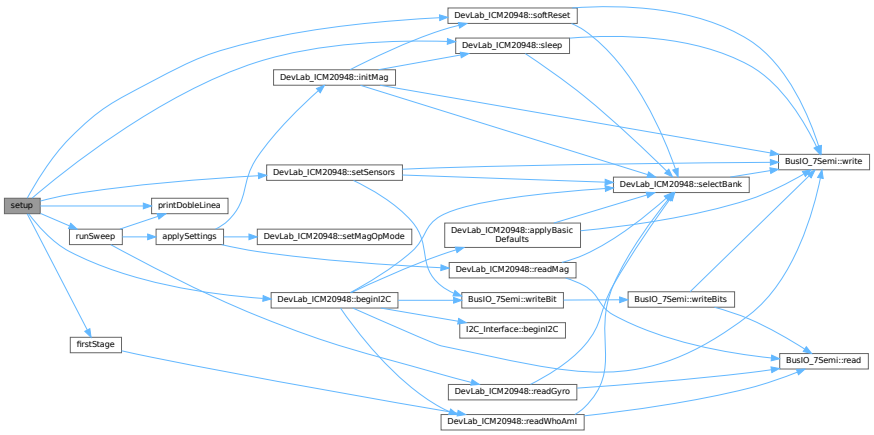
Here is the caller graph for this function:



6.1.3.7 setup()

```
void setup ( )
```

Initialize I2C, reset/wake the IMU, enable gyro, and start the sweep.
Definition at line 203 of file gyr_test.ino.
Here is the call graph for this function:



6.1.4 Variable Documentation

6.1.4.1 configsDLPF

```
*const configDLPF configsDLPF
```

Initial value:

```
= {
  {  0,   0,  229.8f, 1125.0f, "196,6 ; 229,8 ; 1125.0 " },
  {  1,   0,  187.6f, 1125.0f, "151,8 ; 187.6 ; 1125.0 " },
```

```
  {  2,   0,  154.3f,  1125.0f, "119,5 ; 154,3 ; 1125.0 " },
  {  3,   2,   73.3f,  375.0f, "51,2 ; 73,3 ; 375 " },
  {  4,   5,   35.9f,  187.5f, "23,9 ; 35,9 ; 187.5 " },
  {  5,  14,   17.8f,   75.0f, "11,6 ; 17,8 ;  75.0 " },
  {  6,  29,    8.9f,   37.5f, "5,7 ; 8,9 ;  37.5 " },
  {  7,   0,  376.5f, 1125.0f, "361,4 ; 376,5 ; 1125 " },
}
```

Definition at line 41 of file gyr_test.ino.

6.1.4.2 etiquetasGyroAVGCfg

```
const char* etiquetasGyroAVGCfg[] = {"1","2","4","8","16","32","64","128"}
```

Text labels matching gyroAVGCfg.
Definition at line 102 of file gyr_test.ino.

6.1.4.3 etiquetasGyroFSCfg

```
const char* etiquetasGyroFSCfg[] = {"250", "500", "1000", "2000"}
```

Text labels matching gyroFSCfg.
Definition at line 73 of file gyr_test.ino.

6.1.4.4 gyroAVGCfg

```
const ICM20948_Gyro_Average gyroAVGCfg[]
```

Initial value:

```
= {
  ICM20948_Gyro_Average::AVG_1,
  ICM20948_Gyro_Average::AVG_2,
  ICM20948_Gyro_Average::AVG_4,
  ICM20948_Gyro_Average::AVG_8,
  ICM20948_Gyro_Average::AVG_16,
  ICM20948_Gyro_Average::AVG_32,
  ICM20948_Gyro_Average::AVG_64,
  ICM20948_Gyro_Average::AVG_128
}
```

Gyroscope averaging settings available for experiments.
Definition at line 90 of file gyr_test.ino.

6.1.4.5 gyroDLPF

```
const ICM20948_Gyro_DLPF gyroDLPF[]
```

Initial value:

```
= {
  ICM20948_Gyro_DLPF::DLPF_196HZ,
  ICM20948_Gyro_DLPF::DLPF_151HZ,
  ICM20948_Gyro_DLPF::DLPF_119HZ,
  ICM20948_Gyro_DLPF::DLPF_51HZ,
  ICM20948_Gyro_DLPF::DLPF_23HZ,
  ICM20948_Gyro_DLPF::DLPF_11HZ,
  ICM20948_Gyro_DLPF::DLPF_5HZ,
  ICM20948_Gyro_DLPF::DLPF_361HZ
}
```

Gyroscope DLPF enum values matching configsDLPF.
Definition at line 78 of file gyr_test.ino.

6.1.4.6 gyroFSCfg

```
const ICM20948_Gyro_FullScale gyroFSCfg[]
```

Initial value:

```
= {
  ICM20948_Gyro_FullScale::DPS_250,
  ICM20948_Gyro_FullScale::DPS_500,
  ICM20948_Gyro_FullScale::DPS_1000,
  ICM20948_Gyro_FullScale::DPS_2000
}
```

Gyroscope full-scale ranges exercised by the sweep.
Definition at line 65 of file gyr_test.ino.

6.1.4.7 imu

`DevLab_ICM20948 imu`
Global ICM-20948 driver instance.
Definition at line 107 of file gyr_test.ino.

6.1.4.8 N_AVG

`const uint8_t N_AVG = 8`
Number of gyroscope averaging settings.
Definition at line 104 of file gyr_test.ino.

6.1.4.9 N_DLPF

`const uint8_t N_DLPF = sizeof(configsDLPF) / sizeof(configsDLPF[0])`
Number of gyroscope DLPF configurations.
Definition at line 62 of file gyr_test.ino.

6.1.4.10 N_FS

`const uint8_t N_FS = 4`
Number of gyroscope full-scale ranges.
Definition at line 75 of file gyr_test.ino.

6.2 gyr_test.ino

Go to the documentation of this file.

```
00001 /**
00002  * @file gyr_test.ino
00003  * @brief Gyroscope DLPF and full-scale sweep for the 7Semi ICM-20948.
00004  * @details Applies each gyroscope filter/full-scale configuration over I2C and
00005  * prints repeated samples for validation and noise comparison.
00006  */
00007
00008 #include <Wire.h>
00009 #include <DevLab_ICM20948.h>
00010 // ------------------------------------------------------------
00011 // PINES I2C -- ESP32C6 NANO Unit Electronics
00012 // ------------------------------------------------------------
00013 /** @brief I2C SDA pin used by the example board. */
00014 #define SDA_PIN   6
00015 /** @brief I2C SCL pin used by the example board. */
00016 #define SCL_PIN   7
00017 /** @brief I2C bus speed in hertz. */
00018 #define I2C_FREQ  400000UL
00019 /** @brief ICM-20948 I2C address selected by the AD0 pin. */
00020 #define ICM_ADDR  0x69
00021
00022 // ------------------------------------------------------------
00023 // STRUCT
00024 // ------------------------------------------------------------
00025 /** @brief Gyroscope DLPF divider and timing configuration. */
00026 struct configDLPF {
00027   /** @brief Index into the gyroDLPF lookup table. */
00028   uint8_t     dlpf_idx;
00029   /** @brief Sample-rate divider written to the IMU register. */
00030   uint16_t    div;
00031   /** @brief Noise bandwidth in hertz for reporting. */
00032   float       nbw_hz;
00033   /** @brief Output data rate in hertz for timing calculations. */
00034   float       odr_hz;
00035   /** @brief Text label printed with each captured sample. */
00036   const char* nombre;
00037 };
00038 //id,div,nbw_hz,odr_hz,nombre : {DLPF, NBW , ODR}
00039 /*
00040 /** @brief Gyroscope DLPF configurations exercised by the sweep. */
00041 const configDLPF configsDLPF[] = {
00042   {  0,   0,   229.8f, 1125.0f, "196,6 ; 229,8 ; 1125.0 " },
00043   {  1,   0,   187.6f, 1125.0f, "151,8 ; 187.6 ; 1125.0 " },
00044   {  2,   0,   154.3f,  1125.0f, "119,5 ; 154,3 ; 1125.0 " },
00045   {  3,   2,    73.3f,  375.0f, "51,2 ; 73,3 ; 375 " },
00046   {  4,   5,    35.9f,  187.5f, "23,9 ; 35,9 ; 187.5 " },
00047   {  5,  14,    17.8f,   75.0f, "11,6 ; 17,8 ;  75.0 " },
```

```
00048   {  6,  29,    8.9f,   37.5f, "5,7 ; 8,9 ;  37.5 " },
00049   {  7,   0,  376.5f, 1125.0f, "361,4 ; 376,5 ; 1125 " },
00050 };*/
00051 const configDLPF configsDLPF[] = {
00052   {  0,   0,  229.8f, 1125.0f, "196,6 ; 229,8 ; 1125.0 " },
00053   {  1,   0,  187.6f, 1125.0f, "151,8 ; 187.6 ; 1125.0 " },
00054   {  2,   0,  154.3f,  1125.0f, "119,5 ; 154,3 ; 1125.0 " },
00055   {  3,   2,   73.3f,  375.0f, "51,2 ; 73,3 ; 375 " },
00056   {  4,   5,   35.9f,  187.5f, "23,9 ; 35,9 ; 187.5 " },
00057   {  5,  14,   17.8f,   75.0f, "11,6 ; 17,8 ;  75.0 " },
00058   {  6,  29,    8.9f,   37.5f, "5,7 ; 8,9 ;  37.5 " },
00059   {  7,   0,  376.5f, 1125.0f, "361,4 ; 376,5 ; 1125 " },
00060 };
00061 /** @brief Number of gyroscope DLPF configurations. */
00062 const uint8_t N_DLPF = sizeof(configsDLPF) / sizeof(configsDLPF[0]);
00063
00064 /** @brief Gyroscope full-scale ranges exercised by the sweep. */
00065 const ICM20948_Gyro_FullScale gyroFSCfg[] = {
00066   ICM20948_Gyro_FullScale::DPS_250,
00067   ICM20948_Gyro_FullScale::DPS_500,
00068   ICM20948_Gyro_FullScale::DPS_1000,
00069   ICM20948_Gyro_FullScale::DPS_2000
00070 };
00071
00072 /** @brief Text labels matching gyroFSCfg. */
00073 const char* etiquetasGyroFSCfg[] = {"250", "500", "1000", "2000"};
00074 /** @brief Number of gyroscope full-scale ranges. */
00075 const uint8_t N_FS = 4;
00076
00077 /** @brief Gyroscope DLPF enum values matching configsDLPF. */
00078 const ICM20948_Gyro_DLPF gyroDLPF[] = {
00079   ICM20948_Gyro_DLPF::DLPF_196HZ,
00080   ICM20948_Gyro_DLPF::DLPF_151HZ,
00081   ICM20948_Gyro_DLPF::DLPF_119HZ,
00082   ICM20948_Gyro_DLPF::DLPF_51HZ,
00083   ICM20948_Gyro_DLPF::DLPF_23HZ,
00084   ICM20948_Gyro_DLPF::DLPF_11HZ,
00085   ICM20948_Gyro_DLPF::DLPF_5HZ,
00086   ICM20948_Gyro_DLPF::DLPF_361HZ
00087 };
00088
00089 /** @brief Gyroscope averaging settings available for experiments. */
00090 const ICM20948_Gyro_Average gyroAVGCfg[]= {
00091   ICM20948_Gyro_Average::AVG_1,
00092   ICM20948_Gyro_Average::AVG_2,
00093   ICM20948_Gyro_Average::AVG_4,
00094   ICM20948_Gyro_Average::AVG_8,
00095   ICM20948_Gyro_Average::AVG_16,
00096   ICM20948_Gyro_Average::AVG_32,
00097   ICM20948_Gyro_Average::AVG_64,
00098   ICM20948_Gyro_Average::AVG_128
00099 };
00100
00101 /** @brief Text labels matching gyroAVGCfg. */
00102 const char* etiquetasGyroAVGCfg[] = {"1","2","4","8","16","32","64","128"};
00103 /** @brief Number of gyroscope averaging settings. */
00104 const uint8_t N_AVG = 8;
00105
00106 /** @brief Global ICM-20948 driver instance. */
00107 DevLab_ICM20948 imu;
00108
00109 /** @brief Print a separator line to Serial. */
00110 void printLinea()      {
      Serial.println(F("------------------------------------------------------------")); }
00111 /** @brief Print a double separator line to Serial. */
00112 void printDobleLinea() {
      Serial.println(F("============================================================")); }
00113
00114 /**
00115  * @brief Verify the IMU identity register.
00116  */
00117 void firstStage() {
00118   uint8_t who;
00119   if (!imu.readWhoAmI(who) || who != 0xEA) {
00120     Serial.println(F("ERROR: WHO_AM_I mismatch"));
00121     while (1) delay(200);
00122   }
00123   Serial.printf("WHO_AM_I = 0x%02X  OK\n", who);
00124 }
00125
00126 /**
00127  * @brief Apply one gyroscope full-scale and DLPF configuration.
00128  *
00129  * @param fs_idx Index into gyroFSCfg.
00130  * @param cfg DLPF timing and divider configuration.
00131  * @return true if all configuration writes succeeded, otherwise false.
00132  */
```

```
00133 bool applySettings(uint8_t fs_idx, const configDLPF &cfg) {
00134   ICM20948_Gyro_DLPF dlpf_val = gyroDLPF[cfg.dlpf_idx];
00135   bool ok = true;
00136
00137   ok &= imu.setGyroScale(gyroFSCfg[fs_idx]);
00138
00139
00140   // Paso 1: escribe DLPFCFG en bits [5:3]
00141   // SetDLPF false -> Activa DLPF
00142   // SetDLPF true -> Desactiva DLPF
00143   ok &= imu.setDLPF(dlpf_val, false);
00144
00145   //ok &= imu.setGyroAveraging(gyroAVGCfg[avg]);
00146   // Paso 2: activa FCHOICE=1 (DLPF activo) sin tocar DLPFCFG
00147   //ok &= imu.setAccelDLPF(0, true);
00148
00149   // DIV directo al registro, sin conversion interna
00150   ok &= imu.setGyroDivRate(cfg.div);
00151
00152   return ok;
00153 }
00154
00155 /**
00156  * @brief Run the gyroscope validation sweep and print each captured sample.
00157  */
00158 void runSweep(){
00159   uint16_t n = 0;
00160   printDobleLinea();
00161   Serial.println(F("  ICM-20948 | UNIT Electronics"));
00162   Serial.println(F("  MODO VERIFICACION -- FS=+-2g fijo, 3 lecturas por config"));
00163   Serial.println(F("  Sensor PLANO y QUIETO | Az esperado: ~+1.000g"));
00164   printDobleLinea();
00165   for (uint8_t i = 0; i < N_FS ; i++){
00166       for(uint8_t d = 0; d < N_DLPF; d++){
00167
00168         bool cfg_ok = applySettings(i,configsDLPF[d]);
00169         if(!cfg_ok){
00170           Serial.println(F("ERROR: applySettings fallo"));
00171         }
00172
00173         // Esperar que el filtro estabilice (10 periodos del ODR, min 50ms)
00174         uint32_t settle_ms = max((uint32_t)(10000.0f / configsDLPF[d].odr_hz),
00175                                  (uint32_t)50);
00176         delay(settle_ms);
00177         // Tomar 3 lecturas espaciadas por el periodo del ODR
00178         uint32_t periodo_ms = max((uint32_t)(1000.0f / configsDLPF[d].odr_hz),
00179                                   (uint32_t)1);
00180         float gx, gy, gz;
00181         for (uint8_t r = 0; r < 100; r++) {
00182           n++;
00183           delay(periodo_ms);
00184           if (imu.readGyro(gx, gy, gz)) {
00185             Serial.printf("%d ; %s ; %s ; %+.4f ; %+.4f ; %+.4f\n",
00186                           n,
00187                           configsDLPF[d].nombre,   // "246 ; 265 ; 1125"
00188                           etiquetasGyroFSCfg[i],     // "+-2G"
00189                           //etiquetasGyroAVGCfg[j],     // "8"
00190                           gx, gy, gz);
00191           } else {
00192             Serial.println(F("  [?] readGyr() fallo"));
00193           }
00194       }
00195     }
00196
00197   }
00198 }
00199
00200 /**
00201  * @brief Initialize I2C, reset/wake the IMU, enable gyro, and start the sweep.
00202  */
00203 void setup() {
00204   // put your setup code here, to run once:
00205   Serial.begin(115200);
00206   delay(200);
00207
00208     printDobleLinea();
00209   Serial.println(F("  ICM-20948 VALIDACION ACELEROMETRO"));
00210   printDobleLinea();
00211
00212   Wire.begin(SDA_PIN, SCL_PIN);
00213
00214   if (!imu.beginI2C(ICM_ADDR, Wire, I2C_FREQ)) {
00215     Serial.println(F("ERROR: beginI2C() fallo"));
00216     while (1) delay(200);
00217   }
00218
00219   firstStage();
```

```
00220
00221   // softReset pone el chip a dormir
00222   // Despues del reset hay que despertar el dispositivo
00223   imu.softReset();
00224   delay(100);
00225   imu.sleep(false);   // limpia SLEEP bit, CLKSEL=1
00226   delay(50);
00227
00228   if (!imu.setSensors(false, true, false)) {
00229     Serial.println(F("ERROR: setSensors() fallo"));
00230   }
00231
00232   Serial.println(F("  Sensor listo.\n"));
00233   Serial.println(F("  Sensor PLANO y QUIETO sobre la mesa."));
00234   Serial.println(F("  Iniciando en 5 segundos..."));
00235   for (int i = 5; i > 0; i--) {
00236     Serial.printf("  %d...\n", i);
00237     delay(1000);
00238   }
00239
00240   //for (int runs = 0; runs < 20 ; runs++){
00241     runSweep();
00242   //}
00243 }
00244
00245 /**
00246  * @brief Empty loop; this sketch runs the gyroscope sweep once in setup().
00247  */
00248 void loop() {
00249   // put your main code here, to run repeatedly:
00250
00251 }
```

6.3 examples/i2c_accel/i2c_accel.ino File Reference

```
#include <Wire.h>
#include <DevLab_ICM20948.h>
```

Macros

- #define SDA_PIN 6

  I2C SDA pin used by the example board.
- #define SCL_PIN 7

  I2C SCL pin used by the example board.
- #define I2C_FREQ 400000UL

  I2C bus speed in hertz.
- #define ICM_ADDR 0x69

  ICM-20948 I2C address selected by the AD0 pin.

Functions

- void setup ()

  Configure the IMU for accelerometer-only operation over I2C.
- void loop ()

  Read and print accelerometer data in g.

Variables

- DevLab_ICM20948 imu

  Global ICM-20948 driver instance.

6.3.1 Macro Definition Documentation

6.3.1.1 I2C_FREQ

```
#define I2C_FREQ 400000UL
```

I2C bus speed in hertz.
Definition at line 40 of file i2c_accel.ino.

6.3.1.2 ICM_ADDR

```
#define ICM_ADDR 0x69
```

ICM-20948 I2C address selected by the AD0 pin.
Definition at line 42 of file i2c_accel.ino.

6.3.1.3 SCL_PIN

```
#define SCL_PIN 7
```

I2C SCL pin used by the example board.
Definition at line 38 of file i2c_accel.ino.

6.3.1.4 SDA_PIN

```
#define SDA_PIN 6
```

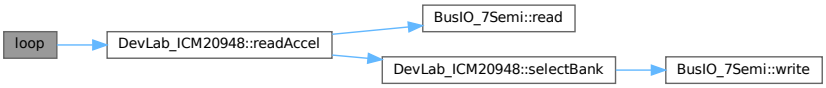
I2C SDA pin used by the example board.
Definition at line 36 of file i2c_accel.ino.

6.3.2 Function Documentation

6.3.2.1 loop()

```
void loop ( )
```

Read and print accelerometer data in g.
Definition at line 121 of file i2c_accel.ino.
Here is the call graph for this function:

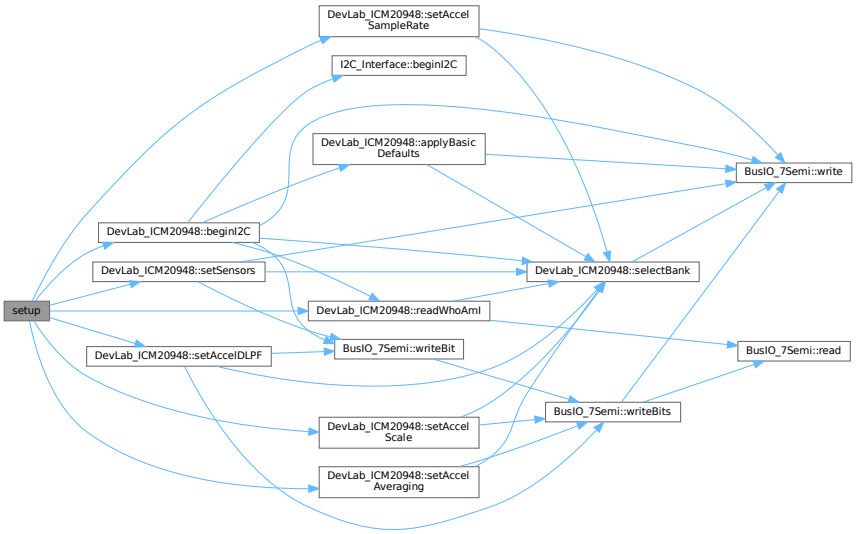


6.3.2.2 setup()

```
void setup ( )
```

Configure the IMU for accelerometer-only operation over I2C.
The function initializes the bus, verifies the device identity, enables the accelerometer, and applies range, DLPF, averaging, and sample-rate settings.
Definition at line 53 of file i2c_accel.ino.

Here is the call graph for this function:



6.3.3 Variable Documentation

6.3.3.1 imu

`DevLab_ICM20948 imu`
Global ICM-20948 driver instance.
Definition at line 45 of file i2c_accel.ino.

6.4 i2c_accel.ino

Go to the documentation of this file.

```
00001 /*************************************************************
00002  * @file    I2C_Accel.ino
00003  * @brief   Minimal I2C bring-up for 7Semi ICM-20948 on UNO/ESP32 +
00004  *          continuous accelerometer readout
00005  *
00006  * Features
00007  * - I2C init on default/custom pins
00008  * - IMU beginI2C() on I2C (addr 0x68/0x69)
00009  * - Manual sensor enable (setSensors)
00010  * - Accel config: scale, DLPF, averaging, sample rate
00011  * - Read accel (g) at ~2 Hz print
00012  *
00013  * Notes
00014  * - WHO_AM_I must be 0xEA
00015  * - Sensors must be explicitly enabled
00016  * - Bank switching handled internally by library
00017  *
00018  * Wiring (Arduino UNO I2C)
00019  * - SDA  -> A4
00020  * - SCL  -> A5
00021  * - VCC  -> 3V3 (or 5V if your board supports it)
00022  * - GND  -> GND
00023  *
00024  * Wiring (ESP32 default I2C)
00025  * - SDA  -> GPIO21
00026  * - SCL  -> GPIO22
00027  * - VCC  -> 3V3
00028  * - GND  -> GND
00029  *************************************************************/
00030
00031 #include <Wire.h>
00032 #include <DevLab_ICM20948.h>
00033
00034 /* ====================== User Config ====================== */
00035 /** @brief I2C SDA pin used by the example board. */
00036 #define SDA_PIN   6
```

```
00037 /** @brief I2C SCL pin used by the example board. */
00038 #define SCL_PIN   7
00039 /** @brief I2C bus speed in hertz. */
00040 #define I2C_FREQ  400000UL
00041 /** @brief ICM-20948 I2C address selected by the AD0 pin. */
00042 #define ICM_ADDR  0x69
00043
00044 /** @brief Global ICM-20948 driver instance. */
00045 DevLab_ICM20948 imu;
00046
00047 /**
00048  * @brief Configure the IMU for accelerometer-only operation over I2C.
00049  *
00050  * The function initializes the bus, verifies the device identity, enables the
00051  * accelerometer, and applies range, DLPF, averaging, and sample-rate settings.
00052  */
00053 void setup() {
00054   Serial.begin(115200);
00055   delay(200);
00056   Serial.println(F("ICM-20948 (I2C) -- Accel Example"));
00057   Wire.begin(SDA_PIN, SCL_PIN);
00058   /* Initialize IMU using I2C. */
00059   if (!imu.beginI2C(ICM_ADDR, Wire, 400000)) {
00060     Serial.println(F("ERROR: ICM-20948 beginI2C() failed."));
00061     while (1) delay(200);
00062   }
00063
00064   Serial.println(F("ICM-20948 ready."));
00065
00066   /* Verify device identity. */
00067   uint8_t who;
00068   if (imu.readWhoAmI(who)) {
00069     Serial.print("WHO_AM_I: 0x");
00070     Serial.println(who, HEX);
00071   }
00072
00073   /* Enable only accelerometer
00074    * - accel = true
00075    * - gyro  = false
00076    * - temp  = false
00077    */
00078   if (!imu.setSensors(true, false, false)) {
00079     Serial.println(F("setSensors failed."));
00080   }
00081
00082   /* ---------------- ACCEL CONFIG ----------------
00083    * - Scale: ±2/±4/±8/±16 g
00084    * - DLPF: noise filtering
00085    * - Averaging: noise reduction
00086    * - Sample rate: output frequency
00087    */
00088
00089   /* Set accelerometer full-scale range. */
00090   if (!imu.setAccelScale(ICM20948_Accel_FullScale::G_4)) {
00091     Serial.println(F("setAccelScale failed."));
00092   }
00093
00094   /* Enable DLPF (recommended)
00095    * - bypass = false → DLPF enabled
00096    */
00097   if (!imu.setAccelDLPF(ACCEL_DLPFCFG_3, false)) {
00098     Serial.println(F("setAccelDLPF failed."));
00099   }
00100
00101   /* Set averaging (noise reduction). */
00102   if (!imu.setAccelAveraging(ICM20948_Accel_Average::AVG_8)) {
00103     Serial.println(F("setAccelAveraging failed."));
00104   }
00105
00106   /* Set output data rate (ODR)
00107    * - Base = 1125 Hz
00108    * - ODR = 1125 / (1 + divider)
00109    * - Example: 225 Hz
00110    */
00111   if (!imu.setAccelSampleRate(225)) {
00112     Serial.println(F("setAccelSampleRate failed."));
00113   }
00114
00115   delay(10);
00116 }
00117
00118 /**
00119  * @brief Read and print accelerometer data in g.
00120  */
00121 void loop() {
00122   float ax, ay, az;
00123
```

```
00124   /* Read accelerometer data
00125    * - Output in g units
00126    * - Returns true on success
00127    */
00128   if (imu.readAccel(ax, ay, az)) {
00129     Serial.print("ACCEL [g]: ");
00130     Serial.print(ax, 3);
00131     Serial.print(", ");
00132     Serial.print(ay, 3);
00133     Serial.print(", ");
00134     Serial.println(az, 3);
00135   } else {
00136     Serial.println(F("ACCEL read failed"));
00137   }
00138
00139   Serial.println(F("------------------------------"));
00140   delay(500);
00141 }
```

6.5 examples/i2c_basic/i2c_basic.ino File Reference

```
#include <Wire.h>
#include <DevLab_ICM20948.h>
```

Macros

- #define SDA_PIN 6

  I2C SDA pin used by the example board.
- #define SCL_PIN 7

  I2C SCL pin used by the example board.
- #define I2C_FREQ 400000UL

  I2C bus speed in hertz.
- #define ICM_ADDR 0x69

  ICM-20948 I2C address selected by the AD0 pin.

Functions

- void setup ()

  Configure Serial, I2C, sensor identity check, and enabled sensors.
- void loop ()

  Read accelerometer, gyroscope, magnetometer, and temperature data.

Variables

- DevLab_ICM20948 imu

  Global ICM-20948 driver instance.

6.5.1 Macro Definition Documentation

6.5.1.1 I2C_FREQ

```
#define I2C_FREQ 400000UL
```

I2C bus speed in hertz.
Definition at line 36 of file i2c_basic.ino.

6.5.1.2 ICM_ADDR

```
#define ICM_ADDR 0x69
```

ICM-20948 I2C address selected by the AD0 pin.
Definition at line 38 of file i2c_basic.ino.

6.5.1.3 SCL_PIN

```
#define SCL_PIN 7
```

I2C SCL pin used by the example board.
Definition at line 34 of file i2c_basic.ino.

6.5.1.4 SDA_PIN

```
#define SDA_PIN 6
```

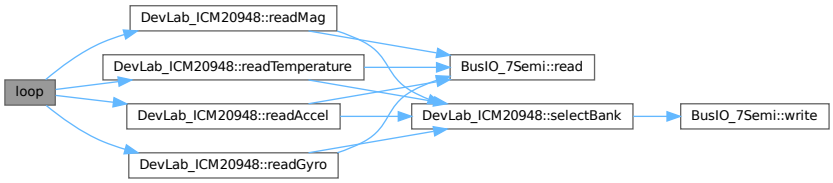
I2C SDA pin used by the example board.
Definition at line 32 of file i2c_basic.ino.

6.5.2 Function Documentation

6.5.2.1 loop()

```
void loop ( )
```

Read accelerometer, gyroscope, magnetometer, and temperature data.
Values are printed periodically to the Serial monitor in engineering units.
Definition at line 90 of file i2c_basic.ino.
Here is the call graph for this function:

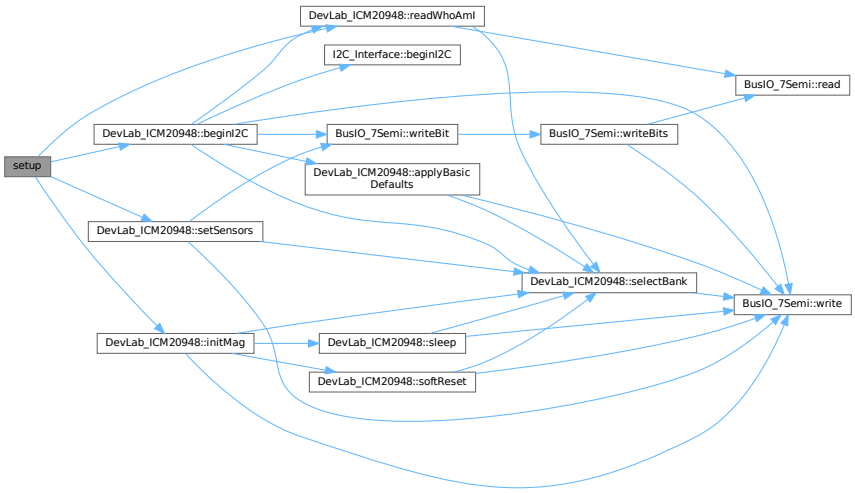


6.5.2.2 setup()

```
void setup ( )
```

Configure Serial, I2C, sensor identity check, and enabled sensors.
The sketch stops in an infinite loop if the IMU cannot be initialized or the WHO_AM_I register does not match the expected ICM-20948 value.
Definition at line 49 of file i2c_basic.ino.

Here is the call graph for this function:



6.5.3 Variable Documentation

6.5.3.1 imu

`DevLab_ICM20948 imu`

Global ICM-20948 driver instance.

Definition at line 41 of file i2c_basic.ino.

6.6 i2c_basic.ino

Go to the documentation of this file.

```
00001 /*************************************************************
00002  * @file    I2C_Basic.ino
00003  * @brief   Minimal I2C bring-up for 7Semi ICM-20948 +
00004  *          basic Accel/Gyro/Temp/Mag readout (updated API).
00005  *
00006  * Features
00007  * - I2C init (UNO / ESP32)
00008  * - beginI2C() initialization
00009  * - Manual sensor enable (setSensors)
00010  * - Read Accel / Gyro / Temp / Mag
00011  *
00012  * Notes
00013  * - WHO_AM_I must be 0xEA
00014  * - Sensors must be explicitly enabled
00015  * - Magnetometer requires initMag()
00016  *
00017  * Wiring (Arduino UNO I2C)
00018  * - SDA -> A4
00019  * - SCL -> A5
00020  * - VCC -> 3V3 (or 5V if supported)
00021  * - GND -> GND
00022  *
00023  * Wiring (ESP32 default I2C)
00024  * - SDA -> GPIO21
00025  * - SCL -> GPIO22
00026  *************************************************************/
00027 #include <Wire.h>
00028 #include <DevLab_ICM20948.h>
00029
00030 /* ====================== User Config ====================== */
00031 /** @brief I2C SDA pin used by the example board. */
00032 #define SDA_PIN   6
00033 /** @brief I2C SCL pin used by the example board. */
00034 #define SCL_PIN   7
00035 /** @brief I2C bus speed in hertz. */
00036 #define I2C_FREQ  400000UL
00037 /** @brief ICM-20948 I2C address selected by the AD0 pin. */
00038 #define ICM_ADDR  0x69
```

```
00039
00040 /** @brief Global ICM-20948 driver instance. */
00041 DevLab_ICM20948 imu;
00042
00043 /**
00044  * @brief Configure Serial, I2C, sensor identity check, and enabled sensors.
00045  *
00046  * The sketch stops in an infinite loop if the IMU cannot be initialized or
00047  * the WHO_AM_I register does not match the expected ICM-20948 value.
00048  */
00049 void setup() {
00050   Serial.begin(115200);
00051   delay(200);
00052   Serial.println(F("ICM-20948 -- I2C Basic"));
00053   Wire.begin(SDA_PIN, SCL_PIN);
00054   /* Initialize IMU. */
00055   if (!imu.beginI2C(ICM_ADDR, Wire, 400000)) {
00056     Serial.println(F("ERROR: beginI2C() failed"));
00057     while (1) delay(200);
00058   }
00059
00060   /* WHO_AM_I check. */
00061   uint8_t who;
00062   if (!imu.readWhoAmI(who) || who != 0xEA) {
00063     Serial.println(F("ERROR: WHO_AM_I mismatch"));
00064     while (1) delay(200);
00065   }
00066
00067   Serial.print(F("WHO_AM_I = 0x"));
00068   Serial.println(who, HEX);
00069
00070   /* Enable all sensors. */
00071   if (!imu.setSensors(true, true, true)) {
00072     Serial.println(F("ERROR: setSensors failed"));
00073   }
00074
00075   /* Initialize magnetometer. */
00076   if (!imu.initMag()) {
00077     Serial.println(F("Mag init failed"));
00078   } else {
00079     Serial.println(F("Mag initialized"));
00080   }
00081
00082   Serial.println(F("Ready.\n"));
00083 }
00084
00085 /**
00086  * @brief Read accelerometer, gyroscope, magnetometer, and temperature data.
00087  *
00088  * Values are printed periodically to the Serial monitor in engineering units.
00089  */
00090 void loop() {
00091   float ax, ay, az;
00092   float gx, gy, gz;
00093   float mx, my, mz;
00094   float tC;
00095
00096   /* Read accelerometer. */
00097   if (imu.readAccel(ax, ay, az)) {
00098     Serial.print(F("ACC [g]: "));
00099     Serial.print(ax, 3); Serial.print(", ");
00100     Serial.print(ay, 3); Serial.print(", ");
00101     Serial.println(az, 3);
00102   } else {
00103     Serial.println(F("ACC read failed"));
00104   }
00105
00106   /* Read gyroscope. */
00107   if (imu.readGyro(gx, gy, gz)) {
00108     Serial.print(F("GYR [dps]: "));
00109     Serial.print(gx, 2); Serial.print(", ");
00110     Serial.print(gy, 2); Serial.print(", ");
00111     Serial.println(gz, 2);
00112   } else {
00113     Serial.println(F("GYR read failed"));
00114   }
00115
00116   /* Read magnetometer. */
00117   if (imu.readMag(mx, my, mz)) {
00118     Serial.print(F("MAG [uT]: "));
00119     Serial.print(mx, 2); Serial.print(", ");
00120     Serial.print(my, 2); Serial.print(", ");
00121     Serial.println(mz, 2);
00122   } else {
00123     Serial.println(F("MAG read failed"));
00124   }
00125
```

```
00126   /* Read temperature. */
00127   if (imu.readTemperature(tC)) {
00128     Serial.print(F("TMP [C]: "));
00129     Serial.println(tC, 2);
00130   } else {
00131     Serial.println(F("TMP read failed"));
00132   }
00133
00134   Serial.println(F("----------------------------"));
00135   delay(500);
00136 }
00137
00138 /* Note on magnetometer:
00139  * - Uses internal I2C master (AK09916 via ICM20948)
00140  * - initMag() must be called before readMag()
00141  * - If values are zero:
00142  *   → check initMag()
00143  *   → verify power + pull-ups
00144  */
```

6.7 examples/i2c_gyro/i2c_gyro.ino File Reference

```
#include <Wire.h>
#include <DevLab_ICM20948.h>
```

Macros

- #define SDA_PIN 6

  I2C SDA pin used by the example board.
- #define SCL_PIN 7

  I2C SCL pin used by the example board.
- #define I2C_FREQ 400000UL

  I2C bus speed in hertz.
- #define ICM_ADDR 0x69

  ICM-20948 I2C address selected by the AD0 pin.

Functions

- void setup ()

  Configure the IMU for gyroscope-only operation over I2C.
- void loop ()

  Read and print gyroscope data in degrees per second.

Variables

- DevLab_ICM20948 imu

  Global ICM-20948 driver instance.

6.7.1 Macro Definition Documentation

6.7.1.1 I2C_FREQ

```
#define I2C_FREQ 400000UL
```

I2C bus speed in hertz.
Definition at line 45 of file i2c_gyro.ino.

6.7.1.2 ICM_ADDR

```
#define ICM_ADDR 0x69
```

ICM-20948 I2C address selected by the AD0 pin.
Definition at line 47 of file i2c_gyro.ino.

6.7.1.3 SCL_PIN

```
#define SCL_PIN 7
```

I2C SCL pin used by the example board.
Definition at line 43 of file i2c_gyro.ino.

6.7.1.4 SDA_PIN

```
#define SDA_PIN 6
```

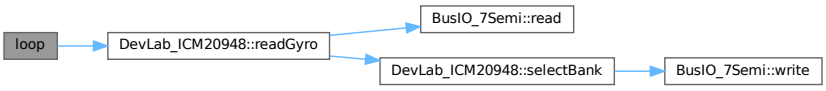
I2C SDA pin used by the example board.
Definition at line 41 of file i2c_gyro.ino.

6.7.2 Function Documentation

6.7.2.1 loop()

```
void loop ( )
```

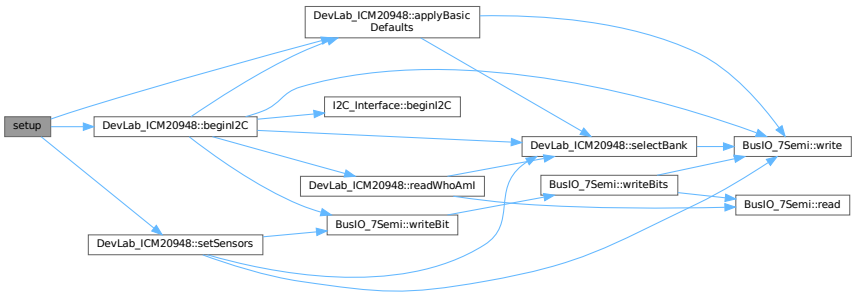
Read and print gyroscope data in degrees per second.
Definition at line 125 of file i2c_gyro.ino.
Here is the call graph for this function:



6.7.2.2 setup()

```
void setup ( )
```

Configure the IMU for gyroscope-only operation over I2C.
The function initializes the I2C interface, applies safe defaults, enables the gyroscope, and configures DLPF, full-scale range, and output data rate.
Definition at line 58 of file i2c_gyro.ino.
Here is the call graph for this function:



6.7.3 Variable Documentation

6.7.3.1 imu

```
DevLab_ICM20948 imu
```

Global ICM-20948 driver instance.
Definition at line 50 of file i2c_gyro.ino.

6.8 i2c_gyro.ino

Go to the documentation of this file.

```
00001 /*****************************************************************
00002  * @file    I2C_Gyro.ino
00003  * @brief   Minimal I2C bring-up for 7Semi ICM-20948 on ESP32/UNO +
00004  *          continuous gyroscope readout.
00005  *
00006  * Features
00007  * - I2C init on default/custom pins
00008  * - IMU begin() on I2C (addr 0x68/0x69)
00009  * - Safe defaults (applyBasicDefaults)
00010  * - Optional sensor gating: gyro only
00011  * - Gyro config: DLPF, full-scale, ODR, AVG
00012  * - Read gyro (dps) at ~2 Hz print
00013  *
00014  * Wiring (Arduino UNO, I2C)
00015  * - SDA  -> A4
00016  * - SCL  -> A5
00017  * - VCC  -> 3V3 (or 5V if your board supports it)
00018  * - GND  -> GND
00019  *
00020  * Wiring (ESP32, I2C)
00021  * - SDA  -> GPIO21 (default)
00022  * - SCL  -> GPIO22 (default)
00023  * - VCC  -> 3V3
00024  * - GND  -> GND
00025  *
00026  * Address
00027  * - Default ICM-20948 I2C address is 0x68 (AD0=LOW). If AD0=HIGH, use 0x69.
00028  *****************************************************************/
00029
00030 #include <Wire.h>
00031 #include <DevLab_ICM20948.h>
00032
00033 /* ====================== User Config ======================
00034  * - Edit pins/addr if needed
00035  * - UNO uses fixed SDA=A4, SCL=A5 (no need to call Wire.begin with pins)
00036  * - ESP32 can use default Wire (SDA=21, SCL=22) or custom pins via Wire.begin(sda,scl)
00037  */
00038
00039 /* ====================== User Config ====================== */
00040 /** @brief I2C SDA pin used by the example board. */
00041 #define SDA_PIN   6
00042 /** @brief I2C SCL pin used by the example board. */
00043 #define SCL_PIN   7
00044 /** @brief I2C bus speed in hertz. */
00045 #define I2C_FREQ  400000UL
00046 /** @brief ICM-20948 I2C address selected by the AD0 pin. */
00047 #define ICM_ADDR  0x69
00048
00049 /** @brief Global ICM-20948 driver instance. */
00050 DevLab_ICM20948 imu;
00051
00052 /**
00053  * @brief Configure the IMU for gyroscope-only operation over I2C.
00054  *
00055  * The function initializes the I2C interface, applies safe defaults, enables
00056  * the gyroscope, and configures DLPF, full-scale range, and output data rate.
00057  */
00058 void setup() {
00059   Serial.begin(115200);
00060   delay(200);
00061   Serial.println(F("ICM-20948 -- I2C Basic"));
00062   Wire.begin(SDA_PIN, SCL_PIN);
00063   /* Initialize IMU. */
00064   if (!imu.beginI2C(ICM_ADDR, Wire, 400000)) {
00065     Serial.println(F("ERROR: beginI2C() failed"));
00066     while (1) delay(200);
00067   }
00068
00069   /* Apply safe defaults. */
00070   if (!imu.applyBasicDefaults()) {
00071     Serial.println(F("ERROR: applyBasicDefaults() failed."));
00072     while (1) delay(200);
00073   }
00074
00075   Serial.println(F("ICM-20948 ready."));
00076
00077   /* Optional: gate sensors
00078    * - setSensors(accel_on, gyro_on, temp_on)
00079    * - Here: enable only gyro (accel/temp off)
00080    */
00081   if (!imu.setSensors(/*accel*/ false, /*gyro*/ true, /*temp*/ false)) {
00082     Serial.println(F("setSensors failed."));
00083   }
```

```
00084
00085   /* ----------------- GYRO DLPF Config (reference) ----------------
00086    * GyroConfig(DLPFCFG, FS_SEL, FCHOICE, XGYRO, YGYRO, ZGYRO, AVGCFG)
00087    * - DLPF     : GYRO_DLPFCFG_0..7 (use 3 or 4 for balanced settings)
00088    * - FS_SEL   : dps250 / dps500 / dps1000 / dps2000
00089    * - FCHOICE  : false → DLPF path (recommended), true → bypass (very wide BW)
00090    * - XGYRO/YGYRO/ZGYRO : per-axis enable (true = enable axis)
00091    * - AVGCFG   : 0..7 internal averaging (higher = more smoothing, more delay)
00092    * FCHOICE=0 (DLPF on):
00093    * - GYRO_DLPFCFG_1 : 3dB ≈ 196.6 Hz, NBW ≈ 229.8 Hz
00094    * - GYRO_DLPFCFG_2 : 3dB ≈ 151.8 Hz, NBW ≈ 187.6 Hz
00095    * - GYRO_DLPFCFG_3 : 3dB ≈ 119.5 Hz, NBW ≈ 154.3 Hz   ← good default
00096    * - GYRO_DLPFCFG_4 : 3dB ≈ 51.2  Hz, NBW ≈ 73.3  Hz
00097    * - GYRO_DLPFCFG_5 : 3dB ≈ 23.9  Hz, NBW ≈ 35.9  Hz
00098    * - GYRO_DLPFCFG_6 : 3dB ≈ 5.7   Hz, NBW ≈ 8.3   Hz
00099    * - GYRO_DLPFCFG_7 : 3dB ≈ 473   Hz, NBW ≈ 499   Hz
00100    *
00101    * FCHOICE=1 (Bypass) : very wide (use with care; highest noise)
00102    */
00103   if (!imu.GyroConfig(/*DLPFCFG*/ GYRO_DLPFCFG_4,
00104                       /*FS_SEL*/   dps2000,
00105                       /*FCHOICE*/  true,       // DLPF ON (recommended)
00106                       /*X*/ true, /*Y*/ true, /*Z*/ true,
00107                       /*AVGCFG*/   0)) {
00108     Serial.println(F("GyroConfig failed."));
00109   }
00110
00111   /* Set gyroscope output data rate (ODR)
00112    * - Gyro_SMPLRT(rate_hz)
00113    * - Base (DLPF path) = 1100 Hz
00114    * - Valid range: ~4.3-1100 Hz
00115    * - Example: 1000 Hz
00116    */
00117   if (!imu.Gyro_SMPLRT(1000)) {
00118     Serial.println(F("Gyro_SMPLRT failed."));
00119   }
00120 }
00121
00122 /**
00123  * @brief Read and print gyroscope data in degrees per second.
00124  */
00125 void loop() {
00126   float gx, gy, gz;  // gyro X/Y/Z in dps
00127
00128   /* - readGyro(x,y,z)
00129    * - Returns:
00130    *   - true on success;
00131    * - Output:
00132    *   - gx/gy/gz in dps
00133    */
00134   if (imu.readGyro(gx, gy, gz)) {
00135     Serial.print("GYRO [dps]: ");
00136     Serial.print(gx, 3);
00137     Serial.print(", ");
00138     Serial.print(gy, 3);
00139     Serial.print(", ");
00140     Serial.println(gz, 3);
00141   } else {
00142     Serial.println(F("GYRO read failed"));
00143   }
00144
00145   Serial.println(F("------------------------------"));
00146   delay(500);
00147 }
```

6.9 examples/i2c_magneto/i2c_magneto.ino File Reference

```
#include <Wire.h>
#include <DevLab_ICM20948.h>
```

Macros

- #define SDA_PIN 6

  I2C SDA pin used by the example board.
- #define SCL_PIN 7

  I2C SCL pin used by the example board.
- #define I2C_FREQ 400000UL

I2C bus speed in hertz.

- #define ICM_ADDR 0x69

  ICM-20948 I2C address selected by the AD0 pin.

Functions

- void setup ()

  Configure I2C, verify the IMU, and initialize the magnetometer.

- void loop ()

  Read and print magnetometer data in microtesla.

Variables

- DevLab_ICM20948 imu

  Global ICM-20948 driver instance.

6.9.1 Macro Definition Documentation

6.9.1.1 I2C_FREQ

```
#define I2C_FREQ 400000UL
```

I2C bus speed in hertz.
Definition at line 42 of file i2c_magneto.ino.

6.9.1.2 ICM_ADDR

```
#define ICM_ADDR 0x69
```

ICM-20948 I2C address selected by the AD0 pin.
Definition at line 44 of file i2c_magneto.ino.

6.9.1.3 SCL_PIN

```
#define SCL_PIN 7
```

I2C SCL pin used by the example board.
Definition at line 40 of file i2c_magneto.ino.

6.9.1.4 SDA_PIN

```
#define SDA_PIN 6
```

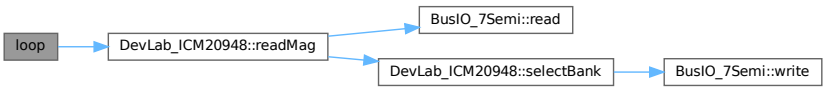
I2C SDA pin used by the example board.
Definition at line 38 of file i2c_magneto.ino.

6.9.2 Function Documentation

6.9.2.1 loop()

```
void loop ( )
```

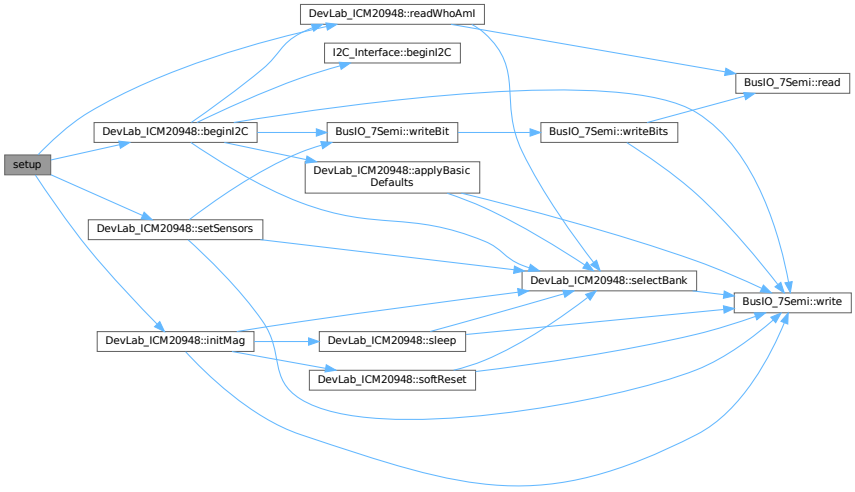
Read and print magnetometer data in microtesla.
Definition at line 97 of file i2c_magneto.ino.
Here is the call graph for this function:



6.9.2.2 setup()

```
void setup ( )
```

Configure I2C, verify the IMU, and initialize the magnetometer.
The magnetometer is accessed through the ICM-20948 internal I2C master, so initMag() must succeed before loop() can read magnetic field data.
Definition at line 55 of file i2c_magneto.ino.
Here is the call graph for this function:



6.9.3 Variable Documentation

6.9.3.1 imu

```
DevLab_ICM20948 imu
```

Global ICM-20948 driver instance.
Definition at line 47 of file i2c_magneto.ino.

6.10 i2c_magneto.ino

Go to the documentation of this file.

```
00001 /*****************************************************************
00002  * @file    I2C_Magneto.ino
00003  * @brief   Minimal I2C bring-up for 7Semi ICM-20948 +
00004  *          AK09916 magnetometer readout (updated API).
00005  *
00006  * Features
00007  * - I2C init on default/custom pins
00008  * - IMU beginI2C() initialization
00009  * - Manual sensor enable
00010  * - Magnetometer initialization (initMag)
00011  * - Read magnetometer (uT) at ~2 Hz
00012  *
00013  * Wiring (Arduino UNO, I2C)
00014  * - SDA  -> A4
00015  * - SCL  -> A5
00016  * - VCC  -> 3V3 (or 5V if board supports it)
00017  * - GND  -> GND
00018  *
00019  * Wiring (ESP32, I2C default)
00020  * - SDA  -> GPIO21
00021  * - SCL  -> GPIO22
00022  * - VCC  -> 3V3
00023  * - GND  -> GND
00024  *
00025  * Notes
00026  * - WHO_AM_I must be 0xEA
00027  * - initMag() must be called before readMag()
00028  * - Magnetometer runs via internal I2C master
```

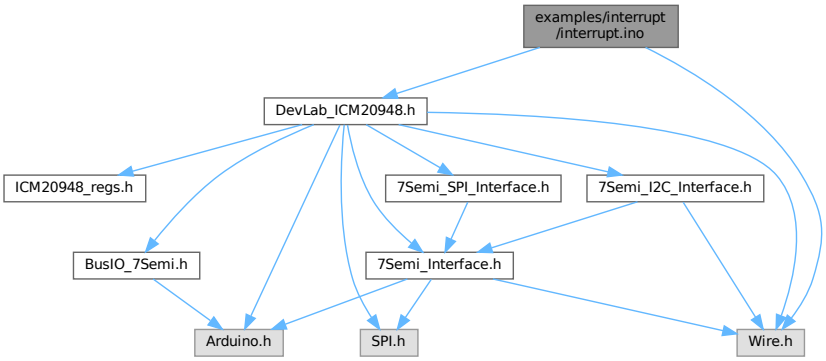
```
00029  *
00030  * Author  : 7Semi
00031  * License : MIT
00032  *****************************************************************/
00033 #include <Wire.h>
00034 #include <DevLab_ICM20948.h>
00035
00036 /* ====================== User Config ====================== */
00037 /** @brief I2C SDA pin used by the example board. */
00038 #define SDA_PIN   6
00039 /** @brief I2C SCL pin used by the example board. */
00040 #define SCL_PIN   7
00041 /** @brief I2C bus speed in hertz. */
00042 #define I2C_FREQ  400000UL
00043 /** @brief ICM-20948 I2C address selected by the AD0 pin. */
00044 #define ICM_ADDR  0x69
00045
00046 /** @brief Global ICM-20948 driver instance. */
00047 DevLab_ICM20948 imu;
00048
00049 /**
00050  * @brief Configure I2C, verify the IMU, and initialize the magnetometer.
00051  *
00052  * The magnetometer is accessed through the ICM-20948 internal I2C master, so
00053  * initMag() must succeed before loop() can read magnetic field data.
00054  */
00055 void setup() {
00056   Serial.begin(115200);
00057   delay(200);
00058   Serial.println(F("ICM-20948 (I2C) -- Magnetometer Example"));
00059   Wire.begin(SDA_PIN, SCL_PIN);
00060   /* Initialize IMU. */
00061   if (!imu.beginI2C(ICM_ADDR, Wire, 400000)) {
00062     Serial.println(F("ERROR: beginI2C() failed."));
00063     while (1) delay(200);
00064   }
00065
00066   Serial.println(F("ICM-20948 ready (I2C)."));
00067
00068   /* Verify device. */
00069   uint8_t who;
00070   if (!imu.readWhoAmI(who) || who != 0xEA) {
00071     Serial.println(F("ERROR: WHO_AM_I mismatch"));
00072     while (1) delay(200);
00073   }
00074
00075   Serial.print(F("WHO_AM_I = 0x"));
00076   Serial.println(who, HEX);
00077
00078   /* Enable required sensors (mag uses internal I2C master)
00079    * - accel/gyro not strictly required but safe to keep ON
00080    */
00081   if (!imu.setSensors(true, true, false)) {
00082     Serial.println(F("setSensors failed."));
00083   }
00084
00085   /* Initialize magnetometer. */
00086   if (!imu.initMag()) {
00087     Serial.println(F("ERROR: initMag() failed"));
00088     while (1) delay(200);
00089   }
00090
00091   Serial.println(F("Magnetometer ready"));
00092 }
00093
00094 /**
00095  * @brief Read and print magnetometer data in microtesla.
00096  */
00097 void loop() {
00098   float mx, my, mz;
00099
00100   /* Read magnetometer data
00101    * - Output in microtesla (uT)
00102    * - Returns true on success
00103    */
00104   if (imu.readMag(mx, my, mz)) {
00105     Serial.print(F("MAG [uT]: "));
00106     Serial.print(mx, 2); Serial.print(F(", "));
00107     Serial.print(my, 2); Serial.print(F(", "));
00108     Serial.println(mz, 2);
00109   } else {
00110     Serial.println(F("Mag read failed"));
00111   }
00112
00113   Serial.println(F("------------------------------"));
00114   delay(500);
00115 }
```

6.11 examples/interrupt/interrupt.ino File Reference

I2C interrupt example for the 7Semi ICM-20948.

```
#include <Wire.h>
#include <DevLab_ICM20948.h>
```

Include dependency graph for interrupt.ino:



Macros

- #define SDA_PIN 6

  I2C SDA pin used by the example board.
- #define SCL_PIN 7

  I2C SCL pin used by the example board.
- #define I2C_FREQ 400000UL

  I2C bus speed in hertz.
- #define ICM_ADDR 0x69

  ICM-20948 I2C address selected by the AD0 pin.
- #define PIN_INT D7

  MCU pin connected to the IMU interrupt output.

Functions

- void printLinea ()

  Print a separator line to Serial.
- void printDobleLinea ()

  Print a double separator line to Serial.
- void IRAM_ATTR isrHandler ()

  Interrupt service routine for the IMU data-ready signal.
- void setup ()

  Initialize I2C, verify the IMU, configure interrupts, and attach ISR.
- void loop ()

  Report interrupt events when the ISR count changes.

Variables

- DevLab_ICM20948 imu

  Global ICM-20948 driver instance.
- ICM20948_IntPinConfig pinCfg

  Interrupt pin electrical and latch configuration.

- ICM20948_IntEnableConfig intEn

  Interrupt enable flags used by this example.
- volatile uint32_t isrCount = 0

  Number of interrupt events observed by the ISR.

6.11.1 Detailed Description

I2C interrupt example for the 7Semi ICM-20948.
Configures the IMU raw-data-ready interrupt pin and counts ISR events on an Arduino-compatible board.
Definition in file interrupt.ino.

6.11.2 Macro Definition Documentation

6.11.2.1 I2C_FREQ

```
#define I2C_FREQ 400000UL
```

I2C bus speed in hertz.
Definition at line 16 of file interrupt.ino.

6.11.2.2 ICM_ADDR

```
#define ICM_ADDR 0x69
```

ICM-20948 I2C address selected by the AD0 pin.
Definition at line 18 of file interrupt.ino.

6.11.2.3 PIN_INT

```
#define PIN_INT D7
```

MCU pin connected to the IMU interrupt output.
Definition at line 20 of file interrupt.ino.

6.11.2.4 SCL_PIN

```
#define SCL_PIN 7
```

I2C SCL pin used by the example board.
Definition at line 14 of file interrupt.ino.

6.11.2.5 SDA_PIN

```
#define SDA_PIN 6
```

I2C SDA pin used by the example board.
Definition at line 12 of file interrupt.ino.

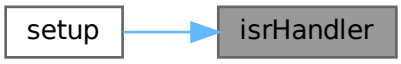
6.11.3 Function Documentation

6.11.3.1 isrHandler()

```
void IRAM_ATTR isrHandler ( )
```

Interrupt service routine for the IMU data-ready signal.
Definition at line 52 of file interrupt.ino.

Here is the caller graph for this function:



6.11.3.2 loop()

```
void loop ( )
```

Report interrupt events when the ISR count changes.
Definition at line 94 of file interrupt.ino.

6.11.3.3 printDobleLinea()

```
void printDobleLinea ( )
```

Print a double separator line to Serial.
Definition at line 28 of file interrupt.ino.

6.11.3.4 printLinea()

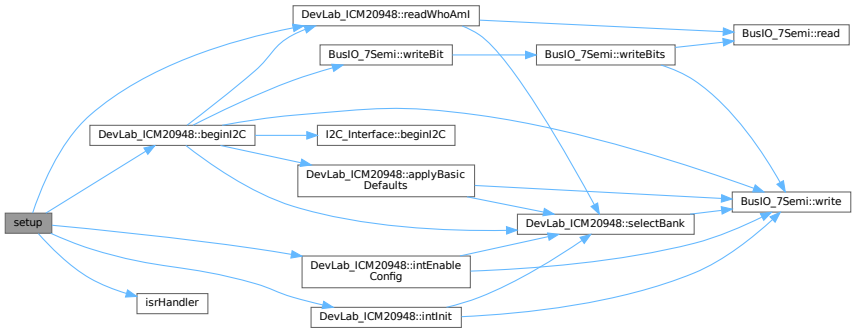
```
void printLinea ( )
```

Print a separator line to Serial.
Definition at line 26 of file interrupt.ino.

6.11.3.5 setup()

```
void setup ( )
```

Initialize I2C, verify the IMU, configure interrupts, and attach ISR.
Definition at line 59 of file interrupt.ino.
Here is the call graph for this function:



6.11.4 Variable Documentation

6.11.4.1 imu

```
DevLab_ICM20948 imu
```

Global ICM-20948 driver instance.
Definition at line 23 of file interrupt.ino.

6.11.4.2 intEn

```
ICM20948_IntEnableConfig intEn
```

Initial value:

```
= {
  .rawDataRdyEn = 1
}
```

Interrupt enable flags used by this example.
Definition at line 42 of file interrupt.ino.

6.11.4.3 isrCount

```
volatile uint32_t isrCount = 0
```

Number of interrupt events observed by the ISR.
Definition at line 47 of file interrupt.ino.

6.11.4.4 pinCfg

```
ICM20948_IntPinConfig pinCfg
```

Initial value:

```
= {
  .activeLevel   = 1,
  .driveMode     = 0,
  .latchMode     = 0,
  .clearMode     = 1,
  .fsyncActLevel = 0,
  .fsyncIntEn    = 0,
  .bypassEn      = 0
}
```

Interrupt pin electrical and latch configuration.
Definition at line 31 of file interrupt.ino.

6.12 interrupt.ino

Go to the documentation of this file.

```
00001 /**
00002  * @file interrupt.ino
00003  * @brief I2C interrupt example for the 7Semi ICM-20948.
00004  * @details Configures the IMU raw-data-ready interrupt pin and counts ISR
00005  * events on an Arduino-compatible board.
00006  */
00007
00008 #include <Wire.h>
00009 #include <DevLab_ICM20948.h>
00010
00011 /** @brief I2C SDA pin used by the example board. */
00012 #define SDA_PIN   6
00013 /** @brief I2C SCL pin used by the example board. */
00014 #define SCL_PIN   7
00015 /** @brief I2C bus speed in hertz. */
00016 #define I2C_FREQ  400000UL
00017 /** @brief ICM-20948 I2C address selected by the AD0 pin. */
00018 #define ICM_ADDR  0x69
00019 /** @brief MCU pin connected to the IMU interrupt output. */
00020 #define PIN_INT   D7   // ajusta a tu pin real
00021
00022 /** @brief Global ICM-20948 driver instance. */
00023 DevLab_ICM20948 imu;
00024
00025 /** @brief Print a separator line to Serial. */
00026 void printLinea()      {
      Serial.println(F("------------------------------------------------------------")); }
00027 /** @brief Print a double separator line to Serial. */
00028 void printDobleLinea() {
      Serial.println(F("============================================================")); }
00029
00030 /** @brief Interrupt pin electrical and latch configuration. */
00031 ICM20948_IntPinConfig pinCfg = {
00032   .activeLevel   = 1,
00033   .driveMode     = 0,
00034   .latchMode     = 0,
00035   .clearMode     = 1,
00036   .fsyncActLevel = 0,
00037   .fsyncIntEn    = 0,
00038   .bypassEn      = 0
00039 };
```

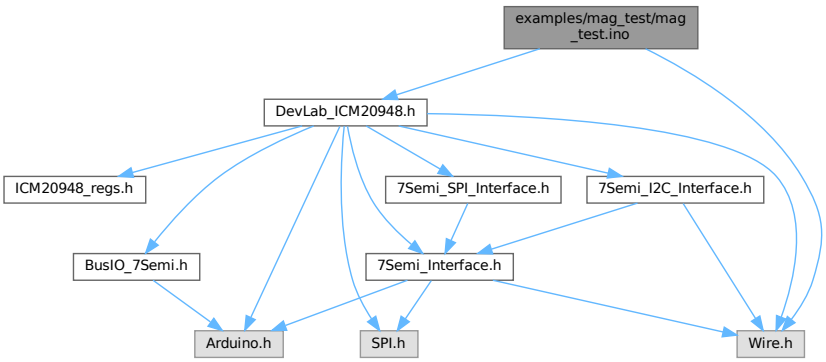
```
00040
00041 /** @brief Interrupt enable flags used by this example. */
00042 ICM20948_IntEnableConfig intEn = {
00043   .rawDataRdyEn = 1   // inicializacion directa
00044 };
00045
00046 /** @brief Number of interrupt events observed by the ISR. */
00047 volatile uint32_t isrCount = 0;
00048
00049 /**
00050  * @brief Interrupt service routine for the IMU data-ready signal.
00051  */
00052 void IRAM_ATTR isrHandler() {
00053   isrCount++;
00054 }
00055
00056 /**
00057  * @brief Initialize I2C, verify the IMU, configure interrupts, and attach ISR.
00058  */
00059 void setup() {
00060   Serial.begin(115200);
00061   delay(200);
00062
00063   Wire.begin(SDA_PIN, SCL_PIN);
00064
00065   if (!imu.beginI2C(ICM_ADDR, Wire, I2C_FREQ)) {
00066     Serial.println(F("ERROR: beginI2C() failed."));
00067     while (1) delay(200);
00068   }
00069   Serial.println(F("ICM-20948 ready."));
00070
00071   uint8_t who;
00072   if (!imu.readWhoAmI(who) || who != 0xEA) {
00073     Serial.println(F("ERROR: WHO_AM_I mismatch"));
00074     while (1) delay(200);
00075   }
00076   Serial.printf("WHO_AM_I = 0x%02X\n", who);
00077
00078   if(!imu.intInit(pinCfg)){
00079     Serial.println(F("ERROR: Initialization interruption failed"));
00080     while(1) delay(200);
00081   };           // nombre correcto
00082   if(!imu.intEnableConfig(intEn)){
00083     Serial.println(F("ERROR: Initialization interruption failed"));
00084     while(1) delay(200);
00085   }
00086
00087   attachInterrupt(digitalPinToInterrupt(PIN_INT), isrHandler, FALLING);
00088   Serial.println(F("Interrupt configured. Waiting..."));
00089 }
00090
00091 /**
00092  * @brief Report interrupt events when the ISR count changes.
00093  */
00094 void loop() {
00095   static uint32_t lastCount = 0;
00096
00097   if (isrCount != lastCount) {
00098     lastCount = isrCount;
00099     Serial.printf(">> ISR disparada! Total: %lu\n", isrCount);
00100   }
00101 }
```

6.13 examples/mag_test/mag_test.ino File Reference

Magnetometer operation-mode sweep for the 7Semi ICM-20948.

```
#include <Wire.h>
#include <DevLab_ICM20948.h>
```

Include dependency graph for mag_test.ino:



Macros

- #define SDA_PIN 6

  I2C SDA pin used by the example board.
- #define SCL_PIN 7

  I2C SCL pin used by the example board.
- #define I2C_FREQ 400000UL

  I2C bus speed in hertz.
- #define ICM_ADDR 0x69

  ICM-20948 I2C address selected by the AD0 pin.

Functions

- void printLinea ()

  Print a separator line to Serial.
- void printDobleLinea ()

  Print a double separator line to Serial.
- void applySettings ()

  Initialize the magnetometer and collect samples for each mode.
- void setup ()

  Initialize I2C, verify the IMU, enable sensors, and start the sweep.
- void loop ()

  Empty loop; this sketch runs the magnetometer sweep once in setup().

Variables

- DevLab_ICM20948 imu

  Global ICM-20948 driver instance.
- const ICM20948_Op_Mode opMode [ ]

  Magnetometer operating modes tested by the sweep.
- const char ∗ etiquetasOpModeCfg [ ]

  Text labels matching the operating modes printed in the sweep.
- const uint8_t N_OPM = 6

  Number of printable magnetometer operation modes.

6.13.1 Detailed Description

Magnetometer operation-mode sweep for the 7Semi ICM-20948.
Initializes the AK09916 magnetometer through the ICM-20948 internal I2C master, iterates through continuous measurement modes, and prints CSV-like magnetic field samples to the Serial monitor.
Definition in file mag_test.ino.

6.13.2 Macro Definition Documentation

6.13.2.1 I2C_FREQ

```
#define I2C_FREQ 400000UL
```

I2C bus speed in hertz.
Definition at line 19 of file mag_test.ino.

6.13.2.2 ICM_ADDR

```
#define ICM_ADDR 0x69
```

ICM-20948 I2C address selected by the AD0 pin.
Definition at line 21 of file mag_test.ino.

6.13.2.3 SCL_PIN

```
#define SCL_PIN 7
```

I2C SCL pin used by the example board.
Definition at line 17 of file mag_test.ino.

6.13.2.4 SDA_PIN

```
#define SDA_PIN 6
```

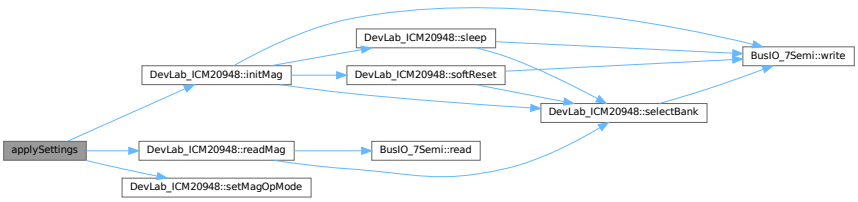
I2C SDA pin used by the example board.
Definition at line 15 of file mag_test.ino.

6.13.3 Function Documentation

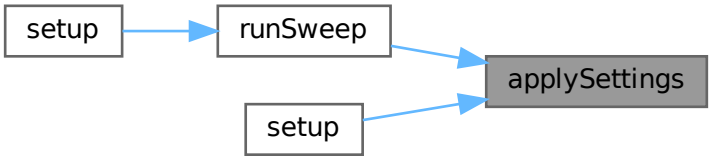
6.13.3.1 applySettings()

```
void applySettings ( )
```

Initialize the magnetometer and collect samples for each mode.
Definition at line 52 of file mag_test.ino.
Here is the call graph for this function:



Here is the caller graph for this function:



6.13.3.2 loop()

`void loop ( )`
Empty loop; this sketch runs the magnetometer sweep once in setup().
Definition at line 137 of file mag_test.ino.

6.13.3.3 printDobleLinea()

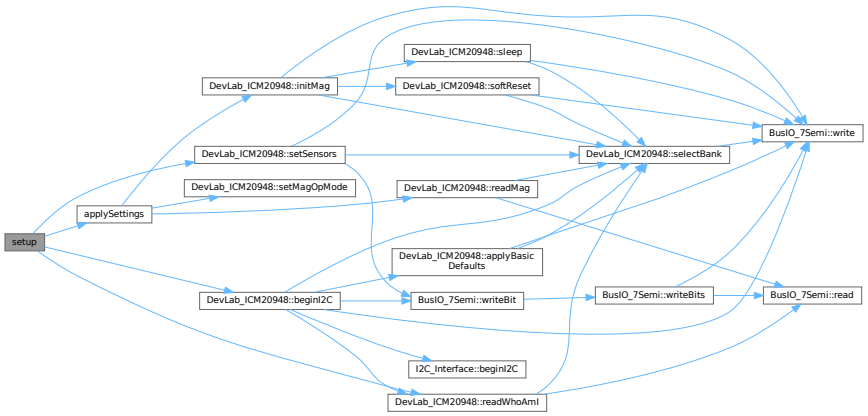
`void printDobleLinea ( )`
Print a double separator line to Serial.
Definition at line 29 of file mag_test.ino.

6.13.3.4 printLinea()

`void printLinea ( )`
Print a separator line to Serial.
Definition at line 27 of file mag_test.ino.

6.13.3.5 setup()

`void setup ( )`
Initialize I2C, verify the IMU, enable sensors, and start the sweep.
Definition at line 90 of file mag_test.ino.
Here is the call graph for this function:



6.13.4 Variable Documentation

6.13.4.1 etiquetasOpModeCfg

```
const char* etiquetasOpModeCfg[]
```

Initial value:

```
= {
  "PowerDown","SingleMeasure","Continuo1","Continuo2","Continuo3","Continuo4"
}
```

Text labels matching the operating modes printed in the sweep.
Definition at line 43 of file mag_test.ino.

6.13.4.2 imu

```
DevLab_ICM20948 imu
```

Global ICM-20948 driver instance.
Definition at line 24 of file mag_test.ino.

6.13.4.3 N_OPM

```
const uint8_t N_OPM = 6
```

Number of printable magnetometer operation modes.
Definition at line 47 of file mag_test.ino.

6.13.4.4 opMode

```
const ICM20948_Op_Mode opMode[]
```

Initial value:

```
= {
  ICM20948_Op_Mode::MODE_POWER_DOWN,
  ICM20948_Op_Mode::MODE_SINGLE_MEASUREMENT,
  ICM20948_Op_Mode::MODE_CONTINUOUS_1,
  ICM20948_Op_Mode::MODE_CONTINUOUS_2,
  ICM20948_Op_Mode::MODE_CONTINUOUS_3,
  ICM20948_Op_Mode::MODE_CONTINUOUS_4,
  ICM20948_Op_Mode::MODE_SELF_TEST
}
```

Magnetometer operating modes tested by the sweep.
Definition at line 32 of file mag_test.ino.

6.14 mag_test.ino

Go to the documentation of this file.

```
00001 /**
00002  * @file mag_test.ino
00003  * @brief Magnetometer operation-mode sweep for the 7Semi ICM-20948.
00004  * @details Initializes the AK09916 magnetometer through the ICM-20948 internal
00005  * I2C master, iterates through continuous measurement modes, and prints CSV-like
00006  * magnetic field samples to the Serial monitor.
00007  */
00008
00009 #include <Wire.h>
00010 #include <DevLab_ICM20948.h>
00011 // ------------------------------------------------------------
00012 // PINES I2C -- ESP32C6 NANO Unit Electronics
00013 // ------------------------------------------------------------
00014 /** @brief I2C SDA pin used by the example board. */
00015 #define SDA_PIN   6
00016 /** @brief I2C SCL pin used by the example board. */
00017 #define SCL_PIN   7
00018 /** @brief I2C bus speed in hertz. */
00019 #define I2C_FREQ  400000UL
00020 /** @brief ICM-20948 I2C address selected by the AD0 pin. */
00021 #define ICM_ADDR  0x69
00022
00023 /** @brief Global ICM-20948 driver instance. */
00024 DevLab_ICM20948 imu;
00025
00026 /** @brief Print a separator line to Serial. */
00027 void printLinea()      {
      Serial.println(F("------------------------------------------------------------")); }
00028 /** @brief Print a double separator line to Serial. */
```

```
00029 void printDobleLinea() {
      Serial.println(F("============================================================")); }
00030
00031 /** @brief Magnetometer operating modes tested by the sweep. */
00032 const ICM20948_Op_Mode opMode[] = {
00033   ICM20948_Op_Mode::MODE_POWER_DOWN,
00034   ICM20948_Op_Mode::MODE_SINGLE_MEASUREMENT,
00035   ICM20948_Op_Mode::MODE_CONTINUOUS_1,
00036   ICM20948_Op_Mode::MODE_CONTINUOUS_2,
00037   ICM20948_Op_Mode::MODE_CONTINUOUS_3,
00038   ICM20948_Op_Mode::MODE_CONTINUOUS_4,
00039   ICM20948_Op_Mode::MODE_SELF_TEST
00040 };
00041
00042 /** @brief Text labels matching the operating modes printed in the sweep. */
00043 const char* etiquetasOpModeCfg[] = {
00044   "PowerDown","SingleMeasure","Continuo1","Continuo2","Continuo3","Continuo4"
00045 };
00046 /** @brief Number of printable magnetometer operation modes. */
00047 const uint8_t N_OPM = 6;
00048
00049 /**
00050  * @brief Initialize the magnetometer and collect samples for each mode.
00051  */
00052 void applySettings(){
00053  /* Initialize magnetometer. */
00054   if (!imu.initMag()) {
00055     Serial.println(F("ERROR: initMag() failed"));
00056     while (1) delay(200);
00057   }
00058   for(int i = 2; i < N_OPM; i++){
00059     imu.setMagOpMode(opMode[i]);
00060      delay(100);
00061     Serial.println(F("Magnetometer ready"));
00062      float mx, my, mz;
00063
00064     /* Read magnetometer data
00065     * - Output in microtesla (uT)
00066     * - Returns true on success
00067     */
00068     for(int r = 0; r  < 500 ; r++){
00069       if (imu.readMag(mx, my, mz)) {
00070         Serial.printf("[%d]",r);
00071         Serial.print(";");
00072         Serial.print(etiquetasOpModeCfg[i]);
00073         Serial.print(";");
00074         Serial.print(mx, 2); Serial.print(F("; "));
00075         Serial.print(my, 2); Serial.print(F("; "));
00076         Serial.println(mz, 2);
00077       } else {
00078         Serial.println(F("Mag read failed"));
00079       }
00080     }
00081     Serial.println(F("------------------------------"));
00082     delay(500);
00083   }
00084
00085 }
00086
00087 /**
00088  * @brief Initialize I2C, verify the IMU, enable sensors, and start the sweep.
00089  */
00090 void setup() {
00091   // put your setup code here, to run once:
00092  Serial.begin(115200);
00093   delay(200);
00094   Serial.println(F("ICM-20948 (I2C) -- Magnetometer Example"));
00095
00096   Wire.begin(SDA_PIN,SCL_PIN);
00097   /* Initialize IMU. */
00098   if (!imu.beginI2C(ICM_ADDR, Wire, 400000)) {
00099     Serial.println(F("ERROR: beginI2C() failed."));
00100     while (1) delay(200);
00101   }
00102
00103   Serial.println(F("ICM-20948 ready (I2C)."));
00104
00105   /* Verify device. */
00106   uint8_t who;
00107   if (!imu.readWhoAmI(who) || who != 0xEA) {
00108     Serial.println(F("ERROR: WHO_AM_I mismatch"));
00109     while (1) delay(200);
00110   }
00111
00112   Serial.print(F("WHO_AM_I = 0x"));
00113   Serial.println(who, HEX);
00114
```

```
00115   /* Enable required sensors (mag uses internal I2C master)
00116    * - accel/gyro not strictly required but safe to keep ON
00117    */
00118   if (!imu.setSensors(true, true, false)) {
00119     Serial.println(F("setSensors failed."));
00120   }
00121
00122   Serial.println(F("  Sensor listo.\n"));
00123   Serial.println(F("  Sensor PLANO y QUIETO sobre la mesa."));
00124   Serial.println(F("  Iniciando en 5 segundos..."));
00125   for (int i = 5; i > 0; i--) {
00126     Serial.printf("  %d...\n", i);
00127     delay(1000);
00128   }
00129
00130   applySettings();
00131
00132 }
00133
00134 /**
00135  * @brief Empty loop; this sketch runs the magnetometer sweep once in setup().
00136  */
00137 void loop() {
00138   // put your main code here, to run repeatedly:
00139
00140 }
```

6.15 examples/spi_accel/spi_accel.ino File Reference

`#include <DevLab_ICM20948.h>`

Macros

- #define SPI_FAST_SPEED 1000000

  SPI clock used during sensor initialization and reads.

Functions

- SPIClass spi_bus (SPI)

  SPI bus object used by the IMU driver.
- void setup ()

  Configure the IMU for accelerometer-only operation over SPI.
- void loop ()

  Read and print accelerometer data in g.

Variables

- DevLab_ICM20948 imu

  Global ICM-20948 driver instance.

6.15.1 Macro Definition Documentation

6.15.1.1 SPI_FAST_SPEED

`#define SPI_FAST_SPEED 1000000`

SPI clock used during sensor initialization and reads.
Definition at line 33 of file spi_accel.ino.

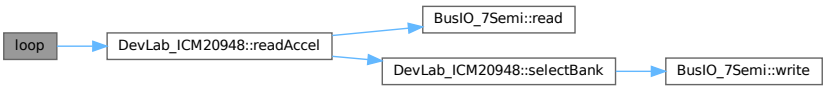
6.15.2 Function Documentation

6.15.2.1 loop()

`void loop ( )`

Read and print accelerometer data in g.
Definition at line 116 of file spi_accel.ino.

Here is the call graph for this function:



6.15.2.2 setup()

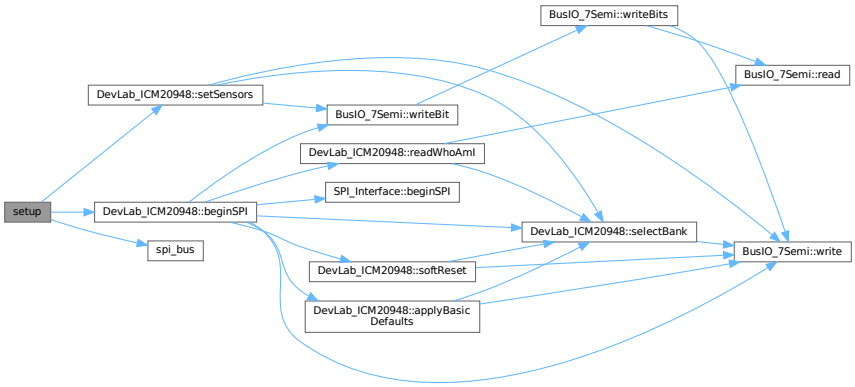
```
void setup ( )
```

Configure the IMU for accelerometer-only operation over SPI.

Note

The sketch expects a board-level CS_PIN definition to select the chip-select pin used by beginSPI().

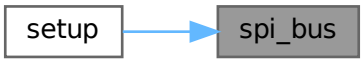
Definition at line 47 of file spi_accel.ino.
Here is the call graph for this function:



6.15.2.3 spi_bus()

```
SPIClass spi_bus (
            SPI  )
```

SPI bus object used by the IMU driver.
Here is the caller graph for this function:



6.15.3 Variable Documentation

6.15.3.1 imu

```
DevLab_ICM20948 imu
```

Global ICM-20948 driver instance.
Definition at line 36 of file spi_accel.ino.

6.16 spi_accel.ino

Go to the documentation of this file.

```
00001 /*****************************************************************
00002  * @file    SPI_Accel.ino
00003  * @brief   Minimal SPI bring-up for 7Semi ICM-20948 on ESP32 +
00004  *          continuous accelerometer readout.
00005  *
00006  * Features
00007  * - SPI init on custom pins (ESP32 VSPI)
00008  * - IMU begin() on SPI (CS pin user-defined)
00009  * - Safe defaults (applyBasicDefaults)
00010  * - Optional sensor gating: accel only
00011  * - Accel config: DLPF, full-scale, ODR, DEC3
00012  * - Read accel (g) at ~2 Hz print
00013  *
00014 * Wiring (Arduino UNO)
00015  * - SCK  -> GPIO13
00016  * - MOSI(SDA) -> GPIO12
00017  * - MISO(SDO) -> GPIO11
00018  * - CS   -> GPIO10 (changeable)
00019  * - VCC  -> 3V3
00020  * - GND  -> GND
00021  * Wiring (ESP32 VSPI default)
00022  * - SCK  -> GPIO18
00023  * - MOSI -> GPIO23
00024  * - MISO -> GPIO19
00025  * - CS   -> GPIO5 (changeable)
00026  * - VCC  -> 3V3
00027  * - GND  -> GND
00028  *****************************************************************/
00029
00030 #include <DevLab_ICM20948.h>
00031
00032 /** @brief SPI clock used during sensor initialization and reads. */
00033 #define SPI_FAST_SPEED 1000000
00034
00035 /** @brief Global ICM-20948 driver instance. */
00036 DevLab_ICM20948 imu;
00037
00038 /** @brief SPI bus object used by the IMU driver. */
00039 SPIClass spi_bus(SPI);   // usa el bus SPI por defecto del Arduino
00040
00041 /**
00042  * @brief Configure the IMU for accelerometer-only operation over SPI.
00043  *
00044  * @note The sketch expects a board-level CS_PIN definition to select the
00045  * chip-select pin used by beginSPI().
00046  */
00047 void setup() {
00048   Serial.begin(115200);
00049   delay(200);
00050
00051
00052   if (!imu.beginSPI(CS_PIN, spi_bus, 1000000)) {
00053     Serial.println("ERROR: beginSPI() failed");
00054     while (1) delay(200);
00055   }
00056
00057
00058   Serial.println(F("ICM-20948 ready."));
00059
00060   /* Optional: gate sensors
00061    * - setSensors(accel_on, gyro_on, temp_on)
00062    * - Here: enable only accel (gyro/temp off)
00063    */
00064   if (!imu.setSensors(/*accel*/ true, /*gyro*/ false, /*temp*/ false)) {
00065     Serial.println(F("setSensors failed."));
00066   }
00067   /* ---------------- ACCEL DLPF Config (reference) ---------------
00068    * ACCEL_DLPFCFG (0..7) -- 3dB & Noise Bandwidth (datasheet)
00069    * DLPF path ODR = 1125 / (1 + ACCEL_SMPLRT_DIV), DIV: 0..4095
00070    *
00071    * FCHOICE=0 (Bypass) : 3dB ≈ 1209 Hz, NBW ≈ 1248 Hz, Rate ≈ 4500 Hz
```

```
00072    * FCHOICE=1 (DLPF on):
00073    * - ACCEL_DLPFCFG_0 : 3dB ≈ 246.0 Hz,  NBW ≈ 265.0 Hz
00074    * - ACCEL_DLPFCFG_1 : 3dB ≈ 246.0 Hz,  NBW ≈ 265.0 Hz
00075    * - ACCEL_DLPFCFG_2 : 3dB ≈ 111.4 Hz,  NBW ≈ 136.0 Hz
00076    * - ACCEL_DLPFCFG_3 : 3dB ≈ 50.4  Hz,  NBW ≈ 68.8  Hz   ← good default
00077    * - ACCEL_DLPFCFG_4 : 3dB ≈ 23.9  Hz,  NBW ≈ 34.4  Hz
00078    * - ACCEL_DLPFCFG_5 : 3dB ≈ 11.5  Hz,  NBW ≈ 17.0  Hz
00079    * - ACCEL_DLPFCFG_6 : 3dB ≈ 5.7   Hz,  NBW ≈ 8.3   Hz
00080    * - ACCEL_DLPFCFG_7 : 3dB ≈ 473   Hz,  NBW ≈ 499   Hz
00081    */
00082
00083   /* Configure accelerometer
00084    * AccelConfigure(DLPF, FS_SEL, dlpf_enable, dec3, stX, stY, stZ)
00085    * - DLPF       : ACCEL_DLPFCFG_0..7 (use 3 for balanced setting)
00086    * - FS_SEL     : 0..3 => ±2/±4/±8/±16 g  (g2=0, g4=1, g8=2, g16=3)
00087    * - dlpf_enable: true to use DLPF path (recommended)
00088    * - dec3       : 0..3 (DEC3_CFG averaging)
00089    *                0 = ACCEL_DEC3_AVG_1_OR_4  (depends on FCHOICE)
00090    *                1 = ACCEL_DEC3_AVG_8
00091    *                2 = ACCEL_DEC3_AVG_16
00092    *                3 = ACCEL_DEC3_AVG_32
00093    * - stX/Y/Z    : enable self-test (false here)
00094    */
00095   if (!imu.AccelConfigure(ACCEL_DLPFCFG_3, /*FS_SEL*/ g4,
00096                           /*dlpf_enable*/ true,  // DLPF really ON
00097                           /*dec3*/ ACCEL_DEC3_AVG_8,
00098                           /*stX*/ false, /*stY*/ false, /*stZ*/ false)) {
00099     Serial.println(F("AccelConfigure failed."));
00100   }
00101
00102   /* Set accelerometer output data rate (ODR)
00103    * - Accel_SMPLRT(rate_hz)
00104    * - Base (DLPF path) = 1125 Hz
00105    * - Valid range: 1-1125 Hz
00106    * - Example: 225 Hz
00107    */
00108   if (!imu.Accel_SMPLRT(225)) {
00109     Serial.println(F("Accel_SMPLRT failed."));
00110   }
00111 }
00112
00113 /**
00114  * @brief Read and print accelerometer data in g.
00115  */
00116 void loop() {
00117   float ax, ay, az;  // accel X/Y/Z in g
00118
00119   /* - readAccel(x,y,z)
00120    * - Returns:
00121    *   - true on success;
00122    * - Output:
00123    *   - ax/ay/az in g
00124    */
00125   if (imu.readAccel(ax, ay, az)) {
00126     Serial.print("ACCEL [g]: ");
00127     Serial.print(ax, 3);
00128     Serial.print(", ");
00129     Serial.print(ay, 3);
00130     Serial.print(", ");
00131     Serial.println(az, 3);
00132   } else {
00133     Serial.println(F("ACCEL read failed"));
00134   }
00135
00136   Serial.println(F("-----------------------------"));
00137   delay(500);
00138 }
```

6.17 examples/spi_basic/spi_basic.ino File Reference

```
#include <DevLab_ICM20948.h>
```

Macros

- #define SPI_FAST_SPEED 1000000

  SPI clock used during sensor initialization and reads.

Functions

- SPIClass spi_bus (SPI)

  SPI bus object used by the IMU driver.
- void setup ()

  Initialize Serial and start the ICM-20948 over SPI.
- void loop ()

  Read and print accelerometer, gyroscope, and temperature data.

Variables

- DevLab_ICM20948 imu

  Global ICM-20948 driver instance.

6.17.1 Macro Definition Documentation

6.17.1.1 SPI_FAST_SPEED

```
#define SPI_FAST_SPEED 1000000
```

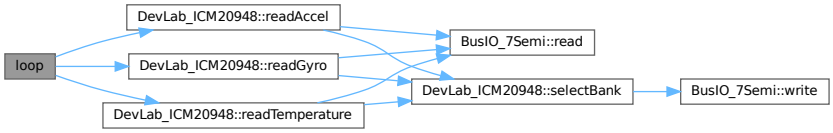
SPI clock used during sensor initialization and reads.
Definition at line 34 of file spi_basic.ino.

6.17.2 Function Documentation

6.17.2.1 loop()

```
void loop ( )
```

Read and print accelerometer, gyroscope, and temperature data.
Definition at line 65 of file spi_basic.ino.
Here is the call graph for this function:



6.17.2.2 setup()

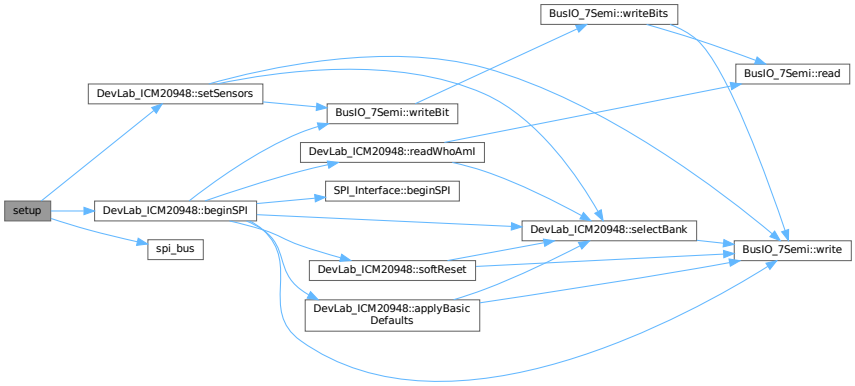
```
void setup ( )
```

Initialize Serial and start the ICM-20948 over SPI.

Note

The sketch expects a board-level CS_PIN definition to select the chip-select pin used by beginSPI().

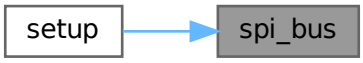
Definition at line 48 of file spi_basic.ino.
Here is the call graph for this function:



6.17.2.3 spi_bus()

```
SPIClass spi_bus (
            SPI  )
```

SPI bus object used by the IMU driver.
Here is the caller graph for this function:



6.17.3 Variable Documentation

6.17.3.1 imu

```
DevLab_ICM20948 imu
```

Global ICM-20948 driver instance.
Definition at line 37 of file spi_basic.ino.

6.18 spi_basic.ino

Go to the documentation of this file.

```
00001 /*************************************************************
00002  * @file    SPI_Basic.ino
00003  * @brief   Minimal SPI bring-up for 7Semi ICM-20948 +
00004  *          basic Accel/Gyro/Temp readout (updated API).
00005  *
00006  * Features
00007  * - SPI init (UNO / ESP32)
00008  * - beginSPI() initialization
00009  * - Manual sensor enable
00010  * - Read Accel / Gyro / Temp
```

```
00011  *
00012  * Notes
00013  * - WHO_AM_I must be 0xEA
00014  * - Sensors must be explicitly enabled
00015  * - SPI speed ~1MHz recommended during init
00016  *
00017  * Wiring (Arduino UNO SPI)
00018  * - SCK  -> D13
00019  * - MOSI -> D11
00020  * - MISO -> D12
00021  * - CS   -> D10 (changeable; update CS_PIN)
00022  * - VCC  -> 3V3
00023  * - GND  -> GND
00024  *
00025  * Wiring (ESP32 VSPI default)
00026  * - SCK  -> D13
00027  * - MOSI -> D11
00028  * - MISO -> D12
00029  * - CS   -> D10
00030  ****************************************************************/
00031 #include <DevLab_ICM20948.h>
00032
00033 /** @brief SPI clock used during sensor initialization and reads. */
00034 #define SPI_FAST_SPEED 1000000
00035
00036 /** @brief Global ICM-20948 driver instance. */
00037 DevLab_ICM20948 imu;
00038
00039 /** @brief SPI bus object used by the IMU driver. */
00040 SPIClass spi_bus(SPI);
00041
00042 /**
00043  * @brief Initialize Serial and start the ICM-20948 over SPI.
00044  *
00045  * @note The sketch expects a board-level CS_PIN definition to select the
00046  * chip-select pin used by beginSPI().
00047  */
00048 void setup() {
00049   Serial.begin(115200);
00050   delay(200);
00051
00052
00053   if (!imu.beginSPI(CS_PIN, spi_bus, 1000000)) {
00054     Serial.println("ERROR: beginSPI() failed");
00055     while (1) delay(200);
00056   }
00057
00058   imu.setSensors(true, true, true);
00059   Serial.println("Ready.");
00060 }
00061
00062 /**
00063  * @brief Read and print accelerometer, gyroscope, and temperature data.
00064  */
00065 void loop() {
00066   float ax, ay, az;
00067   float gx, gy, gz;
00068   float tC;
00069
00070   /* Read accelerometer. */
00071   if (imu.readAccel(ax, ay, az)) {
00072     Serial.print(F("ACC [g]: "));
00073     Serial.print(ax, 3); Serial.print(", ");
00074     Serial.print(ay, 3); Serial.print(", ");
00075     Serial.println(az, 3);
00076   } else {
00077     Serial.println(F("ACC read failed"));
00078   }
00079
00080   /* Read gyroscope. */
00081   if (imu.readGyro(gx, gy, gz)) {
00082     Serial.print(F("GYR [dps]: "));
00083     Serial.print(gx, 2); Serial.print(", ");
00084     Serial.print(gy, 2); Serial.print(", ");
00085     Serial.println(gz, 2);
00086   } else {
00087     Serial.println(F("GYR read failed"));
00088   }
00089
00090   /* Read temperature. */
00091   if (imu.readTemperature(tC)) {
00092     Serial.print(F("TMP [C]: "));
00093     Serial.println(tC, 2);
00094   } else {
00095     Serial.println(F("TMP read failed"));
00096   }
00097
```

```
00098   Serial.println(F("-----------------------------"));
00099   delay(500);
00100 }
```

6.19 examples/spi_gyro/spi_gyro.ino File Reference

```
#include <DevLab_ICM20948.h>
```

Macros

- #define SPI_FAST_SPEED 1000000

  SPI clock used during sensor initialization and reads.

Functions

- SPIClass spi_bus (SPI)

  SPI bus object used by the IMU driver.
- void setup ()

  Configure the IMU for gyroscope-only operation over SPI.
- void loop ()

  Read and print gyroscope data in degrees per second.

Variables

- DevLab_ICM20948 imu

  Global ICM-20948 driver instance.

6.19.1 Macro Definition Documentation

6.19.1.1 SPI_FAST_SPEED

```
#define SPI_FAST_SPEED 1000000
```

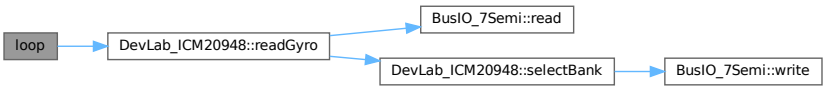
SPI clock used during sensor initialization and reads.
Definition at line 33 of file spi_gyro.ino.

6.19.2 Function Documentation

6.19.2.1 loop()

```
void loop ( )
```

Read and print gyroscope data in degrees per second.
Definition at line 117 of file spi_gyro.ino.
Here is the call graph for this function:



6.19.2.2 setup()

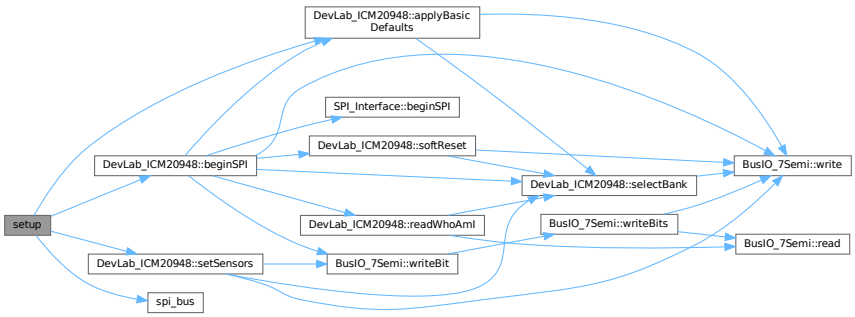
```
void setup ( )
```

Configure the IMU for gyroscope-only operation over SPI.

Note

The sketch expects a board-level CS_PIN definition to select the chip-select pin used by beginSPI().

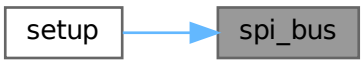
Definition at line 47 of file spi_gyro.ino.
Here is the call graph for this function:



6.19.2.3 spi_bus()

```
SPIClass spi_bus (
            SPI  )
```

SPI bus object used by the IMU driver.
Here is the caller graph for this function:



6.19.3 Variable Documentation

6.19.3.1 imu

```
DevLab_ICM20948 imu
```

Global ICM-20948 driver instance.
Definition at line 36 of file spi_gyro.ino.

6.20 spi_gyro.ino

Go to the documentation of this file.

```
00001 /*****************************************************************
00002  * @file    ICM20948_SPI_Gyro.ino
00003  * @brief   Minimal SPI bring-up for 7Semi ICM-20948 on ESP32/UNO +
00004  *          continuous gyroscope readout.
00005  *
00006  * Features
```

```
00007  * - SPI init on custom pins
00008  * - IMU begin() on SPI (CS pin user-defined)
00009  * - Safe defaults (applyBasicDefaults)
00010  * - Optional sensor gating: gyro only
00011  * - Gyro config: DLPF, full-scale, ODR, AVG
00012  * - Read gyro (dps) at ~2 Hz print
00013  *
00014  * Wiring (Arduino UNO)
00015  * - SCK  -> GPIO13
00016  * - MOSI(SDA) -> GPIO11
00017  * - MISO(SDO) -> GPIO12
00018  * - CS   -> GPIO10 (changeable)
00019  * - VCC  -> 3V3
00020  * - GND  -> GND
00021  * Wiring (ESP32 VSPI default)
00022  * - SCK  -> GPIO18
00023  * - MOSI -> GPIO23
00024  * - MISO -> GPIO19
00025  * - CS   -> GPIO5 (changeable)
00026  * - VCC  -> 3V3
00027  * - GND  -> GND
00028  ************************************************************/
00029
00030 #include <DevLab_ICM20948.h>
00031
00032 /** @brief SPI clock used during sensor initialization and reads. */
00033 #define SPI_FAST_SPEED 1000000
00034
00035 /** @brief Global ICM-20948 driver instance. */
00036 DevLab_ICM20948 imu;
00037
00038 /** @brief SPI bus object used by the IMU driver. */
00039 SPIClass spi_bus(SPI);
00040
00041 /**
00042  * @brief Configure the IMU for gyroscope-only operation over SPI.
00043  *
00044  * @note The sketch expects a board-level CS_PIN definition to select the
00045  * chip-select pin used by beginSPI().
00046  */
00047 void setup() {
00048   Serial.begin(115200);
00049   delay(200);
00050   Serial.println(F("ICM-20948 (SPI) -- Gyro Example"));
00051
00052   /* Init SPI bus on your pins. */
00053   // SPI.begin(SCK_PIN, MISO_PIN, MOSI_PIN, CS_PIN);
00054
00055   if (!imu.beginSPI(CS_PIN, spi_bus, 1000000)) {
00056     Serial.println("ERROR: beginSPI() failed");
00057     while (1) delay(200);
00058   }
00059
00060   /* Apply safe defaults. */
00061   if (!imu.applyBasicDefaults()) {
00062     Serial.println(F("ERROR: applyBasicDefaults() failed."));
00063     while (1) delay(200);
00064   }
00065
00066   Serial.println(F("ICM-20948 ready."));
00067
00068   /* Optional: gate sensors
00069    * - setSensors(accel_on, gyro_on, temp_on)
00070    * - Here: enable only gyro (accel/temp off)
00071    */
00072   if (!imu.setSensors(/*accel*/ false, /*gyro*/ true, /*temp*/ false)) {
00073     Serial.println(F("setSensors failed."));
00074   }
00075
00076   /* ----------------- GYRO DLPF Config (reference) ----------------
00077    * GyroConfig(DLPFCFG, FS_SEL, FCHOICE, XGYRO, YGYRO, ZGYRO, AVGCFG)
00078    * - DLPF     : GYRO_DLPFCFG_0..7 (use 3 for balanced setting)
00079    * - FS_SEL   : dps250 / dps500 / dps1000 / dps2000
00080    * - FCHOICE  : false → DLPF path (recommended), true → bypass (very wide BW)
00081    * - XGYRO/YGYRO/ZGYRO : per-axis enable (true = enable axis)
00082    * - AVGCFG   : 0..7 internal averaging (higher = more smoothing, more delay)
00083    * FCHOICE=0 (DLPF on):
00084    * - GYRO_DLPFCFG_0 : 3dB ≈ 12106 Hz, NBW ≈ 12316 Hz (very wide)
00085    * - GYRO_DLPFCFG_1 : 3dB ≈ 196.6 Hz, NBW ≈ 229.8 Hz
00086    * - GYRO_DLPFCFG_2 : 3dB ≈ 151.8 Hz, NBW ≈ 187.6 Hz
00087    * - GYRO_DLPFCFG_3 : 3dB ≈ 119.5 Hz, NBW ≈ 154.3 Hz   ← good default
00088    * - GYRO_DLPFCFG_4 : 3dB ≈ 51.2  Hz, NBW ≈ 73.3  Hz
00089    * - GYRO_DLPFCFG_5 : 3dB ≈ 23.9  Hz, NBW ≈ 35.9  Hz
00090    * - GYRO_DLPFCFG_6 : 3dB ≈ 5.7   Hz, NBW ≈ 8.3   Hz
00091    * - GYRO_DLPFCFG_7 : 3dB ≈ 473   Hz, NBW ≈ 499   Hz
00092    *
00093    * FCHOICE=1 (Bypass) : very wide (use with care; highest noise)
```

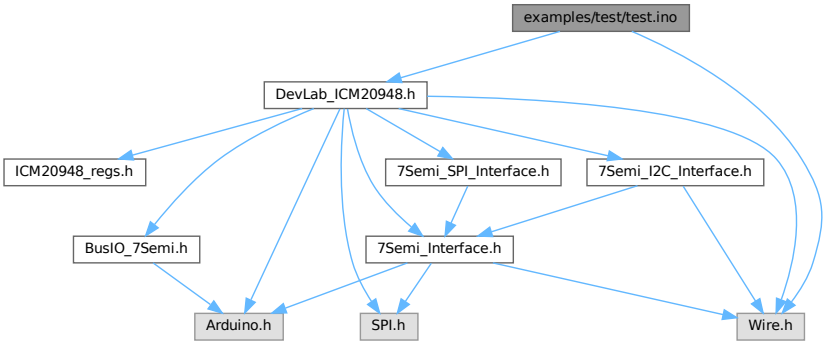
```
00094    */
00095   if (!imu.GyroConfig(/*DLPFCFG*/ GYRO_DLPFCFG_4,
00096                       /*FS_SEL*/ dps2000,
00097                       /*FCHOICE*/ true,  // DLPF ON
00098                       /*X*/ true, /*Y*/ true, /*Z*/ true,
00099                       /*AVGCFG*/ 0)) {
00100     Serial.println(F("GyroConfig failed."));
00101   }
00102
00103   /* Set gyroscope output data rate (ODR)
00104    * - Gyro_SMPLRT(rate_hz)
00105    * - Base (DLPF path) = 1100 Hz
00106    * - Valid range: ~4.3-1100 Hz
00107    * - Example: 1000 Hz
00108    */
00109   if (!imu.Gyro_SMPLRT(1000)) {
00110     Serial.println(F("Gyro_SMPLRT failed."));
00111   }
00112 }
00113
00114 /**
00115  * @brief Read and print gyroscope data in degrees per second.
00116  */
00117 void loop() {
00118   float gx, gy, gz;  // gyro X/Y/Z in dps
00119
00120   /* - readGyro(x,y,z)
00121    * - Returns:
00122    *   - true on success;
00123    * - Output:
00124    *   - gx/gy/gz in dps
00125    */
00126   if (imu.readGyro(gx, gy, gz)) {
00127     Serial.print("GYRO [dps]: ");
00128     Serial.print(gx, 3);
00129     Serial.print(", ");
00130     Serial.print(gy, 3);
00131     Serial.print(", ");
00132     Serial.println(gz, 3);
00133
00134   } else {
00135     Serial.println(F("GYRO read failed"));
00136   }
00137
00138   Serial.println(F("------------------------------"));
00139   delay(500);
00140 }
```

6.21 examples/test/test.ino File Reference

Accelerometer DLPF validation sketch for the 7Semi ICM-20948.

```
#include <Wire.h>
#include <DevLab_ICM20948.h>
```

Include dependency graph for test.ino:



Classes

- struct configDLPF

  Gyroscope DLPF divider and timing configuration.

Macros

- #define SDA_PIN 6

  I2C SDA pin used by the example board.
- #define SCL_PIN 7

  I2C SCL pin used by the example board.
- #define I2C_FREQ 400000UL

  I2C bus speed in hertz.
- #define ICM_ADDR 0x69

  ICM-20948 I2C address selected by the AD0 pin.

Functions

- void printLinea ()

  Print a separator line to Serial.
- void printDobleLinea ()

  Print a double separator line to Serial.
- bool applySettings (uint8_t fs_idx, const configDLPF &cfg)

  Apply one accelerometer full-scale and DLPF configuration.
- void firstStage ()

  Verify the IMU identity register.
- void runSweep ()

  Run the accelerometer DLPF validation sweep and print samples.
- void setup ()

  Initialize I2C, reset/wake the IMU, enable sensors, and start sweep.
- void loop ()

  Optional post-sweep loop for manual accelerometer reads.

Variables

- const configDLPF configsDLPF [ ]

  Accelerometer DLPF configurations exercised by this validation.
- const uint8_t N_DLPF = sizeof(configsDLPF) / sizeof(configsDLPF[0])

  Number of accelerometer DLPF configurations.
- const ICM20948_Accel_FullScale accelCfg [ ] = { ICM20948_Accel_FullScale::G_2, ICM20948_Accel_FullScale::G_4, ICM20948_Accel_FullScale::G_8, ICM20948_Accel_FullScale::G_16}

  Accelerometer full-scale ranges available for validation.
- const char * etiquetasAccelCfg [ ] = {"+-2G","+-4G", "+-8G", "+-16G"}

  Text labels matching accelCfg.
- const uint8_t N_FS = 4

  Number of accelerometer full-scale ranges.
- const ICM20948_Accel_DLPFCFG accelDLPF [ ] = { ICM20948_Accel_DLPFCFG::DLPF0_246HZ, ICM20948_Accel_DLPFCFG::DLPF_246HZ, ICM20948_Accel_DLPFCFG::DLPF_111HZ, ICM20948_Accel_DLPFCFG::DLPF ICM20948_Accel_DLPFCFG::DLPF_24HZ, ICM20948_Accel_DLPFCFG::DLPF_12HZ, ICM20948_Accel_DLPFCFG::DLPF_6 }

  Accelerometer DLPF enum values matching configsDLPF.
- const char * etiquetasAccelDLPFCFG [ ] = { "246/265", "246/265", "111/136", "50.4/68.8","23.9/34.4", "11.↩ 5/17.0", "5.7/8.3","473/499"}

  Text labels describing accelDLPF bandwidth pairs.

- const ICM20948_Accel_Average accelAVG [ ] = {ICM20948_Accel_Average::AVG_1_OR_4 ,ICM20948_Accel_Average::AVG_8 ICM20948_Accel_Average::AVG_16, ICM20948_Accel_Average::AVG_32}

  Accelerometer averaging settings available for validation.
- const char * etiquetasAccelAVG [ ] = {"1 OR 4","8","16","32"}

  Text labels matching accelAVG.
- const uint16_t accelSR [ ] = {0, 1, 3, 5, 7 , 10, 15, 22, 31, 63, 127, 255, 513, 1022, 2044, 4095}

  Raw accelerometer sample-rate divider values for experiments.
- const char * etiquetasSR [ ] = { "1125.0","562.5", "281.3", "187.5","140.6", "102.3", "70.3", "48.9", "35.2", "17.6", "8.8", "4.4", "2.2", "1.1", "0.55", "0.27"}

  Text labels matching accelSR output data rates.
- DevLab_ICM20948 imu

  Global ICM-20948 driver instance.
- uint8_t total_pass = 0

  Count of validation passes reserved for future checks.
- uint8_t total_fail = 0

  Count of validation failures reserved for future checks.
- uint8_t total_warn = 0

  Count of validation warnings reserved for future checks.

6.21.1 Detailed Description

Accelerometer DLPF validation sketch for the 7Semi ICM-20948.
Runs an I2C accelerometer sweep across DLPF settings, prints sample data to Serial, and is intended for bench validation with the sensor flat and stationary.
Definition in file test.ino.

6.21.2 Macro Definition Documentation

6.21.2.1 I2C_FREQ

```
#define I2C_FREQ 400000UL
```

I2C bus speed in hertz.
Definition at line 19 of file test.ino.

6.21.2.2 ICM_ADDR

```
#define ICM_ADDR 0x69
```

ICM-20948 I2C address selected by the AD0 pin.
Definition at line 25 of file test.ino.

6.21.2.3 SCL_PIN

```
#define SCL_PIN 7
```

I2C SCL pin used by the example board.
Definition at line 17 of file test.ino.

6.21.2.4 SDA_PIN

```
#define SDA_PIN 6
```

I2C SDA pin used by the example board.
Definition at line 15 of file test.ino.

6.21.3 Function Documentation

6.21.3.1 applySettings()

```
bool applySettings (
            uint8_t fs_idx,
            const configDLPF & cfg )
```

Apply one accelerometer full-scale and DLPF configuration.

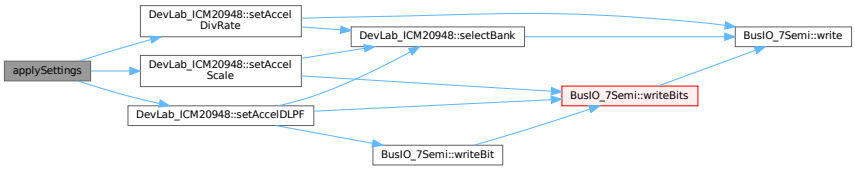
Parameters

fs_idx	Index into accelCfg.
cfg	DLPF timing and divider configuration.

Returns

true if all configuration writes succeeded, otherwise false.

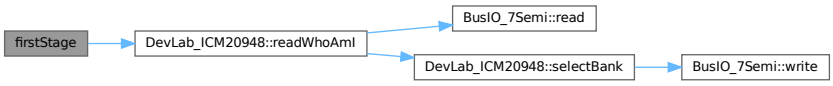
Definition at line 114 of file test.ino.
Here is the call graph for this function:



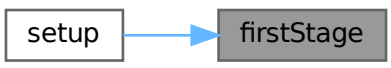
6.21.3.2 firstStage()

```
void firstStage ( )
```

Verify the IMU identity register.
Definition at line 131 of file test.ino.
Here is the call graph for this function:



Here is the caller graph for this function:



6.21.3.3 loop()

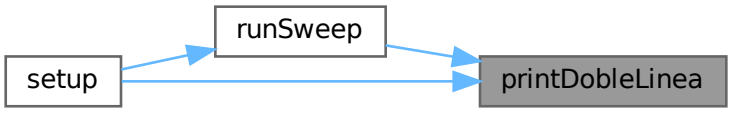
```
void loop ( )
```

Optional post-sweep loop for manual accelerometer reads.
Definition at line 225 of file test.ino.

6.21.3.4 printDobleLinea()

```
void printDobleLinea ( )
```

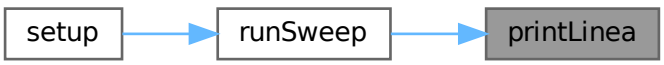
Print a double separator line to Serial.
Definition at line 103 of file test.ino.
Here is the caller graph for this function:



6.21.3.5 printLinea()

```
void printLinea ( )
```

Print a separator line to Serial.
Definition at line 98 of file test.ino.
Here is the caller graph for this function:

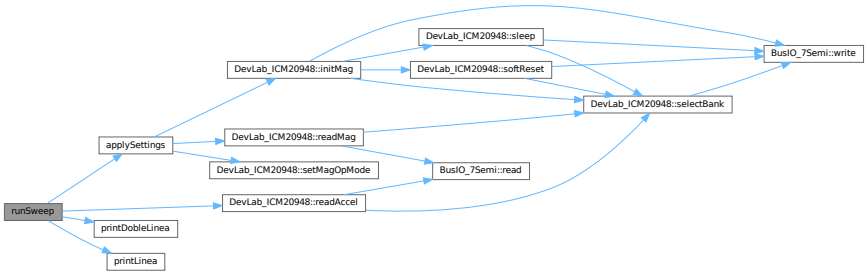


6.21.3.6 runSweep()

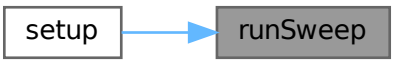
```
void runSweep ( )
```

Run the accelerometer DLPF validation sweep and print samples.
Definition at line 144 of file test.ino.

Here is the call graph for this function:



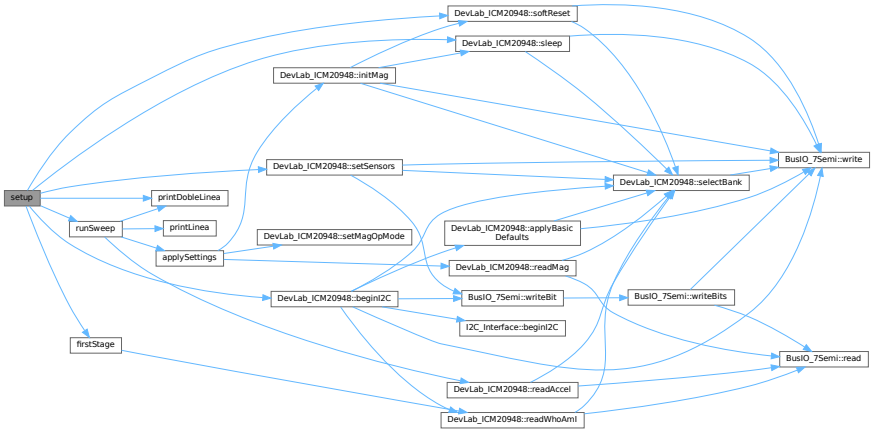
Here is the caller graph for this function:



6.21.3.7 setup()

```
void setup ( )
```

Initialize I2C, reset/wake the IMU, enable sensors, and start sweep.
Definition at line 186 of file test.ino.
Here is the call graph for this function:



6.21.4 Variable Documentation

6.21.4.1 accelAVG

```
const ICM20948_Accel_Average accelAVG[] = {ICM20948_Accel_Average::AVG_1_OR_4 ,ICM20948_Accel_Average::AVG_8,
ICM20948_Accel_Average::AVG_16, ICM20948_Accel_Average::AVG_32}
```

Accelerometer averaging settings available for validation.

Definition at line 72 of file test.ino.

6.21.4.2 accelCfg

```
const ICM20948_Accel_FullScale accelCfg[] = { ICM20948_Accel_FullScale::G_2, ICM20948_Accel_FullScale::G_4,
ICM20948_Accel_FullScale::G_8, ICM20948_Accel_FullScale::G_16}
```

Accelerometer full-scale ranges available for validation.
Definition at line 60 of file test.ino.

6.21.4.3 accelDLPF

```
const ICM20948_Accel_DLPFCFG accelDLPF[] = { ICM20948_Accel_DLPFCFG::DLPF0_246HZ, ICM20948_Accel_DLPFCFG::DLPF
ICM20948_Accel_DLPFCFG::DLPF_111HZ, ICM20948_Accel_DLPFCFG::DLPF_50HZ, ICM20948_Accel_DLPFCFG::DLPF_24HZ,
ICM20948_Accel_DLPFCFG::DLPF_12HZ, ICM20948_Accel_DLPFCFG::DLPF_6HZ,ICM20948_Accel_DLPFCFG::DLPF_473HZ
}
```

Accelerometer DLPF enum values matching configsDLPF.
Definition at line 67 of file test.ino.

6.21.4.4 accelSR

```
const uint16_t accelSR[] = {0, 1, 3, 5, 7 , 10, 15, 22, 31, 63, 127, 255, 513, 1022, 2044,
4095}
```

Raw accelerometer sample-rate divider values for experiments.
Definition at line 77 of file test.ino.

6.21.4.5 configsDLPF

```
const configDLPF configsDLPF[]
```

Initial value:

```
= {

  {  0,   0,   265.0f,  1125.0f,  "DLPF0_246 | NBW=265Hz | ODR=1125Hz (4.2x)" },
  {  1,   0,   265.0f,  1125.0f,  "DLPF_246  | NBW=265Hz | ODR=1125Hz (4.2x)" },
  {  2,   1,   136.0f,   562.5f,  "DLPF_111  | NBW=136Hz | ODR= 562Hz (4.1x)" },
  {  3,   3,    68.8f,   281.3f,  "DLPF_50   | NBW= 69Hz | ODR= 281Hz (4.1x)" },
  {  4,   7,    34.4f,   140.6f,  "DLPF_24   | NBW= 34Hz | ODR= 141Hz (4.1x)" },
  {  5,  15,    17.0f,    70.3f,  "DLPF_12   | NBW= 17Hz | ODR=  70Hz (4.1x)" },
  {  6,  31,     8.3f,    35.2f,  "DLPF_6    | NBW=  8Hz | ODR=  35Hz (4.2x)" },
  {  7,   0,   499.0f,  1125.0f,  "DLPF_473  | NBW=499Hz | ODR=1125Hz (2.3x)" },
}
```

Accelerometer DLPF configurations exercised by this validation.
Definition at line 42 of file test.ino.

6.21.4.6 etiquetasAccelAVG

```
const char* etiquetasAccelAVG[] = {"1 OR 4","8","16","32"}
```

Text labels matching accelAVG.
Definition at line 74 of file test.ino.

6.21.4.7 etiquetasAccelCfg

```
const char* etiquetasAccelCfg[] = {"+-2G","+-4G", "+-8G", "+-16G"}
```

Text labels matching accelCfg.
Definition at line 62 of file test.ino.

6.21.4.8 etiquetasAccelDLPFCFG

```
const char* etiquetasAccelDLPFCFG[] = { "246/265", "246/265", "111/136", "50.4/68.8","23.↩
9/34.4", "11.5/17.0", "5.7/8.3","473/499"}
```

Text labels describing accelDLPF bandwidth pairs.
Definition at line 69 of file test.ino.

6.21.4.9 etiquetasSR

```
const char* etiquetasSR[] = { "1125.0","562.5", "281.3", "187.5","140.6", "102.3", "70.3",
"48.9", "35.2", "17.6", "8.8", "4.4", "2.2", "1.1", "0.55", "0.27"}
```

Text labels matching accelSR output data rates.
Definition at line 79 of file test.ino.

6.21.4.10 imu

```
DevLab_ICM20948 imu
```

Global ICM-20948 driver instance.
Definition at line 84 of file test.ino.

6.21.4.11 N_DLPF

```
const uint8_t N_DLPF = sizeof(configsDLPF) / sizeof(configsDLPF[0])
```

Number of accelerometer DLPF configurations.
Definition at line 57 of file test.ino.

6.21.4.12 N_FS

```
const uint8_t N_FS = 4
```

Number of accelerometer full-scale ranges.
Definition at line 64 of file test.ino.

6.21.4.13 total_fail

```
uint8_t total_fail = 0
```

Count of validation failures reserved for future checks.
Definition at line 89 of file test.ino.

6.21.4.14 total_pass

```
uint8_t total_pass = 0
```

Count of validation passes reserved for future checks.
Definition at line 87 of file test.ino.

6.21.4.15 total_warn

```
uint8_t total_warn = 0
```

Count of validation warnings reserved for future checks.
Definition at line 91 of file test.ino.

6.22 test.ino

Go to the documentation of this file.

```
00001 /**
00002  * @file test.ino
00003  * @brief Accelerometer DLPF validation sketch for the 7Semi ICM-20948.
00004  * @details Runs an I2C accelerometer sweep across DLPF settings, prints sample
00005  * data to Serial, and is intended for bench validation with the sensor flat and
00006  * stationary.
00007  */
00008
00009 #include <Wire.h>
00010 #include <DevLab_ICM20948.h>
00011 // ---------------------------------------------------------
00012 // PINES I2C -- ESP32C6 NANO Unit Electronics
00013 // ---------------------------------------------------------
00014 /** @brief I2C SDA pin used by the example board. */
00015 #define SDA_PIN         6
00016 /** @brief I2C SCL pin used by the example board. */
00017 #define SCL_PIN         7
00018 /** @brief I2C bus speed in hertz. */
00019 #define I2C_FREQ        400000UL   // Fast Mode 400 kHz
```

```
00020
00021 // ------------------------------------------------------------
00022 // DIRECCIONES I2C
00023 // ------------------------------------------------------------
00024 /** @brief ICM-20948 I2C address selected by the AD0 pin. */
00025 #define ICM_ADDR        0x69       // AD0 = LOW (GND)
00026
00027 /** @brief Accelerometer DLPF divider and timing configuration. */
00028 struct configDLPF {
00029   /** @brief Index into the accelDLPF lookup table. */
00030   uint8_t     dlpf_idx;
00031   /** @brief Sample-rate divider written to the IMU register. */
00032   uint16_t    div;       // el valor REAL del registro, 0-4095, sin ambigüedad
00033   /** @brief Noise bandwidth in hertz for reporting. */
00034   float       nbw_hz;
00035   /** @brief Output data rate in hertz for timing calculations. */
00036   float       odr_hz;   // = 1125.0f / (1 + div), para calcular tiempos
00037   /** @brief Text label printed with each captured sample. */
00038   const char* nombre;
00039 };
00040
00041 /** @brief Accelerometer DLPF configurations exercised by this validation. */
00042 const configDLPF configsDLPF[] = {
00043   // idx  div   NBW(Hz)   ODR(Hz)    nombre
00044   {  0,   0,   265.0f,  1125.0f,  "DLPF0_246 | NBW=265Hz | ODR=1125Hz (4.2x)" },
00045   {  1,   0,   265.0f,  1125.0f,  "DLPF_246  | NBW=265Hz | ODR=1125Hz (4.2x)" },
00046   {  2,   1,   136.0f,   562.5f,  "DLPF_111  | NBW=136Hz | ODR= 562Hz (4.1x)" },
00047   {  3,   3,    68.8f,   281.3f,  "DLPF_50   | NBW= 69Hz | ODR= 281Hz (4.1x)" },
00048   {  4,   7,    34.4f,   140.6f,  "DLPF_24   | NBW= 34Hz | ODR= 141Hz (4.1x)" },
00049   {  5,  15,    17.0f,    70.3f,  "DLPF_12   | NBW= 17Hz | ODR=  70Hz (4.1x)" },
00050   {  6,  31,     8.3f,    35.2f,  "DLPF_6    | NBW=  8Hz | ODR=  35Hz (4.2x)" },
00051   {  7,   0,   499.0f,  1125.0f,  "DLPF_473  | NBW=499Hz | ODR=1125Hz (2.3x)" },
00052 };
00053 //-----------------------------------------------------
00054 //  ARRAYS DE CONFIGURACIÓN
00055 // ------------------------------------------------------------
00056 /** @brief Number of accelerometer DLPF configurations. */
00057 const uint8_t N_DLPF = sizeof(configsDLPF) / sizeof(configsDLPF[0]);
00058
00059 /** @brief Accelerometer full-scale ranges available for validation. */
00060 const ICM20948_Accel_FullScale accelCfg[] = { ICM20948_Accel_FullScale::G_2,
      ICM20948_Accel_FullScale::G_4, ICM20948_Accel_FullScale::G_8, ICM20948_Accel_FullScale::G_16};
00061 /** @brief Text labels matching accelCfg. */
00062 const char* etiquetasAccelCfg[] = {"+-2G","+-4G", "+-8G", "+-16G"};
00063 /** @brief Number of accelerometer full-scale ranges. */
00064 const uint8_t N_FS = 4;
00065
00066 /** @brief Accelerometer DLPF enum values matching configsDLPF. */
00067 const ICM20948_Accel_DLPFCFG accelDLPF[] = { ICM20948_Accel_DLPFCFG::DLPF0_246HZ,
      ICM20948_Accel_DLPFCFG::DLPF_246HZ, ICM20948_Accel_DLPFCFG::DLPF_111HZ,
      ICM20948_Accel_DLPFCFG::DLPF_50HZ, ICM20948_Accel_DLPFCFG::DLPF_24HZ,
      ICM20948_Accel_DLPFCFG::DLPF_12HZ, ICM20948_Accel_DLPFCFG::DLPF_6HZ,ICM20948_Accel_DLPFCFG::DLPF_473HZ
      };
00068 /** @brief Text labels describing accelDLPF bandwidth pairs. */
00069 const char* etiquetasAccelDLPFCFG[] = { "246/265", "246/265", "111/136", "50.4/68.8","23.9/34.4",
      "11.5/17.0", "5.7/8.3","473/499"};
00070
00071 /** @brief Accelerometer averaging settings available for validation. */
00072 const ICM20948_Accel_Average accelAVG[] = {ICM20948_Accel_Average::AVG_1_OR_4
      ,ICM20948_Accel_Average::AVG_8, ICM20948_Accel_Average::AVG_16, ICM20948_Accel_Average::AVG_32};
00073 /** @brief Text labels matching accelAVG. */
00074 const char* etiquetasAccelAVG[] = {"1 OR 4","8","16","32"};
00075
00076 /** @brief Raw accelerometer sample-rate divider values for experiments. */
00077 const uint16_t accelSR[] = {0, 1, 3, 5, 7 , 10, 15, 22, 31, 63, 127, 255, 513, 1022, 2044, 4095};
00078 /** @brief Text labels matching accelSR output data rates. */
00079 const char* etiquetasSR[] = { "1125.0","562.5", "281.3", "187.5","140.6", "102.3", "70.3", "48.9",
      "35.2", "17.6", "8.8", "4.4", "2.2", "1.1", "0.55", "0.27"};
00080 // ------------------------------------------------------------
00081 //  INSTANCIA DEL SENSOR
00082 // ------------------------------------------------------------
00083 /** @brief Global ICM-20948 driver instance. */
00084 DevLab_ICM20948 imu;
00085
00086 /** @brief Count of validation passes reserved for future checks. */
00087 uint8_t total_pass = 0;
00088 /** @brief Count of validation failures reserved for future checks. */
00089 uint8_t total_fail = 0;
00090 /** @brief Count of validation warnings reserved for future checks. */
00091 uint8_t total_warn = 0;
00092
00093 // ============================================================
00094 //  UTILIDADES DE IMPRESIÓN
00095 // ============================================================
00096
00097 /** @brief Print a separator line to Serial. */
00098 void printLinea() {
```

```
00099   Serial.println(F("------------------------------------------------------------"));
00100 }
00101
00102 /** @brief Print a double separator line to Serial. */
00103 void printDobleLinea() {
00104   Serial.println(F("============================================================"));
00105 }
00106
00107 /**
00108  * @brief Apply one accelerometer full-scale and DLPF configuration.
00109  *
00110  * @param fs_idx Index into accelCfg.
00111  * @param cfg DLPF timing and divider configuration.
00112  * @return true if all configuration writes succeeded, otherwise false.
00113  */
00114 bool applySettings(uint8_t fs_idx,const configDLPF &cfg){
00115   uint8_t dlpf_val = (uint8_t)accelDLPF[cfg.dlpf_idx];
00116   bool ok = true;
00117
00118   ok &= imu.setAccelScale(accelCfg[fs_idx]);
00119
00120   ok &= imu.setAccelDLPF(dlpf_val,false);
00121
00122   ok &= imu.setAccelDivRate(cfg.div);
00123
00124   return ok;
00125 }
00126
00127
00128 /**
00129  * @brief Verify the IMU identity register.
00130  */
00131 void firstStage(){
00132   uint8_t who;
00133   if (!imu.readWhoAmI(who) || who != 0xEA) {
00134     Serial.println(F("ERROR: WHO_AM_I mismatch"));
00135     while (1) delay(200);
00136   }
00137   Serial.print(F("WHO_AM_I = 0x"));
00138   Serial.println(who, HEX);
00139 }
00140
00141 /**
00142  * @brief Run the accelerometer DLPF validation sweep and print samples.
00143  */
00144 void runSweep(){
00145   printDobleLinea();
00146   Serial.println("Acelerometro ICM-200948 | UNIT Electronics");
00147   Serial.printf("%d DLPFS x %d FS = %d combinaciones\n",N_DLPF,N_FS,N_DLPF * N_FS);
00148   Serial.println(F("Coloque el sensor plano y quieto mirando hacia arriba"));
00149   printDobleLinea();
00150   for(uint8_t d = 0; d < N_DLPF; d++){
00151     Serial.printf(" BLOQUE %d/%d: %s\n", d+1, N_DLPF,configsDLPF[d].nombre);
00152     Serial.printf(" ODR = %.1fHz | Delay = %lums | Periodo=
      %.1fms\n",configsDLPF[d].odr_hz,max((uint32_t)(10000.0f/configsDLPF[d].odr_hz),(uint32_t)50),
00153                   1000.0f/configsDLPF[d].odr_hz);
00154   printDobleLinea();
00155   bool cfg_ok = applySettings(configsDLPF[d].dlpf_idx, configsDLPF[d]);
00156   if (!cfg_ok) {
00157     Serial.println(F("  ERROR: applySettings() fallo"));
00158     continue;
00159   }
00160   // Esperar que el filtro estabilice (10 periodos del ODR, min 50ms)
00161     uint32_t settle_ms = max((uint32_t)(10000.0f / configsDLPF[d].odr_hz),
00162                              (uint32_t)50);
00163     delay(settle_ms);
00164
00165
00166     // Tomar 3 lecturas espaciadas por el periodo del ODR
00167     uint32_t periodo_ms = max((uint32_t)(1000.0f / configsDLPF[d].odr_hz),
00168                               (uint32_t)1);
00169     float ax, ay, az;
00170     for (uint8_t r = 0; r < 3; r++) {
00171       delay(periodo_ms);
00172       if (imu.readAccel(ax, ay, az)) {
00173         Serial.printf("  [%d] Ax=%+.3f  Ay=%+.3f  Az=%+.3f g\n",
00174                       r+1, ax, ay, az);
00175       } else {
00176         Serial.println(F("  [?] readAccel() fallo"));
00177       }
00178     }
00179   printLinea();
00180   }
00181 }
00182
00183 /**
00184  * @brief Initialize I2C, reset/wake the IMU, enable sensors, and start sweep.
```

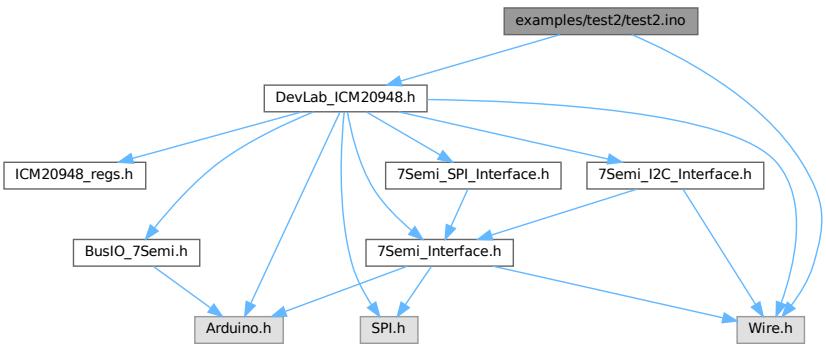
```
00185  */
00186 void setup() {
00187
00188   Serial.begin(115200);
00189   delay(200);
00190   Serial.println(F("ICM-20948 -- I2C Basic"));
00191   printDobleLinea();
00192   Serial.println(F("  ICM-20948 VALIDACION ACELEROMETRO"));
00193   printDobleLinea();
00194   // I2C init (UNO/ESP32 defaults). For ESP32 custom pins: Wire.begin(SDA,SCL);
00195   Wire.begin(SDA_PIN,SCL_PIN);
00196   // Attach IMU
00197   if (!imu.beginI2C(ICM_ADDR,Wire,I2C_FREQ)) {
00198     Serial.println(F("ERROR: begin(I2C) failed"));
00199     while (1) delay(200);
00200   }
00201
00202   firstStage();
00203
00204   imu.softReset();
00205
00206   imu.sleep(false);
00207   delay(50);
00208   /* Enable all sensors. */
00209   if (!imu.setSensors(true, true, true)) {
00210     Serial.println(F("ERROR: setSensors failed"));
00211   }
00212   Serial.println(F("  Sensor listo.\n"));
00213   Serial.println(F("  Sensor PLANO y QUIETO sobre la mesa."));
00214   Serial.println(F("  Iniciando en 5 segundos..."));
00215   for (int i = 5; i > 0; i--) {
00216     Serial.printf("  %d...\n", i);
00217     delay(1000);
00218   }
00219   runSweep();

00220 }
00221
00222 /**
00223  * @brief Optional post-sweep loop for manual accelerometer reads.
00224  */
00225 void loop() {
00226   /*float ax, ay, az;
00227   if (imu.readAccel(ax, ay, az)) {
00228     Serial.printf("X:%.3f  Y:%.3f  Z:%.3f\n", ax, ay, az);
00229   }
00230   delay(200);*/
00231 }
```

6.23 examples/test2/test2.ino File Reference

Extended accelerometer DLPF, full-scale, and averaging sweep.

```
#include <Wire.h>
#include <DevLab_ICM20948.h>
```

Include dependency graph for test2.ino:



Classes

- struct configDLPF

  Gyroscope DLPF divider and timing configuration.

Macros

- #define SDA_PIN 6

  I2C SDA pin used by the example board.
- #define SCL_PIN 7

  I2C SCL pin used by the example board.
- #define I2C_FREQ 400000UL

  I2C bus speed in hertz.
- #define ICM_ADDR 0x69

  ICM-20948 I2C address selected by the AD0 pin.

Functions

- void printLinea ()

  Print a separator line to Serial.
- void printDobleLinea ()

  Print a double separator line to Serial.
- void firstStage ()

  Verify the IMU identity register.
- bool applySettings (uint8_t fs_idx, const configDLPF &cfg, uint8_t avg)

  Apply one accelerometer full-scale, DLPF, and averaging configuration.
- void runSweep ()

  Run the full accelerometer validation matrix and print samples.
- void setup ()

  Initialize I2C, reset/wake the IMU, enable sensors, and start sweep.
- void loop ()

  Optional post-sweep loop for manual accelerometer reads.

Variables

- const configDLPF configsDLPF [ ]

  Accelerometer DLPF configurations exercised by this sweep.
- const uint8_t N_DLPF = sizeof(configsDLPF) / sizeof(configsDLPF[0])

  Number of accelerometer DLPF configurations.
- const ICM20948_Accel_FullScale accelCfg [ ]

  Accelerometer full-scale ranges exercised by the sweep.
- const char * etiquetasAccelCfg [ ] = { "+-2G","+-4G","+-8G","+-16G" }

  Text labels matching accelCfg.
- const uint8_t N_FS = 4

  Number of accelerometer full-scale ranges.
- const ICM20948_Accel_DLPFCFG accelDLPF [ ]

  Accelerometer DLPF enum values matching configsDLPF.
- const ICM20948_Accel_Average accelAVG [ ]

  Accelerometer averaging settings exercised by the sweep.
- const char * etiquetasAccelAVG [ ] = { "14","8","16","32" }

  Text labels matching accelAVG.
- const uint8_t N_AVG = 4

  Number of accelerometer averaging settings.

- const uint16_t accelSR [ ]

  Raw accelerometer sample-rate divider values for experiments.
- const char ∗ etiquetasSR [ ]

  Text labels matching accelSR output data rates.
- DevLab_ICM20948 imu

  Global ICM-20948 driver instance.
- uint8_t total_pass = 0

  Count of validation passes reserved for future checks.
- uint8_t total_fail = 0

  Count of validation failures reserved for future checks.
- uint8_t total_warn = 0

  Count of validation warnings reserved for future checks.

6.23.1 Detailed Description

Extended accelerometer DLPF, full-scale, and averaging sweep.
Runs a larger I2C validation matrix for the 7Semi ICM-20948 accelerometer and prints CSV-like samples for later analysis.
Definition in file test2.ino.

6.23.2 Macro Definition Documentation

6.23.2.1 I2C_FREQ

```
#define I2C_FREQ 400000UL
```

I2C bus speed in hertz.
Definition at line 19 of file test2.ino.

6.23.2.2 ICM_ADDR

```
#define ICM_ADDR 0x69
```

ICM-20948 I2C address selected by the AD0 pin.
Definition at line 21 of file test2.ino.

6.23.2.3 SCL_PIN

```
#define SCL_PIN 7
```

I2C SCL pin used by the example board.
Definition at line 17 of file test2.ino.

6.23.2.4 SDA_PIN

```
#define SDA_PIN 6
```

I2C SDA pin used by the example board.
Definition at line 15 of file test2.ino.

6.23.3 Function Documentation

6.23.3.1 applySettings()

```
bool applySettings (
            uint8_t fs_idx,
            const configDLPF & cfg,
            uint8_t avg )
```

Apply one accelerometer full-scale, DLPF, and averaging configuration.

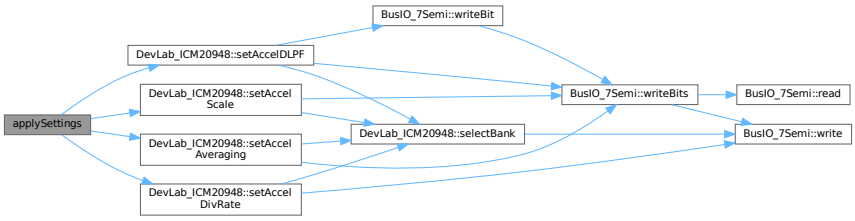
Parameters

fs_idx	Index into accelCfg.
cfg	DLPF timing and divider configuration.
avg	Index into accelAVG.

Returns

true if all configuration writes succeeded, otherwise false.

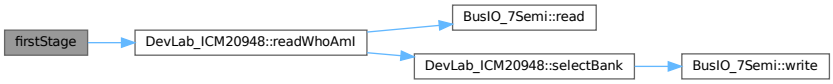
Definition at line 149 of file test2.ino.
Here is the call graph for this function:



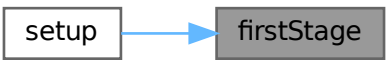
6.23.3.2 firstStage()

```
void firstStage ( )
```

Verify the IMU identity register.
Definition at line 123 of file test2.ino.
Here is the call graph for this function:



Here is the caller graph for this function:



6.23.3.3 loop()

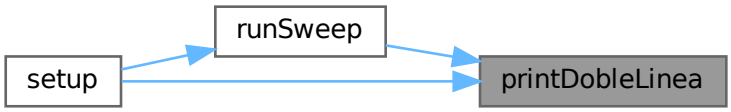
```
void loop ( )
```

Optional post-sweep loop for manual accelerometer reads.
Definition at line 298 of file test2.ino.

6.23.3.4 printDobleLinea()

```
void printDobleLinea ( )
```

Print a double separator line to Serial.
Definition at line 116 of file test2.ino.
Here is the caller graph for this function:



6.23.3.5 printLinea()

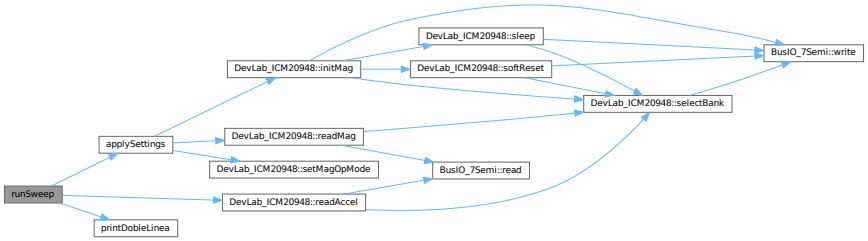
```
void printLinea ( )
```

Print a separator line to Serial.
Definition at line 114 of file test2.ino.

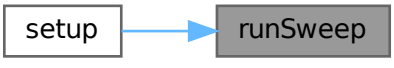
6.23.3.6 runSweep()

```
void runSweep ( )
```

Run the full accelerometer validation matrix and print samples.
Definition at line 183 of file test2.ino.
Here is the call graph for this function:



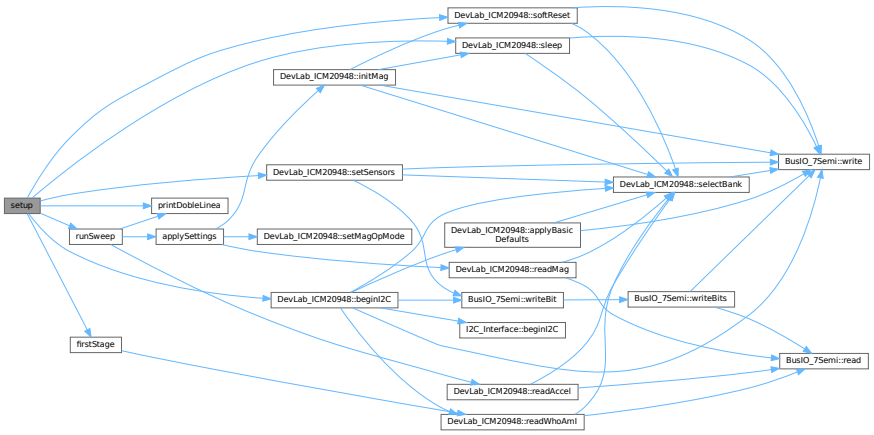
Here is the caller graph for this function:



6.23.3.7 setup()

```
void setup ( )
```

Initialize I2C, reset/wake the IMU, enable sensors, and start sweep.
Definition at line 250 of file test2.ino.
Here is the call graph for this function:



6.23.4 Variable Documentation

6.23.4.1 accelAVG

```
const ICM20948_Accel_Average accelAVG[]
```

Initial value:

```
= {
  ICM20948_Accel_Average::AVG_1_OR_4,
  ICM20948_Accel_Average::AVG_8,
  ICM20948_Accel_Average::AVG_16,
  ICM20948_Accel_Average::AVG_32
}
```

Accelerometer averaging settings exercised by the sweep.
Definition at line 82 of file test2.ino.

6.23.4.2 accelCfg

```
const ICM20948_Accel_FullScale accelCfg[]
```

Initial value:

```
= {
  ICM20948_Accel_FullScale::G_2,
  ICM20948_Accel_FullScale::G_4,
  ICM20948_Accel_FullScale::G_8,
  ICM20948_Accel_FullScale::G_16
}
```

Accelerometer full-scale ranges exercised by the sweep.
Definition at line 58 of file test2.ino.

6.23.4.3 accelDLPF

```
const ICM20948_Accel_DLPFCFG accelDLPF[]
```

Initial value:

```
= {
  ICM20948_Accel_DLPFCFG::DLPF0_246HZ,
  ICM20948_Accel_DLPFCFG::DLPF_246HZ,
  ICM20948_Accel_DLPFCFG::DLPF_111HZ,
  ICM20948_Accel_DLPFCFG::DLPF_50HZ,
  ICM20948_Accel_DLPFCFG::DLPF_24HZ,
  ICM20948_Accel_DLPFCFG::DLPF_12HZ,
  ICM20948_Accel_DLPFCFG::DLPF_6HZ,
  ICM20948_Accel_DLPFCFG::DLPF_473HZ
}
```

Accelerometer DLPF enum values matching configsDLPF.
Definition at line 70 of file test2.ino.

6.23.4.4 accelSR

```
const uint16_t accelSR[]
```

Initial value:

```
= {
  0, 1, 3, 5, 7, 10, 15, 22, 31, 63, 127, 255, 513, 1022, 2044, 4095
}
```

Raw accelerometer sample-rate divider values for experiments.
Definition at line 93 of file test2.ino.

6.23.4.5 configsDLPF

```
const configDLPF configsDLPF[]
```

Initial value:

```
= {
  {  0,   0,  265.0f, 1125.0f, "246 ; 265 ; 1125 " },
  {  1,   0,  265.0f, 1125.0f, "246 ; 265 ; 1125 " },
  {  2,   1,  136.0f,  562.5f, "111 ; 136 ; 562 " },
  {  3,   2,   68.8f,  375.0f, "50 ; 69 ; 375 " },
  {  4,   7,   34.4f,  140.6f, "24 ; 34 ; 141 " },
  {  5,  15,   17.0f,   70.3f, "12 ; 17 ;  70 " },
  {  6,  29,    8.3f,   37.5f, "6 ; 8 ;  35 " },
  {  7,   0,  499.0f, 1125.0f, "473 ; 499 ; 1125 " },
}
```

Accelerometer DLPF configurations exercised by this sweep.
Definition at line 41 of file test2.ino.

6.23.4.6 etiquetasAccelAVG

```
const char* etiquetasAccelAVG[] = { "14","8","16","32" }
```

Text labels matching accelAVG.
Definition at line 89 of file test2.ino.

6.23.4.7 etiquetasAccelCfg

```
const char* etiquetasAccelCfg[] = { "+-2G","+-4G","+-8G","+-16G" }
```

Text labels matching accelCfg.
Definition at line 65 of file test2.ino.

6.23.4.8 etiquetasSR

```
const char* etiquetasSR[]
```

Initial value:

```
= {
  "1125.0","562.5","281.3","187.5","140.6","102.3","70.3",
  "48.9","35.2","17.6","8.8","4.4","2.2","1.1","0.55","0.27"
}
```

Text labels matching accelSR output data rates.
Definition at line 97 of file test2.ino.

6.23.4.9 imu

```
DevLab_ICM20948 imu
```

Global ICM-20948 driver instance.
Definition at line 104 of file test2.ino.

6.23.4.10 N_AVG

```
const uint8_t N_AVG = 4
```

Number of accelerometer averaging settings.
Definition at line 91 of file test2.ino.

6.23.4.11 N_DLPF

```
const uint8_t N_DLPF = sizeof(configsDLPF) / sizeof(configsDLPF[0])
```
Number of accelerometer DLPF configurations.
Definition at line 52 of file test2.ino.

6.23.4.12 N_FS

```
const uint8_t N_FS = 4
```
Number of accelerometer full-scale ranges.
Definition at line 67 of file test2.ino.

6.23.4.13 total_fail

```
uint8_t total_fail = 0
```
Count of validation failures reserved for future checks.
Definition at line 108 of file test2.ino.

6.23.4.14 total_pass

```
uint8_t total_pass = 0
```
Count of validation passes reserved for future checks.
Definition at line 106 of file test2.ino.

6.23.4.15 total_warn

```
uint8_t total_warn = 0
```
Count of validation warnings reserved for future checks.
Definition at line 110 of file test2.ino.

6.24 test2.ino

Go to the documentation of this file.

```
00001 /**
00002  * @file test2.ino
00003  * @brief Extended accelerometer DLPF, full-scale, and averaging sweep.
00004  * @details Runs a larger I2C validation matrix for the 7Semi ICM-20948
00005  * accelerometer and prints CSV-like samples for later analysis.
00006  */
00007
00008 #include <Wire.h>
00009 #include <DevLab_ICM20948.h>
00010
00011 // ------------------------------------------------------------
00012 // PINES I2C -- ESP32C6 NANO Unit Electronics
00013 // ------------------------------------------------------------
00014 /** @brief I2C SDA pin used by the example board. */
00015 #define SDA_PIN   6
00016 /** @brief I2C SCL pin used by the example board. */
00017 #define SCL_PIN   7
00018 /** @brief I2C bus speed in hertz. */
00019 #define I2C_FREQ  400000UL
00020 /** @brief ICM-20948 I2C address selected by the AD0 pin. */
00021 #define ICM_ADDR  0x69
00022
00023 // ------------------------------------------------------------
00024 // STRUCT
00025 // ------------------------------------------------------------
00026 /** @brief Accelerometer DLPF divider and timing configuration. */
00027 struct configDLPF {
00028   /** @brief Index into the accelDLPF lookup table. */
00029   uint8_t     dlpf_idx;
00030   /** @brief Sample-rate divider written to the IMU register. */
00031   uint16_t    div;
00032   /** @brief Noise bandwidth in hertz for reporting. */
00033   float       nbw_hz;
00034   /** @brief Output data rate in hertz for timing calculations. */
00035   float       odr_hz;
00036   /** @brief Text label printed with each captured sample. */
00037   const char* nombre;
00038 };
00039 //id,div,nbw_hz,odr_hz,nombre : {DLPF, NBW , ODR}
```

```
00040 /** @brief Accelerometer DLPF configurations exercised by this sweep. */
00041 const configDLPF configsDLPF[] = {
00042   {  0,   0,  265.0f, 1125.0f, "246 ; 265 ; 1125 " },
00043   {  1,   0,  265.0f, 1125.0f, "246 ; 265 ; 1125 " },
00044   {  2,   1,  136.0f,  562.5f, "111 ; 136 ; 562 " },
00045   {  3,   2,   68.8f,  375.0f, "50 ; 69 ; 375 " },
00046   {  4,   7,   34.4f,  140.6f, "24 ; 34 ; 141 " },
00047   {  5,  15,   17.0f,   70.3f, "12 ; 17 ;  70 " },
00048   {  6,  29,    8.3f,   37.5f, "6 ; 8 ;  35 " },
00049   {  7,   0,  499.0f, 1125.0f, "473 ; 499 ; 1125 " },
00050 };
00051 /** @brief Number of accelerometer DLPF configurations. */
00052 const uint8_t N_DLPF = sizeof(configsDLPF) / sizeof(configsDLPF[0]);
00053
00054 // ---------------------------------------------------------
00055 // ARRAYS ORIGINALES
00056 // ---------------------------------------------------------
00057 /** @brief Accelerometer full-scale ranges exercised by the sweep. */
00058 const ICM20948_Accel_FullScale accelCfg[] = {
00059   ICM20948_Accel_FullScale::G_2,
00060   ICM20948_Accel_FullScale::G_4,
00061   ICM20948_Accel_FullScale::G_8,
00062   ICM20948_Accel_FullScale::G_16
00063 };
00064 /** @brief Text labels matching accelCfg. */
00065 const char* etiquetasAccelCfg[] = { "+-2G","+-4G","+-8G","+-16G" };
00066 /** @brief Number of accelerometer full-scale ranges. */
00067 const uint8_t N_FS = 4;
00068
00069 /** @brief Accelerometer DLPF enum values matching configsDLPF. */
00070 const ICM20948_Accel_DLPFCFG accelDLPF[] = {
00071   ICM20948_Accel_DLPFCFG::DLPF0_246HZ,
00072   ICM20948_Accel_DLPFCFG::DLPF_246HZ,
00073   ICM20948_Accel_DLPFCFG::DLPF_111HZ,
00074   ICM20948_Accel_DLPFCFG::DLPF_50HZ,
00075   ICM20948_Accel_DLPFCFG::DLPF_24HZ,
00076   ICM20948_Accel_DLPFCFG::DLPF_12HZ,
00077   ICM20948_Accel_DLPFCFG::DLPF_6HZ,
00078   ICM20948_Accel_DLPFCFG::DLPF_473HZ
00079 };
00080
00081 /** @brief Accelerometer averaging settings exercised by the sweep. */
00082 const ICM20948_Accel_Average accelAVG[] = {
00083   ICM20948_Accel_Average::AVG_1_OR_4,
00084   ICM20948_Accel_Average::AVG_8,
00085   ICM20948_Accel_Average::AVG_16,
00086   ICM20948_Accel_Average::AVG_32
00087 };
00088 /** @brief Text labels matching accelAVG. */
00089 const char* etiquetasAccelAVG[] = { "14","8","16","32" };
00090 /** @brief Number of accelerometer averaging settings. */
00091 const uint8_t N_AVG = 4;
00092 /** @brief Raw accelerometer sample-rate divider values for experiments. */
00093 const uint16_t accelSR[] = {
00094   0, 1, 3, 5, 7, 10, 15, 22, 31, 63, 127, 255, 513, 1022, 2044, 4095
00095 };
00096 /** @brief Text labels matching accelSR output data rates. */
00097 const char* etiquetasSR[] = {
00098   "1125.0","562.5","281.3","187.5","140.6","102.3","70.3",
00099   "48.9","35.2","17.6","8.8","4.4","2.2","1.1","0.55","0.27"
00100 };
00101
00102 // ---------------------------------------------------------
00103 /** @brief Global ICM-20948 driver instance. */
00104 DevLab_ICM20948 imu;
00105 /** @brief Count of validation passes reserved for future checks. */
00106 uint8_t total_pass = 0;
00107 /** @brief Count of validation failures reserved for future checks. */
00108 uint8_t total_fail = 0;
00109 /** @brief Count of validation warnings reserved for future checks. */
00110 uint8_t total_warn = 0;
00111
00112
00113 /** @brief Print a separator line to Serial. */
00114 void printLinea()      {
      Serial.println(F("------------------------------------------------------------")); }
00115 /** @brief Print a double separator line to Serial. */
00116 void printDobleLinea() {
      Serial.println(F("============================================================")); }
00117 // ---------------------------------------------------------
00118 // WHO_AM_I
00119 // ---------------------------------------------------------
00120 /**
00121  * @brief Verify the IMU identity register.
00122  */
00123 void firstStage() {
00124   uint8_t who;
```

```
00125   if (!imu.readWhoAmI(who) || who != 0xEA) {
00126     Serial.println(F("ERROR: WHO_AM_I mismatch"));
00127     while (1) delay(200);
00128   }
00129   Serial.printf("WHO_AM_I = 0x%02X  OK\n", who);
00130 }
00131
00132 // ------------------------------------------------------------
00133 // APLICAR CONFIGURACION
00134 //
00135 // FIX 1: Se agregan ambas llamadas a setAccelDLPF:
00136 //   Llamada 1 (bypass=false): escribe bits DLPFCFG, deja FCHOICE=0
00137 //   Llamada 2 (bypass=true):  activa FCHOICE=1 sin tocar DLPFCFG
00138 //
00139 // FIX 2: setAccelDivRate escribe el DIV directo al registro
00140 // ------------------------------------------------------------
00141 /**
00142  * @brief Apply one accelerometer full-scale, DLPF, and averaging configuration.
00143  *
00144  * @param fs_idx Index into accelCfg.
00145  * @param cfg DLPF timing and divider configuration.
00146  * @param avg Index into accelAVG.
00147  * @return true if all configuration writes succeeded, otherwise false.
00148  */
00149 bool applySettings(uint8_t fs_idx, const configDLPF &cfg,uint8_t avg) {
00150   uint8_t dlpf_val = (uint8_t)accelDLPF[cfg.dlpf_idx];
00151   bool ok = true;
00152
00153   ok &= imu.setAccelScale(accelCfg[fs_idx]);
00154
00155
00156   // Paso 1: escribe DLPFCFG en bits [5:3]
00157   ok &= imu.setAccelDLPF(dlpf_val, false);
00158
00159   ok &= imu.setAccelAveraging(accelAVG[avg]);
00160   // Paso 2: activa FCHOICE=1 (DLPF activo) sin tocar DLPFCFG
00161   //ok &= imu.setAccelDLPF(0, true);
00162
00163   // DIV directo al registro, sin conversion interna
00164   ok &= imu.setAccelDivRate(cfg.div);
00165
00166   return ok;
00167 }
00168
00169 // ------------------------------------------------------------
00170 // SWEEP -- MODO VERIFICACION
00171 //
00172 // Por cada configuracion DLPF:
00173 //   1. Aplica la config (FS fijo en +-2g para verificacion)
00174 //   2. Espera settle del filtro
00175 //   3. Imprime 3 lecturas reales
00176 //
00177 // Cuando las lecturas muestren Az ~1g en reposo,
00178 // el sensor esta funcionando y se puede activar el sweep completo.
00179 // ------------------------------------------------------------
00180 /**
00181  * @brief Run the full accelerometer validation matrix and print samples.
00182  */
00183 void runSweep() {
00184
00185   uint16_t n = 0;
00186   printDobleLinea();
00187   Serial.println(F("  ICM-20948 | UNIT Electronics"));
00188   Serial.println(F("  MODO VERIFICACION -- FS=+-2g fijo, 3 lecturas por config"));
00189   Serial.println(F("  Sensor PLANO y QUIETO | Az esperado: ~+1.000g"));
00190   printDobleLinea();
00191   for(uint8_t i = 0; i < N_AVG; i++){
00192     for(uint8_t j = 0; j < N_FS;j++){
00193       for (uint8_t d = 0; d < N_DLPF; d++) {
00194
00195         //Serial.println();
00196         //printLinea();
00197         /*Serial.printf("  Config %d/%d: %s\n", d+1, N_DLPF, configsDLPF[d].nombre);
00198         Serial.printf("  ODR=%.1fHz | Periodo=%.1fms\n",
00199                       configsDLPF[d].odr_hz,
00200                       1000.0f / configsDLPF[d].odr_hz);*/
00201
00202         // --- FIX: aplicar la configuracion antes de leer ---
00203         bool cfg_ok = applySettings(j, configsDLPF[d],i);  // FS=0 → +-2g
00204         if (!cfg_ok) {
00205           Serial.println(F("  ERROR: applySettings() fallo"));
00206           continue;
00207         }
00208
00209         // Esperar que el filtro estabilice (10 periodos del ODR, min 50ms)
00210         uint32_t settle_ms = max((uint32_t)(10000.0f / configsDLPF[d].odr_hz),
00211                                  (uint32_t)50);
```

```
00212         delay(settle_ms);
00213
00214         // Tomar 3 lecturas espaciadas por el periodo del ODR
00215         uint32_t periodo_ms = max((uint32_t)(1000.0f / configsDLPF[d].odr_hz),
00216                                   (uint32_t)1);
00217         float ax, ay, az;
00218         for (uint8_t r = 0; r < 100; r++) {
00219           n++;
00220           delay(periodo_ms);
00221           if (imu.readAccel(ax, ay, az)) {
00222             Serial.printf("%d ; %s ; %s ; %s ; %+.4f ; %+.4f ; %+.4f\n",
00223                           n,
00224                           configsDLPF[d].nombre,   // "246 ; 265 ; 1125"
00225                           etiquetasAccelCfg[i],     // "+-2G"
00226                           etiquetasAccelAVG[j],     // "8"
00227                           ax, ay, az);
00228           } else {
00229             Serial.println(F("  [?] readAccel() fallo"));
00230           }
00231         }
00232       }
00233     }
00234   }
00235   Serial.println();
00236   printDobleLinea();
00237   Serial.println(F("  VERIFICACION COMPLETA"));
00238   Serial.println(F("  Si Az ~ +1.0g en todos los bloques:"));
00239   Serial.println(F("  el sensor funciona correctamente."));
00240   Serial.println(F("  Activar sweep completo cuando este listo."));
00241   printDobleLinea();
00242 }
00243
00244 // ------------------------------------------------------------
00245 // SETUP
00246 // ------------------------------------------------------------
00247 /**
00248  * @brief Initialize I2C, reset/wake the IMU, enable sensors, and start sweep.
00249  */
00250 void setup() {
00251   Serial.begin(115200);
00252   delay(200);
00253   Serial.println(F("ICM-20948 -- Verificacion"));
00254
00255   printDobleLinea();
00256   Serial.println(F("  ICM-20948 VALIDACION ACELEROMETRO"));
00257   printDobleLinea();
00258
00259   Wire.begin(SDA_PIN, SCL_PIN);
00260
00261   if (!imu.beginI2C(ICM_ADDR, Wire, I2C_FREQ)) {
00262     Serial.println(F("ERROR: beginI2C() fallo"));
00263     while (1) delay(200);
00264   }
00265
00266   firstStage();
00267
00268   // softReset pone el chip a dormir
00269   // Despues del reset hay que despertar el dispositivo
00270   imu.softReset();
00271   delay(100);
00272   imu.sleep(false);   // limpia SLEEP bit, CLKSEL=1
00273   delay(50);
00274
00275   if (!imu.setSensors(true, true, true)) {
00276     Serial.println(F("ERROR: setSensors() fallo"));
00277   }
00278
00279   Serial.println(F("  Sensor listo.\n"));
00280   Serial.println(F("  Sensor PLANO y QUIETO sobre la mesa."));
00281   Serial.println(F("  Iniciando en 5 segundos..."));
00282   for (int i = 5; i > 0; i--) {
00283     Serial.printf("  %d...\n", i);
00284     delay(1000);
00285   }
00286
00287   //for (int runs = 0; runs < 20 ; runs++){
00288     runSweep();
00289   //}
00290 }
00291
00292 // ------------------------------------------------------------
00293 // LOOP -- lectura simple post-sweep
00294 // ------------------------------------------------------------
00295 /**
00296  * @brief Optional post-sweep loop for manual accelerometer reads.
00297  */
00298 void loop() {
```

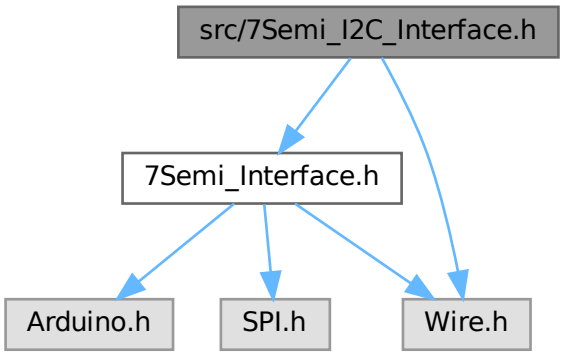
```
00299   /*float ax, ay, az;
00300   if (imu.readAccel(ax, ay, az)) {
00301     Serial.printf("X:%+.3f  Y:%+.3f  Z:%+.3f\n", ax, ay, az);
00302   }
00303   delay(500);*/
00304 }
```

6.25 README.md File Reference

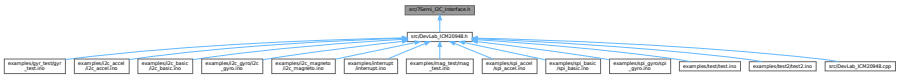
6.26 src/7Semi_I2C_Interface.h File Reference

```
#include "7Semi_Interface.h"
#include <Wire.h>
```

Include dependency graph for 7Semi_I2C_Interface.h:



This graph shows which files directly or indirectly include this file:



Classes

- class I2C_Interface

6.27 7Semi_I2C_Interface.h

Go to the documentation of this file.

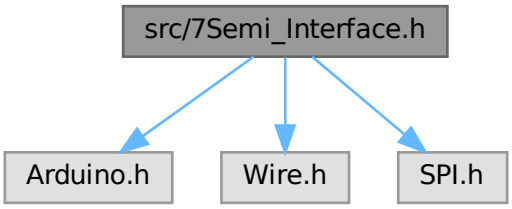
```
00001 #pragma once
00002 #include "7Semi_Interface.h"
00003 #include <Wire.h>
00004
00005 class I2C_Interface : public Interface_7Semi
00006 {
00007 public:
00008     TwoWire *i2c = nullptr;
00009     uint8_t address = 0;
00010
00011     bool beginI2C(uint8_t addr, TwoWire &wire, uint32_t speed,
00012                   uint8_t sda = 255, uint8_t scl = 255)
00013     {
00014         address = addr;
```

```
00015         i2c = &wire;
00016
00017 #if defined(ESP32)
00018         if (sda != 255 && scl != 255)
00019             i2c->begin(sda, scl);
00020         else
00021             i2c->begin();
00022 #else
00023         i2c->begin();
00024 #endif
00025         // i2c->setClock(speed);
00026
00027         // Probe device
00028         i2c->beginTransmission(address);
00029         return (i2c->endTransmission() == 0);
00030     }
00031
00032     int8_t read(uint8_t reg, uint8_t *data, uint32_t len) override
00033     {
00034         if (!i2c)
00035             return -1;
00036         i2c->beginTransmission(address);
00037         i2c->write(reg );
00038
00039         if (i2c->endTransmission(false) != 0)
00040             return -1;
00041
00042         delay(1);
00043
00044         if (len > 255)
00045             return -3;
00046         uint8_t received = i2c->requestFrom(address, (uint8_t)len);
00047         if (received != len)
00048             return -2;
00049
00050         for (uint32_t i = 0; i < len; i++)
00051         {
00052             if (!i2c->available())
00053                 return -4;
00054             data[i] = i2c->read();
00055         }
00056         // Serial.print("R-Reg: ");
00057         // Serial.print(reg, HEX);
00058         // Serial.print(" |Data: ");
00059         // for (int i = 0; i < len; i++)
00060         // {
00061         //     Serial.print(" ");
00062         //     Serial.print(data[i], HEX);
00063         // }
00064         // Serial.println();
00065
00066         return 0;
00067     }
00068
00069     bool beginSPI(uint8_t, SPIClass &, uint32_t,
00070               uint8_t, uint8_t, uint8_t) override
00071     {
00072         return false;
00073     }
00074
00075     int8_t write(uint8_t reg, const uint8_t *data, uint32_t len) override
00076     {
00077         if (!i2c)
00078             return -1;
00079
00080         i2c->beginTransmission(address);
00081         i2c->write(reg);
00082         for (uint32_t i = 0; i < len; i++)
00083             i2c->write(data[i]);
00084
00085         uint8_t status = i2c->endTransmission();
00086
00087         // Serial.print("W-Reg: ");
00088         // Serial.print(reg, HEX);
00089         // Serial.print(" |Data: ");
00090         // for (int i = 0; i < len; i++)
00091         // {
00092         //     Serial.print(" ");
00093         //     Serial.print(data[i], HEX);
00094         // }
00095         // Serial.println();
00096         return (status == 0) ? 0 : -status;
00097     }
00098 };
```

6.28 src/7Semi_Interface.h File Reference

```
#include <Arduino.h>
#include <Wire.h>
#include <SPI.h>
```

Include dependency graph for 7Semi_Interface.h:



This graph shows which files directly or indirectly include this file:

Classes

- class Interface_7Semi

6.29 7Semi_Interface.h

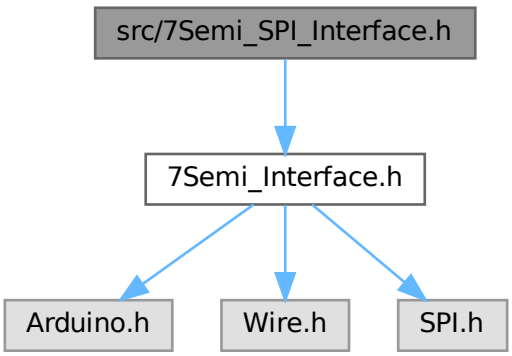
Go to the documentation of this file.

```
00001 // #pragma once
00002 #ifndef INTERFACE_7SEMI_H
00003 #define INTERFACE_7SEMI_H
00004
00005 #include <Arduino.h>
00006 #include <Wire.h>
00007 #include <SPI.h>
00008
00009 /**
00010  * 7Semi Universal Interface Layer
00011  *
00012  * - Abstract communication layer for I2C / SPI
00013  * - Ensures sensor drivers remain hardware independent
00014  */
00015 class Interface_7Semi {
00016 public:
00017
00018     virtual ~Interface_7Semi() {}
00019
00020     /**
00021      * beginI2C()
00022      *
00023      * - Initializes I2C peripheral
00024      * - Configures SDA / SCL pins if supported
00025      * - Sets communication clock
00026      *
00027      * Returns:
00028      * - true  → Initialization successful
00029      * - false → Initialization failed
00030      */
00031     virtual bool beginI2C(
```

```
00032         uint8_t address,
00033         TwoWire &wire,
00034         uint32_t speed,
00035         uint8_t sda = 255,
00036         uint8_t scl = 255) = 0;
00037
00038     virtual bool beginSPI(
00039         uint8_t cs_pin,
00040         SPIClass &wire,
00041         uint32_t speed,
00042         uint8_t csk = 255,
00043         uint8_t miso = 255,
00044         uint8_t mosi = 255) = 0;
00045
00046
00047     /**
00048      * read()
00049      *
00050      * - Reads data from device register
00051      *
00052      * Parameters:
00053      * - reg  : Register address
00054      * - data : Output buffer
00055      * - len  : Number of bytes
00056      *
00057      * Returns:
00058      * - 0  → Success
00059      * - <0 → Error
00060      */
00061     virtual int8_t read(
00062         uint8_t reg,
00063         uint8_t *data,
00064         uint32_t len) = 0;
00065
00066     /**
00067      * write()
00068      *
00069      * - Writes data to device register
00070      *
00071      * Returns:
00072      * - 0  → Success
00073      * - <0 → Error
00074      */
00075     virtual int8_t write(
00076         uint8_t reg,
00077         const uint8_t *data,
00078         uint32_t len) = 0;
00079
00080 protected:
00081     /** Delay wrapper */
00082     static void delay_us(uint32_t us)
00083     {
00084         delayMicroseconds(us);
00085     }
00086 };
00087 #endif
```

6.30 src/7Semi_SPI_Interface.h File Reference

`#include "7Semi_Interface.h"`
Include dependency graph for 7Semi_SPI_Interface.h:



This graph shows which files directly or indirectly include this file:

Classes

- class SPI_Interface

6.31 7Semi_SPI_Interface.h

Go to the documentation of this file.

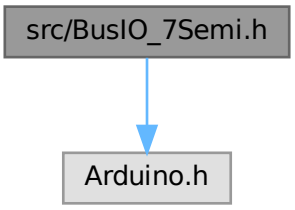
```
00001 #pragma once
00002 #include "7Semi_Interface.h"
00003
00004 /**
00005  * 7Semi SPI Interface Implementation
00006  */
00007 class SPI_Interface : public Interface_7Semi
00008 {
00009
00010 public:
00011     SPIClass *spi = nullptr;
00012     uint8_t cs = 255;
00013     uint32_t speed = 1000000;
00014
00015     /**
00016      * beginSPI()
00017      *
00018      * - Initialize SPI interface
00019      * - Configure CS and SPI bus
00020      */
00021     bool beginSPI(uint8_t csPin,
00022                   SPIClass &spiPort,
00023                   uint32_t spiSpeed,
00024                   uint8_t sck = 255,
00025                   uint8_t miso = 255,
00026                   uint8_t mosi = 255) override
00027     {
00028         if (csPin == 255)
00029             return false;
```

```
00030
00031         spi = &spiPort;
00032         cs = csPin;
00033         speed = spiSpeed;
00034
00035         pinMode(cs, OUTPUT);
00036         digitalWrite(cs, HIGH);
00037
00038 #if defined(ESP32)
00039         if (sck != 255 && miso != 255 && mosi != 255)
00040             spi->begin(sck, miso, mosi, cs);
00041         else
00042             spi->begin();
00043 #else
00044         spi->begin();
00045 #endif
00046
00047         delay(1);
00048         return true;
00049     }
00050
00051     bool beginI2C(uint8_t, TwoWire &, uint32_t,
00052                   uint8_t, uint8_t) override
00053     {
00054         return false;
00055     }
00056
00057     /**
00058      * read()
00059      *
00060      * - Read multiple bytes from register
00061      */
00062     int8_t read(uint8_t reg, uint8_t *data, uint32_t len) override
00063     {
00064         if (!spi || !data || len == 0)
00065             return -1;
00066
00067         SPISettings settings(speed, MSBFIRST, SPI_MODE0);
00068
00069         spi->beginTransaction(settings);
00070         digitalWrite(cs, LOW);
00071         delayMicroseconds(10);
00072
00073         /**
00074          * Send register address (read)
00075          */
00076         spi->transfer(reg | 0x80);
00077
00078         delayMicroseconds(10);
00079
00080         /**
00081          * Read data
00082          */
00083         for (uint32_t i = 0; i < len; i++)
00084             data[i] = spi->transfer(0x00);
00085
00086         digitalWrite(cs, HIGH);
00087         delayMicroseconds(10);
00088         spi->endTransaction();
00089
00090         // Serial.print("R-Reg: ");
00091         // Serial.print(reg, HEX);
00092         // Serial.print(" |Data: ");
00093         // for (int i = 0; i < len; i++)
00094         // {
00095         //     Serial.print(" ");
00096         //     Serial.print(data[i], HEX);
00097         // }
00098         // Serial.println();
00099
00100         return 0;
00101     }
00102
00103     /**
00104      * write()
00105      *
00106      * - Write multiple bytes to register
00107      */
00108     int8_t write(uint8_t reg, const uint8_t *data, uint32_t len) override
00109     {
00110         if (!spi || !data || len == 0)
00111             return -1;
00112
00113         SPISettings settings(speed, MSBFIRST, SPI_MODE0);
00114
00115         spi->beginTransaction(settings);
00116         digitalWrite(cs, LOW);
```

```
00117         delayMicroseconds(10);
00118
00119         /**
00120          * Send register address (write)
00121          */
00122         spi->transfer(reg & 0x7F);
00123
00124         /**
00125          * Write data
00126          */
00127         for (uint32_t i = 0; i < len; i++)
00128             spi->transfer(data[i]);
00129
00130         digitalWrite(cs, HIGH);
00131         delayMicroseconds(10);
00132         spi->endTransaction();
00133
00134         // Serial.print("W-Reg: ");
00135         // Serial.print(reg, HEX);
00136         // Serial.print(" |Data: ");
00137         // for (int i = 0; i < len; i++)
00138         // {
00139         //     Serial.print(" ");
00140         //     Serial.print(data[i], HEX);
00141         // }
00142         // Serial.println();
00143
00144         return 0;
00145     }
00146 };
```

6.32 src/BusIO_7Semi.h File Reference

`#include <Arduino.h>`
Include dependency graph for BusIO_7Semi.h:



This graph shows which files directly or indirectly include this file:

Classes

- class BusIO_7Semi< Bus >

6.33 BusIO_7Semi.h

Go to the documentation of this file.

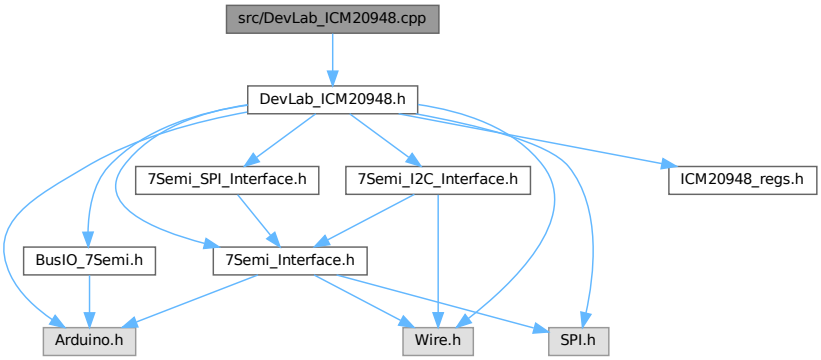
```
00001 #pragma once
00002 #include <Arduino.h>
```

```
00003
00004 template <typename Bus>
00005 class BusIO_7Semi
00006 {
00007
00008 public:
00009     /**
00010      * Constructor
00011      *
00012      * - Stores interface reference
00013      */
00014     BusIO_7Semi(Bus &busRef) : bus(busRef) {}
00015
00016     /** read 8-bit */
00017     inline bool read(uint8_t reg, uint8_t &value)
00018     {
00019         return (bus.read(reg, &value, 1) == 0);
00020     }
00021
00022     /** read 16-bit */
00023     inline bool read(uint8_t reg, uint16_t &value)
00024     {
00025         uint8_t data[2];
00026         if (bus.read(reg, data, 2) != 0)
00027             return false;
00028         value = (data[0] << 8) | data[1];
00029         return true;
00030     }
00031
00032     /** read burst */
00033     inline bool read(uint8_t reg, uint8_t *data, uint32_t len)
00034     {
00035         return (bus.read(reg, data, len) == 0);
00036     }
00037
00038     /** write 8-bit */
00039     inline bool write(uint8_t reg, uint8_t value)
00040     {
00041         return (bus.write(reg, &value, 1) == 0);
00042     }
00043
00044     /** write 16-bit */
00045     inline bool write(uint8_t reg, uint16_t value)
00046     {
00047         uint8_t data[2] = {(uint8_t)(value >> 8), (uint8_t)(value & 0xFF)};
00048         return (bus.write(reg, data, 2) == 0);
00049     }
00050
00051     /** write burst */
00052     inline bool write(uint8_t reg, const uint8_t *data, uint32_t len)
00053     {
00054         return (bus.write(reg, data, len) == 0);
00055     }
00056
00057     /**
00058      * readBits (8-bit register)
00059      */
00060     bool readBits(uint8_t reg, uint8_t pos, uint8_t len, uint8_t &value)
00061     {
00062         if (len == 0 || len > 8 || (pos + len) > 8)
00063             return false;
00064
00065         uint8_t reg_val;
00066         if (!read(reg, reg_val))
00067             return false;
00068
00069         uint8_t mask = ((uint8_t)1 << len) - 1;
00070
00071         value = (reg_val >> pos) & mask;
00072         return true;
00073     }
00074
00075     /**
00076      * readBits (16-bit register)
00077      */
00078     bool readBits(uint8_t reg, uint8_t pos, uint8_t len, uint16_t &value)
00079     {
00080         if (len == 0 || len > 16 || (pos + len) > 16)
00081             return false;
00082
00083         uint8_t reg_val;
00084         if (!read(reg, reg_val))
00085             return false;
00086
00087         uint16_t mask = ((uint16_t)1 << len) - 1;
00088
00089         value = (reg_val >> pos) & mask;
```

```
00090         return true;
00091     }
00092
00093     /**
00094      * writeBits (8-bit register)
00095      *
00096      * - Modifies specific bits in 8-bit register
00097      */
00098     bool writeBits(uint8_t reg, uint8_t pos, uint8_t len, uint8_t value)
00099     {
00100         if (len == 0 || len > 8 || (pos + len) > 8)
00101             return false;
00102
00103         uint8_t reg_val;
00104         if (!read(reg, reg_val))
00105             return false;
00106
00107         uint8_t mask = ((uint8_t)1 << len) - 1;
00108
00109         reg_val &= ~(mask << pos);
00110         reg_val |= (value & mask) << pos;
00111
00112         return write(reg, reg_val);
00113     }
00114
00115     /**
00116      * writeBits (16-bit register)
00117      *
00118      * - Modifies specific bits in 16-bit register
00119      */
00120     bool writeBits(uint8_t reg, uint8_t pos, uint8_t len, uint16_t value)
00121     {
00122         if (len == 0 || len > 16 || (pos + len) > 16)
00123             return false;
00124
00125         uint8_t reg_val;
00126         if (!read(reg, reg_val))
00127             return false;
00128
00129         uint16_t mask = ((uint16_t)1 << len) - 1;
00130
00131         reg_val &= ~(mask << pos);
00132         reg_val |= (value & mask) << pos;
00133
00134         return write(reg, reg_val);
00135     }
00136
00137     bool readBit(uint8_t reg, uint8_t pos, uint8_t &value)
00138     {
00139         return readBits(reg, pos, 1, value);
00140     }
00141
00142     bool readBit(uint8_t reg, uint8_t pos, uint16_t &value)
00143     {
00144         return readBits(reg, pos, 1, value);
00145     }
00146
00147     bool writeBit(uint8_t reg, uint8_t pos, uint8_t value)
00148     {
00149         return writeBits(reg, pos, 1, (uint8_t)value );
00150     }
00151     bool writeBit(uint8_t reg, uint8_t pos, uint16_t value)
00152     {
00153         return writeBits(reg, pos, 1, value);
00154     }
00155
00156 private:
00157     Bus &bus;
00158 };
```

6.34 src/DevLab_ICM20948.cpp File Reference

`#include "DevLab_ICM20948.h"`
Include dependency graph for DevLab_ICM20948.cpp:



6.35 DevLab_ICM20948.cpp

Go to the documentation of this file.

```
00001 #include "DevLab_ICM20948.h"
00002
00003 /** - Construct with default I2C address; call begin() to attach bus */
00004 DevLab_ICM20948::DevLab_ICM20948() {}
00005
00006 bool DevLab_ICM20948::beginI2C(uint8_t address, TwoWire &i2cPort, uint32_t i2cSpeed)
00007 {
00008   // Free previous BusIO instance
00009   if (bus)
00010   {
00011     delete bus;
00012     bus = nullptr;
00013   }
00014
00015   // Assign I2C interface implementation
00016   iface = &i2c;
00017
00018   // Initialize I2C (400kHz) and probe device
00019   if (!i2c.beginI2C(address, i2cPort, i2cSpeed))
00020     return false;
00021
00022   // Create BusIO abstraction layer
00023   bus = new BusIO_7Semi<Interface_7Semi>(*iface);
00024   if (!bus)
00025     return false;
00026
00027   // Allow device to stabilize after power-up
00028   delay(100);
00029
00030   // Read WHO_AM_I register
00031   uint8_t who_am_i;
00032   if (!readWhoAmI(who_am_i))
00033     return false;
00034
00035   // Validate device ID (expected 0xEA)
00036   if (who_am_i != WHO_AM_I_VAL)
00037     return false;
00038
00039   if (!selectBank(2))
00040     return false;
00041
00042   if (!bus->write(ODR_ALIGN_EN, (uint8_t)0x01))
00043     return false;
00044
00045   // Enable I2C interface (I2C_IF_DIS = 0)
00046   if (!bus->writeBit(USER_CTRL, 4, (uint8_t)0x00))
00047     return false;
00048
00049   // Set clock source to PLL (auto select best clock)
00050   if (!bus->write(PWR_MGMT_1, (uint8_t)0x01))
```

```
00051     return false;
00052
00053   if(!applyBasicDefaults())
00054     return false;
00055
00056   // Initialization successful
00057   return true;
00058 }
00059
00060 bool DevLab_ICM20948::beginSPI(uint8_t csPin, SPIClass &spiPort, uint32_t spiSpeed)
00061 {
00062 // Free previous BusIO instance
00063   if (bus)
00064   {
00065     delete bus;
00066     bus = nullptr;
00067   }
00068
00069   // Assign SPI interface implementation
00070   iface = &spi;
00071
00072   // Initialize SPI (1MHz) and probe device
00073   if (!spi.beginSPI(csPin, spiPort, spiSpeed))
00074     return false;
00075
00076   // Create BusIO abstraction layer
00077   bus = new BusIO_7Semi<Interface_7Semi>(*iface);
00078   if (!bus)
00079     return false;
00080
00081     softReset();
00082
00083   // Allow device to stabilize after power-up
00084   delay(100);
00085
00086
00087   // Read WHO_AM_I register
00088   uint8_t who_am_i;
00089
00090   if (!readWhoAmI(who_am_i))
00091     return false;
00092
00093   // Validate device ID (expected 0xEA)
00094   if (who_am_i != WHO_AM_I_VAL)
00095     return false;
00096
00097   if (!selectBank(2))
00098     return false;
00099
00100   if (!bus->write(ODR_ALIGN_EN, (uint8_t)0x01))
00101     return false;
00102
00103   // Enable I2C interface (I2C_IF_DIS = 0)
00104   if (!bus->writeBit(USER_CTRL, 4, (uint8_t)0x01))
00105     return false;
00106
00107   // Wake device from sleep mode (SLEEP = 0)
00108   if (!bus->write(PWR_MGMT_1,(uint8_t)0x01))
00109     return false;
00110
00111   if(!applyBasicDefaults())
00112     return false;
00113
00114   // Initialization successful
00115   return true;
00116 }
00117
00118 bool DevLab_ICM20948::readWhoAmI(uint8_t &whoAmI)
00119 {
00120   // Select USER BANK 0
00121   if (!selectBank(0))
00122     return false;
00123
00124   // Read WHO_AM_I register
00125   if (!bus->read(WHO_AM_I, whoAmI))
00126     return false;
00127
00128   // Success
00129   return true;
00130 }
00131
00132 bool DevLab_ICM20948::selectBank(uint8_t bank)
00133 {
00134   if (bank > 3)
00135     return false;
00136
00137   // Mask bank value to valid range (0-3) and shift to bits [5:4]
```

```
00138   uint8_t v = (bank & 0x03) << 4;
00139
00140   // Write bank selection to REG_BANK_SEL register
00141   return bus->write(REG_BANK_SEL, v);
00142 }
00143
00144 bool DevLab_ICM20948::softReset()
00145 {
00146   // Select USER BANK 0
00147   if (!selectBank(0))
00148     return false;
00149
00150   // Set DEVICE_RESET bit (bit 7)
00151   if (!bus->write(PWR_MGMT_1, (uint8_t)0x80))
00152     return false;
00153
00154   // Allow device to reset
00155   delay(100);
00156
00157   selectBank(2);
00158   bus->write(ODR_ALIGN_EN, (uint8_t)0x01);
00159
00160   // Success
00161   return true;
00162 }
00163
00164 bool DevLab_ICM20948::sleep(bool en)
00165 {
00166   // Select USER BANK 0
00167   if (!selectBank(0))
00168     return false;
00169  uint8_t v = 1;
00170     if(en)
00171    v|= 1<<6;
00172
00173   // Set or clear SLEEP bit (bit 6)
00174   if (!bus->write(PWR_MGMT_1, v))
00175     return false;
00176
00177   // Success
00178   return true;
00179 }
00180
00181 bool DevLab_ICM20948::applyBasicDefaults()
00182 {
00183   // Select USER BANK 0
00184   if (!selectBank(0))
00185     return false;
00186
00187   // Wake device and set clock source (CLKSEL=1, SLEEP=0)
00188   if (!bus->write(PWR_MGMT_1, (uint8_t)0x01))
00189     return false;
00190
00191   if (!bus->write(PWR_MGMT_2, (uint8_t)0x00))
00192     return false;
00193
00194
00195   // Allow device to stabilize
00196   delay(10);
00197
00198   // Disable duty-cycling (LP mode off)
00199   if (!bus->write(LP_CONFIG, (uint8_t)0x00))
00200     return false;
00201
00202   // Select USER BANK 2
00203   if (!selectBank(2))
00204     return false;
00205
00206   // Configure gyro (DLPF enabled, FS = 2000 dps)
00207   if (!bus->write(GYRO_CONFIG_1, (uint8_t)0x1F))
00208     return false;
00209
00210   // Set gyro sample rate divider (SRD = 0 → max rate)
00211   if (!bus->write(GYRO_SMPLRT_DIV, (uint8_t)0x00))
00212     return false;
00213
00214   // Configure accel (DLPF enabled, FS = 16g)
00215   if (!bus->write(ACCEL_CONFIG, (uint8_t)0x1F))
00216     return false;
00217
00218   // Set accel sample rate divider high byte
00219   if (!bus->write(ACCEL_SMPLRT_DIV_1, (uint8_t)0x00))
00220     return false;
00221
00222   // Set accel sample rate divider low byte
00223   if (!bus->write(ACCEL_SMPLRT_DIV_2, (uint8_t)0x00))
00224     return false;
```

```
00225
00226   // Select USER BANK 0
00227   if (!selectBank(0))
00228     return false;
00229
00230   // Configure interrupt pin (open-drain, active-low, latch)
00231   if (!bus->write(INT_PIN_CFG, (uint8_t)0x30))
00232     return false;
00233
00234   // Configuration successful
00235   return true;
00236 }
00237
00238 bool DevLab_ICM20948::readAccel(float &x, float &y, float &z)
00239 {
00240   // Select USER BANK 0
00241   if (!selectBank(0))
00242     return false;
00243
00244   // Read 6 bytes (X, Y, Z)
00245   uint8_t raw[6];
00246   if (!bus->read(ACCEL_XOUT_H, raw, 6))
00247     return false;
00248
00249   // Convert raw data to g
00250   x = (int16_t)((raw[0] « 8) | raw[1]) / mg_per_lsb;
00251   y = (int16_t)((raw[2] « 8) | raw[3]) / mg_per_lsb;
00252   z = (int16_t)((raw[4] « 8) | raw[5]) / mg_per_lsb;
00253
00254   // Success
00255   return true;
00256 }
00257
00258 bool DevLab_ICM20948::readGyro(float &x, float &y, float &z)
00259 {
00260   // Select USER BANK 0
00261   if (!selectBank(0))
00262     return false;
00263
00264   // Read 6 bytes (X, Y, Z)
00265   uint8_t raw[6];
00266   if (!bus->read(GYRO_XOUT_H, raw, 6))
00267     return false;
00268
00269   // Convert raw data to dps
00270   x = (int16_t)((raw[0] « 8) | raw[1]) / degree_per_second;
00271   y = (int16_t)((raw[2] « 8) | raw[3]) / degree_per_second;
00272   z = (int16_t)((raw[4] « 8) | raw[5]) / degree_per_second;
00273
00274   // Success
00275   return true;
00276 }
00277
00278 bool DevLab_ICM20948::readTemperature(float &temperature)
00279 {
00280   // Select USER BANK 0
00281   if (!selectBank(0))
00282     return false;
00283
00284   // Initialize accumulator
00285   temperature = 0;
00286
00287   // Read and average 5 samples
00288   for (int i = 0; i < 5; i++)
00289   {
00290     // Read temperature registers
00291     uint8_t raw[2];
00292     if (!bus->read(TEMP_OUT_H, raw, 2))
00293       return false;
00294
00295     // Convert raw to signed value
00296     int16_t temp = (int16_t)((raw[0] « 8) | raw[1]);
00297
00298     // Convert to °C and accumulate
00299     temperature += temp / 333.87f + 21.0f;
00300   }
00301
00302   // Compute average temperature
00303   temperature /= 5.0f;
00304
00305   // Success
00306   return true;
00307 }
00308
00309
00310 bool DevLab_ICM20948::readMag(float &x, float &y, float &z)
00311 {
```

```
00312   // Select USER BANK 0
00313   if (!selectBank(0))
00314     return false;
00315
00316   // Read 8 bytes (ST1 + XYZ + ST2)
00317   uint8_t buf[8];
00318   if (!bus->read(EXT_SLV_SENS_DATA_00, buf, 8))
00319     return false;
00320
00321   int16_t mx = (int16_t)(buf[1] | (buf[2] << 8));
00322   int16_t my = (int16_t)(buf[3] | (buf[4] << 8));
00323   int16_t mz = (int16_t)(buf[5] | (buf[6] << 8));
00324
00325   // Convert to μT (AK09916 sensitivity)
00326   x = mx * 0.15f;
00327   y = my * 0.15f;
00328   z = mz * 0.15f;
00329
00330   // Success
00331   return true;
00332 }
00333
00334 bool DevLab_ICM20948::initMag()
00335 {
00336   if (!bus)
00337     return false;
00338
00339   softReset();
00340
00341   sleep(false);
00342
00343   if (!selectBank(0))
00344     return false;
00345   // Enable I2C master mode and disable I2C slave mode
00346   if (!bus->write(USER_CTRL, (uint8_t)USER_CTRL_I2C_MST_EN))
00347     return false;
00348
00349   if (!selectBank(3))
00350     return false;
00351
00352   if (!bus->write(I2C_MST_CTRL, (uint8_t)0x07))
00353     return false;
00354
00355   delay(10);
00356
00357   //Soft Reset
00358   writeSlave4(AK_CNTL3, 0x01);
00359
00360   delay(100);
00361
00362   uint8_t i, who;
00363   for (i = 0; i < 10; i++)
00364   {
00365     readSlave4(AK_WIA2, who);
00366     if (who == AK_WIA2_VAL)
00367     {
00368       //Mode 3 (continuous measurement 100Hz)
00369       writeSlave4(AK_CNTL2, 0x08);
00370
00371       delay(10);
00372
00373       // ---- Setup auto-read (SLV0) ----
00374       if (!selectBank(3))
00375         return false;
00376
00377       if (!bus->write(I2C_SLV0_ADDR, (uint8_t)(AK09916_I2C_ADDR | 0x80)))
00378         return false;
00379
00380       if (!bus->write(I2C_SLV0_REG, (uint8_t)AK_ST1))
00381         return false;
00382
00383       if (!bus->write(I2C_SLV0_CTRL, (uint8_t)(I2C_SLVx_EN | 8)))
00384         return false;
00385
00386       return true;
00387     }
00388
00389      writeSlave4(AK_CNTL2, 0x08);
00390
00391       delay(10);
00392
00393       // ---- Setup auto-read (SLV0) ----
00394       if (!selectBank(3))
00395         return false;
00396
00397       if (!bus->write(I2C_SLV0_ADDR, (uint8_t)(AK09916_I2C_ADDR | 0x80)))
00398         return false;
```

```
00399
00400       if (!bus->write(I2C_SLV0_REG, (uint8_t)AK_ST1))
00401         return false;
00402
00403       if (!bus->write(I2C_SLV0_CTRL, (uint8_t)(I2C_SLVx_EN | 8)))
00404         return false;
00405
00406     return false;
00407   }
00408 }
00409
00410 bool DevLab_ICM20948::setMagOpMode(ICM20948_Op_Mode opMode)
00411 {
00412   // Write operation mode to magnetometer
00413   writeSlave4(AK_CNTL2, (uint8_t)opMode);
00414
00415   return true;
00416 }
00417
00418 bool DevLab_ICM20948::writeSlave4(uint8_t reg, uint8_t value)
00419 {
00420   // Select USER BANK 3
00421   if (!selectBank(3))
00422     return false;
00423
00424   // 7 - 1:r , 0:W
00425   //[6:0]: I2C slave address (AK09916_I2C_ADDR)
00426   // Set slave address (write)
00427   if (!bus->write(I2C_SLV4_ADDR, (uint8_t)AK09916_I2C_ADDR))
00428     return false;
00429
00430   // Set data to write to slave
00431   if (!bus->write(I2C_SLV4_DO, value))
00432     return false;
00433
00434   // Set target register
00435   if (!bus->write(I2C_SLV4_REG, reg))
00436     return false;
00437
00438   // Start transaction (EN bit)
00439   if (!bus->write(I2C_SLV4_CTRL, (uint8_t)128))
00440     return false;
00441   // Wait for completion (with timeout)
00442   uint32_t start = millis();
00443   uint8_t status;
00444
00445   while (millis() - start < 10)
00446   {
00447     if (bus->read(I2C_SLV4_CTRL, status))
00448       if (!(status & 0x80))
00449         return true;
00450   }
00451
00452   // Timeout
00453   return false;
00454 }
00455
00456 bool DevLab_ICM20948::readSlave4(uint8_t reg, uint8_t &value)
00457 {
00458   // Select USER BANK 3
00459   if (!selectBank(3))
00460     return false;
00461
00462   // Set slave address (read)
00463   if (!bus->write(I2C_SLV4_ADDR, (uint8_t)(AK09916_I2C_ADDR | 0x80)))
00464     return false;
00465
00466   // Set target register
00467   if (!bus->write(I2C_SLV4_REG, reg))
00468     return false;
00469
00470   // Start transaction
00471   if (!bus->write(I2C_SLV4_CTRL, (uint8_t)128))
00472     return false;
00473   // Wait for completion
00474   uint32_t start = millis();
00475   uint8_t status;
00476   while (millis() - start < 10)
00477   {
00478     if (bus->read(I2C_SLV4_CTRL, status))
00479       if (!(status & 0x80))
00480       { if (!bus->read(I2C_SLV4_DI, value))
00481     return false;
00482     return true;
00483 }
00484   }
00485     return false;
```

```
00486 }
00487
00488 bool DevLab_ICM20948::setLowPower(bool enable)
00489 {
00490   // Select USER BANK 0
00491   if (!selectBank(0))
00492     return false;
00493
00494   // Set or clear LP_EN bit (bit 5)
00495   if (!bus->writeBit(PWR_MGMT_1, 5, (uint8_t)enable))
00496     return false;
00497
00498   // Success
00499   return true;
00500 }
00501
00502 bool DevLab_ICM20948::getLowPower(bool &enable)
00503 {
00504   // Select USER BANK 0
00505   if (!selectBank(0))
00506     return false;
00507
00508   // Read LP_EN bit (bit 5)
00509   uint8_t v = 0;
00510   if (!bus->readBit(PWR_MGMT_1, 5, v))
00511     return false;
00512
00513   // Convert bit value to boolean
00514   enable = (v == 1);
00515
00516   // Success
00517   return true;
00518 }
00519
00520 bool DevLab_ICM20948::setClock(ICM20948_Clock_Source clock)
00521 {
00522   // Select USER BANK 0
00523   if (!selectBank(0))
00524     return false;
00525
00526   // Set CLKSEL bits [2:0]
00527   if (!bus->write(PWR_MGMT_1, (uint8_t)clock))
00528     return false;
00529
00530   // Success
00531   return true;
00532 }
00533
00534 bool DevLab_ICM20948::getClock(uint8_t &clock)
00535 {
00536   // Select USER BANK 0
00537   if (!selectBank(0))
00538     return false;
00539
00540   // Read CLKSEL bits [2:0]
00541   if (!bus->read(PWR_MGMT_1, clock))
00542     return false;
00543
00544     clock &= 0x07;
00545
00546   // Success
00547   return true;
00548 }
00549
00550 bool DevLab_ICM20948::setGyroSampleRate(float sampleRate)
00551 {
00552   // Validate sample rate range (approx 4.3 Hz to 1100 Hz)
00553   if ((sampleRate < 4.3f) || (sampleRate > 1100.0f))
00554     return false;
00555
00556   // Compute divider value
00557   uint8_t v = (uint8_t)((1100.0f / sampleRate) - 1.0f);
00558
00559   // Select USER BANK 2
00560   if (!selectBank(2))
00561     return false;
00562
00563   // Write sample rate divider
00564   if (!bus->write(GYRO_SMPLRT_DIV, v))
00565     return false;
00566
00567   // Success
00568   return true;
00569 }
00570
00571 bool DevLab_ICM20948::getGyroSampleRate(float &sampleRate)
00572 {
```

```
00573   // Validate bus pointer
00574   if (!bus)
00575     return false;
00576
00577   // Select USER BANK 2
00578   if (!selectBank(2))
00579     return false;
00580
00581   // Read sample rate divider
00582   uint8_t v = 0;
00583   if (!bus->read(GYRO_SMPLRT_DIV, v))
00584     return false;
00585
00586   // Compute sample rate
00587   sampleRate = 1100.0f / (1.0f + v);
00588
00589   // Success
00590   return true;
00591 }
00592
00593 bool DevLab_ICM20948::setDLPF(ICM20948_Gyro_DLPF dlpf, bool bypass)
00594 {
00595   // Validate bus pointer
00596   if (!bus)
00597     return false;
00598
00599   // Select USER BANK 2
00600   if (!selectBank(2))
00601     return false;
00602
00603   // Enable or disable DLPF bypass (bit 0)
00604   if (!bus->writeBit(GYRO_CONFIG_1, 0, (uint8_t)bypass))
00605     return false;
00606
00607   // If bypass enabled, skip DLPF configuration
00608   if (bypass)
00609     return true;
00610
00611   // Set DLPF configuration bits [5:3]
00612   if (!bus->writeBits(GYRO_CONFIG_1, 3, 3, (uint8_t)dlpf))
00613     return false;
00614
00615   // Success
00616   return true;
00617 }
00618
00619 bool DevLab_ICM20948::getDLPF(uint8_t &dlpf, bool &bypass)
00620 {
00621   // Validate bus pointer
00622   if (!bus)
00623     return false;
00624
00625   // Select USER BANK 2
00626   if (!selectBank(2))
00627     return false;
00628
00629   // Read GYRO_CONFIG_1 register
00630   uint8_t v = 0;
00631   if (!bus->read(GYRO_CONFIG_1, v))
00632     return false;
00633
00634   // Extract bypass bit (bit 0)
00635   bypass = (v & 0x01);
00636
00637   // Extract DLPF bits [5:3]
00638   dlpf = (v » 3) & 0x07;
00639
00640   // Success
00641   return true;
00642 }
00643
00644 bool DevLab_ICM20948::setGyroScale(ICM20948_Gyro_FullScale fullScale)
00645 {
00646   // Validate bus pointer
00647   if (!bus)
00648     return false;
00649
00650   // Select USER BANK 2
00651   if (!selectBank(2))
00652     return false;
00653
00654   // Set full-scale range bits [2:1]
00655   if (!bus->writeBits(GYRO_CONFIG_1, 1, 2, (uint8_t)fullScale))
00656     return false;
00657
00658   // Update internal scale (LSB per g)
00659    degree_per_second = 32768.0f / (250.0f * (1 « ((uint8_t)fullScale)));
```

```
00660
00661   // Success
00662   return true;
00663 }
00664
00665 bool DevLab_ICM20948::getGyroScale(uint8_t &fullScale)
00666 {
00667   // Validate bus pointer
00668   if (!bus)
00669     return false;
00670
00671   // Select USER BANK 2
00672   if (!selectBank(2))
00673     return false;
00674
00675   // Read full-scale range bits [2:1]
00676   if (!bus->readBits(GYRO_CONFIG_1, 1, 2, fullScale))
00677     return false;
00678
00679   // Success
00680   return true;
00681 }
00682
00683 bool DevLab_ICM20948::selfTestGyro(bool x, bool y, bool z)
00684 {
00685   // Validate bus pointer
00686   if (!bus)
00687     return false;
00688
00689   // Select USER BANK 2
00690   if (!selectBank(2))
00691     return false;
00692
00693   // Set Z-axis self-test (bit 3)
00694   if (!bus->writeBit(GYRO_CONFIG_2, 3, (uint8_t)z))
00695     return false;
00696
00697   // Set Y-axis self-test (bit 4)
00698   if (!bus->writeBit(GYRO_CONFIG_2, 4, (uint8_t)y))
00699     return false;
00700
00701   // Set X-axis self-test (bit 5)
00702   if (!bus->writeBit(GYRO_CONFIG_2, 5, (uint8_t)x))
00703     return false;
00704
00705   // Success
00706   return true;
00707 }
00708
00709 bool DevLab_ICM20948::setAccelSampleRate(uint16_t sampleRate)
00710 {
00711   // Validate bus pointer
00712   if (!bus)
00713     return false;
00714
00715   // Select USER BANK 2
00716   if (!selectBank(2))
00717     return false;
00718
00719   // Validate input
00720   if (sampleRate == 0)
00721     return false;
00722
00723   // Clamp maximum rate
00724   if (sampleRate >= 1125)
00725     return false;
00726
00727   // Compute divider (rounded)
00728   uint16_t div = (1125.0f / sampleRate) - 1;
00729
00730   if (div > 4095u)
00731     div = 4095u;
00732
00733   // Split divider into high and low bytes
00734   uint8_t msb = (uint8_t)((div » 8) & 0x0F);
00735   uint8_t lsb = (uint8_t)(div & 0xFF);
00736
00737   // Write divider high byte
00738   if (!bus->write(ACCEL_SMPLRT_DIV_1, msb))
00739     return false;
00740
00741   // Write divider low byte
00742   if (!bus->write(ACCEL_SMPLRT_DIV_2, lsb))
00743     return false;
00744
00745   // Success
00746   return true;
```

```
00747 }
00748 bool DevLab_ICM20948::setAccelDivRate(uint16_t divisor)
00749 {
00750   if (!bus)
00751     return false;
00752
00753   // Select USER BANK 2
00754   if (!selectBank(2))
00755     return false;
00756
00757   // Validate input
00758   if (divisor == 0)
00759     return false;
00760
00761   if (divisor > 4095u)
00762     divisor = 4095u;
00763
00764   // Split divider into high and low bytes
00765   uint8_t msb = (uint8_t)((divisor » 8) & 0x0F);
00766   uint8_t lsb = (uint8_t)(divisor & 0xFF);
00767
00768   // Write divider high byte
00769   if (!bus->write(ACCEL_SMPLRT_DIV_1, msb))
00770     return false;
00771
00772   // Write divider low byte
00773   if (!bus->write(ACCEL_SMPLRT_DIV_2, lsb))
00774     return false;
00775
00776   // Success
00777   return true;
00778 }
00779
00780 bool DevLab_ICM20948::setGyroDivRate(uint8_t divisor)
00781 {
00782   if (!bus)
00783     return false;
00784   // Select USER BANK 2
00785   if (!selectBank(2))
00786     return false;
00787
00788   // Write sample rate divider
00789   if (!bus->write(GYRO_SMPLRT_DIV, divisor))
00790     return false;
00791
00792   // Success
00793   return true;
00794 }
00795
00796 bool DevLab_ICM20948::getAccelSampleRate(float &sampleRate)
00797 {
00798   // Validate bus pointer
00799   if (!bus)
00800     return false;
00801
00802   // Select USER BANK 2
00803   if (!selectBank(2))
00804     return false;
00805
00806   // Read divider high byte
00807   uint8_t msb = 0;
00808   if (!bus->read(ACCEL_SMPLRT_DIV_1, msb))
00809     return false;
00810
00811   // Read divider low byte
00812   uint8_t lsb = 0;
00813   if (!bus->read(ACCEL_SMPLRT_DIV_2, lsb))
00814     return false;
00815
00816   // Combine 12-bit divider
00817   uint16_t div = ((msb & 0x0F) « 8) | lsb;
00818
00819   // Compute sample rate
00820   sampleRate = 1125.0f / (1.0f + div);
00821
00822   // Success
00823   return true;
00824 }
00825
00826 bool DevLab_ICM20948::selfTestAccel(bool x, bool y, bool z)
00827 {
00828   // Validate bus pointer
00829   if (!bus)
00830     return false;
00831
00832   // Select USER BANK 2
00833   if (!selectBank(2))
```

```
00834     return false;
00835
00836   if (x)
00837   { // Set X-axis self-test (bit 7)
00838     if (!bus->writeBit(ACCEL_CONFIG_2, 7, (uint8_t)x))
00839       return false;
00840   }
00841
00842   if (y)
00843   { // Set Y-axis self-test (bit 6)
00844     if (!bus->writeBit(ACCEL_CONFIG_2, 6, (uint8_t)y))
00845       return false;
00846   }
00847
00848   if (z)
00849   { // Set Z-axis self-test (bit 5)
00850     if (!bus->writeBit(ACCEL_CONFIG_2, 5, (uint8_t)z))
00851       return false;
00852   }
00853   // Success
00854   return true;
00855 }
00856
00857 bool DevLab_ICM20948::setAccelScale(ICM20948_Accel_FullScale fullScale)
00858 {
00859   // Validate bus pointer
00860   if (!bus)
00861     return false;
00862
00863   // Select USER BANK 2
00864   if (!selectBank(2))
00865     return false;
00866
00867   // Set full-scale bits [2:1]
00868   if (!bus->writeBits(ACCEL_CONFIG, 1, 2, (uint8_t)fullScale))
00869     return false;
00870
00871   // Update internal scale (LSB per g)
00872   mg_per_lsb = 16384.0f / (1 « (uint8_t)fullScale);
00873
00874   // Success
00875   return true;
00876 }
00877
00878 bool DevLab_ICM20948::getAccelScale(uint8_t &fullScale)
00879 {
00880   // Validate bus pointer
00881   if (!bus)
00882     return false;
00883
00884   // Select USER BANK 2
00885   if (!selectBank(2))
00886     return false;
00887
00888   // Read full-scale bits [2:1]
00889   if (!bus->readBits(ACCEL_CONFIG, 1, 2, fullScale))
00890     return false;
00891
00892   // Success
00893   return true;
00894 }
00895
00896
00897 bool DevLab_ICM20948::setAccelDLPF(uint8_t dlpf, bool bypass)
00898 {
00899   // Validate bus pointer
00900   if (!bus)
00901     return false;
00902
00903   // Select USER BANK 2
00904   if (!selectBank(2))
00905     return false;
00906
00907   // Set or clear DLPF bypass (bit 0)
00908   if (!bus->writeBit(ACCEL_CONFIG, 0, (uint8_t)bypass))
00909     return false;
00910
00911   // If bypass enabled, skip DLPF configuration
00912   if (bypass)
00913     return true;
00914
00915   // Limit DLPF value to valid range
00916   dlpf &= 0x07;
00917
00918   // Set DLPF configuration bits [5:3]
00919   if (!bus->writeBits(ACCEL_CONFIG, 3, 3, dlpf))
00920     return false;
```

```
00921
00922   // Success
00923   return true;
00924 }
00925
00926 bool DevLab_ICM20948::getAccelDLPF(uint8_t &dlpf, bool &bypass)
00927 {
00928   // Validate bus pointer
00929   if (!bus)
00930     return false;
00931
00932   // Select USER BANK 2
00933   if (!selectBank(2))
00934     return false;
00935
00936   // Read ACCEL_CONFIG register
00937   uint8_t v = 0;
00938   if (!bus->read(ACCEL_CONFIG, v))
00939     return false;
00940
00941   // Extract bypass bit (bit 0)
00942   bypass = (v & 0x01);
00943
00944   // Extract DLPF bits [5:3]
00945   dlpf = (v » 3) & 0x07;
00946
00947   // Success
00948   return true;
00949 }
00950
00951
00952 bool DevLab_ICM20948::setAccelAveraging(ICM20948_Accel_Average avg)
00953 {
00954   // Validate bus pointer
00955   if (!bus)
00956     return false;
00957
00958   // Select USER BANK 2
00959   if (!selectBank(2))
00960     return false;
00961
00962   // Set DEC3 averaging bits [1:0]
00963   if (!bus->writeBits(ACCEL_CONFIG_2, 0, 2, (uint8_t)avg))
00964     return false;
00965
00966   // Success
00967   return true;
00968 }
00969
00970 bool DevLab_ICM20948::setGyroAveraging(ICM20948_Gyro_Average avg)
00971 {
00972   // Validate bus pointer
00973   if (!bus)
00974     return false;
00975
00976   // Select USER BANK 2
00977   if (!selectBank(2))
00978     return false;
00979
00980   // Set DEC3 averaging bits [1:0]
00981   if (!bus->writeBits(GYRO_CONFIG_2, 0, 2, (uint8_t)avg))
00982     return false;
00983
00984   // Success
00985   return true;
00986 }
00987
00988 bool DevLab_ICM20948::getAccelAveraging(uint8_t &avg)
00989 {
00990   // Validate bus pointer
00991   if (!bus)
00992     return false;
00993
00994   // Select USER BANK 2
00995   if (!selectBank(2))
00996     return false;
00997
00998   // Read DEC3 averaging bits [1:0]
00999   if (!bus->readBits(ACCEL_CONFIG_2, 0, 2, avg))
01000     return false;
01001
01002   // Success
01003   return true;
01004 }
01005
01006 bool DevLab_ICM20948::setSensors(bool accel_on, bool gyro_on, bool temp_on)
01007 {
```

```
01008   // Validate bus pointer
01009   if (!bus)
01010     return false;
01011
01012   // Select USER BANK 0
01013   if (!selectBank(0))
01014     return false;
01015
01016   // Build PWR_MGMT_2 mask
01017   uint8_t v = 0;
01018
01019   // Disable accel axes if not enabled
01020   if (!accel_on)
01021     v |= 0x38; // bits [5:3]
01022
01023   // Disable gyro axes if not enabled
01024   if (!gyro_on)
01025     v |= 0x07; // bits [2:0]
01026
01027   // Write sensor enable/disable mask
01028   if (!bus->write(PWR_MGMT_2, v))
01029     return false;
01030
01031   // Set or clear TEMP_DIS bit (bit 3)
01032   if (!bus->writeBit(PWR_MGMT_1, 3, (uint8_t)!temp_on))
01033     return false;
01034
01035   // Success
01036   return true;
01037 }
01038
01039 bool DevLab_ICM20948::getSensors(bool &accel_on, bool &gyro_on, bool &temp_on)
01040 {
01041   // Validate bus pointer
01042   if (!bus)
01043     return false;
01044
01045   // Select USER BANK 0
01046   if (!selectBank(0))
01047     return false;
01048
01049   // Read PWR_MGMT_2 register
01050   uint8_t v = 0;
01051   if (!bus->read(PWR_MGMT_2, v))
01052     return false;
01053
01054   // Decode gyro state (bits [2:0])
01055   gyro_on = ((v & 0x07) == 0);
01056
01057   // Decode accel state (bits [5:3])
01058   accel_on = ((v & 0x38) == 0);
01059
01060   // Read TEMP_DIS bit (bit 3)
01061   uint8_t t = 0;
01062   if (!bus->readBit(PWR_MGMT_1, 3, t))
01063     return false;
01064
01065   // Decode temperature state
01066   temp_on = (t == 0);
01067
01068   // Success
01069   return true;
01070 }
01071
01072 bool DevLab_ICM20948::setGyroOffset(uint16_t offsetX, uint16_t offsetY, uint16_t offsetZ)
01073 {
01074   // Validate bus pointer
01075   if (!bus)
01076     return false;
01077
01078   // Select USER BANK 2
01079   if (!selectBank(2))
01080     return false;
01081
01082   // Pack offset values into byte array
01083   uint8_t data[6] = {
01084       (uint8_t)(offsetX » 8),
01085       (uint8_t)(offsetX & 0xFF),
01086       (uint8_t)(offsetY » 8),
01087       (uint8_t)(offsetY & 0xFF),
01088       (uint8_t)(offsetZ » 8),
01089       (uint8_t)(offsetZ & 0xFF)};
01090
01091   // Write offset registers
01092   if (!bus->write(XG_OFFS_USRH, data, 6))
01093     return false;
01094
```

```
01095   // Success
01096   return true;
01097 }
01098
01099 bool DevLab_ICM20948::getGyroOffset(int16_t &offsetX, int16_t &offsetY, int16_t &offsetZ)
01100 {
01101   // Validate bus pointer
01102   if (!bus)
01103     return false;
01104
01105   // Select USER BANK 2
01106   if (!selectBank(2))
01107     return false;
01108
01109   // Read offset registers
01110   uint8_t data[6];
01111   if (!bus->read(XG_OFFS_USRH, data, 6))
01112     return false;
01113
01114   // Convert to signed values
01115   offsetX = (int16_t)((data[0] << 8) | data[1]);
01116   offsetY = (int16_t)((data[2] << 8) | data[3]);
01117   offsetZ = (int16_t)((data[4] << 8) | data[5]);
01118
01119   // Success
01120   return true;
01121 }
01122
01123
01124 bool DevLab_ICM20948::setAccelOffset(int16_t offsetX, int16_t offsetY, int16_t offsetZ)
01125 {
01126   // Validate bus pointer
01127   if (!bus)
01128     return false;
01129
01130   // Select USER BANK 1 (accel offsets are here)
01131   if (!selectBank(1))
01132     return false;
01133
01134   // Pack offset values into byte array
01135   uint8_t data[6] = {
01136       (uint8_t)(offsetX >> 8),
01137       (uint8_t)(offsetX & 0xFF),
01138       (uint8_t)(offsetY >> 8),
01139       (uint8_t)(offsetY & 0xFF),
01140       (uint8_t)(offsetZ >> 8),
01141       (uint8_t)(offsetZ & 0xFF)};
01142
01143   // Write offset registers
01144   if (!bus->write(XA_OFFS_H, data, 6))
01145     return false;
01146
01147   // Success
01148   return true;
01149 }
01150
01151 bool DevLab_ICM20948::getAccelOffset(int16_t &offsetX, int16_t &offsetY, int16_t &offsetZ)
01152 {
01153   // Validate bus pointer
01154   if (!bus)
01155     return false;
01156
01157   // Select USER BANK 1
01158   if (!selectBank(1))
01159     return false;
01160
01161   // Read offset registers
01162   uint8_t data[6];
01163   if (!bus->read(XA_OFFS_H, data, 6))
01164     return false;
01165
01166   // Convert to signed values
01167   offsetX = (int16_t)((data[0] << 8) | data[1]);
01168   offsetY = (int16_t)((data[2] << 8) | data[3]);
01169   offsetZ = (int16_t)((data[4] << 8) | data[5]);
01170
01171   // Success
01172   return true;
01173 }
01174
01175
01176 bool DevLab_ICM20948::intInit(const ICM20948_IntPinConfig &cfg){
01177   // Validate bus pointer
01178   if (!bus)
01179     return false;
01180
01181   if(!selectBank(0))
```

```
01182     return false;
01183
01184   uint8_t reg = (cfg.activeLevel   « 7) |
01185                 (cfg.driveMode     « 6) |
01186                 (cfg.latchMode     « 5) |
01187                 (cfg.clearMode     « 4) |
01188                 (cfg.fsyncActLevel « 3) |
01189                 (cfg.fsyncIntEn    « 2) |
01190                 (cfg.bypassEn      « 1);
01191   // Initialize INT1
01192   // Configure interrupt pin (open-drain, active-low, latch)
01193   if (!bus->write(INT_PIN_CFG, reg))
01194     return false;
01195
01196   // Configuration successful
01197   return true;
01198 }
01199
01200 bool DevLab_ICM20948::intEnableConfig(const ICM20948_IntEnableConfig &cfg)
01201 {
01202   if (!bus)           return false;
01203   if (!selectBank(0)) return false;
01204
01205   // INT_ENABLE
01206   uint8_t reg0 = (cfg.wofEn       « 7) |
01207                  (cfg.womIntEn    « 3) |
01208                  (cfg.pllRdyEn    « 2) |
01209                  (cfg.dmpInt1En   « 1) |
01210                  (cfg.i2cMstIntEn « 0);
01211   if (!bus->write(INT_ENABLE, reg0)) return false;
01212
01213   // INT_ENABLE_1
01214   if (!bus->write(INT_ENABLE_1, (uint8_t)(cfg.rawDataRdyEn & 0x01))) return false;
01215
01216   // INT_ENABLE_2 -- construye máscara desde el array
01217   uint8_t ovfMask = 0;
01218   for (uint8_t i = 0; i < 5; i++)
01219     ovfMask |= (cfg.fifoOvfEn[i] ? BIT(i) : 0);
01220   if (!bus->write(INT_ENABLE_2, ovfMask)) return false;
01221
01222   // INT_ENABLE_3 -- igual
01223   uint8_t wmMask = 0;
01224   for (uint8_t i = 0; i < 5; i++)
01225     wmMask |= (cfg.fifoWmEn[i] ? BIT(i) : 0);
01226   if (!bus->write(INT_ENABLE_3, wmMask)) return false;
01227
01228   return true;
01229 }
01230
01231 bool DevLab_ICM20948::checkIntStatus(ICM20948_IntStatus &status)
01232 {
01233   if (!bus)           return false;
01234   if (!selectBank(0)) return false;
01235
01236   // Leer los 4 registros de status en una sola operación de burst
01237   uint8_t raw[4];
01238   if (!bus->read(INT_STATUS, raw, 4)) return false;
01239
01240   // INT_STATUS (0x19)
01241   status.womInt    = (raw[0] & STS_WOM_INT)     ? 1 : 0;
01242   status.pllRdyInt = (raw[0] & STS_PLL_RDY_INT) ? 1 : 0;
01243   status.dmpInt1   = (raw[0] & STS_DMP_INT1)    ? 1 : 0;
01244   status.i2cMstInt = (raw[0] & STS_I2C_MST_INT) ? 1 : 0;
01245
01246   // INT_STATUS_1 (0x1A)
01247   status.rawDataRdy = (raw[1] & STS_RAW_DATA_0_RDY_INT) ? 1 : 0;
01248
01249   // INT_STATUS_2 (0x1B) -- FIFO overflow canal por canal
01250   for (uint8_t i = 0; i < 5; i++)
01251     status.fifoOvf[i] = (raw[2] & BIT(i)) ? 1 : 0;
01252
01253   // INT_STATUS_3 (0x1C) -- FIFO watermark canal por canal
01254   for (uint8_t i = 0; i < 5; i++)
01255     status.fifoWm[i] = (raw[3] & BIT(i)) ? 1 : 0;
01256
01257   return true;
01258 }
01259
01260 bool DevLab_ICM20948::auxMasterEnable(uint8_t clkFreq)
01261 {
01262     // BANK 0: asegurarse que BYPASS_EN=0 e I2C_MST_EN=1
01263     if (!selectBank(0)) return false;
01264
01265     // Limpiar BYPASS_EN (bit 1 de INT_PIN_CFG)
01266     if (!bus->writeBit(INT_PIN_CFG, 1, (uint8_t)0)) return false;
01267
01268     // Activar I2C_MST_EN (bit 5 de USER_CTRL)
```

```
01269     if (!bus->writeBit(USER_CTRL, 5, (uint8_t)1)) return false;
01270
01271     // BANK 3: configurar clock del master auxiliar
01272     if (!selectBank(3)) return false;
01273
01274     // clkFreq típico: 0x07 = ~345.6 kHz  |  0x0D = ~400 kHz
01275     if (!bus->writeBits(I2C_MST_CTRL, 0, 4, clkFreq)) return false;
01276
01277     // Volver a BANK 0
01278     return selectBank(0);
01279 }
01280
01281 bool DevLab_ICM20948::auxWriteByte(uint8_t slaveAddr, uint8_t reg, uint8_t data)
01282 {
01283     if (!selectBank(3)) return false;
01284
01285     // Dirección del slave, bit 7=0 para escritura
01286     if (!bus->write(I2C_SLV4_ADDR, (uint8_t)(slaveAddr & 0x7F))) return false;
01287
01288     // Registro destino que queremos escribir en el slave
01289     if (!bus->write(I2C_SLV4_REG, reg)) return false;
01290
01291     // Dato a escribir
01292     if (!bus->write(I2C_SLV4_DO, data)) return false;
01293
01294     // Disparar transacción (I2C_SLV4_EN, se auto-limpia al terminar)
01295     if (!bus->write(I2C_SLV4_CTRL, (uint8_t)I2C_SLVx_EN)) return false;
01296
01297     // Esperar a que complete -- verificar SLV4_DONE en BANK 0
01298     if (!selectBank(0)) return false;
01299
01300     uint8_t status = 0;
01301     uint8_t timeout = 50;
01302     do {
01303         delay(1);
01304         if (!bus->read(I2C_MST_STATUS, status)) return false;
01305         if (--timeout == 0) return false;  // timeout
01306     } while (!(status & MST_SLV4_DONE));
01307
01308     // Verificar que no hubo NACK
01309     return !(status & MST_SLV4_NACK);
01310 }
01311
01312 bool DevLab_ICM20948::auxReadByte(uint8_t slaveAddr, uint8_t reg, uint8_t &data)
01313 {
01314     if (!selectBank(3)) return false;
01315
01316     // Dirección del slave, bit 7=1 para lectura
01317     if (!bus->write(I2C_SLV4_ADDR, (uint8_t)((slaveAddr & 0x7F) | I2C_SLVx_RNW))) return false;
01318
01319     // Registro origen que queremos leer del slave
01320     if (!bus->write(I2C_SLV4_REG, reg)) return false;
01321
01322     // Disparar transacción (I2C_SLV4_EN, se auto-limpia al terminar)
01323     if (!bus->write(I2C_SLV4_CTRL, (uint8_t)I2C_SLVx_EN)) return false;
01324
01325     // Esperar a que complete -- verificar SLV4_DONE en BANK 0
01326     if (!selectBank(0)) return false;
01327
01328     uint8_t status = 0;
01329     uint8_t timeout = 50;
01330     do {
01331         delay(1);
01332         if (!bus->read(I2C_MST_STATUS, status)) return false;
01333         if (--timeout == 0) return false;  // timeout
01334     } while (!(status & MST_SLV4_DONE));
01335
01336     // Verificar que no hubo NACK
01337     if (status & MST_SLV4_NACK) return false;
01338
01339     // Leer el dato recibido del slave (BANK 3)
01340     if (!selectBank(3)) return false;
01341     if (!bus->read(I2C_SLV4_DI, data)) return false;
01342
01343     return selectBank(0);
01344 }
01345
01346 bool DevLab_ICM20948::auxWriteCommand(uint8_t slaveAddr, uint8_t cmd)
01347 {
01348     if (!selectBank(3)) return false;
01349
01350     // Dirección del slave, bit 7=0 para escritura
01351     if (!bus->write(I2C_SLV4_ADDR, (uint8_t)(slaveAddr & 0x7F))) return false;
01352
01353     // Byte de comando a enviar (único byte de la transacción)
01354     if (!bus->write(I2C_SLV4_DO, cmd)) return false;
01355
```

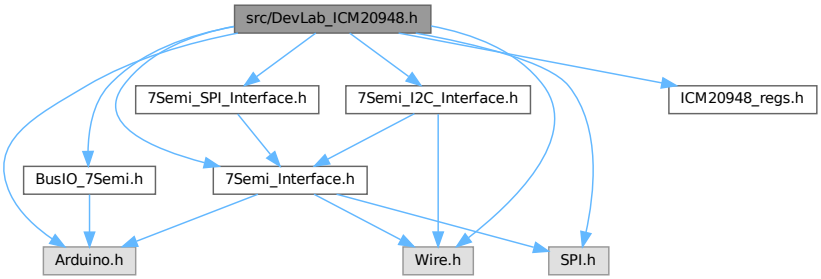
```
01356     // EN + REG_DIS: el SLV4 envía solo I2C_SLV4_DO, sin byte de registro
01357     if (!bus->write(I2C_SLV4_CTRL, (uint8_t)(I2C_SLVx_EN | I2C_SLVx_REG_DIS))) return false;
01358
01359     // Esperar a que complete -- verificar SLV4_DONE en BANK 0
01360     if (!selectBank(0)) return false;
01361
01362     uint8_t status = 0;
01363     uint8_t timeout = 50;
01364     do {
01365         delay(1);
01366         if (!bus->read(I2C_MST_STATUS, status)) return false;
01367         if (--timeout == 0) return false;  // timeout
01368     } while (!(status & MST_SLV4_DONE));
01369
01370     // Verificar que no hubo NACK
01371     return !(status & MST_SLV4_NACK);
01372 }
01373
01374   bool DevLab_ICM20948::auxConfigSlave(uint8_t slaveAddr, uint8_t reg, uint8_t numBytes)
01375 {
01376     if (!selectBank(3)) return false;
01377
01378     // Slave 0: dirección + bit READ
01379     if (!bus->write(I2C_SLV0_ADDR, (uint8_t)(slaveAddr | I2C_SLVx_RNW))) return false;
01380
01381     // Registro de inicio
01382     if (!bus->write(I2C_SLV0_REG, reg)) return false;
01383
01384     // Habilitar + número de bytes
01385     if (!bus->write(I2C_SLV0_CTRL, (uint8_t)(I2C_SLVx_EN | (numBytes & 0x0F)))) return false;
01386
01387     return selectBank(0);
01388 }
01389
01390 // Para leer los datos que el ICM leyó automáticamente:
01391 bool DevLab_ICM20948::auxReadSensorData(uint8_t *buf, uint8_t len)
01392 {
01393     if (!selectBank(0)) return false;
01394     return bus->read(EXT_SLV_SENS_DATA_00, buf, len);
01395 }
01396
01397 bool DevLab_ICM20948::auxReadResponse(uint8_t slaveAddr, uint8_t &data)
01398 {
01399     if (!selectBank(3)) return false;
01400
01401     // Slave address con bit READ, sin byte de registro (REG_DIS)
01402     // Genera: [START, addr+R, lee 1 byte, STOP]
01403     if (!bus->write(I2C_SLV4_ADDR, (uint8_t)((slaveAddr & 0x7F) | I2C_SLVx_RNW))) return false;
01404     if (!bus->write(I2C_SLV4_CTRL, (uint8_t)(I2C_SLVx_EN | I2C_SLVx_REG_DIS))) return false;
01405
01406     if (!selectBank(0)) return false;
01407
01408     uint8_t status = 0;
01409     uint8_t timeout = 50;
01410     do {
01411         delay(1);
01412         if (!bus->read(I2C_MST_STATUS, status)) return false;
01413         if (--timeout == 0) return false;
01414     } while (!(status & MST_SLV4_DONE));
01415
01416     if (status & MST_SLV4_NACK) return false;
01417
01418     if (!selectBank(3)) return false;
01419     if (!bus->read(I2C_SLV4_DI, data)) return false;
01420
01421     return selectBank(0);
01422 }
01423
01424 bool DevLab_ICM20948::auxRead12bit(uint16_t &raw)
01425 {
01426     uint8_t buf[2];
01427     if (!auxReadSensorData(buf, 2)) return false;
01428
01429     // LSB primero (little-endian): buf[0] = byte bajo, buf[1] = byte alto
01430     raw = (uint16_t)(buf[0] | (buf[1] << 8)) & 0x0FFF;
01431
01432     return true;
01433 }
```

6.36 src/DevLab_ICM20948.h File Reference

```
#include <Arduino.h>
#include <Wire.h>
```

```
#include <SPI.h>
#include "ICM20948_regs.h"
#include "7Semi_Interface.h"
#include "7Semi_I2C_Interface.h"
#include "7Semi_SPI_Interface.h"
#include "BusIO_7Semi.h"
```

Include dependency graph for DevLab_ICM20948.h:



This graph shows which files directly or indirectly include this file:

Classes

- struct ICM20948_IntPinConfig
- struct ICM20948_IntEnableConfig
- struct ICM20948_IntStatus
- class DevLab_ICM20948

Enumerations

- enum class ICM20948_Op_Mode : uint8_t {
  MODE_POWER_DOWN = 0x00 , MODE_SINGLE_MEASUREMENT = 0x01 , MODE_CONTINUOUS_1 = 0x02 , MODE_CONTINUOUS_2 = 0x04 ,
  MODE_CONTINUOUS_3 = 0x06 , MODE_CONTINUOUS_4 = 0x08 , MODE_SELF_TEST = 0x10 }
- enum class ICM20948_Clock_Source : uint8_t {
  INTERNAL_20MHZ_0 = 0x00 , AUTO_PLL_1 = 0x01 , AUTO_PLL_2 = 0x02 , AUTO_PLL_3 = 0x03 ,
  AUTO_PLL_4 = 0x04 , AUTO_PLL_5 = 0x05 , INTERNAL_20MHZ_6 = 0x06 , STOP_CLOCK = 0x07 }
- enum class ICM20948_Gyro_DLPF : uint8_t {
  DLPF_196HZ = 0x00 , DLPF_151HZ = 0x01 , DLPF_119HZ = 0x02 , DLPF_51HZ = 0x03 ,
  DLPF_23HZ = 0x04 , DLPF_11HZ = 0x05 , DLPF_5HZ = 0x06 , DLPF_361HZ = 0x07 }
- enum class ICM20948_Gyro_Average : uint8_t {
  AVG_1 = 0x00 , AVG_2 = 0x01 , AVG_4 = 0x02 , AVG_8 = 0x03 ,
  AVG_16 = 0x04 , AVG_32 = 0x05 , AVG_64 = 0x06 , AVG_128 = 0x07 }
- enum class ICM20948_Gyro_FullScale : uint8_t { DPS_250 = 0x00 , DPS_500 = 0x01 , DPS_1000 = 0x02 , DPS_2000 = 0x03 }
- enum class ICM20948_Accel_FullScale : uint8_t { G_2 = 0x00 , G_4 = 0x01 , G_8 = 0x02 , G_16 = 0x03 }
- enum class ICM20948_Accel_Average : uint8_t { AVG_1_OR_4 = 0x00 , AVG_8 = 0x01 , AVG_16 = 0x02 , AVG_32 = 0x03 }
- enum class ICM20948_Accel_DLPFCFG : uint8_t {
  DLPF0_246HZ = 0x00 , DLPF_246HZ = 0x01 , DLPF_111HZ = 0x02 , DLPF_50HZ = 0x03 ,
  DLPF_24HZ = 0x04 , DLPF_12HZ = 0x05 , DLPF_6HZ = 0x06 , DLPF_473HZ = 0x07 }

6.36.1 Enumeration Type Documentation

6.36.1.1 ICM20948_Accel_Average

```
enum class ICM20948_Accel_Average : uint8_t [strong]
```

Enumerator

AVG_1_OR_4	
AVG_8	
AVG_16	
AVG_32	

Definition at line 76 of file DevLab_ICM20948.h.

6.36.1.2 ICM20948_Accel_DLPFCFG

```
enum class ICM20948_Accel_DLPFCFG : uint8_t [strong]
```

Enumerator

DLPF0_246HZ	
DLPF_246HZ	
DLPF_111HZ	
DLPF_50HZ	
DLPF_24HZ	
DLPF_12HZ	
DLPF_6HZ	
DLPF_473HZ	

Definition at line 84 of file DevLab_ICM20948.h.

6.36.1.3 ICM20948_Accel_FullScale

```
enum class ICM20948_Accel_FullScale : uint8_t [strong]
```

Enumerator

G_2	
G_4	
G_8	
G_16	

Definition at line 68 of file DevLab_ICM20948.h.

6.36.1.4 ICM20948_Clock_Source

```
enum class ICM20948_Clock_Source : uint8_t [strong]
```

Enumerator

INTERNAL_20MHZ_0	
AUTO_PLL_1	
AUTO_PLL_2	
AUTO_PLL_3	

Enumerator

AUTO_PLL_4	
AUTO_PLL_5	
INTERNAL_20MHZ↩_6	
STOP_CLOCK	

Definition at line 25 of file DevLab_ICM20948.h.

6.36.1.5 ICM20948_Gyro_Average

```
enum class ICM20948_Gyro_Average : uint8_t [strong]
```

Enumerator

AVG_1	
AVG_2	
AVG_4	
AVG_8	
AVG_16	
AVG_32	
AVG_64	
AVG_128	

Definition at line 49 of file DevLab_ICM20948.h.

6.36.1.6 ICM20948_Gyro_DLPF

```
enum class ICM20948_Gyro_DLPF : uint8_t [strong]
```

Enumerator

DLPF_196HZ	
DLPF_151HZ	
DLPF_119HZ	
DLPF_51HZ	
DLPF_23HZ	
DLPF_11HZ	
DLPF_5HZ	
DLPF_361HZ	

Definition at line 37 of file DevLab_ICM20948.h.

6.36.1.7 ICM20948_Gyro_FullScale

```
enum class ICM20948_Gyro_FullScale : uint8_t [strong]
```

Enumerator

DPS_250	
DPS_500	
DPS_1000	
DPS_2000	

Definition at line 60 of file DevLab_ICM20948.h.

6.36.1.8 ICM20948_Op_Mode

```
enum class ICM20948_Op_Mode :  uint8_t  [strong]
```

Enumerator

MODE_POWER_DOWN	
MODE_SINGLE_MEASUREMENT	
MODE_CONTINUOUS_1	
MODE_CONTINUOUS_2	
MODE_CONTINUOUS_3	
MODE_CONTINUOUS_4	
MODE_SELF_TEST	

Definition at line 14 of file DevLab_ICM20948.h.

6.37 DevLab_ICM20948.h

Go to the documentation of this file.

```
00001 #ifndef ICM20948_DEVLAB_H
00002 #define ICM20948_DEVLAB_H
00003
00004 #include <Arduino.h>
00005 #include <Wire.h>
00006 #include <SPI.h>
00007 #include "ICM20948_regs.h"
00008
00009 #include "7Semi_Interface.h"
00010 #include "7Semi_I2C_Interface.h"
00011 #include "7Semi_SPI_Interface.h"
00012 #include "BusIO_7Semi.h"
00013
00014 enum class ICM20948_Op_Mode : uint8_t
00015 {
00016   MODE_POWER_DOWN = 0x00,
00017   MODE_SINGLE_MEASUREMENT = 0x01,
00018   MODE_CONTINUOUS_1 = 0x02,
00019   MODE_CONTINUOUS_2 = 0x04,
00020   MODE_CONTINUOUS_3 = 0x06,
00021   MODE_CONTINUOUS_4 = 0x08,
00022   MODE_SELF_TEST = 0x10
00023 };
00024
00025 enum class ICM20948_Clock_Source : uint8_t
00026 {
00027   INTERNAL_20MHZ_0 = 0x00, // Internal oscillator
00028   AUTO_PLL_1 = 0x01,       // Auto select (PLL if ready)
00029   AUTO_PLL_2 = 0x02,
00030   AUTO_PLL_3 = 0x03,
00031   AUTO_PLL_4 = 0x04,
00032   AUTO_PLL_5 = 0x05,
00033   INTERNAL_20MHZ_6 = 0x06, // Internal oscillator
00034   STOP_CLOCK = 0x07        // Stops clock, timing generator reset
00035 };
00036
00037 enum class ICM20948_Gyro_DLPF : uint8_t
00038 {
00039   DLPF_196HZ = 0x00, // BW ≈ 196.6 Hz, NBW ≈ 229.8 Hz
00040   DLPF_151HZ = 0x01, // BW ≈ 151.8 Hz, NBW ≈ 187.6 Hz
00041   DLPF_119HZ = 0x02, // BW ≈ 119.5 Hz, NBW ≈ 154.3 Hz
00042   DLPF_51HZ = 0x03,  // BW ≈ 51.2 Hz,  NBW ≈ 73.3 Hz
00043   DLPF_23HZ = 0x04,  // BW ≈ 23.9 Hz,  NBW ≈ 35.9 Hz
00044   DLPF_11HZ = 0x05,  // BW ≈ 11.6 Hz,  NBW ≈ 17.8 Hz
00045   DLPF_5HZ = 0x06,   // BW ≈ 5.7 Hz,   NBW ≈ 8.9 Hz
00046   DLPF_361HZ = 0x07  // BW ≈ 361.4 Hz, NBW ≈ 376.5 Hz
00047 };
00048
00049 enum class ICM20948_Gyro_Average : uint8_t
00050 {
00051   AVG_1 = 0x00,      // 1x Average (no averaging)
00052   AVG_2 = 0x01,      // 2x Average
00053   AVG_4 = 0x02,      // 4x Average
00054   AVG_8 = 0x03,      // 8x Average
```

```
00055   AVG_16 = 0x04,     // Average 16 samples
00056   AVG_32 = 0x05,      // Average 32 samples
00057   AVG_64 = 0x06,     // Average 64 samples
00058   AVG_128 = 0x07     // Average 128 samples
00059 };
00060 enum class ICM20948_Gyro_FullScale : uint8_t
00061 {
00062   DPS_250 = 0x00,  // ±250 dps
00063   DPS_500 = 0x01,  // ±500 dps
00064   DPS_1000 = 0x02, // ±1000 dps
00065   DPS_2000 = 0x03  // ±2000 dps
00066 };
00067
00068 enum class ICM20948_Accel_FullScale : uint8_t
00069 {
00070   G_2 = 0x00, // ±2g
00071   G_4 = 0x01, // ±4g
00072   G_8 = 0x02, // ±8g
00073   G_16 = 0x03 // ±16g
00074 };
00075
00076 enum class ICM20948_Accel_Average : uint8_t
00077 {
00078   AVG_1_OR_4 = 0x00, // Depends on ACCEL_FCHOICE
00079   AVG_8 = 0x01,      // Average 8 samples
00080   AVG_16 = 0x02,     // Average 16 samples
00081   AVG_32 = 0x03      // Average 32 samples
00082 };
00083
00084 enum class ICM20948_Accel_DLPFCFG : uint8_t
00085 {
00086   DLPF0_246HZ = 0x00, // BW ≈ 246 Hz, NBW ≈ 265 Hz
00087   DLPF_246HZ = 0x01, // BW ≈ 246 Hz, NBW ≈ 265 Hz
00088   DLPF_111HZ = 0x02, // BW ≈ 111 Hz, NBW ≈ 136 Hz
00089   DLPF_50HZ = 0x03,  // BW ≈ 50 Hz,  NBW ≈ 68.8 Hz
00090   DLPF_24HZ = 0x04,  // BW ≈ 24 Hz,  NBW ≈ 34.4 Hz
00091   DLPF_12HZ = 0x05,  // BW ≈ 12 Hz,  NBW ≈ 17 Hz
00092   DLPF_6HZ = 0x06,   // BW ≈ 6 Hz,   NBW ≈ 8.3 Hz
00093   DLPF_473HZ = 0x07, // BW ≈ 473 Hz, NBW ≈ 499 Hz
00094 };
00095
00096 /**
00097  * ICM20948_IntPinConfig
00098  *
00099  * - Physical configuration of the INT pin (INT_PIN_CFG register, BANK 0)
00100  * - Controls electrical behavior and clear condition of the interrupt line
00101  * - Pass to intInit() to apply settings
00102  */
00103 struct ICM20948_IntPinConfig
00104 {
00105   uint8_t activeLevel    : 1;  // bit 7 | 0 = active high,       1 = active low
00106   uint8_t driveMode      : 1;  // bit 6 | 0 = push-pull,         1 = open-drain
00107   uint8_t latchMode      : 1;  // bit 5 | 0 = pulse 50μs,        1 = latch held
00108   uint8_t clearMode      : 1;  // bit 4 | 0 = read INT_STATUS,   1 = any read
00109   uint8_t fsyncActLevel  : 1;  // bit 3 | 0 = FSYNC active high, 1 = FSYNC active low
00110   uint8_t fsyncIntEn     : 1;  // bit 2 | 0 = FSYNC disabled,    1 = FSYNC as interrupt
00111   uint8_t bypassEn       : 1;  // bit 1 | 0 = normal,            1 = I2C bypass mode
00112 };
00113
00114 /**
00115  * ICM20948_IntEnableConfig
00116  *
00117  * - Selects which interrupt sources are routed to the INT pin
00118  * - Covers INT_ENABLE (0x10), INT_ENABLE_1 (0x11), INT_ENABLE_2 (0x12), INT_ENABLE_3 (0x13)
00119  * - FIFO overflow and watermark fields are per-channel arrays [0]-[4]
00120  * - Pass to intEnableConfig() to write all four registers in one call
00121  */
00122 struct ICM20948_IntEnableConfig
00123 {
00124   // --- INT_ENABLE (0x10) ---
00125   uint8_t wofEn        : 1;  // Wake on FSYNC
00126   uint8_t womIntEn     : 1;  // Wake on motion
00127   uint8_t pllRdyEn     : 1;  // PLL ready
00128   uint8_t dmpInt1En    : 1;  // DMP interrupt
00129   uint8_t i2cMstIntEn  : 1;  // I2C master interrupt
00130
00131   // --- INT_ENABLE_1 (0x11) ---
00132   uint8_t rawDataRdyEn : 1;  // Raw data ready
00133
00134   // --- INT_ENABLE_2 (0x12) --- canal por canal
00135   uint8_t fifoOvfEn[5];      // [0]-[4]: 1 = overflow interrupt ON para ese canal
00136
00137   // --- INT_ENABLE_3 (0x13) --- canal por canal
00138   uint8_t fifoWmEn[5];       // [0]-[4]: 1 = watermark interrupt ON para ese canal
00139 };
00140
00141 /**
```

```
00142  * ICM20948_IntStatus
00143  *
00144  * - Holds the parsed state of the four interrupt status registers
00145  * - Covers INT_STATUS (0x19), INT_STATUS_1 (0x1A), INT_STATUS_2 (0x1B), INT_STATUS_3 (0x1C)
00146  * - Populated by checkIntStatus() via a single 4-byte burst read
00147  * - Reading these registers clears the interrupt flags (when clearMode = 0 in IntPinConfig)
00148  */
00149 struct ICM20948_IntStatus
00150 {
00151   // --- INT_STATUS (0x19) ---
00152   uint8_t womInt     : 1;  // Wake on motion ocurrió
00153   uint8_t pllRdyInt  : 1;  // PLL listo
00154   uint8_t dmpInt1    : 1;  // DMP generó interrupción
00155   uint8_t i2cMstInt  : 1;  // I2C master generó interrupción
00156
00157   // --- INT_STATUS_1 (0x1A) ---
00158   uint8_t rawDataRdy : 1;  // Datos de sensores listos para leer
00159
00160   // --- INT_STATUS_2 (0x1B) ---
00161   uint8_t fifoOvf[5];      // [0]-[4]: FIFO overflow por canal
00162
00163   // --- INT_STATUS_3 (0x1C) ---
00164   uint8_t fifoWm[5];       // [0]-[4]: FIFO watermark por canal
00165 };
00166
00167 /**
00168  * Class
00169  * - Manages ICM-20948 over I2C (SPI path disabled in this cut)
00170  * - Caches scale factors for LSB->physical conversions
00171  */
00172 class DevLab_ICM20948
00173 {
00174 public:
00175
00176   DevLab_ICM20948();
00177
00178   /**
00179    * beginI2C
00180    *
00181    * - Initialize ICM20948 using I2C interface
00182    * - Sets up communication layer (Interface + BusIO)
00183    * - Verifies device identity (WHO_AM_I)
00184    * - Configures basic power and clock settings
00185    *
00186    * Returns:
00187    * - true  → Device initialized successfully
00188    * - false → Communication or configuration failed
00189    */
00190   bool beginI2C(uint8_t address = 0x69, TwoWire &i2cPort = Wire, uint32_t i2cSpeed = 400000);
00191
00192   bool beginSPI(uint8_t csPin, SPIClass &spiPort = SPI, uint32_t spiSpeed = 1000000);
00193
00194   /**
00195    * readWhoAmI
00196    *
00197    * - Read WHO_AM_I register (device ID)
00198    * - Used during initialization to verify ICM20948
00199    * - Expected value: 0xEA
00200    *
00201    * Returns:
00202    * - true  → Read successful
00203    * - false → Operation failed
00204    */
00205   bool readWhoAmI(uint8_t &whoAmI);
00206
00207   /**
00208    * selectBank
00209    *
00210    * - Select USER BANK (0-3)
00211    * - Used to access different register groups inside ICM20948
00212    *
00213    * Returns:
00214    * - true  → Bank switch successful
00215    * - false → Write failed
00216    */
00217   bool selectBank(uint8_t bank);
00218
00219   /**
00220    * softReset
00221    *
00222    * - Perform software reset of ICM20948
00223    * - Resets all internal registers to default state
00224    *
00225    * Flow:
00226    * - Select USER BANK 0
00227    * - Set DEVICE_RESET bit
00228    * - Wait for device to restart
```

```
00229     *
00230     * Returns:
00231     * - true  → Reset successful
00232     * - false → Operation failed
00233     */
00234    bool softReset();
00235
00236    /**
00237     * sleep
00238     *
00239     * - Enable or disable sleep mode
00240     * - Maintains clock source (CLKSEL = 1)
00241     *
00242     * Flow:
00243     * - Select USER BANK 0
00244     * - Set or clear SLEEP bit
00245     *
00246     * Returns:
00247     * - true  → Operation successful
00248     * - false → Operation failed
00249     */
00250    bool sleep(bool en);
00251
00252    /**
00253     * applyBasicDefaults
00254     *
00255     * - Apply basic sensor configuration
00256     * - Configure power, filters, ranges, and sampling rates
00257     *
00258     * Returns:
00259     * - true  → Configuration successful
00260     * - false → Any register write failed
00261     */
00262    bool applyBasicDefaults();
00263
00264    /**
00265     * readAccel
00266     *
00267     * - Read accelerometer data (X, Y, Z)
00268     * - Convert raw values to g using LSB scale
00269     *
00270     * Returns:
00271     * - true  → Read successful
00272     * - false → Read failed
00273     */
00274    bool readAccel(float &x, float &y, float &z);
00275
00276    /**
00277     * readGyro
00278     *
00279     * - Read gyroscope data (X, Y, Z)
00280     * - Convert raw values to dps using LSB scale
00281     *
00282     * Returns:
00283     * - true  → Read successful
00284     * - false → Read failed
00285     */
00286    bool readGyro(float &x, float &y, float &z);
00287
00288    /**
00289     * readTemperature
00290     *
00291     * - Read temperature sensor
00292     * - Convert raw values to °C
00293     *
00294     * Returns:
00295     * - true  → Read successful
00296     * - false → Read failed
00297     */
00298    bool readTemperature(float &temperature);
00299
00300    /**
00301     * readMag
00302     *
00303     * - Read magnetometer data via internal I2C master
00304     * - Data comes from EXT_SLV_SENS_DATA registers
00305     * - Convert raw values to μT
00306     *
00307     * Returns:
00308     * - true  → Read successful
00309     * - false → Read failed
00310     */
00311    bool readMag(float &x, float &y, float &z);
00312
00313    /**
00314     * initMag
00315     *
```

```
00316     * - Initialize AK09916 magnetometer using internal I2C master
00317     * - Verifies WHO_AM_I and enables continuous mode
00318     * - Configures SLV0 for automatic data read
00319     *
00320     * Returns:
00321     * - true  → Initialization successful
00322     * - false → Operation failed
00323     */
00324    bool initMag();
00325
00326    /**
00327     * setMagOpMode
00328     *
00329     * - Set magnetometer operation mode
00330     *
00331     * Returns:
00332     * - true  → Mode set successfully
00333     * - false → Write failed
00334     */
00335    bool setMagOpMode(ICM20948_Op_Mode opMode);
00336
00337    /**
00338     * setLowPower
00339     *
00340     * - Enable or disable low power mode
00341     * - Controls LP_EN bit in PWR_MGMT_1
00342     *
00343     * Returns:
00344     * - true  → Operation successful
00345     * - false → Operation failed
00346     */
00347    bool setLowPower(bool enable);
00348
00349    /**
00350     * getLowPower
00351     *
00352     * - Read low power mode status
00353     * - Returns state of LP_EN bit
00354     *
00355     * Returns:
00356     * - true  → Read successful
00357     * - false → Operation failed
00358     */
00359    bool getLowPower(bool &enable);
00360
00361    /**
00362     * setClock
00363     *
00364     * - Set clock source (CLKSEL [2:0])
00365     * - Selects internal clock for sensor operation
00366     *
00367     * Returns:
00368     * - true  → Operation successful
00369     * - false → Operation failed
00370     */
00371    bool setClock(ICM20948_Clock_Source clock);
00372
00373    /**
00374     * getClock
00375     *
00376     * - Read current clock source (CLKSEL [2:0])
00377     *
00378     * Returns:
00379     * - true  → Read successful
00380     * - false → Operation failed
00381     */
00382    bool getClock(uint8_t &clock);
00383
00384    /**
00385     * setGyroSampleRate
00386     *
00387     * - Set gyroscope sample rate
00388     * - Uses internal divider (SRD) based on 1100 Hz base rate
00389     *
00390     * Returns:
00391     * - true  → Operation successful
00392     * - false → Invalid input or write failed
00393     */
00394    bool setGyroSampleRate(float sampleRate);
00395
00396    /**
00397     * getGyroSampleRate
00398     *
00399     * - Get current gyroscope sample rate
00400     * - Computes value from divider register
00401     *
00402     * Returns:
```

```
00403    * - true  → Read successful
00404    * - false → Operation failed
00405    */
00406   bool getGyroSampleRate(float &sampleRate);
00407   bool setGyroDivRate(uint8_t divisor);
00408   /**
00409    * setDLPF
00410    *
00411    * - Configure gyroscope digital low-pass filter (DLPF)
00412    * - Option to bypass filter (full bandwidth)
00413    *
00414    * Returns:
00415    * - true  → Operation successful
00416    * - false → Operation failed
00417    */
00418   bool setDLPF(ICM20948_Gyro_DLPF dlpf, bool bypass);
00419
00420   /**
00421    * getDLPF
00422    *
00423    * - Read gyroscope DLPF configuration
00424    * - Returns filter setting and bypass state
00425    *
00426    * Returns:
00427    * - true  → Read successful
00428    * - false → Operation failed
00429    */
00430   bool getDLPF(uint8_t &dlpf, bool &bypass);
00431
00432   /**
00433    * setGyroScale
00434    *
00435    * - Set gyroscope full-scale range (FS_SEL)
00436    * - Controls sensitivity (dps range)
00437    *
00438    * Flow:
00439    * - Validate bus pointer
00440    * - Select USER BANK 2
00441    * - Write FS_SEL bits [2:1]
00442    *
00443    * Returns:
00444    * - true  → Operation successful
00445    * - false → Operation failed
00446    */
00447   bool setGyroScale(ICM20948_Gyro_FullScale fullScale);
00448   bool setGyroAveraging(ICM20948_Gyro_Average avg);
00449   /**
00450    * getGyroScale
00451    *
00452    * - Get current gyroscope full-scale range (FS_SEL
00453    *
00454    * Returns:
00455    * - true  → Read successful
00456    * - false → Operation failed
00457    */
00458   bool getGyroScale(uint8_t &fullScale);
00459
00460   /**
00461    * selfTestGyro
00462    *
00463    * - Enable or disable gyroscope self-test per axis
00464    *
00465    * Returns:
00466    * - true  → Operation successful
00467    * - false → Operation failed
00468    */
00469   bool selfTestGyro(bool x, bool y, bool z);
00470
00471   bool setAccelDivRate(uint16_t divisor);
00472   /**
00473    * setAccelSampleRate
00474    *
00475    * - Set accelerometer output data rate (ODR)
00476    * - Uses 1125 Hz base rate with 12-bit divider
00477    *
00478    * Returns:
00479    * - true  → Operation successful
00480    * - false → Operation failed
00481    */
00482   bool setAccelSampleRate(uint16_t sampleRate);
00483
00484   /**
00485    * getAccelSampleRate
00486    *
00487    * - Get accelerometer output data rate (ODR)
00488    * - Computes value from divider registers
00489    *
```

```
00490     * Returns:
00491     * - true  → Read successful
00492     * - false → Operation failed
00493     */
00494    bool getAccelSampleRate(float &sampleRate);
00495
00496    /**
00497     * selfTestAccel
00498     *
00499     * - Enable or disable accelerometer self-test per axis
00500     *
00501     * Returns:
00502     * - true  → Operation successful
00503     * - false → Operation failed
00504     */
00505    bool selfTestAccel(bool x, bool y, bool z);
00506
00507    /**
00508     * setAccelScale
00509     *
00510     * - Set accelerometer full-scale range (FS_SEL)
00511     * - Controls measurement range (±2g, ±4g, ±8g, ±16g)
00512     *
00513     * Returns:
00514     * - true  → Operation successful
00515     * - false → Operation failed
00516     */
00517    bool setAccelScale(ICM20948_Accel_FullScale fullScale);
00518
00519    /**
00520     * getAccelScale
00521     *
00522     * - Get current accelerometer full-scale range (FS_SEL)
00523     *
00524     * Returns:
00525     * - true  → Read successful
00526     * - false → Operation failed
00527     */
00528    bool getAccelScale(uint8_t &fullScale);
00529
00530    /**
00531     * setAccelDLPF
00532     *
00533     * - Configure accelerometer digital low-pass filter (DLPF)
00534     * - Option to bypass filter (full bandwidth)
00535     *
00536     * Flow:
00537     * - Validate bus pointer
00538     * - Select USER BANK 2
00539     * - Set or clear bypass bit
00540     * - Configure DLPF bits if not bypassed
00541     *
00542     * Returns:
00543     * - true  → Operation successful
00544     * - false → Operation failed
00545     */
00546    bool setAccelDLPF(uint8_t dlpf, bool bypass);
00547
00548    /**
00549     * getAccelDLPF
00550     *
00551     * - Read accelerometer DLPF configuration
00552     * - Returns filter setting and bypass state
00553     *
00554     * Flow:
00555     * - Validate bus pointer
00556     * - Select USER BANK 2
00557     * - Read ACCEL_CONFIG register
00558     * - Extract bypass and DLPF bits
00559     *
00560     * Returns:
00561     * - true  → Read successful
00562     * - false → Operation failed
00563     */
00564    bool getAccelDLPF(uint8_t &dlpf, bool &bypass);
00565
00566    /**
00567     * setAccelAveraging
00568     *
00569     * - Configure accelerometer averaging / decimation (DEC3)
00570     * - Controls internal averaging of accel samples
00571     *
00572     * Flow:
00573     * - Validate bus pointer
00574     * - Select USER BANK 2
00575     * - Write DEC3 bits [1:0]
00576     *
```

```
00577     * Returns:
00578     * - true  → Operation successful
00579     * - false → Operation failed
00580     */
00581    bool setAccelAveraging(ICM20948_Accel_Average avg);
00582
00583    /**
00584     * getAccelAveraging
00585     *
00586     * - Read accelerometer averaging / decimation setting (DEC3)
00587     *
00588     * Flow:
00589     * - Validate bus pointer
00590     * - Select USER BANK 2
00591     * - Read DEC3 bits [1:0]
00592     *
00593     * Returns:
00594     * - true  → Read successful
00595     * - false → Operation failed
00596     */
00597    bool getAccelAveraging(uint8_t &avg);
00598
00599    /**
00600     * setSensors
00601     *
00602     * - Enable or disable accel, gyro, and temperature sensor
00603     * - Controls power gating via PWR_MGMT_2 and TEMP_DIS
00604     *
00605     * Flow:
00606     * - Validate bus pointer
00607     * - Select USER BANK 0
00608     * - Build PWR_MGMT_2 mask
00609     * - Configure temperature enable/disable
00610     *
00611     * Returns:
00612     * - true  → Operation successful
00613     * - false → Operation failed
00614     */
00615    bool setSensors(bool accel_on, bool gyro_on, bool temp_on);
00616
00617    /**
00618     * getSensors
00619     *
00620     * - Read accel, gyro, and temperature enable state
00621     *
00622     * Flow:
00623     * - Validate bus pointer
00624     * - Select USER BANK 0
00625     * - Read PWR_MGMT_2 register
00626     * - Read TEMP_DIS bit
00627     * - Decode enable states
00628     *
00629     * Returns:
00630     * - true  → Read successful
00631     * - false → Operation failed
00632     */
00633    bool getSensors(bool &accel_on, bool &gyro_on, bool &temp_on);
00634
00635    /**
00636     * setGyroOffset
00637     *
00638     * - Set gyroscope offset values for X, Y, Z axes
00639     * - Writes offset registers in USER BANK 2
00640     *
00641     * Flow:
00642     * - Validate bus pointer
00643     * - Select USER BANK 2
00644     * - Pack offsets into byte array
00645     * - Write to offset registers
00646     *
00647     * Returns:
00648     * - true  → Operation successful
00649     * - false → Operation failed
00650     */
00651    bool setGyroOffset(uint16_t offsetX, uint16_t offsetY, uint16_t offsetZ);
00652
00653    /**
00654     * getGyroOffset
00655     *
00656     * - Read gyroscope offset values for X, Y, Z axes
00657     * - Reads offset registers from USER BANK 2
00658     *
00659     * Flow:
00660     * - Validate bus pointer
00661     * - Select USER BANK 2
00662     * - Read offset registers
00663     * - Convert to signed values
```

```
00664    *
00665    * Returns:
00666    * - true  → Read successful
00667    * - false → Operation failed
00668    */
00669   bool getGyroOffset(int16_t &offsetX, int16_t &offsetY, int16_t &offsetZ);
00670
00671   /**
00672    * setAccelOffset
00673    *
00674    * - Set accelerometer offset values for X, Y, Z axes
00675    * - Writes offset registers in USER BANK 1
00676    *
00677    * Flow:
00678    * - Validate bus pointer
00679    * - Select USER BANK 1
00680    * - Pack offsets into byte array
00681    * - Write to offset registers
00682    *
00683    * Returns:
00684    * - true  → Operation successful
00685    * - false → Operation failed
00686    */
00687   bool setAccelOffset(int16_t offsetX, int16_t offsetY, int16_t offsetZ);
00688
00689   /**
00690    * getAccelOffset
00691    *
00692    * - Read accelerometer offset values for X, Y, Z axes
00693    * - Reads offset registers from USER BANK 1
00694    *
00695    * Flow:
00696    * - Validate bus pointer
00697    * - Select USER BANK 1
00698    * - Read offset registers
00699    * - Convert to signed values
00700    *
00701    * Returns:
00702    * - true  → Read successful
00703    * - false → Operation failed
00704    */
00705   bool getAccelOffset(int16_t &offsetX, int16_t &offsetY, int16_t &offsetZ);
00706
00707   /**
00708    * intInit
00709    *
00710    * - Configure the physical behavior of the INT pin (INT_PIN_CFG register, BANK 0)
00711    * - Sets active level, drive mode (push-pull / open-drain), latch vs pulse mode,
00712    *   clear condition, FSYNC level, FSYNC interrupt enable, and I2C bypass
00713    *
00714    * Parameters:
00715    * - cfg: ICM20948_IntPinConfig struct with the desired pin settings
00716    *
00717    * Returns:
00718    * - true  → Configuration written successfully
00719    * - false → Operation failed
00720    */
00721   bool intInit(const ICM20948_IntPinConfig &cfg);
00722
00723   /**
00724    * intEnableConfig
00725    *
00726    * - Enable or disable individual interrupt sources routed to the INT pin
00727    * - Writes INT_ENABLE (0x10), INT_ENABLE_1 (0x11), INT_ENABLE_2 (0x12),
00728    *   and INT_ENABLE_3 (0x13) in BANK 0
00729    * - Must call intInit() first to configure the pin electrical behavior
00730    *
00731    * Parameters:
00732    * - cfg: ICM20948_IntEnableConfig struct with the desired interrupt sources enabled
00733    *
00734    * Returns:
00735    * - true  → All four registers written successfully
00736    * - false → Operation failed
00737    */
00738   bool intEnableConfig(const ICM20948_IntEnableConfig &cfg);
00739
00740   /**
00741    * checkIntStatus
00742    *
00743    * - Read and parse all four interrupt status registers in a single burst read
00744    * - Covers INT_STATUS (0x19), INT_STATUS_1 (0x1A), INT_STATUS_2 (0x1B),
00745    *   INT_STATUS_3 (0x1C) from BANK 0
00746    * - Reading clears the interrupt flags when clearMode = 0 in ICM20948_IntPinConfig
00747    * - Typically called inside the INT pin ISR or polled in the main loop
00748    *
00749    * Parameters:
00750    * - status: ICM20948_IntStatus struct populated with the current interrupt flags
```

```
00751    *
00752    * Returns:
00753    * - true  → Status read successfully
00754    * - false → Operation failed
00755    */
00756   bool checkIntStatus(ICM20948_IntStatus &status);
00757
00758   /**
00759    * auxMasterEnable
00760    *
00761    * - Enable the ICM20948 internal I2C master for auxiliary bus
00762    * - Clears BYPASS_EN so the aux bus is controlled by the ICM
00763    * - Sets I2C_MST_EN in USER_CTRL
00764    * - Configures auxiliary master clock frequency
00765    *
00766    * Parameters:
00767    * - clkFreq: clock divider value (e.g. 0x07 ≈ 345.6 kHz, 0x0D ≈ 400 kHz)
00768    *
00769    * Returns:
00770    * - true  → Master enabled successfully
00771    * - false → Operation failed
00772    */
00773   bool auxMasterEnable(uint8_t clkFreq);
00774
00775   /**
00776    * auxWriteByte
00777    *
00778    * - Write a single byte to a register of an auxiliary I2C slave via SLV4
00779    * - Uses one-shot SLV4 transaction: polls SLV4_DONE, checks for NACK
00780    * - Suitable for configuration writes (not continuous streaming)
00781    *
00782    * Parameters:
00783    * - slaveAddr: 7-bit I2C address of the auxiliary slave
00784    * - reg:       target register address on the slave
00785    * - data:      byte to write
00786    *
00787    * Returns:
00788    * - true  → Write acknowledged by slave
00789    * - false → Timeout or NACK
00790    */
00791   bool auxWriteByte(uint8_t slaveAddr, uint8_t reg, uint8_t data);
00792
00793   /**
00794    * auxReadByte
00795    *
00796    * - Read a single byte from a register of an auxiliary I2C slave via SLV4
00797    * - Generates a combined write-then-read transaction (reg byte + repeated START)
00798    * - Suitable for slaves that follow standard I2C register-read protocol
00799    *
00800    * Note: slaves that need separate write/read transactions (e.g. command-response
00801    * firmware) should use auxWriteCommand + auxReadResponse instead.
00802    *
00803    * Parameters:
00804    * - slaveAddr: 7-bit I2C address of the auxiliary slave
00805    * - reg:       register address to read from
00806    * - data:      output byte received from slave
00807    *
00808    * Returns:
00809    * - true  → Read successful
00810    * - false → Timeout or NACK
00811    */
00812   bool auxReadByte(uint8_t slaveAddr, uint8_t reg, uint8_t &data);
00813
00814   /**
00815    * auxWriteCommand
00816    *
00817    * - Send a single command byte to an auxiliary I2C slave via SLV4
00818    * - Uses REG_DIS: generates [START, addr+W, cmd_byte, STOP] with no register prefix
00819    * - Designed for slaves that use a command-response protocol (no register addressing)
00820    *
00821    * Parameters:
00822    * - slaveAddr: 7-bit I2C address of the auxiliary slave
00823    * - cmd:       command byte to send
00824    *
00825    * Returns:
00826    * - true  → Command acknowledged by slave
00827    * - false → Timeout or NACK
00828    */
00829   bool auxWriteCommand(uint8_t slaveAddr, uint8_t cmd);
00830
00831   /**
00832    * auxConfigSlave
00833    *
00834    * - Configure SLV0 for automatic periodic reads from an auxiliary I2C slave
00835    * - The ICM20948 master will read numBytes starting at reg on every ODR cycle
00836    * - Data is deposited into EXT_SLV_SENS_DATA_00 and subsequent registers
00837    * - Call once during initialization; read results with auxReadSensorData
```

```
00838     *
00839     * Parameters:
00840     * - slaveAddr: 7-bit I2C address of the auxiliary slave
00841     * - reg:       starting register address to read from on the slave
00842     * - numBytes:  number of bytes to read per cycle (1-15)
00843     *
00844     * Returns:
00845     * - true  → SLV0 configured successfully
00846     * - false → Operation failed
00847     */
00848    bool auxConfigSlave(uint8_t slaveAddr, uint8_t reg, uint8_t numBytes);
00849
00850    /**
00851     * auxReadSensorData
00852     *
00853     * - Read bytes from EXT_SLV_SENS_DATA registers (BANK 0)
00854     * - Returns data collected by the ICM20948 master from SLV0-SLV3 on the last cycle
00855     * - Must call auxConfigSlave first to set up the automatic read
00856     *
00857     * Parameters:
00858     * - buf: output buffer to store the received bytes
00859     * - len: number of bytes to read (must match numBytes configured in auxConfigSlave)
00860     *
00861     * Returns:
00862     * - true  → Read successful
00863     * - false → Bus error
00864     */
00865    bool auxReadSensorData(uint8_t *buf, uint8_t len);
00866
00867    /**
00868     * auxReadResponse
00869     *
00870     * - Read a single response byte from an auxiliary I2C slave via SLV4
00871     * - Generates a pure read transaction: [START, addr+R, 1 byte, STOP]
00872     * - No register byte is sent (REG_DIS); slave must already be ready to transmit
00873     * - Use after auxWriteCommand to complete a command-response exchange with slaves
00874     *   that require separate write and read transactions (e.g. PY32F0 firmware slaves)
00875     *
00876     * Parameters:
00877     * - slaveAddr: 7-bit I2C address of the auxiliary slave
00878     * - data:      output byte received from slave
00879     *
00880     * Returns:
00881     * - true  → Response received successfully
00882     * - false → Timeout or NACK
00883     */
00884    bool auxReadResponse(uint8_t slaveAddr, uint8_t &data);
00885
00886    /**
00887     * auxRead12bit
00888     *
00889     * - Read a 12-bit value previously configured via auxConfigSlave (SLV0, numBytes = 2)
00890     * - Assumes little-endian packing: first byte = LSB, second byte = MSB
00891     * - Result is masked to 12 bits (0-4095)
00892     *
00893     * Returns:
00894     * - true  → Read successful
00895     * - false → Read failed
00896     */
00897    bool auxRead12bit(uint16_t &raw);
00898  private:
00899    I2C_Interface i2c;
00900    SPI_Interface spi;
00901
00902    Interface_7Semi *iface = nullptr;
00903
00904    BusIO_7Semi<Interface_7Semi> *bus = nullptr;
00905
00906    float mg_per_lsb = 16384.0f;        // LSB/g at ±16g
00907    float degree_per_second = 131.072f; // LSB/dps at ±2000 dps
00908
00909    bool writeSlave4(uint8_t reg, uint8_t value);
00910    bool readSlave4(uint8_t reg, uint8_t &value);
00911  };
00912
00913
00914  #endif /* ICM20948_DEVLAB_H */
```

6.38 src/ICM20948_platform.h File Reference

Classes

- struct icm_plat_t

6.39 ICM20948_platform.h

Go to the documentation of this file.

```
00001 #ifndef ICM20948_PLATFORM_H
00002 #define ICM20948_PLATFORM_H
00003
00004 // #include <stdint.h>
00005 // #include <stddef.h>
00006
00007 /**
00008  * 7Semi comment style
00009  * - Platform glue for bus IO, delays, and timing
00010  * - Implement these for your system (I2C or SPI)
00011  */
00012
00013 typedef struct {
00014   /**
00015      * - Bus read: read `len` bytes from `reg` into `buf`
00016      * - Return 0 on success, nonzero on error
00017      */
00018   int (*read)(uint8_t reg, uint8_t *buf, size_t len, void *user);
00019   /**
00020      * - Bus write: write `len` bytes from `buf` to `reg`
00021      * - Return 0 on success, nonzero on error
00022      */
00023   int (*write)(uint8_t reg, const uint8_t *buf, size_t len, void *user);
00024   /**
00025      * - Optional: SPI transfers may need register R/W bit mangling
00026      * - For I2C, leave as NULL
00027      */
00028   uint8_t (*spi_reg_rw_bits)(uint8_t reg, int is_write);
00029   /**
00030      * - Millisecond delay
00031      */
00032   void (*delay_ms)(uint32_t ms);
00033   /**
00034      * - User context pointer passed to read/write
00035      */
00036   void *user;
00037   /**
00038      * - If using SPI: set chip select active (1) or inactive (0)
00039      * - For I2C: leave as NULL
00040      */
00041   void (*spi_cs)(int level, void *user);
00042   /**
00043      * - Device address (I2C) or ignored for SPI if you wrap inside user
00044      */
00045   uint8_t i2c_addr;
00046 } icm_plat_t;
00047
00048 #endif
```

6.40 src/ICM20948_regs.h File Reference

This graph shows which files directly or indirectly include this file:

Macros

- #define WHO_AM_I 0x00 /* WHO_AM_I[7:0] */
- #define WHO_AM_I_VAL 0xEA
- #define USER_CTRL 0x03 /* DMP_EN FIFO_EN I2C_MST_EN I2C_IF_DIS DMP_RST SRAM_RST I2C↩ _MST_RST - */
- #define USER_CTRL_DMP_EN BIT(7)
- #define USER_CTRL_FIFO_EN BIT(6)
- #define USER_CTRL_I2C_MST_EN BIT(5)
- #define USER_CTRL_I2C_IF_DIS BIT(4)
- #define USER_CTRL_DMP_RST BIT(3)
- #define USER_CTRL_SRAM_RST BIT(2)

- #define USER_CTRL_I2C_MST_RST BIT(1)
- #define LP_CONFIG 0x05 /* I2C_MST_CYCLE ACCEL_CYCLE GYRO_CYCLE - */
- #define LP_I2C_MST_CYCLE BIT(6)
- #define LP_ACCEL_CYCLE BIT(5)
- #define LP_GYRO_CYCLE BIT(4)
- #define PWR_MGMT_1 0x06 /* DEVICE_RESET SLEEP LP_EN - TEMP_DIS CLKSEL[2:0] */
- #define PWR_DEVICE_RESET BIT(7)
- #define PWR_SLEEP BIT(6)
- #define PWR_LP_EN BIT(5)
- #define PWR_TEMP_DIS BIT(3)
- #define PWR_CLKSEL_MASK 0x07
- #define PWR_CLKSEL_INT_20MHZ 0x01
- #define PWR_CLKSEL_AUTO 0x01 /* typical: auto selects best source */
- #define PWR_MGMT_2 0x07 /* - DISABLE_ACCEL DISABLE_GYRO */
- #define PWR_DISABLE_ACCEL BIT(3)
- #define PWR_DISABLE_GYRO BIT(0)
- #define INT_PIN_CFG 0x0F /* INT1_ACTL INT1_OPEN INT1_LATCH_INT_EN INT_ANYRD_2CLEAR ACTL_FSYNC FSYNC_INT_MODE_EN BYPASS_EN - */
- #define INT1_ACTL BIT(7) /* active low */
- #define INT1_OPEN BIT(6) /* open-drain */
- #define INT1_LATCH_INT_EN BIT(5) /* latch until status read */
- #define INT_ANYRD_2CLEAR BIT(4)
- #define INT_ACTL_FSYNC BIT(3)
- #define FSYNC_INT_MODE_EN BIT(2)
- #define BYPASS_EN BIT(1) /* I2C bypass to aux devices */
- #define INT_ENABLE 0x10 /* REG_WOF_EN - WOM_INT_EN PLL_RDY_EN DMP_INT1_EN I2C_MST_↩ INT_EN */
- #define INT_REG_WOF_EN BIT(7)
- #define INT_WOM_INT_EN BIT(3)
- #define INT_PLL_RDY_EN BIT(2)
- #define INT_DMP_INT1_EN BIT(1)
- #define INT_I2C_MST_INT_EN BIT(0)
- #define INT_ENABLE_1 0x11 /* RAW_DATA_0_RDY_EN */
- #define INT_RAW_DATA_0_RDY_EN BIT(0)
- #define INT_ENABLE_2 0x12 /* FIFO_OVERFLOW_EN[4:0] */
- #define INT_ENABLE_3 0x13 /* FIFO_WM_EN[4:0] */
- #define I2C_MST_STATUS 0x17 /* PASS_THROUGH, *_DONE, *_NACK, LOST_ARB */
- #define MST_PASS_THROUGH BIT(7)
- #define MST_SLV4_DONE BIT(6)
- #define MST_LOST_ARB BIT(5)
- #define MST_SLV4_NACK BIT(4)
- #define MST_SLV3_NACK BIT(3)
- #define MST_SLV2_NACK BIT(2)
- #define MST_SLV1_NACK BIT(1)
- #define MST_SLV0_NACK BIT(0)
- #define INT_STATUS 0x19 /* WOM_INT PLL_RDY_INT DMP_INT1 I2C_MST_INT */
- #define STS_WOM_INT BIT(3)
- #define STS_PLL_RDY_INT BIT(2)
- #define STS_DMP_INT1 BIT(1)
- #define STS_I2C_MST_INT BIT(0)
- #define INT_STATUS_1 0x1A /* RAW_DATA_0_RDY_INT */
- #define STS_RAW_DATA_0_RDY_INT BIT(0)
- #define INT_STATUS_2 0x1B /* FIFO_OVERFLOW_INT[4:0] */
- #define INT_STATUS_3 0x1C /* FIFO_WM_INT[4:0] */
- #define DELAY_TIMEH 0x28

- #define DELAY_TIMEL 0x29
- #define ACCEL_XOUT_H 0x2D
- #define ACCEL_XOUT_L 0x2E
- #define ACCEL_YOUT_H 0x2F
- #define ACCEL_YOUT_L 0x30
- #define ACCEL_ZOUT_H 0x31
- #define ACCEL_ZOUT_L 0x32
- #define GYRO_XOUT_H 0x33
- #define GYRO_XOUT_L 0x34
- #define GYRO_YOUT_H 0x35
- #define GYRO_YOUT_L 0x36
- #define GYRO_ZOUT_H 0x37
- #define GYRO_ZOUT_L 0x38
- #define TEMP_OUT_H 0x39
- #define TEMP_OUT_L 0x3A
- #define EXT_SLV_SENS_DATA_00 0x3B
- #define EXT_SLV_SENS_DATA_23 0x52 /* contiguous range 0x3B..0x52 */
- #define FIFO_EN_1 0x66 /* SLV3..SLV0 FIFO_EN */
- #define FIFO_EN_2 0x67 /* ACCEL GYRO_Z/Y/X TEMP FIFO_EN */
- #define FIFO_RST 0x68 /* FIFO_RESET[4:0] */
- #define FIFO_MODE 0x69 /* FIFO_MODE[4:0] */
- #define FIFO_COUNTH 0x70
- #define FIFO_COUNTL 0x71
- #define FIFO_R_W 0x72
- #define FIFO_CFG 0x76
- #define REG_BANK_SEL 0x7F /* USER_BANK[1:0] */
- #define SELF_TEST_X_GYRO 0x02 /* XG_ST_DATA[7:0] */
- #define SELF_TEST_Y_GYRO 0x03 /* YG_ST_DATA[7:0] */
- #define SELF_TEST_Z_GYRO 0x04 /* ZG_ST_DATA[7:0] */
- #define SELF_TEST_X_ACCEL 0x0E /* XA_ST_DATA[7:0] */
- #define SELF_TEST_Y_ACCEL 0x0F /* YA_ST_DATA[7:0] */
- #define SELF_TEST_Z_ACCEL 0x10 /* ZA_ST_DATA[7:0] */
- #define XA_OFFS_H 0x14 /* XA_OFFS[14:7] */
- #define XA_OFFS_L 0x15 /* XA_OFFS[6:0] */
- #define YA_OFFS_H 0x17 /* YA_OFFS[14:7] */
- #define YA_OFFS_L 0x18 /* YA_OFFS[6:0] */
- #define ZA_OFFS_H 0x1A /* ZA_OFFS[14:7] */
- #define ZA_OFFS_L 0x1B /* ZA_OFFS[6:0] */
- #define TIMEBASE_CORRECTION_PLL 0x28 /* TBC_PLL[7:0] */
- #define BANK1_REG_BANK_SEL 0x7F /* mirror of REG_BANK_SEL */
- #define GYRO_SMPLRT_DIV 0x00 /* GYRO_SMPLRT_DIV[7:0] */
- #define GYRO_CONFIG_1 0x01 /* GYRO_DLPFCFG[2:0] GYRO_FS_SEL[1:0] GYRO_FCHOICE */
- #define GYRO_FCHOICE BIT(0) /* when 0: DLPF on, when 1: off (per datasheet) */
- #define GYRO_FS_SEL_SHIFT 1
- #define GYRO_FS_SEL_MASK (0x3u << GYRO_FS_SEL_SHIFT)
- #define GYRO_FS_250DPS (0u << GYRO_FS_SEL_SHIFT)
- #define GYRO_FS_500DPS (1u << GYRO_FS_SEL_SHIFT)
- #define GYRO_FS_1000DPS (2u << GYRO_FS_SEL_SHIFT)
- #define GYRO_FS_2000DPS (3u << GYRO_FS_SEL_SHIFT)
- #define GYRO_DLPFCFG_SHIFT 5
- #define GYRO_DLPFCFG_MASK (0x7u << GYRO_DLPFCFG_SHIFT)
- #define GYRO_CONFIG_2 0x02 /* XGYRO_CTEN YGYRO_CTEN ZGYRO_CTEN GYRO_AVGCFG[2:0] */
- #define XGYRO_CTEN BIT(5)
- #define YGYRO_CTEN BIT(4)
- #define ZGYRO_CTEN BIT(3)

- #define GYRO_AVGCFG_SHIFT 0
- #define GYRO_AVGCFG_MASK (0x7u << GYRO_AVGCFG_SHIFT)
- #define XG_OFFS_USRH 0x03
- #define XG_OFFS_USRL 0x04
- #define YG_OFFS_USRH 0x05
- #define YG_OFFS_USRL 0x06
- #define ZG_OFFS_USRH 0x07
- #define ZG_OFFS_USRL 0x08
- #define ODR_ALIGN_EN 0x09 /* ODR_ALIGN_EN */
- #define ODR_ALIGN_EN_BIT BIT(0)
- #define ACCEL_SMPLRT_DIV_1 0x10 /* ACCEL_SMPLRT_DIV[11:8] */
- #define ACCEL_SMPLRT_DIV_2 0x11 /* ACCEL_SMPLRT_DIV[7:0] */
- #define ACCEL_INTEL_CTRL 0x12 /* ACCEL_INTEL_EN ACCEL_INTEL_MODE_INT */
- #define ACCEL_INTEL_EN BIT(7)
- #define ACCEL_INTEL_MODE_INT BIT(6)
- #define ACCEL_WOM_THR 0x13 /* WOM_THRESHOLD[7:0] */
- #define ACCEL_CONFIG 0x14 /* ACCEL_DLPFCFG[2:0] ACCEL_FS_SEL[1:0] ACCEL_FCHOICE */
- #define ACCEL_FCHOICE BIT(0)
- #define ACCEL_FS_SEL_SHIFT 1
- #define ACCEL_FS_SEL_MASK (0x3u << ACCEL_FS_SEL_SHIFT)
- #define ACCEL_FS_2G (0u << ACCEL_FS_SEL_SHIFT)
- #define ACCEL_FS_4G (1u << ACCEL_FS_SEL_SHIFT)
- #define ACCEL_FS_8G (2u << ACCEL_FS_SEL_SHIFT)
- #define ACCEL_FS_16G (3u << ACCEL_FS_SEL_SHIFT)
- #define ACCEL_DLPFCFG_SHIFT 5
- #define ACCEL_DLPFCFG_MASK (0x7u << ACCEL_DLPFCFG_SHIFT)
- #define ACCEL_CONFIG_2 0x15 /* AX_ST_EN_REG AY_ST_EN_REG AZ_ST_EN_REG DEC3_CFG[1:0] */
- #define AX_ST_EN_REG BIT(7)
- #define AY_ST_EN_REG BIT(6)
- #define AZ_ST_EN_REG BIT(5)
- #define DEC3_CFG_SHIFT 0
- #define DEC3_CFG_MASK (0x3u << DEC3_CFG_SHIFT)
- #define FSYNC_CONFIG 0x52 /* DELAY_TIME_EN WOF_DEGLITCH_EN WOF_EDGE_INT EXT_SYNC↩ _SET[3:0] */
- #define DELAY_TIME_EN BIT(7)
- #define WOF_DEGLITCH_EN BIT(6)
- #define WOF_EDGE_INT BIT(5)
- #define EXT_SYNC_SET_MASK 0x0F
- #define TEMP_CONFIG 0x53 /* TEMP_DLPFCFG[2:0] */
- #define TEMP_DLPFCFG_SHIFT 5
- #define TEMP_DLPFCFG_MASK (0x7u << TEMP_DLPFCFG_SHIFT)
- #define MOD_CTRL_USR 0x54 /* REG_LP_DMP_EN */
- #define REG_LP_DMP_EN BIT(7)
- #define BANK2_REG_BANK_SEL 0x7F /* mirror of REG_BANK_SEL */
- #define I2C_MST_ODR_CONFIG 0x00 /* I2C_MST_ODR_CONFIG[3:0] */
- #define MST_ODR_CFG_MASK 0x0F
- #define I2C_MST_CTRL 0x01 /* MULT_MST_EN - I2C_MST_P_NSR I2C_MST_CLK[3:0] */
- #define MULT_MST_EN BIT(7)
- #define I2C_MST_P_NSR BIT(4)
- #define I2C_MST_CLK_MASK 0x0F
- #define I2C_MST_DELAY_CTRL 0x02 /* DELAY_ES_SHADOW + I2C_SLVx_DELAY_EN bits */
- #define DELAY_ES_SHADOW BIT(7)
- #define I2C_SLV4_DELAY_EN BIT(4)
- #define I2C_SLV3_DELAY_EN BIT(3)

- #define I2C_SLV2_DELAY_EN BIT(2)
- #define I2C_SLV1_DELAY_EN BIT(1)
- #define I2C_SLV0_DELAY_EN BIT(0)
- #define I2C_SLV0_ADDR 0x03 /* RNW + ID[6:0] */
- #define I2C_SLVx_RNW BIT(7)
- #define I2C_SLV0_REG 0x04
- #define I2C_SLV0_CTRL 0x05 /* EN BYTE_SW REG_DIS GRP LENG[3:0] */
- #define I2C_SLVx_EN BIT(7)
- #define I2C_SLVx_BYTE_SW BIT(6)
- #define I2C_SLVx_REG_DIS BIT(5)
- #define I2C_SLVx_GRP BIT(4)
- #define I2C_SLVx_LENG_MASK 0x0F
- #define I2C_SLV0_DO 0x06
- #define I2C_SLV1_ADDR 0x07
- #define I2C_SLV1_REG 0x08
- #define I2C_SLV1_CTRL 0x09
- #define I2C_SLV1_DO 0x0A
- #define I2C_SLV2_ADDR 0x0B
- #define I2C_SLV2_REG 0x0C
- #define I2C_SLV2_CTRL 0x0D
- #define I2C_SLV2_DO 0x0E
- #define I2C_SLV3_ADDR 0x0F
- #define I2C_SLV3_REG 0x10
- #define I2C_SLV3_CTRL 0x11
- #define I2C_SLV3_DO 0x12
- #define I2C_SLV4_ADDR 0x13
- #define I2C_SLV4_REG 0x14
- #define I2C_SLV4_CTRL 0x15 /* EN BYTE_SW REG_DIS DLY[4:0] */
- #define I2C_SLV4_DLY_MASK 0x1F
- #define I2C_SLV4_DO 0x16
- #define I2C_SLV4_DI 0x17
- #define BANK3_REG_BANK_SEL 0x7F /* mirror of REG_BANK_SEL */
- #define REG_BANK_SEL_USER_BANK_SHIFT 4
- #define REG_BANK_SEL_USER_BANK_MASK (0x3u << REG_BANK_SEL_USER_BANK_SHIFT)
- #define USER_BANK_0 BANK0
- #define USER_BANK_1 BANK1
- #define USER_BANK_2 BANK2
- #define USER_BANK_3 BANK3
- #define AK09916_I2C_ADDR 0x0C
- #define AK_WIA2 0x01
- #define AK_ST1 0x10
- #define AK_HXL 0x11
- #define AK_ST2 0x18
- #define AK_CNTL2 0x31
- #define AK_CNTL3 0x32
- #define AK_WIA2_VAL 0x09 /* AK09916C WIA2 expected */
- #define INTERNAL_20MHZ 0x00 /* 0: Internal 20 MHz RC */
- #define AUTO_SEL 0x01 /* 1–5: Auto/PLL preferred (use 1 as default) */
- #define CLK_STOP 0x07 /* 7: Stop clock / timing gen reset */
- #define g2 0
- #define g4 1
- #define g8 2
- #define g16 3
- #define ACCEL_FCHOICE_BYPASS 0
- #define ACCEL_FCHOICE_DLPF 1

- #define ACCEL_DLPFCFG_0 0
- #define ACCEL_DLPFCFG_1 1
- #define ACCEL_DLPFCFG_2 2
- #define ACCEL_DLPFCFG_3 3
- #define ACCEL_DLPFCFG_4 4
- #define ACCEL_DLPFCFG_5 5
- #define ACCEL_DLPFCFG_6 6
- #define ACCEL_DLPFCFG_7 7
- #define ACCEL_DEC3_AVG_4 0u /* averages 1 or 4 (depends on FCHOICE) */
- #define ACCEL_DEC3_AVG_8 1u
- #define ACCEL_DEC3_AVG_16 2u
- #define ACCEL_DEC3_AVG_32 3u
- #define ACCEL_SMPLRT_DIV_MIN 0u
- #define ACCEL_SMPLRT_DIV_MAX 4095u
- #define ACCEL_DLPF_BASE_HZ 1125.0f
- #define ACCEL_DLPF_ODR_HZ(div) (ACCEL_DLPF_BASE_HZ / (1.0f + (float)(div)))
- #define ACCEL_BYPASS_3DB_BW_HZ 1209.0f
- #define ACCEL_BYPASS_NBW_HZ 1248.0f
- #define ACCEL_BYPASS_RATE_HZ 4500.0f
- #define ACCEL_DLPF0_3DB_BW_HZ 246.0f
- #define ACCEL_DLPF0_NBW_HZ 265.0f
- #define ACCEL_DLPF1_3DB_BW_HZ 246.0f
- #define ACCEL_DLPF1_NBW_HZ 265.0f
- #define ACCEL_DLPF2_3DB_BW_HZ 111.4f
- #define ACCEL_DLPF2_NBW_HZ 136.0f
- #define ACCEL_DLPF3_3DB_BW_HZ 50.4f
- #define ACCEL_DLPF3_NBW_HZ 68.8f
- #define ACCEL_DLPF4_3DB_BW_HZ 23.9f
- #define ACCEL_DLPF4_NBW_HZ 34.4f
- #define ACCEL_DLPF5_3DB_BW_HZ 11.5f
- #define ACCEL_DLPF5_NBW_HZ 17.0f
- #define ACCEL_DLPF6_3DB_BW_HZ 5.7f
- #define ACCEL_DLPF6_NBW_HZ 8.3f
- #define ACCEL_DLPF7_3DB_BW_HZ 473.0f
- #define ACCEL_DLPF7_NBW_HZ 499.0f
- #define dps250 0
- #define dps500 1
- #define dps1000 2
- #define dps2000 3
- #define GYRO_FCHOICE_BYPASS 0
- #define GYRO_FCHOICE_DLPF 1
- #define GYRO_DLPFCFG_0 0
- #define GYRO_DLPFCFG_1 1
- #define GYRO_DLPFCFG_2 2
- #define GYRO_DLPFCFG_3 3
- #define GYRO_DLPFCFG_4 4
- #define GYRO_DLPFCFG_5 5
- #define GYRO_DLPFCFG_6 6
- #define GYRO_DLPFCFG_7 7
- #define GYRO_SMPLRT_MIN_HZ 4.3f
- #define GYRO_DLPF_BASE_HZ 1100.0f
- #define GYRO_DLPF_ODR_HZ(rate_request) (rate_request) /* symbolic; your driver computes DIV = round(1100/rate)-1 */
- #define GYRO_BW_ULTRA_WIDE 0 /* bypass path */
- #define GYRO_BW_WIDE 1 /* e.g., CFG 0/1 */

- #define GYRO_BW_MEDIUM 2 /* e.g., CFG 2/3 */
- #define GYRO_BW_NARROW 3 /* e.g., CFG 4/5/6/7 */
- #define Hz2 0
- #define Hz6 1
- #define Hz8 2
- #define Hz10 3
- #define Hz15 4
- #define Hz20 5
- #define Hz25 6
- #define Hz30 7
- #define LowPower 0
- #define Regular 1
- #define EnhancedRegular 2
- #define HighAccuracy 3

6.40.1 Macro Definition Documentation

6.40.1.1 ACCEL_BYPASS_3DB_BW_HZ

```
#define ACCEL_BYPASS_3DB_BW_HZ 1209.0f
```

Definition at line 369 of file ICM20948_regs.h.

6.40.1.2 ACCEL_BYPASS_NBW_HZ

```
#define ACCEL_BYPASS_NBW_HZ 1248.0f
```

Definition at line 370 of file ICM20948_regs.h.

6.40.1.3 ACCEL_BYPASS_RATE_HZ

```
#define ACCEL_BYPASS_RATE_HZ 4500.0f
```

Definition at line 371 of file ICM20948_regs.h.

6.40.1.4 ACCEL_CONFIG

```
#define ACCEL_CONFIG 0x14 /* ACCEL_DLPFCFG[2:0] ACCEL_FS_SEL[1:0] ACCEL_FCHOICE */
```

Definition at line 209 of file ICM20948_regs.h.

6.40.1.5 ACCEL_CONFIG_2

```
#define ACCEL_CONFIG_2 0x15 /* AX_ST_EN_REG AY_ST_EN_REG AZ_ST_EN_REG DEC3_CFG[1:0] */
```

Definition at line 220 of file ICM20948_regs.h.

6.40.1.6 ACCEL_DEC3_AVG_16

```
#define ACCEL_DEC3_AVG_16 2u
```

Definition at line 356 of file ICM20948_regs.h.

6.40.1.7 ACCEL_DEC3_AVG_32

```
#define ACCEL_DEC3_AVG_32 3u
```

Definition at line 357 of file ICM20948_regs.h.

6.40.1.8 ACCEL_DEC3_AVG_4

```
#define ACCEL_DEC3_AVG_4 0u /* averages 1 or 4 (depends on FCHOICE) */
```

Definition at line 354 of file ICM20948_regs.h.

6.40.1.9 ACCEL_DEC3_AVG_8

```
#define ACCEL_DEC3_AVG_8 1u
```
Definition at line 355 of file ICM20948_regs.h.

6.40.1.10 ACCEL_DLPF0_3DB_BW_HZ

```
#define ACCEL_DLPF0_3DB_BW_HZ 246.0f
```
Definition at line 376 of file ICM20948_regs.h.

6.40.1.11 ACCEL_DLPF0_NBW_HZ

```
#define ACCEL_DLPF0_NBW_HZ 265.0f
```
Definition at line 377 of file ICM20948_regs.h.

6.40.1.12 ACCEL_DLPF1_3DB_BW_HZ

```
#define ACCEL_DLPF1_3DB_BW_HZ 246.0f
```
Definition at line 378 of file ICM20948_regs.h.

6.40.1.13 ACCEL_DLPF1_NBW_HZ

```
#define ACCEL_DLPF1_NBW_HZ 265.0f
```
Definition at line 379 of file ICM20948_regs.h.

6.40.1.14 ACCEL_DLPF2_3DB_BW_HZ

```
#define ACCEL_DLPF2_3DB_BW_HZ 111.4f
```
Definition at line 380 of file ICM20948_regs.h.

6.40.1.15 ACCEL_DLPF2_NBW_HZ

```
#define ACCEL_DLPF2_NBW_HZ 136.0f
```
Definition at line 381 of file ICM20948_regs.h.

6.40.1.16 ACCEL_DLPF3_3DB_BW_HZ

```
#define ACCEL_DLPF3_3DB_BW_HZ 50.4f
```
Definition at line 382 of file ICM20948_regs.h.

6.40.1.17 ACCEL_DLPF3_NBW_HZ

```
#define ACCEL_DLPF3_NBW_HZ 68.8f
```
Definition at line 383 of file ICM20948_regs.h.

6.40.1.18 ACCEL_DLPF4_3DB_BW_HZ

```
#define ACCEL_DLPF4_3DB_BW_HZ 23.9f
```
Definition at line 384 of file ICM20948_regs.h.

6.40.1.19 ACCEL_DLPF4_NBW_HZ

```
#define ACCEL_DLPF4_NBW_HZ 34.4f
```
Definition at line 385 of file ICM20948_regs.h.

6.40.1.20 ACCEL_DLPF5_3DB_BW_HZ

```
#define ACCEL_DLPF5_3DB_BW_HZ 11.5f
```
Definition at line 386 of file ICM20948_regs.h.

6.40.1.21 ACCEL_DLPF5_NBW_HZ

```
#define ACCEL_DLPF5_NBW_HZ 17.0f
```
Definition at line 387 of file ICM20948_regs.h.

6.40.1.22 ACCEL_DLPF6_3DB_BW_HZ

```
#define ACCEL_DLPF6_3DB_BW_HZ 5.7f
```
Definition at line 388 of file ICM20948_regs.h.

6.40.1.23 ACCEL_DLPF6_NBW_HZ

```
#define ACCEL_DLPF6_NBW_HZ 8.3f
```
Definition at line 389 of file ICM20948_regs.h.

6.40.1.24 ACCEL_DLPF7_3DB_BW_HZ

```
#define ACCEL_DLPF7_3DB_BW_HZ 473.0f
```
Definition at line 390 of file ICM20948_regs.h.

6.40.1.25 ACCEL_DLPF7_NBW_HZ

```
#define ACCEL_DLPF7_NBW_HZ 499.0f
```
Definition at line 391 of file ICM20948_regs.h.

6.40.1.26 ACCEL_DLPF_BASE_HZ

```
#define ACCEL_DLPF_BASE_HZ 1125.0f
```
Definition at line 362 of file ICM20948_regs.h.

6.40.1.27 ACCEL_DLPF_ODR_HZ

```
#define ACCEL_DLPF_ODR_HZ(
              div ) (ACCEL_DLPF_BASE_HZ / (1.0f + (float)(div)))
```
Definition at line 363 of file ICM20948_regs.h.

6.40.1.28 ACCEL_DLPFCFG_0

```
#define ACCEL_DLPFCFG_0 0
```
ACCEL_DLPFCFG (0..7) — choose cutoff/noise BW set (when DLPF is ON). Tip: start with ACCEL_DLPFCFG_3 for a good noise/latency tradeoff.
Definition at line 345 of file ICM20948_regs.h.

6.40.1.29 ACCEL_DLPFCFG_1

```
#define ACCEL_DLPFCFG_1 1
```
Definition at line 346 of file ICM20948_regs.h.

6.40.1.30 ACCEL_DLPFCFG_2

```
#define ACCEL_DLPFCFG_2 2
```
Definition at line 347 of file ICM20948_regs.h.

6.40.1.31 ACCEL_DLPFCFG_3

```
#define ACCEL_DLPFCFG_3 3
```
Definition at line 348 of file ICM20948_regs.h.

6.40.1.32 ACCEL_DLPFCFG_4

```
#define ACCEL_DLPFCFG_4 4
```
Definition at line 349 of file ICM20948_regs.h.

6.40.1.33 ACCEL_DLPFCFG_5

```
#define ACCEL_DLPFCFG_5 5
```
Definition at line 350 of file ICM20948_regs.h.

6.40.1.34 ACCEL_DLPFCFG_6

```
#define ACCEL_DLPFCFG_6 6
```
Definition at line 351 of file ICM20948_regs.h.

6.40.1.35 ACCEL_DLPFCFG_7

```
#define ACCEL_DLPFCFG_7 7
```
Definition at line 352 of file ICM20948_regs.h.

6.40.1.36 ACCEL_DLPFCFG_MASK

```
#define ACCEL_DLPFCFG_MASK (0x7u << ACCEL_DLPFCFG_SHIFT)
```
Definition at line 218 of file ICM20948_regs.h.

6.40.1.37 ACCEL_DLPFCFG_SHIFT

```
#define ACCEL_DLPFCFG_SHIFT 5
```
Definition at line 217 of file ICM20948_regs.h.

6.40.1.38 ACCEL_FCHOICE

```
#define ACCEL_FCHOICE BIT(0)
```
Definition at line 210 of file ICM20948_regs.h.

6.40.1.39 ACCEL_FCHOICE_BYPASS

```
#define ACCEL_FCHOICE_BYPASS 0
```
Path select for accel:

- ACCEL_FCHOICE_BYPASS : DLPF bypassed (very wide BW, fastest)
- ACCEL_FCHOICE_DLPF : DLPF enabled (use ACCEL_DLPFCFG + SMPLRT_DIV)

Definition at line 337 of file ICM20948_regs.h.

6.40.1.40 ACCEL_FCHOICE_DLPF

```
#define ACCEL_FCHOICE_DLPF 1
```
Definition at line 338 of file ICM20948_regs.h.

6.40.1.41 ACCEL_FS_16G

```
#define ACCEL_FS_16G (3u << ACCEL_FS_SEL_SHIFT)
```
Definition at line 216 of file ICM20948_regs.h.

6.40.1.42 ACCEL_FS_2G

```
#define ACCEL_FS_2G (0u << ACCEL_FS_SEL_SHIFT)
```
Definition at line 213 of file ICM20948_regs.h.

6.40.1.43 ACCEL_FS_4G

```
#define ACCEL_FS_4G (1u << ACCEL_FS_SEL_SHIFT)
```
Definition at line 214 of file ICM20948_regs.h.

6.40.1.44 ACCEL_FS_8G

```
#define ACCEL_FS_8G (2u << ACCEL_FS_SEL_SHIFT)
```
Definition at line 215 of file ICM20948_regs.h.

6.40.1.45 ACCEL_FS_SEL_MASK

```
#define ACCEL_FS_SEL_MASK (0x3u << ACCEL_FS_SEL_SHIFT)
```
Definition at line 212 of file ICM20948_regs.h.

6.40.1.46 ACCEL_FS_SEL_SHIFT

```
#define ACCEL_FS_SEL_SHIFT 1
```
Definition at line 211 of file ICM20948_regs.h.

6.40.1.47 ACCEL_INTEL_CTRL

```
#define ACCEL_INTEL_CTRL 0x12 /* ACCEL_INTEL_EN ACCEL_INTEL_MODE_INT */
```
Definition at line 203 of file ICM20948_regs.h.

6.40.1.48 ACCEL_INTEL_EN

```
#define ACCEL_INTEL_EN BIT(7)
```
Definition at line 204 of file ICM20948_regs.h.

6.40.1.49 ACCEL_INTEL_MODE_INT

```
#define ACCEL_INTEL_MODE_INT BIT(6)
```
Definition at line 205 of file ICM20948_regs.h.

6.40.1.50 ACCEL_SMPLRT_DIV_1

```
#define ACCEL_SMPLRT_DIV_1 0x10 /* ACCEL_SMPLRT_DIV[11:8] */
```
Definition at line 200 of file ICM20948_regs.h.

6.40.1.51 ACCEL_SMPLRT_DIV_2

```
#define ACCEL_SMPLRT_DIV_2 0x11 /* ACCEL_SMPLRT_DIV[7:0] */
```
Definition at line 201 of file ICM20948_regs.h.

6.40.1.52 ACCEL_SMPLRT_DIV_MAX

```
#define ACCEL_SMPLRT_DIV_MAX 4095u
```
Definition at line 361 of file ICM20948_regs.h.

6.40.1.53 ACCEL_SMPLRT_DIV_MIN

```
#define ACCEL_SMPLRT_DIV_MIN 0u
```
Definition at line 360 of file ICM20948_regs.h.

6.40.1.54 ACCEL_WOM_THR

```
#define ACCEL_WOM_THR 0x13 /* WOM_THRESHOLD[7:0] */
```
Definition at line 207 of file ICM20948_regs.h.

6.40.1.55 ACCEL_XOUT_H

```
#define ACCEL_XOUT_H 0x2D
```
Definition at line 111 of file ICM20948_regs.h.

6.40.1.56 ACCEL_XOUT_L

```
#define ACCEL_XOUT_L 0x2E
```
Definition at line 112 of file ICM20948_regs.h.

6.40.1.57 ACCEL_YOUT_H

```
#define ACCEL_YOUT_H 0x2F
```
Definition at line 113 of file ICM20948_regs.h.

6.40.1.58 ACCEL_YOUT_L

```
#define ACCEL_YOUT_L 0x30
```
Definition at line 114 of file ICM20948_regs.h.

6.40.1.59 ACCEL_ZOUT_H

```
#define ACCEL_ZOUT_H 0x31
```
Definition at line 115 of file ICM20948_regs.h.

6.40.1.60 ACCEL_ZOUT_L

```
#define ACCEL_ZOUT_L 0x32
```
Definition at line 116 of file ICM20948_regs.h.

6.40.1.61 AK09916_I2C_ADDR

```
#define AK09916_I2C_ADDR 0x0C
```
Definition at line 310 of file ICM20948_regs.h.

6.40.1.62 AK_CNTL2

```
#define AK_CNTL2 0x31
```
Definition at line 315 of file ICM20948_regs.h.

6.40.1.63 AK_CNTL3

```
#define AK_CNTL3 0x32
```
Definition at line 316 of file ICM20948_regs.h.

6.40.1.64 AK_HXL

```
#define AK_HXL 0x11
```
Definition at line 313 of file ICM20948_regs.h.

6.40.1.65 AK_ST1

```
#define AK_ST1 0x10
```
Definition at line 312 of file ICM20948_regs.h.

6.40.1.66 AK_ST2

```
#define AK_ST2 0x18
```
Definition at line 314 of file ICM20948_regs.h.

6.40.1.67 AK_WIA2

```
#define AK_WIA2 0x01
```
Definition at line 311 of file ICM20948_regs.h.

6.40.1.68 AK_WIA2_VAL

```
#define AK_WIA2_VAL 0x09 /* AK09916C WIA2 expected */
```
Definition at line 317 of file ICM20948_regs.h.

6.40.1.69 AUTO_SEL

```
#define AUTO_SEL 0x01 /* 1-5:  Auto/PLL preferred (use 1 as default) */
```
Definition at line 320 of file ICM20948_regs.h.

6.40.1.70 AX_ST_EN_REG

```
#define AX_ST_EN_REG BIT(7)
```
Definition at line 221 of file ICM20948_regs.h.

6.40.1.71 AY_ST_EN_REG

```
#define AY_ST_EN_REG BIT(6)
```
Definition at line 222 of file ICM20948_regs.h.

6.40.1.72 AZ_ST_EN_REG

```
#define AZ_ST_EN_REG BIT(5)
```
Definition at line 223 of file ICM20948_regs.h.

6.40.1.73 BANK1_REG_BANK_SEL

```
#define BANK1_REG_BANK_SEL 0x7F /* mirror of REG_BANK_SEL */
```
Definition at line 163 of file ICM20948_regs.h.

6.40.1.74 BANK2_REG_BANK_SEL

```
#define BANK2_REG_BANK_SEL 0x7F /* mirror of REG_BANK_SEL */
```
Definition at line 240 of file ICM20948_regs.h.

6.40.1.75 BANK3_REG_BANK_SEL

```
#define BANK3_REG_BANK_SEL 0x7F /* mirror of REG_BANK_SEL */
```
Definition at line 296 of file ICM20948_regs.h.

6.40.1.76 BYPASS_EN

```
#define BYPASS_EN BIT(1) /* I2C bypass to aux devices */
```
Definition at line 71 of file ICM20948_regs.h.

6.40.1.77 CLK_STOP

```
#define CLK_STOP 0x07 /* 7:  Stop clock / timing gen reset */
```
Definition at line 321 of file ICM20948_regs.h.

6.40.1.78 DEC3_CFG_MASK

```
#define DEC3_CFG_MASK (0x3u << DEC3_CFG_SHIFT)
```
Definition at line 225 of file ICM20948_regs.h.

6.40.1.79 DEC3_CFG_SHIFT

```
#define DEC3_CFG_SHIFT 0
```
Definition at line 224 of file ICM20948_regs.h.

6.40.1.80 DELAY_ES_SHADOW

```
#define DELAY_ES_SHADOW BIT(7)
```
Definition at line 256 of file ICM20948_regs.h.

6.40.1.81 DELAY_TIME_EN

```
#define DELAY_TIME_EN BIT(7)
```
Definition at line 228 of file ICM20948_regs.h.

6.40.1.82 DELAY_TIMEH

```
#define DELAY_TIMEH 0x28
```
Definition at line 107 of file ICM20948_regs.h.

6.40.1.83 DELAY_TIMEL

```
#define DELAY_TIMEL 0x29
```
Definition at line 108 of file ICM20948_regs.h.

6.40.1.84 dps1000

```
#define dps1000 2
```
Definition at line 398 of file ICM20948_regs.h.

6.40.1.85 dps2000

```
#define dps2000 3
```
Definition at line 399 of file ICM20948_regs.h.

6.40.1.86 dps250

```
#define dps250 0
```

- ±250/±500/±1000/±2000 dps (maps to FS_SEL 0..3)

Definition at line 396 of file ICM20948_regs.h.

6.40.1.87 dps500

```
#define dps500 1
```
Definition at line 397 of file ICM20948_regs.h.

6.40.1.88 EnhancedRegular

```
#define EnhancedRegular 2
```
Definition at line 456 of file ICM20948_regs.h.

6.40.1.89 EXT_SLV_SENS_DATA_00

```
#define EXT_SLV_SENS_DATA_00 0x3B
```
Definition at line 127 of file ICM20948_regs.h.

6.40.1.90 EXT_SLV_SENS_DATA_23

```
#define EXT_SLV_SENS_DATA_23 0x52 /* contiguous range 0x3B..0x52 */
```
Definition at line 128 of file ICM20948_regs.h.

6.40.1.91 EXT_SYNC_SET_MASK

```
#define EXT_SYNC_SET_MASK 0x0F
```
Definition at line 231 of file ICM20948_regs.h.

6.40.1.92 FIFO_CFG

```
#define FIFO_CFG 0x76
```
Definition at line 138 of file ICM20948_regs.h.

6.40.1.93 FIFO_COUNTH

```
#define FIFO_COUNTH 0x70
```
Definition at line 134 of file ICM20948_regs.h.

6.40.1.94 FIFO_COUNTL

```
#define FIFO_COUNTL 0x71
```
Definition at line 135 of file ICM20948_regs.h.

6.40.1.95 FIFO_EN_1

```
#define FIFO_EN_1 0x66 /* SLV3..SLV0 FIFO_EN */
```
Definition at line 130 of file ICM20948_regs.h.

6.40.1.96 FIFO_EN_2

```
#define FIFO_EN_2 0x67 /* ACCEL GYRO_Z/Y/X TEMP FIFO_EN */
```
Definition at line 131 of file ICM20948_regs.h.

6.40.1.97 FIFO_MODE

```
#define FIFO_MODE 0x69 /* FIFO_MODE[4:0] */
```
Definition at line 133 of file ICM20948_regs.h.

6.40.1.98 FIFO_R_W

```
#define FIFO_R_W 0x72
```
Definition at line 136 of file ICM20948_regs.h.

6.40.1.99 FIFO_RST

```
#define FIFO_RST 0x68 /* FIFO_RESET[4:0] */
```
Definition at line 132 of file ICM20948_regs.h.

6.40.1.100 FSYNC_CONFIG

```
#define FSYNC_CONFIG 0x52 /* DELAY_TIME_EN WOF_DEGLITCH_EN WOF_EDGE_INT EXT_SYNC_SET[3:0] */
```
Definition at line 227 of file ICM20948_regs.h.

6.40.1.101 FSYNC_INT_MODE_EN

```
#define FSYNC_INT_MODE_EN BIT(2)
```
Definition at line 70 of file ICM20948_regs.h.

6.40.1.102 g16

```
#define g16 3
```
Definition at line 329 of file ICM20948_regs.h.

6.40.1.103 g2

```
#define g2 0
```

- ±2g / ±4g / ±8g / ±16g (maps to FS_SEL 0..3)

Definition at line 326 of file ICM20948_regs.h.

6.40.1.104 g4

```
#define g4 1
```
Definition at line 327 of file ICM20948_regs.h.

6.40.1.105 g8

```
#define g8 2
```
Definition at line 328 of file ICM20948_regs.h.

6.40.1.106 GYRO_AVGCFG_MASK

```
#define GYRO_AVGCFG_MASK (0x7u << GYRO_AVGCFG_SHIFT)
```
Definition at line 188 of file ICM20948_regs.h.

6.40.1.107 GYRO_AVGCFG_SHIFT

```
#define GYRO_AVGCFG_SHIFT 0
```
Definition at line 187 of file ICM20948_regs.h.

6.40.1.108 GYRO_BW_MEDIUM

```
#define GYRO_BW_MEDIUM 2 /* e.g., CFG 2/3 */
```
Definition at line 435 of file ICM20948_regs.h.

6.40.1.109 GYRO_BW_NARROW

```
#define GYRO_BW_NARROW 3 /* e.g., CFG 4/5/6/7 */
```
Definition at line 436 of file ICM20948_regs.h.

6.40.1.110 GYRO_BW_ULTRA_WIDE

```
#define GYRO_BW_ULTRA_WIDE 0 /* bypass path */
```
Definition at line 433 of file ICM20948_regs.h.

6.40.1.111 GYRO_BW_WIDE

```
#define GYRO_BW_WIDE 1 /* e.g., CFG 0/1 */
```
Definition at line 434 of file ICM20948_regs.h.

6.40.1.112 GYRO_CONFIG_1

```
#define GYRO_CONFIG_1 0x01 /* GYRO_DLPFCFG[2:0] GYRO_FS_SEL[1:0] GYRO_FCHOICE */
```
Definition at line 172 of file ICM20948_regs.h.

6.40.1.113 GYRO_CONFIG_2

```
#define GYRO_CONFIG_2 0x02 /* XGYRO_CTEN YGYRO_CTEN ZGYRO_CTEN GYRO_AVGCFG[2:0] */
```
Definition at line 183 of file ICM20948_regs.h.

6.40.1.114 GYRO_DLPF_BASE_HZ

```
#define GYRO_DLPF_BASE_HZ 1100.0f
```
Definition at line 426 of file ICM20948_regs.h.

6.40.1.115 GYRO_DLPF_ODR_HZ

```
#define GYRO_DLPF_ODR_HZ(
            rate_request ) (rate_request) /* symbolic; your driver computes DIV = round(1100/rate)-1
*/
```

Definition at line 427 of file ICM20948_regs.h.

6.40.1.116 GYRO_DLPFCFG_0

```
#define GYRO_DLPFCFG_0 0
```

GYRO_DLPFCFG (0..7) — choose cutoff/noise BW set (when DLPF is ON). Tip: start with GYRO_DLPFCFG_3 or _2 for balanced noise/latency.
Definition at line 415 of file ICM20948_regs.h.

6.40.1.117 GYRO_DLPFCFG_1

```
#define GYRO_DLPFCFG_1 1
```

Definition at line 416 of file ICM20948_regs.h.

6.40.1.118 GYRO_DLPFCFG_2

```
#define GYRO_DLPFCFG_2 2
```

Definition at line 417 of file ICM20948_regs.h.

6.40.1.119 GYRO_DLPFCFG_3

```
#define GYRO_DLPFCFG_3 3
```

Definition at line 418 of file ICM20948_regs.h.

6.40.1.120 GYRO_DLPFCFG_4

```
#define GYRO_DLPFCFG_4 4
```

Definition at line 419 of file ICM20948_regs.h.

6.40.1.121 GYRO_DLPFCFG_5

```
#define GYRO_DLPFCFG_5 5
```

Definition at line 420 of file ICM20948_regs.h.

6.40.1.122 GYRO_DLPFCFG_6

```
#define GYRO_DLPFCFG_6 6
```

Definition at line 421 of file ICM20948_regs.h.

6.40.1.123 GYRO_DLPFCFG_7

```
#define GYRO_DLPFCFG_7 7
```

Definition at line 422 of file ICM20948_regs.h.

6.40.1.124 GYRO_DLPFCFG_MASK

```
#define GYRO_DLPFCFG_MASK (0x7u << GYRO_DLPFCFG_SHIFT)
```

Definition at line 181 of file ICM20948_regs.h.

6.40.1.125 GYRO_DLPFCFG_SHIFT

```
#define GYRO_DLPFCFG_SHIFT 5
```

Definition at line 180 of file ICM20948_regs.h.

6.40.1.126 GYRO_FCHOICE

```
#define GYRO_FCHOICE BIT(0) /* when 0:  DLPF on, when 1:  off (per datasheet) */
```

Definition at line 173 of file ICM20948_regs.h.

6.40.1.127 GYRO_FCHOICE_BYPASS

```
#define GYRO_FCHOICE_BYPASS 0
```

Path select for gyro:

- GYRO_FCHOICE_BYPASS : DLPF bypassed (widest BW, fastest)
- GYRO_FCHOICE_DLPF : DLPF enabled (use GYRO_DLPFCFG + SMPLRT_DIV)

Definition at line 407 of file ICM20948_regs.h.

6.40.1.128 GYRO_FCHOICE_DLPF

```
#define GYRO_FCHOICE_DLPF 1
```

Definition at line 408 of file ICM20948_regs.h.

6.40.1.129 GYRO_FS_1000DPS

```
#define GYRO_FS_1000DPS (2u << GYRO_FS_SEL_SHIFT)
```

Definition at line 178 of file ICM20948_regs.h.

6.40.1.130 GYRO_FS_2000DPS

```
#define GYRO_FS_2000DPS (3u << GYRO_FS_SEL_SHIFT)
```

Definition at line 179 of file ICM20948_regs.h.

6.40.1.131 GYRO_FS_250DPS

```
#define GYRO_FS_250DPS (0u << GYRO_FS_SEL_SHIFT)
```

Definition at line 176 of file ICM20948_regs.h.

6.40.1.132 GYRO_FS_500DPS

```
#define GYRO_FS_500DPS (1u << GYRO_FS_SEL_SHIFT)
```

Definition at line 177 of file ICM20948_regs.h.

6.40.1.133 GYRO_FS_SEL_MASK

```
#define GYRO_FS_SEL_MASK (0x3u << GYRO_FS_SEL_SHIFT)
```

Definition at line 175 of file ICM20948_regs.h.

6.40.1.134 GYRO_FS_SEL_SHIFT

```
#define GYRO_FS_SEL_SHIFT 1
```

Definition at line 174 of file ICM20948_regs.h.

6.40.1.135 GYRO_SMPLRT_DIV

```
#define GYRO_SMPLRT_DIV 0x00 /* GYRO_SMPLRT_DIV[7:0] */
```

- Gyro/Accel configuration, offsets, FSYNC, temperature filter

Definition at line 170 of file ICM20948_regs.h.

6.40.1.136 GYRO_SMPLRT_MIN_HZ

```
#define GYRO_SMPLRT_MIN_HZ 4.3f
```

Definition at line 425 of file ICM20948_regs.h.

6.40.1.137 GYRO_XOUT_H

```
#define GYRO_XOUT_H 0x33
```
Definition at line 117 of file ICM20948_regs.h.

6.40.1.138 GYRO_XOUT_L

```
#define GYRO_XOUT_L 0x34
```
Definition at line 118 of file ICM20948_regs.h.

6.40.1.139 GYRO_YOUT_H

```
#define GYRO_YOUT_H 0x35
```
Definition at line 119 of file ICM20948_regs.h.

6.40.1.140 GYRO_YOUT_L

```
#define GYRO_YOUT_L 0x36
```
Definition at line 120 of file ICM20948_regs.h.

6.40.1.141 GYRO_ZOUT_H

```
#define GYRO_ZOUT_H 0x37
```
Definition at line 121 of file ICM20948_regs.h.

6.40.1.142 GYRO_ZOUT_L

```
#define GYRO_ZOUT_L 0x38
```
Definition at line 122 of file ICM20948_regs.h.

6.40.1.143 HighAccuracy

```
#define HighAccuracy 3
```
Definition at line 457 of file ICM20948_regs.h.

6.40.1.144 Hz10

```
#define Hz10 3
```
Definition at line 443 of file ICM20948_regs.h.

6.40.1.145 Hz15

```
#define Hz15 4
```
Definition at line 444 of file ICM20948_regs.h.

6.40.1.146 Hz2

```
#define Hz2 0
```

- Friendly ODR presets you can map to dividers

Definition at line 440 of file ICM20948_regs.h.

6.40.1.147 Hz20

```
#define Hz20 5
```
Definition at line 445 of file ICM20948_regs.h.

6.40.1.148 Hz25

```
#define Hz25 6
```
Definition at line 446 of file ICM20948_regs.h.

6.40.1.149 Hz30

```
#define Hz30 7
```
Definition at line 447 of file ICM20948_regs.h.

6.40.1.150 Hz6

```
#define Hz6 1
```
Definition at line 441 of file ICM20948_regs.h.

6.40.1.151 Hz8

```
#define Hz8 2
```
Definition at line 442 of file ICM20948_regs.h.

6.40.1.152 I2C_MST_CLK_MASK

```
#define I2C_MST_CLK_MASK 0x0F
```
Definition at line 253 of file ICM20948_regs.h.

6.40.1.153 I2C_MST_CTRL

```
#define I2C_MST_CTRL 0x01 /* MULT_MST_EN - I2C_MST_P_NSR I2C_MST_CLK[3:0] */
```
Definition at line 250 of file ICM20948_regs.h.

6.40.1.154 I2C_MST_DELAY_CTRL

```
#define I2C_MST_DELAY_CTRL 0x02 /* DELAY_ES_SHADOW + I2C_SLVx_DELAY_EN bits */
```
Definition at line 255 of file ICM20948_regs.h.

6.40.1.155 I2C_MST_ODR_CONFIG

```
#define I2C_MST_ODR_CONFIG 0x00 /* I2C_MST_ODR_CONFIG[3:0] */
```

- I2C master (aux) interface and slave windows

Definition at line 247 of file ICM20948_regs.h.

6.40.1.156 I2C_MST_P_NSR

```
#define I2C_MST_P_NSR BIT(4)
```
Definition at line 252 of file ICM20948_regs.h.

6.40.1.157 I2C_MST_STATUS

```
#define I2C_MST_STATUS 0x17 /* PASS_THROUGH, *_DONE, *_NACK, LOST_ARB */
```
Definition at line 86 of file ICM20948_regs.h.

6.40.1.158 I2C_SLV0_ADDR

```
#define I2C_SLV0_ADDR 0x03 /* RNW + ID[6:0] */
```
Definition at line 263 of file ICM20948_regs.h.

6.40.1.159 I2C_SLV0_CTRL

```
#define I2C_SLV0_CTRL 0x05 /* EN BYTE_SW REG_DIS GRP LENG[3:0] */
```
Definition at line 266 of file ICM20948_regs.h.

6.40.1.160 I2C_SLV0_DELAY_EN

```
#define I2C_SLV0_DELAY_EN BIT(0)
```
Definition at line 261 of file ICM20948_regs.h.

6.40.1.161 I2C_SLV0_DO

```
#define I2C_SLV0_DO 0x06
```
Definition at line 272 of file ICM20948_regs.h.

6.40.1.162 I2C_SLV0_REG

```
#define I2C_SLV0_REG 0x04
```
Definition at line 265 of file ICM20948_regs.h.

6.40.1.163 I2C_SLV1_ADDR

```
#define I2C_SLV1_ADDR 0x07
```
Definition at line 274 of file ICM20948_regs.h.

6.40.1.164 I2C_SLV1_CTRL

```
#define I2C_SLV1_CTRL 0x09
```
Definition at line 276 of file ICM20948_regs.h.

6.40.1.165 I2C_SLV1_DELAY_EN

```
#define I2C_SLV1_DELAY_EN BIT(1)
```
Definition at line 260 of file ICM20948_regs.h.

6.40.1.166 I2C_SLV1_DO

```
#define I2C_SLV1_DO 0x0A
```
Definition at line 277 of file ICM20948_regs.h.

6.40.1.167 I2C_SLV1_REG

```
#define I2C_SLV1_REG 0x08
```
Definition at line 275 of file ICM20948_regs.h.

6.40.1.168 I2C_SLV2_ADDR

```
#define I2C_SLV2_ADDR 0x0B
```
Definition at line 279 of file ICM20948_regs.h.

6.40.1.169 I2C_SLV2_CTRL

```
#define I2C_SLV2_CTRL 0x0D
```
Definition at line 281 of file ICM20948_regs.h.

6.40.1.170 I2C_SLV2_DELAY_EN

```
#define I2C_SLV2_DELAY_EN BIT(2)
```
Definition at line 259 of file ICM20948_regs.h.

6.40.1.171 I2C_SLV2_DO

```
#define I2C_SLV2_DO 0x0E
```
Definition at line 282 of file ICM20948_regs.h.

6.40.1.172 I2C_SLV2_REG

```
#define I2C_SLV2_REG 0x0C
```
Definition at line 280 of file ICM20948_regs.h.

6.40.1.173 I2C_SLV3_ADDR

```
#define I2C_SLV3_ADDR 0x0F
```
Definition at line 284 of file ICM20948_regs.h.

6.40.1.174 I2C_SLV3_CTRL

```
#define I2C_SLV3_CTRL 0x11
```
Definition at line 286 of file ICM20948_regs.h.

6.40.1.175 I2C_SLV3_DELAY_EN

```
#define I2C_SLV3_DELAY_EN BIT(3)
```
Definition at line 258 of file ICM20948_regs.h.

6.40.1.176 I2C_SLV3_DO

```
#define I2C_SLV3_DO 0x12
```
Definition at line 287 of file ICM20948_regs.h.

6.40.1.177 I2C_SLV3_REG

```
#define I2C_SLV3_REG 0x10
```
Definition at line 285 of file ICM20948_regs.h.

6.40.1.178 I2C_SLV4_ADDR

```
#define I2C_SLV4_ADDR 0x13
```
Definition at line 289 of file ICM20948_regs.h.

6.40.1.179 I2C_SLV4_CTRL

```
#define I2C_SLV4_CTRL 0x15 /* EN BYTE_SW REG_DIS DLY[4:0] */
```
Definition at line 291 of file ICM20948_regs.h.

6.40.1.180 I2C_SLV4_DELAY_EN

```
#define I2C_SLV4_DELAY_EN BIT(4)
```
Definition at line 257 of file ICM20948_regs.h.

6.40.1.181 I2C_SLV4_DI

```
#define I2C_SLV4_DI 0x17
```
Definition at line 294 of file ICM20948_regs.h.

6.40.1.182 I2C_SLV4_DLY_MASK

```
#define I2C_SLV4_DLY_MASK 0x1F
```
Definition at line 292 of file ICM20948_regs.h.

6.40.1.183 I2C_SLV4_DO

```
#define I2C_SLV4_DO 0x16
```
Definition at line 293 of file ICM20948_regs.h.

6.40.1.184 I2C_SLV4_REG

```
#define I2C_SLV4_REG 0x14
```
Definition at line 290 of file ICM20948_regs.h.

6.40.1.185 I2C_SLVx_BYTE_SW

```
#define I2C_SLVx_BYTE_SW BIT(6)
```
Definition at line 268 of file ICM20948_regs.h.

6.40.1.186 I2C_SLVx_EN

```
#define I2C_SLVx_EN BIT(7)
```
Definition at line 267 of file ICM20948_regs.h.

6.40.1.187 I2C_SLVx_GRP

```
#define I2C_SLVx_GRP BIT(4)
```
Definition at line 270 of file ICM20948_regs.h.

6.40.1.188 I2C_SLVx_LENG_MASK

```
#define I2C_SLVx_LENG_MASK 0x0F
```
Definition at line 271 of file ICM20948_regs.h.

6.40.1.189 I2C_SLVx_REG_DIS

```
#define I2C_SLVx_REG_DIS BIT(5)
```
Definition at line 269 of file ICM20948_regs.h.

6.40.1.190 I2C_SLVx_RNW

```
#define I2C_SLVx_RNW BIT(7)
```
Definition at line 264 of file ICM20948_regs.h.

6.40.1.191 INT1_ACTL

```
#define INT1_ACTL BIT(7) /* active low */
```
Definition at line 65 of file ICM20948_regs.h.

6.40.1.192 INT1_LATCH_INT_EN

```
#define INT1_LATCH_INT_EN BIT(5) /* latch until status read */
```
Definition at line 67 of file ICM20948_regs.h.

6.40.1.193 INT1_OPEN

```
#define INT1_OPEN BIT(6) /* open-drain */
```
Definition at line 66 of file ICM20948_regs.h.

6.40.1.194 INT_ACTL_FSYNC

```
#define INT_ACTL_FSYNC BIT(3)
```
Definition at line 69 of file ICM20948_regs.h.

6.40.1.195 INT_ANYRD_2CLEAR

```
#define INT_ANYRD_2CLEAR BIT(4)
```
Definition at line 68 of file ICM20948_regs.h.

6.40.1.196 INT_DMP_INT1_EN

```
#define INT_DMP_INT1_EN BIT(1)
```
Definition at line 77 of file ICM20948_regs.h.

6.40.1.197 INT_ENABLE

```
#define INT_ENABLE 0x10 /* REG_WOF_EN - WOM_INT_EN PLL_RDY_EN DMP_INT1_EN I2C_MST_INT_EN */
```
Definition at line 73 of file ICM20948_regs.h.

6.40.1.198 INT_ENABLE_1

```
#define INT_ENABLE_1 0x11 /* RAW_DATA_0_RDY_EN */
```
Definition at line 80 of file ICM20948_regs.h.

6.40.1.199 INT_ENABLE_2

```
#define INT_ENABLE_2 0x12 /* FIFO_OVERFLOW_EN[4:0] */
```
Definition at line 83 of file ICM20948_regs.h.

6.40.1.200 INT_ENABLE_3

```
#define INT_ENABLE_3 0x13 /* FIFO_WM_EN[4:0] */
```
Definition at line 84 of file ICM20948_regs.h.

6.40.1.201 INT_I2C_MST_INT_EN

```
#define INT_I2C_MST_INT_EN BIT(0)
```
Definition at line 78 of file ICM20948_regs.h.

6.40.1.202 INT_PIN_CFG

```
#define INT_PIN_CFG 0x0F /* INT1_ACTL INT1_OPEN INT1_LATCH_INT_EN INT_ANYRD_2CLEAR ACTL_FSYNC
FSYNC_INT_MODE_EN BYPASS_EN - */
```
Definition at line 64 of file ICM20948_regs.h.

6.40.1.203 INT_PLL_RDY_EN

```
#define INT_PLL_RDY_EN BIT(2)
```
Definition at line 76 of file ICM20948_regs.h.

6.40.1.204 INT_RAW_DATA_0_RDY_EN

```
#define INT_RAW_DATA_0_RDY_EN BIT(0)
```
Definition at line 81 of file ICM20948_regs.h.

6.40.1.205 INT_REG_WOF_EN

```
#define INT_REG_WOF_EN BIT(7)
```
Definition at line 74 of file ICM20948_regs.h.

6.40.1.206 INT_STATUS

```
#define INT_STATUS 0x19 /* WOM_INT PLL_RDY_INT DMP_INT1 I2C_MST_INT */
```
Definition at line 96 of file ICM20948_regs.h.

6.40.1.207 INT_STATUS_1

```
#define INT_STATUS_1 0x1A /* RAW_DATA_0_RDY_INT */
```
Definition at line 102 of file ICM20948_regs.h.

6.40.1.208 INT_STATUS_2

```
#define INT_STATUS_2 0x1B /* FIFO_OVERFLOW_INT[4:0] */
```
Definition at line 104 of file ICM20948_regs.h.

6.40.1.209 INT_STATUS_3

```
#define INT_STATUS_3 0x1C /* FIFO_WM_INT[4:0] */
```
Definition at line 105 of file ICM20948_regs.h.

6.40.1.210 INT_WOM_INT_EN

```
#define INT_WOM_INT_EN BIT(3)
```
Definition at line 75 of file ICM20948_regs.h.

6.40.1.211 INTERNAL_20MHZ

```
#define INTERNAL_20MHZ 0x00 /* 0:  Internal 20 MHz RC */
```
Definition at line 319 of file ICM20948_regs.h.

6.40.1.212 LowPower

```
#define LowPower 0
```
Generic power or fusion quality modes (BNO-style). Map to sensor-specific sequences inside your driver.
Definition at line 454 of file ICM20948_regs.h.

6.40.1.213 LP_ACCEL_CYCLE

```
#define LP_ACCEL_CYCLE BIT(5)
```
Definition at line 47 of file ICM20948_regs.h.

6.40.1.214 LP_CONFIG

```
#define LP_CONFIG 0x05 /* I2C_MST_CYCLE ACCEL_CYCLE GYRO_CYCLE - */
```
Definition at line 45 of file ICM20948_regs.h.

6.40.1.215 LP_GYRO_CYCLE

```
#define LP_GYRO_CYCLE BIT(4)
```
Definition at line 48 of file ICM20948_regs.h.

6.40.1.216 LP_I2C_MST_CYCLE

```
#define LP_I2C_MST_CYCLE BIT(6)
```
Definition at line 46 of file ICM20948_regs.h.

6.40.1.217 MOD_CTRL_USR

```
#define MOD_CTRL_USR 0x54 /* REG_LP_DMP_EN */
```
Definition at line 237 of file ICM20948_regs.h.

6.40.1.218 MST_LOST_ARB

```
#define MST_LOST_ARB BIT(5)
```
Definition at line 89 of file ICM20948_regs.h.

6.40.1.219 MST_ODR_CFG_MASK

```
#define MST_ODR_CFG_MASK 0x0F
```
Definition at line 248 of file ICM20948_regs.h.

6.40.1.220 MST_PASS_THROUGH

```
#define MST_PASS_THROUGH BIT(7)
```
Definition at line 87 of file ICM20948_regs.h.

6.40.1.221 MST_SLV0_NACK

```
#define MST_SLV0_NACK BIT(0)
```
Definition at line 94 of file ICM20948_regs.h.

6.40.1.222 MST_SLV1_NACK

```
#define MST_SLV1_NACK BIT(1)
```
Definition at line 93 of file ICM20948_regs.h.

6.40.1.223 MST_SLV2_NACK

```
#define MST_SLV2_NACK BIT(2)
```
Definition at line 92 of file ICM20948_regs.h.

6.40.1.224 MST_SLV3_NACK

```
#define MST_SLV3_NACK BIT(3)
```
Definition at line 91 of file ICM20948_regs.h.

6.40.1.225 MST_SLV4_DONE

```
#define MST_SLV4_DONE BIT(6)
```
Definition at line 88 of file ICM20948_regs.h.

6.40.1.226 MST_SLV4_NACK

```
#define MST_SLV4_NACK BIT(4)
```
Definition at line 90 of file ICM20948_regs.h.

6.40.1.227 MULT_MST_EN

```
#define MULT_MST_EN BIT(7)
```
Definition at line 251 of file ICM20948_regs.h.

6.40.1.228 ODR_ALIGN_EN

```
#define ODR_ALIGN_EN 0x09 /* ODR_ALIGN_EN */
```
Definition at line 197 of file ICM20948_regs.h.

6.40.1.229 ODR_ALIGN_EN_BIT

```
#define ODR_ALIGN_EN_BIT BIT(0)
```
Definition at line 198 of file ICM20948_regs.h.

6.40.1.230 PWR_CLKSEL_AUTO

```
#define PWR_CLKSEL_AUTO 0x01 /* typical:  auto selects best source */
```
Definition at line 57 of file ICM20948_regs.h.

6.40.1.231 PWR_CLKSEL_INT_20MHZ

```
#define PWR_CLKSEL_INT_20MHZ 0x01
```
Definition at line 56 of file ICM20948_regs.h.

6.40.1.232 PWR_CLKSEL_MASK

```
#define PWR_CLKSEL_MASK 0x07
```
Definition at line 55 of file ICM20948_regs.h.

6.40.1.233 PWR_DEVICE_RESET

```
#define PWR_DEVICE_RESET BIT(7)
```
Definition at line 51 of file ICM20948_regs.h.

6.40.1.234 PWR_DISABLE_ACCEL

```
#define PWR_DISABLE_ACCEL BIT(3)
```
Definition at line 60 of file ICM20948_regs.h.

6.40.1.235 PWR_DISABLE_GYRO

```
#define PWR_DISABLE_GYRO BIT(0)
```
Definition at line 61 of file ICM20948_regs.h.

6.40.1.236 PWR_LP_EN

```
#define PWR_LP_EN BIT(5)
```
Definition at line 53 of file ICM20948_regs.h.

6.40.1.237 PWR_MGMT_1

```
#define PWR_MGMT_1 0x06 /* DEVICE_RESET SLEEP LP_EN - TEMP_DIS CLKSEL[2:0] */
```
Definition at line 50 of file ICM20948_regs.h.

6.40.1.238 PWR_MGMT_2

```
#define PWR_MGMT_2 0x07 /* - DISABLE_ACCEL DISABLE_GYRO */
```
Definition at line 59 of file ICM20948_regs.h.

6.40.1.239 PWR_SLEEP

```
#define PWR_SLEEP BIT(6)
```
Definition at line 52 of file ICM20948_regs.h.

6.40.1.240 PWR_TEMP_DIS

```
#define PWR_TEMP_DIS BIT(3)
```
Definition at line 54 of file ICM20948_regs.h.

6.40.1.241 REG_BANK_SEL

```
#define REG_BANK_SEL 0x7F /* USER_BANK[1:0] */
```
Definition at line 140 of file ICM20948_regs.h.

6.40.1.242 REG_BANK_SEL_USER_BANK_MASK

```
#define REG_BANK_SEL_USER_BANK_MASK (0x3u << REG_BANK_SEL_USER_BANK_SHIFT)
```
Definition at line 303 of file ICM20948_regs.h.

6.40.1.243 REG_BANK_SEL_USER_BANK_SHIFT

```
#define REG_BANK_SEL_USER_BANK_SHIFT 4
```
Definition at line 302 of file ICM20948_regs.h.

6.40.1.244 REG_LP_DMP_EN

```
#define REG_LP_DMP_EN BIT(7)
```
Definition at line 238 of file ICM20948_regs.h.

6.40.1.245 Regular

```
#define Regular 1
```
Definition at line 455 of file ICM20948_regs.h.

6.40.1.246 SELF_TEST_X_ACCEL

```
#define SELF_TEST_X_ACCEL 0x0E /* XA_ST_DATA[7:0] */
```
Definition at line 150 of file ICM20948_regs.h.

6.40.1.247 SELF_TEST_X_GYRO

```
#define SELF_TEST_X_GYRO 0x02 /* XG_ST_DATA[7:0] */
```

- Self-test / accel offsets / timebase

Definition at line 147 of file ICM20948_regs.h.

6.40.1.248 SELF_TEST_Y_ACCEL

```
#define SELF_TEST_Y_ACCEL 0x0F /* YA_ST_DATA[7:0] */
```
Definition at line 151 of file ICM20948_regs.h.

6.40.1.249 SELF_TEST_Y_GYRO

```
#define SELF_TEST_Y_GYRO 0x03 /* YG_ST_DATA[7:0] */
```
Definition at line 148 of file ICM20948_regs.h.

6.40.1.250 SELF_TEST_Z_ACCEL

```
#define SELF_TEST_Z_ACCEL 0x10 /* ZA_ST_DATA[7:0] */
```
Definition at line 152 of file ICM20948_regs.h.

6.40.1.251 SELF_TEST_Z_GYRO

```
#define SELF_TEST_Z_GYRO 0x04 /* ZG_ST_DATA[7:0] */
```
Definition at line 149 of file ICM20948_regs.h.

6.40.1.252 STS_DMP_INT1

```
#define STS_DMP_INT1 BIT(1)
```
Definition at line 99 of file ICM20948_regs.h.

6.40.1.253 STS_I2C_MST_INT

```
#define STS_I2C_MST_INT BIT(0)
```
Definition at line 100 of file ICM20948_regs.h.

6.40.1.254 STS_PLL_RDY_INT

```
#define STS_PLL_RDY_INT BIT(2)
```
Definition at line 98 of file ICM20948_regs.h.

6.40.1.255 STS_RAW_DATA_0_RDY_INT

```
#define STS_RAW_DATA_0_RDY_INT BIT(0)
```
Definition at line 103 of file ICM20948_regs.h.

6.40.1.256 STS_WOM_INT

```
#define STS_WOM_INT BIT(3)
```
Definition at line 97 of file ICM20948_regs.h.

6.40.1.257 TEMP_CONFIG

```
#define TEMP_CONFIG 0x53 /* TEMP_DLPFCFG[2:0] */
```
Definition at line 233 of file ICM20948_regs.h.

6.40.1.258 TEMP_DLPFCFG_MASK

```
#define TEMP_DLPFCFG_MASK (0x7u << TEMP_DLPFCFG_SHIFT)
```
Definition at line 235 of file ICM20948_regs.h.

6.40.1.259 TEMP_DLPFCFG_SHIFT

```
#define TEMP_DLPFCFG_SHIFT 5
```
Definition at line 234 of file ICM20948_regs.h.

6.40.1.260 TEMP_OUT_H

```
#define TEMP_OUT_H 0x39
```
Definition at line 123 of file ICM20948_regs.h.

6.40.1.261 TEMP_OUT_L

```
#define TEMP_OUT_L 0x3A
```
Definition at line 124 of file ICM20948_regs.h.

6.40.1.262 TIMEBASE_CORRECTION_PLL

```
#define TIMEBASE_CORRECTION_PLL 0x28 /* TBC_PLL[7:0] */
```
Definition at line 161 of file ICM20948_regs.h.

6.40.1.263 USER_BANK_0

```
#define USER_BANK_0 BANK0
```
Definition at line 304 of file ICM20948_regs.h.

6.40.1.264 USER_BANK_1

```
#define USER_BANK_1 BANK1
```
Definition at line 305 of file ICM20948_regs.h.

6.40.1.265 USER_BANK_2

```
#define USER_BANK_2 BANK2
```
Definition at line 306 of file ICM20948_regs.h.

6.40.1.266 USER_BANK_3

```
#define USER_BANK_3 BANK3
```
Definition at line 307 of file ICM20948_regs.h.

6.40.1.267 USER_CTRL

```
#define USER_CTRL 0x03 /* DMP_EN FIFO_EN I2C_MST_EN I2C_IF_DIS DMP_RST SRAM_RST I2C_MST_RST -
*/
```
Definition at line 35 of file ICM20948_regs.h.

6.40.1.268 USER_CTRL_DMP_EN

```
#define USER_CTRL_DMP_EN BIT(7)
```
Definition at line 37 of file ICM20948_regs.h.

6.40.1.269 USER_CTRL_DMP_RST

```
#define USER_CTRL_DMP_RST BIT(3)
```
Definition at line 41 of file ICM20948_regs.h.

6.40.1.270 USER_CTRL_FIFO_EN

```
#define USER_CTRL_FIFO_EN BIT(6)
```
Definition at line 38 of file ICM20948_regs.h.

6.40.1.271 USER_CTRL_I2C_IF_DIS

```
#define USER_CTRL_I2C_IF_DIS BIT(4)
```
Definition at line 40 of file ICM20948_regs.h.

6.40.1.272 USER_CTRL_I2C_MST_EN

```
#define USER_CTRL_I2C_MST_EN BIT(5)
```
Definition at line 39 of file ICM20948_regs.h.

6.40.1.273 USER_CTRL_I2C_MST_RST

```
#define USER_CTRL_I2C_MST_RST BIT(1)
```
Definition at line 43 of file ICM20948_regs.h.

6.40.1.274 USER_CTRL_SRAM_RST

```
#define USER_CTRL_SRAM_RST BIT(2)
```
Definition at line 42 of file ICM20948_regs.h.

6.40.1.275 WHO_AM_I

```
#define WHO_AM_I 0x00 /* WHO_AM_I[7:0] */
```
7Semi comment style

- Full ICM-20948 register map (Banks 0–3) + key bit fields
- Simple, portable #defines for firmware use
- Datasheet naming kept where practical; fields grouped by bank

Notes

- WHO_AM_I expected value: 0xEA
- Select register bank via REG_BANK_SEL in any bank
- Use BANK(n) helper or write raw values 0x00/0x10/0x20/0x30
- Identity / power / interrupts / sensor data

Definition at line 32 of file ICM20948_regs.h.

6.40.1.276 WHO_AM_I_VAL

```
#define WHO_AM_I_VAL 0xEA
```
Definition at line 33 of file ICM20948_regs.h.

6.40.1.277 WOF_DEGLITCH_EN

```
#define WOF_DEGLITCH_EN BIT(6)
```
Definition at line 229 of file ICM20948_regs.h.

6.40.1.278 WOF_EDGE_INT

```
#define WOF_EDGE_INT BIT(5)
```
Definition at line 230 of file ICM20948_regs.h.

6.40.1.279 XA_OFFS_H

```
#define XA_OFFS_H 0x14 /* XA_OFFS[14:7] */
```
Definition at line 154 of file ICM20948_regs.h.

6.40.1.280 XA_OFFS_L

```
#define XA_OFFS_L 0x15 /* XA_OFFS[6:0] */
```
Definition at line 155 of file ICM20948_regs.h.

6.40.1.281 XG_OFFS_USRH

```
#define XG_OFFS_USRH 0x03
```
Definition at line 190 of file ICM20948_regs.h.

6.40.1.282 XG_OFFS_USRL

```
#define XG_OFFS_USRL 0x04
```
Definition at line 191 of file ICM20948_regs.h.

6.40.1.283 XGYRO_CTEN

```
#define XGYRO_CTEN BIT(5)
```
Definition at line 184 of file ICM20948_regs.h.

6.40.1.284 YA_OFFS_H

```
#define YA_OFFS_H 0x17 /* YA_OFFS[14:7] */
```
Definition at line 156 of file ICM20948_regs.h.

6.40.1.285 YA_OFFS_L

```
#define YA_OFFS_L 0x18 /* YA_OFFS[6:0] */
```
Definition at line 157 of file ICM20948_regs.h.

6.40.1.286 YG_OFFS_USRH

```
#define YG_OFFS_USRH 0x05
```
Definition at line 192 of file ICM20948_regs.h.

6.40.1.287 YG_OFFS_USRL

```
#define YG_OFFS_USRL 0x06
```
Definition at line 193 of file ICM20948_regs.h.

6.40.1.288 YGYRO_CTEN

```
#define YGYRO_CTEN BIT(4)
```
Definition at line 185 of file ICM20948_regs.h.

6.40.1.289 ZA_OFFS_H

```
#define ZA_OFFS_H 0x1A /* ZA_OFFS[14:7] */
```
Definition at line 158 of file ICM20948_regs.h.

6.40.1.290 ZA_OFFS_L

```
#define ZA_OFFS_L 0x1B /* ZA_OFFS[6:0] */
```
Definition at line 159 of file ICM20948_regs.h.

6.40.1.291 ZG_OFFS_USRH

```
#define ZG_OFFS_USRH 0x07
```
Definition at line 194 of file ICM20948_regs.h.

6.40.1.292 ZG_OFFS_USRL

```
#define ZG_OFFS_USRL 0x08
```
Definition at line 195 of file ICM20948_regs.h.

6.40.1.293 ZGYRO_CTEN

```
#define ZGYRO_CTEN BIT(3)
```
Definition at line 186 of file ICM20948_regs.h.

6.41 ICM20948_regs.h

Go to the documentation of this file.

```
00001 #ifndef ICM20948_REGS_H
00002 #define ICM20948_REGS_H
00003
00004 /**
00005  * 7Semi comment style
00006  * - Full ICM-20948 register map (Banks 0-3) + key bit fields
00007  * - Simple, portable #defines for firmware use
00008  * - Datasheet naming kept where practical; fields grouped by bank
00009  *
00010  * Notes
00011  * - WHO_AM_I expected value: 0xEA
00012  * - Select register bank via REG_BANK_SEL in any bank
00013  * - Use BANK(n) helper or write raw values 0x00/0x10/0x20/0x30
00014  */
00015
00016 /* -------- Common helpers -------- */
00017 #ifndef BIT
00018 #define BIT(n) (1u << (n))
00019 #endif
00020
00021 // #define BANK(n) ((uint8_t)((n) << 4))
00022 // #define BANK0 BANK(0)
00023 // #define BANK1 BANK(1)
00024 // #define BANK2 BANK(2)
00025 // #define BANK3 BANK(3)
00026
00027 /* ============================================================================ */
00028 /*                                  BANK 0                                      */
00029 /* ============================================================================ */
00030
00031 /** - Identity / power / interrupts / sensor data */
00032 #define WHO_AM_I 0x00 /* WHO_AM_I[7:0] */
00033 #define WHO_AM_I_VAL 0xEA
00034
00035 #define USER_CTRL 0x03 /* DMP_EN FIFO_EN I2C_MST_EN I2C_IF_DIS DMP_RST SRAM_RST I2C_MST_RST - */
00036 /* bits */
00037 #define USER_CTRL_DMP_EN BIT(7)
00038 #define USER_CTRL_FIFO_EN BIT(6)
00039 #define USER_CTRL_I2C_MST_EN BIT(5)
00040 #define USER_CTRL_I2C_IF_DIS BIT(4)
00041 #define USER_CTRL_DMP_RST BIT(3)
00042 #define USER_CTRL_SRAM_RST BIT(2)
00043 #define USER_CTRL_I2C_MST_RST BIT(1)
00044
00045 #define LP_CONFIG 0x05 /* I2C_MST_CYCLE ACCEL_CYCLE GYRO_CYCLE - */
00046 #define LP_I2C_MST_CYCLE BIT(6)
00047 #define LP_ACCEL_CYCLE BIT(5)
00048 #define LP_GYRO_CYCLE BIT(4)
00049
00050 #define PWR_MGMT_1 0x06 /* DEVICE_RESET SLEEP LP_EN - TEMP_DIS CLKSEL[2:0] */
00051 #define PWR_DEVICE_RESET BIT(7)
00052 #define PWR_SLEEP BIT(6)
00053 #define PWR_LP_EN BIT(5)
```

```
00054 #define PWR_TEMP_DIS BIT(3)
00055 #define PWR_CLKSEL_MASK 0x07
00056 #define PWR_CLKSEL_INT_20MHZ 0x01
00057 #define PWR_CLKSEL_AUTO 0x01 /* typical: auto selects best source */
00058
00059 #define PWR_MGMT_2 0x07 /* - DISABLE_ACCEL DISABLE_GYRO */
00060 #define PWR_DISABLE_ACCEL BIT(3)
00061 #define PWR_DISABLE_GYRO BIT(0)
00062
00063 //Interrupt Configs register
00064 #define INT_PIN_CFG 0x0F         /* INT1_ACTL INT1_OPEN INT1_LATCH_INT_EN INT_ANYRD_2CLEAR ACTL_FSYNC
      FSYNC_INT_MODE_EN BYPASS_EN - */
00065 #define INT1_ACTL BIT(7)         /* active low */
00066 #define INT1_OPEN BIT(6)         /* open-drain */
00067 #define INT1_LATCH_INT_EN BIT(5) /* latch until status read */
00068 #define INT_ANYRD_2CLEAR BIT(4)
00069 #define INT_ACTL_FSYNC BIT(3)
00070 #define FSYNC_INT_MODE_EN BIT(2)
00071 #define BYPASS_EN BIT(1) /* I2C bypass to aux devices */
00072
00073 #define INT_ENABLE 0x10 /* REG_WOF_EN - WOM_INT_EN PLL_RDY_EN DMP_INT1_EN I2C_MST_INT_EN */
00074 #define INT_REG_WOF_EN BIT(7)
00075 #define INT_WOM_INT_EN BIT(3)
00076 #define INT_PLL_RDY_EN BIT(2)
00077 #define INT_DMP_INT1_EN BIT(1)
00078 #define INT_I2C_MST_INT_EN BIT(0)
00079
00080 #define INT_ENABLE_1 0x11 /* RAW_DATA_0_RDY_EN */
00081 #define INT_RAW_DATA_0_RDY_EN BIT(0)
00082
00083 #define INT_ENABLE_2 0x12 /* FIFO_OVERFLOW_EN[4:0] */
00084 #define INT_ENABLE_3 0x13 /* FIFO_WM_EN[4:0] */
00085
00086 #define I2C_MST_STATUS 0x17 /* PASS_THROUGH, *_DONE, *_NACK, LOST_ARB */
00087 #define MST_PASS_THROUGH BIT(7)
00088 #define MST_SLV4_DONE BIT(6)
00089 #define MST_LOST_ARB BIT(5)
00090 #define MST_SLV4_NACK BIT(4)
00091 #define MST_SLV3_NACK BIT(3)
00092 #define MST_SLV2_NACK BIT(2)
00093 #define MST_SLV1_NACK BIT(1)
00094 #define MST_SLV0_NACK BIT(0)
00095
00096 #define INT_STATUS 0x19 /* WOM_INT PLL_RDY_INT DMP_INT1 I2C_MST_INT */
00097 #define STS_WOM_INT BIT(3)
00098 #define STS_PLL_RDY_INT BIT(2)
00099 #define STS_DMP_INT1 BIT(1)
00100 #define STS_I2C_MST_INT BIT(0)
00101
00102 #define INT_STATUS_1 0x1A /* RAW_DATA_0_RDY_INT */
00103 #define STS_RAW_DATA_0_RDY_INT BIT(0)
00104 #define INT_STATUS_2 0x1B /* FIFO_OVERFLOW_INT[4:0] */
00105 #define INT_STATUS_3 0x1C /* FIFO_WM_INT[4:0] */
00106
00107 #define DELAY_TIMEH 0x28
00108 #define DELAY_TIMEL 0x29
00109
00110 /* - Sensor outputs */
00111 #define ACCEL_XOUT_H 0x2D
00112 #define ACCEL_XOUT_L 0x2E
00113 #define ACCEL_YOUT_H 0x2F
00114 #define ACCEL_YOUT_L 0x30
00115 #define ACCEL_ZOUT_H 0x31
00116 #define ACCEL_ZOUT_L 0x32
00117 #define GYRO_XOUT_H 0x33
00118 #define GYRO_XOUT_L 0x34
00119 #define GYRO_YOUT_H 0x35
00120 #define GYRO_YOUT_L 0x36
00121 #define GYRO_ZOUT_H 0x37
00122 #define GYRO_ZOUT_L 0x38
00123 #define TEMP_OUT_H 0x39
00124 #define TEMP_OUT_L 0x3A
00125
00126 /* - External sensor shadow data (aux I2C) */
00127 #define EXT_SLV_SENS_DATA_00 0x3B
00128 #define EXT_SLV_SENS_DATA_23 0x52 /* contiguous range 0x3B..0x52 */
00129
00130 #define FIFO_EN_1 0x66 /* SLV3..SLV0 FIFO_EN */
00131 #define FIFO_EN_2 0x67 /* ACCEL GYRO_Z/Y/X TEMP FIFO_EN */
00132 #define FIFO_RST 0x68  /* FIFO_RESET[4:0] */
00133 #define FIFO_MODE 0x69 /* FIFO_MODE[4:0] */
00134 #define FIFO_COUNTH 0x70
00135 #define FIFO_COUNTL 0x71
00136 #define FIFO_R_W 0x72
00137 // #define DATA_RDY_STATUS 0x74 /* WOF_STATUS RAW_DATA_RDY[3:0] */
00138 #define FIFO_CFG 0x76
00139
```

```
00140 #define REG_BANK_SEL 0x7F /* USER_BANK[1:0] */
00141
00142 /* ============================================================================ */
00143 /*                                BANK 1                                        */
00144 /* ============================================================================ */
00145
00146 /** - Self-test / accel offsets / timebase */
00147 #define SELF_TEST_X_GYRO 0x02  /* XG_ST_DATA[7:0] */
00148 #define SELF_TEST_Y_GYRO 0x03  /* YG_ST_DATA[7:0] */
00149 #define SELF_TEST_Z_GYRO 0x04  /* ZG_ST_DATA[7:0] */
00150 #define SELF_TEST_X_ACCEL 0x0E /* XA_ST_DATA[7:0] */
00151 #define SELF_TEST_Y_ACCEL 0x0F /* YA_ST_DATA[7:0] */
00152 #define SELF_TEST_Z_ACCEL 0x10 /* ZA_ST_DATA[7:0] */
00153
00154 #define XA_OFFS_H 0x14 /* XA_OFFS[14:7] */
00155 #define XA_OFFS_L 0x15 /* XA_OFFS[6:0] */
00156 #define YA_OFFS_H 0x17 /* YA_OFFS[14:7] */
00157 #define YA_OFFS_L 0x18 /* YA_OFFS[6:0] */
00158 #define ZA_OFFS_H 0x1A /* ZA_OFFS[14:7] */
00159 #define ZA_OFFS_L 0x1B /* ZA_OFFS[6:0] */
00160
00161 #define TIMEBASE_CORRECTION_PLL 0x28 /* TBC_PLL[7:0] */
00162
00163 #define BANK1_REG_BANK_SEL 0x7F /* mirror of REG_BANK_SEL */
00164
00165 /* ============================================================================ */
00166 /*                                BANK 2                                        */
00167 /* ============================================================================ */
00168
00169 /** - Gyro/Accel configuration, offsets, FSYNC, temperature filter */
00170 #define GYRO_SMPLRT_DIV 0x00 /* GYRO_SMPLRT_DIV[7:0] */
00171
00172 #define GYRO_CONFIG_1 0x01  /* GYRO_DLPFCFG[2:0] GYRO_FS_SEL[1:0] GYRO_FCHOICE */
00173 #define GYRO_FCHOICE BIT(0) /* when 0: DLPF on, when 1: off (per datasheet) */
00174 #define GYRO_FS_SEL_SHIFT 1
00175 #define GYRO_FS_SEL_MASK (0x3u « GYRO_FS_SEL_SHIFT)
00176 #define GYRO_FS_250DPS (0u « GYRO_FS_SEL_SHIFT)
00177 #define GYRO_FS_500DPS (1u « GYRO_FS_SEL_SHIFT)
00178 #define GYRO_FS_1000DPS (2u « GYRO_FS_SEL_SHIFT)
00179 #define GYRO_FS_2000DPS (3u « GYRO_FS_SEL_SHIFT)
00180 #define GYRO_DLPFCFG_SHIFT 5
00181 #define GYRO_DLPFCFG_MASK (0x7u « GYRO_DLPFCFG_SHIFT)
00182
00183 #define GYRO_CONFIG_2 0x02 /* XGYRO_CTEN YGYRO_CTEN ZGYRO_CTEN GYRO_AVGCFG[2:0] */
00184 #define XGYRO_CTEN BIT(5)
00185 #define YGYRO_CTEN BIT(4)
00186 #define ZGYRO_CTEN BIT(3)
00187 #define GYRO_AVGCFG_SHIFT 0
00188 #define GYRO_AVGCFG_MASK (0x7u « GYRO_AVGCFG_SHIFT)
00189
00190 #define XG_OFFS_USRH 0x03
00191 #define XG_OFFS_USRL 0x04
00192 #define YG_OFFS_USRH 0x05
00193 #define YG_OFFS_USRL 0x06
00194 #define ZG_OFFS_USRH 0x07
00195 #define ZG_OFFS_USRL 0x08
00196
00197 #define ODR_ALIGN_EN 0x09 /* ODR_ALIGN_EN */
00198 #define ODR_ALIGN_EN_BIT BIT(0)
00199
00200 #define ACCEL_SMPLRT_DIV_1 0x10 /* ACCEL_SMPLRT_DIV[11:8] */
00201 #define ACCEL_SMPLRT_DIV_2 0x11 /* ACCEL_SMPLRT_DIV[7:0] */
00202
00203 #define ACCEL_INTEL_CTRL 0x12 /* ACCEL_INTEL_EN ACCEL_INTEL_MODE_INT */
00204 #define ACCEL_INTEL_EN BIT(7)
00205 #define ACCEL_INTEL_MODE_INT BIT(6)
00206
00207 #define ACCEL_WOM_THR 0x13 /* WOM_THRESHOLD[7:0] */
00208
00209 #define ACCEL_CONFIG 0x14 /* ACCEL_DLPFCFG[2:0] ACCEL_FS_SEL[1:0] ACCEL_FCHOICE */
00210 #define ACCEL_FCHOICE BIT(0)
00211 #define ACCEL_FS_SEL_SHIFT 1
00212 #define ACCEL_FS_SEL_MASK (0x3u « ACCEL_FS_SEL_SHIFT)
00213 #define ACCEL_FS_2G (0u « ACCEL_FS_SEL_SHIFT)
00214 #define ACCEL_FS_4G (1u « ACCEL_FS_SEL_SHIFT)
00215 #define ACCEL_FS_8G (2u « ACCEL_FS_SEL_SHIFT)
00216 #define ACCEL_FS_16G (3u « ACCEL_FS_SEL_SHIFT)
00217 #define ACCEL_DLPFCFG_SHIFT 5
00218 #define ACCEL_DLPFCFG_MASK (0x7u « ACCEL_DLPFCFG_SHIFT)
00219
00220 #define ACCEL_CONFIG_2 0x15 /* AX_ST_EN_REG AY_ST_EN_REG AZ_ST_EN_REG DEC3_CFG[1:0] */
00221 #define AX_ST_EN_REG BIT(7)
00222 #define AY_ST_EN_REG BIT(6)
00223 #define AZ_ST_EN_REG BIT(5)
00224 #define DEC3_CFG_SHIFT 0
00225 #define DEC3_CFG_MASK (0x3u « DEC3_CFG_SHIFT)
00226
```

```
00227 #define FSYNC_CONFIG 0x52 /* DELAY_TIME_EN WOF_DEGLITCH_EN WOF_EDGE_INT EXT_SYNC_SET[3:0] */
00228 #define DELAY_TIME_EN BIT(7)
00229 #define WOF_DEGLITCH_EN BIT(6)
00230 #define WOF_EDGE_INT BIT(5)
00231 #define EXT_SYNC_SET_MASK 0x0F
00232
00233 #define TEMP_CONFIG 0x53 /* TEMP_DLPFCFG[2:0] */
00234 #define TEMP_DLPFCFG_SHIFT 5
00235 #define TEMP_DLPFCFG_MASK (0x7u « TEMP_DLPFCFG_SHIFT)
00236
00237 #define MOD_CTRL_USR 0x54 /* REG_LP_DMP_EN */
00238 #define REG_LP_DMP_EN BIT(7)
00239
00240 #define BANK2_REG_BANK_SEL 0x7F /* mirror of REG_BANK_SEL */
00241
00242 /* ============================================================================ */
00243 /*                                BANK 3                                        */
00244 /* ============================================================================ */
00245
00246 /** - I2C master (aux) interface and slave windows */
00247 #define I2C_MST_ODR_CONFIG 0x00 /* I2C_MST_ODR_CONFIG[3:0] */
00248 #define MST_ODR_CFG_MASK 0x0F
00249
00250 #define I2C_MST_CTRL 0x01 /* MULT_MST_EN - I2C_MST_P_NSR I2C_MST_CLK[3:0] */
00251 #define MULT_MST_EN BIT(7)
00252 #define I2C_MST_P_NSR BIT(4)
00253 #define I2C_MST_CLK_MASK 0x0F
00254
00255 #define I2C_MST_DELAY_CTRL 0x02 /* DELAY_ES_SHADOW + I2C_SLVx_DELAY_EN bits */
00256 #define DELAY_ES_SHADOW BIT(7)
00257 #define I2C_SLV4_DELAY_EN BIT(4)
00258 #define I2C_SLV3_DELAY_EN BIT(3)
00259 #define I2C_SLV2_DELAY_EN BIT(2)
00260 #define I2C_SLV1_DELAY_EN BIT(1)
00261 #define I2C_SLV0_DELAY_EN BIT(0)
00262
00263 #define I2C_SLV0_ADDR 0x03 /* RNW + ID[6:0] */
00264 #define I2C_SLVx_RNW BIT(7)
00265 #define I2C_SLV0_REG 0x04
00266 #define I2C_SLV0_CTRL 0x05 /* EN BYTE_SW REG_DIS GRP LENG[3:0] */
00267 #define I2C_SLVx_EN BIT(7)
00268 #define I2C_SLVx_BYTE_SW BIT(6)
00269 #define I2C_SLVx_REG_DIS BIT(5)
00270 #define I2C_SLVx_GRP BIT(4)
00271 #define I2C_SLVx_LENG_MASK 0x0F
00272 #define I2C_SLV0_DO 0x06
00273
00274 #define I2C_SLV1_ADDR 0x07
00275 #define I2C_SLV1_REG 0x08
00276 #define I2C_SLV1_CTRL 0x09
00277 #define I2C_SLV1_DO 0x0A
00278
00279 #define I2C_SLV2_ADDR 0x0B
00280 #define I2C_SLV2_REG 0x0C
00281 #define I2C_SLV2_CTRL 0x0D
00282 #define I2C_SLV2_DO 0x0E
00283
00284 #define I2C_SLV3_ADDR 0x0F
00285 #define I2C_SLV3_REG 0x10
00286 #define I2C_SLV3_CTRL 0x11
00287 #define I2C_SLV3_DO 0x12
00288
00289 #define I2C_SLV4_ADDR 0x13
00290 #define I2C_SLV4_REG 0x14
00291 #define I2C_SLV4_CTRL 0x15 /* EN BYTE_SW REG_DIS DLY[4:0] */
00292 #define I2C_SLV4_DLY_MASK 0x1F
00293 #define I2C_SLV4_DO 0x16
00294 #define I2C_SLV4_DI 0x17
00295
00296 #define BANK3_REG_BANK_SEL 0x7F /* mirror of REG_BANK_SEL */
00297
00298 /* ============================================================================ */
00299 /*                          REG_BANK_SEL (all banks)                            */
00300 /* ============================================================================ */
00301
00302 #define REG_BANK_SEL_USER_BANK_SHIFT 4
00303 #define REG_BANK_SEL_USER_BANK_MASK (0x3u « REG_BANK_SEL_USER_BANK_SHIFT)
00304 #define USER_BANK_0 BANK0
00305 #define USER_BANK_1 BANK1
00306 #define USER_BANK_2 BANK2
00307 #define USER_BANK_3 BANK3
00308
00309 /* AK09916 magnetometer registers (connected internally) */
00310 #define AK09916_I2C_ADDR 0x0C
00311 #define AK_WIA2 0x01
00312 #define AK_ST1 0x10
00313 #define AK_HXL 0x11
```

```
00314 #define AK_ST2 0x18
00315 #define AK_CNTL2 0x31
00316 #define AK_CNTL3 0x32
00317 #define AK_WIA2_VAL 0x09 /* AK09916C WIA2 expected */
00318
00319 #define INTERNAL_20MHZ 0x00 /* 0: Internal 20 MHz RC */
00320 #define AUTO_SEL 0x01       /* 1-5: Auto/PLL preferred (use 1 as default) */
00321 #define CLK_STOP 0x07       /* 7: Stop clock / timing gen reset */
00322
00323
00324 /* ------------------ Accelerometer Range ------------------ */
00325 /** - ±2g / ±4g / ±8g / ±16g (maps to FS_SEL 0..3) */
00326 #define g2    0
00327 #define g4    1
00328 #define g8    2
00329 #define g16   3
00330
00331 /* ------------- Accelerometer Filter Path (FCHOICE) ------------- */
00332 /**
00333  * Path select for accel:
00334  * - ACCEL_FCHOICE_BYPASS : DLPF bypassed (very wide BW, fastest)
00335  * - ACCEL_FCHOICE_DLPF   : DLPF enabled (use ACCEL_DLPFCFG + SMPLRT_DIV)
00336  */
00337 #define ACCEL_FCHOICE_BYPASS   0
00338 #define ACCEL_FCHOICE_DLPF     1
00339
00340 /* ---------------- Accelerometer DLPF Config (CFG) ---------------- */
00341 /**
00342  * ACCEL_DLPFCFG (0..7) -- choose cutoff/noise BW set (when DLPF is ON).
00343  * Tip: start with ACCEL_DLPFCFG_3 for a good noise/latency tradeoff.
00344  */
00345 #define ACCEL_DLPFCFG_0        0
00346 #define ACCEL_DLPFCFG_1        1
00347 #define ACCEL_DLPFCFG_2        2
00348 #define ACCEL_DLPFCFG_3        3
00349 #define ACCEL_DLPFCFG_4        4
00350 #define ACCEL_DLPFCFG_5        5
00351 #define ACCEL_DLPFCFG_6        6
00352 #define ACCEL_DLPFCFG_7        7
00353
00354 #define ACCEL_DEC3_AVG_4       0u  /* averages 1 or 4 (depends on FCHOICE) */
00355 #define ACCEL_DEC3_AVG_8       1u
00356 #define ACCEL_DEC3_AVG_16      2u
00357 #define ACCEL_DEC3_AVG_32      3u
00358
00359 /* ---- Accel DLPF ODR helper (Hz) : base 1125 Hz, DIV 0..4095 ---- */
00360 #define ACCEL_SMPLRT_DIV_MIN     0u
00361 #define ACCEL_SMPLRT_DIV_MAX     4095u
00362 #define ACCEL_DLPF_BASE_HZ       1125.0f
00363 #define ACCEL_DLPF_ODR_HZ(div)   (ACCEL_DLPF_BASE_HZ / (1.0f + (float)(div)))
00364
00365 /* ---- Accel BYPASS quick reference (FCHOICE=0) ----
00366  * 3dB ≈ 1209 Hz, NBW ≈ 1248 Hz, output rate ≈ 4500 Hz
00367  * (No divider/ODR control in bypass path.)
00368  */
00369 #define ACCEL_BYPASS_3DB_BW_HZ   1209.0f
00370 #define ACCEL_BYPASS_NBW_HZ      1248.0f
00371 #define ACCEL_BYPASS_RATE_HZ     4500.0f
00372
00373 /* (Optional) Accel DLPF nominal bandwidth references (FCHOICE=1)
00374  * Keep here for quick docs; exact values depend on datasheet rev.
00375  */
00376 #define ACCEL_DLPF0_3DB_BW_HZ    246.0f
00377 #define ACCEL_DLPF0_NBW_HZ       265.0f
00378 #define ACCEL_DLPF1_3DB_BW_HZ    246.0f
00379 #define ACCEL_DLPF1_NBW_HZ       265.0f
00380 #define ACCEL_DLPF2_3DB_BW_HZ    111.4f
00381 #define ACCEL_DLPF2_NBW_HZ       136.0f
00382 #define ACCEL_DLPF3_3DB_BW_HZ    50.4f
00383 #define ACCEL_DLPF3_NBW_HZ       68.8f
00384 #define ACCEL_DLPF4_3DB_BW_HZ    23.9f
00385 #define ACCEL_DLPF4_NBW_HZ       34.4f
00386 #define ACCEL_DLPF5_3DB_BW_HZ    11.5f
00387 #define ACCEL_DLPF5_NBW_HZ       17.0f
00388 #define ACCEL_DLPF6_3DB_BW_HZ    5.7f
00389 #define ACCEL_DLPF6_NBW_HZ       8.3f
00390 #define ACCEL_DLPF7_3DB_BW_HZ    473.0f
00391 #define ACCEL_DLPF7_NBW_HZ       499.0f
00392
00393
00394 /* -------------------- Gyroscope Range -------------------- */
00395 /** - ±250/±500/±1000/±2000 dps (maps to FS_SEL 0..3) */
00396 #define dps250   0
00397 #define dps500   1
00398 #define dps1000  2
00399 #define dps2000  3
00400
```

```
00401 /* --------------- Gyroscope Filter Path (FCHOICE) --------------- */
00402 /**
00403  * Path select for gyro:
00404  * - GYRO_FCHOICE_BYPASS : DLPF bypassed (widest BW, fastest)
00405  * - GYRO_FCHOICE_DLPF   : DLPF enabled (use GYRO_DLPFCFG + SMPLRT_DIV)
00406  */
00407 #define GYRO_FCHOICE_BYPASS   0
00408 #define GYRO_FCHOICE_DLPF     1
00409
00410 /* ----------------- Gyroscope DLPF Config (CFG) ----------------- */
00411 /**
00412  * GYRO_DLPFCFG (0..7) -- choose cutoff/noise BW set (when DLPF is ON).
00413  * Tip: start with GYRO_DLPFCFG_3 or _2 for balanced noise/latency.
00414  */
00415 #define GYRO_DLPFCFG_0        0
00416 #define GYRO_DLPFCFG_1        1
00417 #define GYRO_DLPFCFG_2        2
00418 #define GYRO_DLPFCFG_3        3
00419 #define GYRO_DLPFCFG_4        4
00420 #define GYRO_DLPFCFG_5        5
00421 #define GYRO_DLPFCFG_6        6
00422 #define GYRO_DLPFCFG_7        7
00423
00424 /* ---- Gyro DLPF ODR helper (Hz) : base 1100 Hz, guard ≥~4.3 Hz ---- */
00425 #define GYRO_SMPLRT_MIN_HZ      4.3f
00426 #define GYRO_DLPF_BASE_HZ       1100.0f
00427 #define GYRO_DLPF_ODR_HZ(rate_request)  (rate_request)  /* symbolic; your driver computes DIV =
      round(1100/rate)-1 */
00428
00429 /* (Optional) Gyro BW "buckets" to tag configs in UI/logs (symbolic)
00430  * Use these if you want a quick human-readable label for NBW tiers.
00431  * Exact Hz depend on datasheet table; map per your implementation.
00432  */
00433 #define GYRO_BW_ULTRA_WIDE   0   /* bypass path */
00434 #define GYRO_BW_WIDE         1   /* e.g., CFG 0/1 */
00435 #define GYRO_BW_MEDIUM       2   /* e.g., CFG 2/3 */
00436 #define GYRO_BW_NARROW       3   /* e.g., CFG 4/5/6/7 */
00437
00438 /* ---------------- Convenience: presets (symbolic) --------------- */
00439 /** - Friendly ODR presets you can map to dividers */
00440 #define Hz2    0
00441 #define Hz6    1
00442 #define Hz8    2
00443 #define Hz10   3
00444 #define Hz15   4
00445 #define Hz20   5
00446 #define Hz25   6
00447 #define Hz30   7
00448
00449 /* ------------------ Power / Fusion Modes ----------------- */
00450 /**
00451  * Generic power or fusion quality modes (BNO-style).
00452  * Map to sensor-specific sequences inside your driver.
00453  */
00454 #define LowPower          0
00455 #define Regular           1
00456 #define EnhancedRegular   2
00457 #define HighAccuracy      3
00458
00459 #endif /* ICM20948_REGS_H */
```

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